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GROUP SPECIFICATION

Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Or-Vnfm reference point - Interface and Information Model Specification

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Network Functions Virtualisation (NFV).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies the interfaces supported over the Or-Vnfm reference point of the Network Functions Virtualisation Management and Orchestration (NFV-MANO) architectural framework ETSI GS NFV 006 [i.12] as well as the information elements exchanged over those interfaces.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI Docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI GS NFV-IFA 006: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Vi-Vnfm reference point - Interface and Information Model Specification".

[2] ETSI GS NFV-IFA 010: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Functional requirements specification".

[3] ETSI GS NFV-IFA 011: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; VNF Descriptor and Packaging Specification".

[4] Recommendation ITU-T X.733: "Information technology - Open Systems Interconnection - Systems Management: Alarm reporting function".

[5] ETSI GS NFV-IFA 027: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Performance Measurements Specification".

[6] ETSI GS NFV-IFA 048: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Policy Information Model Specification".

[7] ETSI GS NFV-IFA 045: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Fault and alarms modelling specification".

[8] ETSI GS NFV-IFA 013: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Os-Ma-nfvo reference point - Interface and Information Model Specification".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI GS NFV 002: "Network Functions Virtualisation (NFV); Architectural Framework".

[i.2] ETSI GS NFV 003: "Network Functions Virtualisation (NFV); Terminology for Main Concepts in NFV".

[i.3] ISO/IEC 9646-7: "Information technology - Open Systems Interconnection -- Conformance testing methodology and framework -- Part 7: Implementation Conformance Statements".

[i.4] ETSI GS NFV-IFA 005: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Or-Vi reference point - Interface and Information Model Specification".

[i.5] ETSI GS NFV-IFA 008: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Ve-Vnfm reference point - Interface and Information Model Specification".

[i.6] ETSI GS NFV-IFA 009: "Network Functions Virtualisation (NFV); Management and Orchestration; Report on Architectural Options".

[i.7] Void.

[i.8] Void.

[i.9] Void.

[i.10] ETSI GS NFV-REL 006: "Network Functions Virtualisation (NFV) Release 3; Reliability; Maintaining Service Availability and Continuity Upon Software Modification".

[i.11] ETSI GS NFV-IFA 032: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Interface and Information Model Specification for Multi-Site Connectivity Services".

[i.12] ETSI GS NFV 006: "Network Functions Virtualisation (NFV) Release 4; Management and Orchestration; Architectural Framework Specification".

[i.13] ETSI GS NFV-IFA 040: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Requirements for service interfaces and object model for OS container management and orchestration specification".

[i.14] ETSI GS NFV-IFA 031: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Requirements and interfaces specification for management of NFV-MANO".

[i.15] ETSI GS NFV-IFA 026: "Network Functions Virtualisation (NFV) Release 5; Management and Orchestration; Security Architecture enhancements for NFV Specification".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GS NFV 003 [i.2] and the following apply:

compute MCIO: MCIO which declarative descriptor specifies compute infrastructure resource requests

deployable module: set of optional VDUs, specified in the VNFD for a particular VNF deployment flavour, which are used to create VNFC instances only if selected by the consumer in a VNF lifecycle management operation

NOTE: This definition is from ETSI GS NFV-IFA 010 [2].

network MCIO: MCIO which declarative descriptor specifies network infrastructure resource requests

storage MCIO: MCIO which declarative descriptor specifies storage infrastructure resource requests

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI GS NFV 003 [i.2] and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI GS NFV 003 [i.2].

CMF	Certificate Management Function
CP	Connection Point
CPD	Connection Point Descriptor
CSR	Certificate Signing Request
DF	Deployment Flavour
FB	Functional Block
NFVI-Node	Network Functions Virtualisation Infrastructure Node
NFVI-PoP	Network Functions Virtualisation Infrastructure Point of Presence
VDU	VNF Deployment Unit
VL	Virtual Link
VLD	Virtual Link Descriptor

4 Overview of interfaces and information elements associated to the Or-Vnfm reference point

4.1 Introduction

This clause provides an overview of interfaces and information elements associated to the Or-Vnfm reference point.

The Or-Vnfm reference point is used for exchanges between Network Functions Virtualisation Orchestrator (NFVO) and Virtualised Network Function Manager (VNFM), and supports the following interfaces:

- Virtualised Network Function (VNF) Package Management (produced by NFVO, consumed by VNFM).
- VNF Lifecycle Operation Granting (produced by NFVO, consumed by VNFM).
- Virtualised Resources Management (produced by NFVO, consumed by VNFM).
- Virtualised Resources Quota Available Notification (produced by NFVO, consumed by VNFM).
- VNF Lifecycle Management (produced by VNFM, consumed by NFVO).
- VNF Performance Management (produced by VNFM, consumed by NFVO).
- VNF Fault Management (produced by VNFM, consumed by NFVO).
- VNF Indicator (produced by VNFM, consumed by NFVO).
- Policy Management (produced by VNFM, consumed by NFVO).
- VNF Snapshot Package Management (produced by NFVO, consumed by VNFM).

The information elements exchanged by the interfaces above are also part of the present document.

4.2 Relation to other NFV Group Specifications

The present document is referencing information from the following NFV Group Specifications:

- Report on Architectural Options ETSI GS NFV-IFA 009 [i.6]:
  - This report describes architectural options that can influence the way some of the Or-Vnfm interfaces are used or might even suggest the need for extension.

- Functional Requirements Specification ETSI GS NFV-IFA 010 [2]:
  - Interfaces associated with the Or-Vnfm reference point are based on the functional requirements specified in ETSI GS NFV-IFA 010 [2] for the NFVO and VNFM Functional Blocks (FBs).
- Vi-Vnfm reference point - Interface and Information Model Specification ETSI GS NFV-IFA 006 [1]:
  - The interfaces related to Virtualised Resources Management defined in ETSI GS NFV-IFA 006 [1] are also used on the Or-Vnfm reference point.
- Ve-Vnfm reference point - Interface and Information Model Specification ETSI GS NFV-IFA 008 [i.5]:
  - VNF Fault Management, VNF Performance Management and VNF Indicator interfaces defined in ETSI GS NFV-IFA 008 [i.5] are also used on the Or-Vnfm reference point.
- VNF Packaging Specification ETSI GS NFV-IFA 011 [3]:
  - The specification of the Virtualised Network Function Descriptor (VNFD) in ETSI GS NFV-IFA 011 [3] defines information elements that are also relevant in the present document.
- Os-Ma-nfvo reference point - Interface and Information Model Specification ETSI GS NFV-IFA 013 [8]:
  - The VNF Package Management interface defined in ETSI GS NFV-IFA 013 [8] is also used on the Or-Vnfm reference point.

Information about the reference points in the ETSI NFV architecture can be found in ETSI GS NFV 002 [i.1].

4.3 Conventions

The following notations, defined in ISO/IEC 9646-7 [i.3], are used for the qualifier column of interface information elements:

- M mandatory - the capability is required to be supported;
- O optional - the capability may be supported or not;
- CM conditional mandatory - the capability is required to be supported and is conditional on the support of some condition. This condition shall be specified in the Description column;
- CO conditional optional - the capability may be supported or not and is conditional on the support of some condition. This condition shall be specified in the Description column.

The following notation is used for parameters that represent identifiers, and for attributes that represent identifiers in information elements and notifications:

- If parameters are referring to an identifier of an actual object, their type is "Identifier".
- If an object (information element or notification) contains an attribute that identifies the object, the type of that attribute is "Identifier" and the description states that the attribute is the identifier of that particular notification or information element.

EXAMPLE 1: Identifier "resourceId" of the "NetworkSubnet information element" has type "Identifier" and description "Identifier of this NetworkSubnet information element".

- If an object (information element or notification) contains an attribute that references another object or objects defined in an ETSI NFV GS, the type of the attribute is "Identifier", followed by the list of objects it references.

EXAMPLE 2: "Identifier (Reference to Vnfc)" or "Identifier (Reference to Vnfc, VirtualLink or VirtualStorage)".

If the type of a parameter or attribute has been marked as "Not specified" in the "Content" column, this means that its specification is part of the protocol design/data model design.

5 Reference point and interface requirements

5.1 Introduction

This clause defines or references requirements applicable to interfaces in the specific context of the Or-Vnfm reference point.

5.2 Or-Vnfm reference point requirements

Table 5.2-1 specifies requirements applicable to the Or-Vnfm reference point.

Table 5.2-1: Or-Vnfm reference point requirements

Numbering	Requirement
Or-Vnfm.001	The Or-Vnfm reference point shall support the VNF Package Management interface produced by the NFVO.
Or-Vnfm.002	The Or-Vnfm reference point shall support the VNF Lifecycle Operation Granting interface produced by the NFVO.
Or-Vnfm.003	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Information Management interfaces produced by the NFVO.
Or-Vnfm.004	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Management interfaces produced by the NFVO.
Or-Vnfm.005	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Change Notification interfaces produced by the NFVO.
Or-Vnfm.006	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Reservation interfaces produced by the NFVO.
Or-Vnfm.007	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Reservation Change Notification interface produced by the NFVO.
Or-Vnfm.008	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Performance Management interface produced by the NFVO.
Or-Vnfm.009	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Fault Management interface produced by the NFVO.
Or-Vnfm.010	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Quota Management interfaces produced by the NFVO.
Or-Vnfm.011	When VNF-related resource management in indirect mode is applicable, the Or-Vnfm reference point shall support the Virtualised Resources Quota Change Notification interface produced by the NFVO.
Or-Vnfm.012	The Or-Vnfm reference point shall support the VNF Lifecycle Management interface produced by the VNFM.
Or-Vnfm.013	Void.
Or-Vnfm.014	The Or-Vnfm reference point shall support the VNF Performance Management interface produced by the VNFM.
Or-Vnfm.015	The Or-Vnfm reference point shall support the VNF Fault Management interface produced by the VNFM.
Or-Vnfm.016	Void.
Or-Vnfm.017	The Or-Vnfm reference point shall support the VNF Indicator interface produced by the VNFM.
Or-Vnfm.018	The Or-Vnfm reference point should support the Virtualised Resources Quota Available Notification interface produced by the NFVO.
Or-Vnfm.019	The Or-Vnfm reference point shall support the Policy Management interface produced by the VNFM.
Or-Vnfm.020	The Or-Vnfm reference point shall support the VNF Snapshot Package Management interface produced by the NFVO.

5.3 Interface requirements

5.3.1 VNF Package Management interface requirements

Table 5.3.1-1 specifies requirements applicable to the VNF Package Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.1-1: VNF Package Management interface requirements

Numbering	Requirement
Or-Vnfm.VnfPkgm.001	The VNF Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support querying VNF Package information (see note).
Or-Vnfm.VnfPkgm.002	The VNF Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support providing notifications as a result of changes on VNF Package states, and managing subscriptions to such notifications.
Or-Vnfm.VnfPkgm.003	The VNF Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support providing notifications about the on-boarding of VNF Packages, and managing subscriptions to such notifications.
Or-Vnfm.VnfPkgm.004	The VNF Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support fetching a VNF Package, or selected artifacts contained in a VNF Package.
NOTE: VNF Package information can include information such as release date, vendor info, manifest, VNFD, SW image meta-data, files contained in the VNF Package, etc.	

5.3.2 VNF Lifecycle Operation Granting interface requirements

Table 5.3.2-1 specifies requirements applicable to the VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.2-1: VNF Lifecycle Operation Granting interface requirements

Numbering	Requirement
Or-Vnfm.VnfLcog.001	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall support granting lifecycle operations.
Or-Vnfm.VnfLcog.002	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall support indicating the type of the lifecycle event for which a granting is being requested for a VNF instance, together with an identifier of the lifecycle operation occurrence.
Or-Vnfm.VnfLcog.003	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM to indicate the virtualised resources impacted by the VNF lifecycle operation (e.g. allocated or released).
Or-Vnfm.VnfLcog.004	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM obtaining information about the identification and configuration information to access the Virtualised Infrastructure Manager (VIM) or the Container Infrastructure Service Management (CISM).
Or-Vnfm.VnfLcog.005	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM obtaining, if a reservation is applicable, resource reservation identification information applicable for consuming virtualised resources as part of the lifecycle operation.
Or-Vnfm.VnfLcog.006	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM to provide information to identify the VNF Instance and VNFD for the intended lifecycle operation.
Or-Vnfm.VnfLcog.007	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM obtaining, when the VDUs of the VNF are realized by a set of OS containers, namespace identification information applicable for allocating virtualised resources as part of the lifecycle operation.
Or-Vnfm.VnfLcog.008	The VNF Lifecycle Operation Granting interface produced by the NFVO on the Or-Vnfm reference point shall enable the VNFM to indicate the PaaS Services impacted by the VNF lifecycle operation (e.g. that are expected to be allocated, or can be released).

5.3.3 Virtualised Resources Management interfaces requirements

5.3.3.1 Virtualised Resources Information Management interfaces requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Information Management interfaces as produced by the VIM on the Vi-Vnfm reference point are produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.1-1 specifies requirements applicable to the Virtualised Resources Information Management interfaces produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.1-1: Virtualised Resources Information Management interfaces requirements

Numbering	Requirement
Or-Vnfm.Vrim.01	The Virtualised Resources Information Management interfaces produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to invoke the virtualised resources information management operations towards the appropriate VIM (see notes 1 and 2).
NOTE 1: The Virtualised Resources Information Management interface requirements defined in clause 5.3.2 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.2 Virtualised Resources Management interfaces requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Management interfaces as produced by the VIM on the Vi-Vnfm reference point are produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.2-1 specifies requirements applicable to the Virtualised Resources Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.2-1: Virtualised Resources Management interfaces requirements

Numbering	Requirement
Or-Vnfm.Vrm.01	The Virtualised Resources Management interfaces produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to invoke the virtualised resources management operations towards the appropriate VIM (see notes 1 and 2).
NOTE 1: The Virtualised Resources Management interfaces requirements defined in clause 5.3.3 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.3 Virtualised Resources Reservation Management interface requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Reservation Management interface as produced by the VIM on the Vi-Vnfm reference point is produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.3-1 specifies requirements applicable to the Virtualised Resources Reservation Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.3-1: Virtualised Resources Reservation Management interface requirements

Numbering	Requirement
Or-Vnfm.Vrrm.01	The Virtualised Resources Reservation Management interface produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to invoke the virtualised resources reservation management operations towards the appropriate VIM (see notes 1 and 2).
NOTE 1: The Virtualised Resources Reservation Management interfaces requirements defined in clause 5.3.4 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.4 Virtualised Resources Reservation Change Notification interface requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Reservation Change Notification interface as produced by the VIM on the Vi-Vnfm reference point is produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.4-1 specifies requirements applicable to the Virtualised Resources Reservation Change Notification interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.4-1: Virtualised Resources Reservation Change Notification interface requirements

Numbering	Requirement
Or-Vnfm.Vrrcn.01	The Virtualised Resources Reservation Change Notification interface produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to identify the original provider of notifications, and to allow the VNFM to uniquely determine the virtualised resource reservation(s) to which a change notification applies (see notes 1 and 2).
NOTE 1: The Virtualised Resources Reservation Change Notification interface requirements defined in clause 5.3.6 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.5 Virtualised Resources Change Notification interfaces requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Change Notification interface as produced by the VIM on the Vi-Vnfm reference point are produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.5-1 specifies requirements applicable to the Virtualised Resources Change Notification interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.5-1: Virtualised Resources Change Notification interface requirements

Numbering	Requirement
Or-Vnfm.Vrcn.01	The Virtualised Resources Change Notification interfaces produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to identify the original provider of notifications, and to allow the VNFM to uniquely determine the virtualised resource(s) to which a change notification applies (see notes 1 and 2).
NOTE 1: The Virtualised Resources Change Notification interface requirements defined in clause 5.3.5 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.6 Virtualised Resources Performance Management interface requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Performance Management interface as produced by the VIM on the Vi-Vnfm reference point is produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.6-1 specifies requirements applicable to the Virtualised Resources Performance Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.6-1: Virtualised Resources Performance Management interface requirements

Numbering	Requirement
Or-Vnfm.Vrpm.01	The Virtualised Resources Performance Management interface produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to identify the original provider of PM information, and to allow the VNFM to uniquely determine the virtualised resource(s) to which such PM information applies (see notes 1 and 2).
NOTE 1: The Virtualised Resources Performance Management interface requirements defined in clause 5.3.8 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.7 Virtualised Resources Fault Management interface requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Fault Management interface as produced by the VIM on the Vi-Vnfm reference point is produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.7-1 specifies requirements applicable to the Virtualised Resources Fault Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.7-1: Virtualised Resources Fault Management interface requirements

Numbering	Requirement
Or-Vnfm.Vrfm.01	The Virtualised Resources Fault Management interface produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to identify the original provider of alarms, and to allow the VNFM to uniquely determine the virtualised resource(s) to which an alarm applies (see notes 1 and 2).
NOTE 1: The Virtualised Resources Fault Management interface requirements defined in clause 5.3.9 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources.	

5.3.3.8 Virtualised Resources Quota Management interfaces requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Quota Management interfaces as produced by the VIM on the Vi-Vnfm reference point are produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.8-1 specifies requirements applicable to the Virtualised Resources Quota Management interfaces produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.8-1: Virtualised Resources Quota Management interface requirements

Numbering	Requirement
Or-Vnfm.Vrqm.01	The Virtualised Resources Quota Management interfaces produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to invoke the virtualised resources quota management operations towards the appropriate VIM (see notes 1 and 2).
NOTE 1: The Virtualised Resources Quota Management interfaces requirements defined in clause 5.3.7 in ETSI GS NFV-IFA 006 [1] are applicable in the present clause too, in addition to the requirement(s) above. NOTE 2: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources quota.	

5.3.3.9 Virtualised Resources Quota Change Notification interface requirements

When VNF-related resource management in indirect mode is applicable, the Virtualised Resources Quota Change Notification interface as produced by the VIM on the Vi-Vnfm reference point is produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.9-1 specifies requirements applicable to the Virtualised Resources Quota Change Notification interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.9-1: Virtualised Resources Quota Change Notification interface requirements

Numbering	Requirement
Or-Vnfm.Vrcn.01	The Virtualised Resources Quota Change Notification interface produced by the NFVO on the Or-Vnfm reference point shall support notification of changes related to virtualised resource quotas.
Or-Vnfm.Vrcn.02	The Virtualised Resources Quota Change Notification interface produced by the NFVO on the Or-Vnfm reference point shall support the NFVO receiving indication information to enable the NFVO to identify the original provider of notifications, and to allow the VNFM to uniquely determine the virtualised resources quota to which a change notification applies (see note).
NOTE: The indication information is used by the NFVO to determine the entity responsible for the management of the virtualised resources quota.	

5.3.3.10 Virtualised Resources Quota Available Notification interface requirements

Table 5.3.3.10-1 specifies requirements applicable to the Virtualised Resources Quota Available Notification interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.3.10-1: Virtualised Resources Quota Available Notification interface requirements

Numbering	Requirement
Or-Vnfm.Vrqan.01	The Virtualised Resources Quota Available Notification interface produced by the NFVO on the Or-Vnfm reference point should support the capability to notify the availability of virtualised resource quota(s) applicable to this VNFM or the VNF(s) which the VNFM manages and to manage subscriptions to notifications about the availability of such quota.

5.3.4 VNF Lifecycle Management interface requirements

Table 5.3.4-1 specifies requirements applicable to the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point.

Table 5.3.4-1: VNF Lifecycle Management interface requirements

Numbering	Requirement
Or-Vnfm.VnfLcm.001	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support instantiating a VNF.
Or-Vnfm.VnfLcm.002	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support terminating a VNF instance.
Or-Vnfm.VnfLcm.003	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support scaling a VNF instance.
Or-Vnfm.VnfLcm.004	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support querying information about a VNF instance.
Or-Vnfm.VnfLcm.005	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support requesting VNF healing.
Or-Vnfm.VnfLcm.006	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support requesting to change the state of a VNF instance (see note 1).
Or-Vnfm.VnfLcm.007	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support querying the status of a VNF lifecycle management operation.
Or-Vnfm.VnfLcm.008	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support changing the Deployment Flavour (DF) of a VNF instance.
Or-Vnfm.VnfLcm.009	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support modifying information about a VNF instance (see note 2).
Or-Vnfm.VnfLcm.010	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support creating a VNF instance identifier and the associated instance of a VNF information element.
Or-Vnfm.VnfLcm.011	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support deleting a VNF instance identifier and the associated instance of a VNF information element.
Or-Vnfm.VnfLcm.012	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the NFVO notifications about changes of a VNF instance that are related to VNF lifecycle management operation occurrences, further referred to as VNF lifecycle management operation occurrence notifications.

Numbering	Requirement
Or-Vnfm.VnfLcm.013	VNF lifecycle management operation occurrence notifications provided on the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall contain information about the type of VNF lifecycle management operation, the identification of the VNF instance, and the identification of the lifecycle management operation occurrence.
Or-Vnfm.VnfLcm.014	VNF lifecycle management operation occurrence notifications provided on the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall contain information about the addition/deletion of VNFCs, and about the changes on virtualised resources associated to Virtualised Network Function Component(s) (VNFC(s)) as result of the VNF lifecycle management operation occurrence.
Or-Vnfm.VnfLcm.015	VNF lifecycle management operation occurrence notifications provided on the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall contain information about the virtual networks and Connection Points (CPs) that are added/deleted as part of the VNF lifecycle management operation occurrence (see note 3).
Or-Vnfm.VnfLcm.016	VNF lifecycle management operation occurrence notifications provided on the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support indicating the start of the lifecycle management operation occurrence the end and the results of the lifecycle management operation occurrence including any error produced from the lifecycle management operation occurrence.
Or-Vnfm.VnfLcm.017	VNF lifecycle management operation occurrence notifications provided on the VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support indicating updates to the VNF instance information including configurable properties.
Or-Vnfm.VnfLcm.018	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the NFVO notifications about creation and deletion of a VNF identifier and the associated instance of a VNF information element, further referred to as VNF identifier creation/deletion notifications.
Or-Vnfm.VnfLcm.019	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support managing subscriptions to VNF lifecycle management operation occurrence notifications and to VNF identifier creation/deletion notifications.
Or-Vnfm.VnfLcm.020	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the VNFM configuration parameters for a VNF instance. See note 4.
Or-Vnfm.VnfLcm.021	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support requesting to change the external connectivity of a VNF instance.
Or-Vnfm.VnfLcm.022	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support the capability to invoke VNF error handling operations after the VNF life cycle operation occurrence fails. See notes 5 and 6.
Or-Vnfm.VnfLcm.023	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support creating VNF Snapshots.
Or-Vnfm.VnfLcm.024	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support reverting to VNF Snapshots.
Or-Vnfm.VnfLcm.025	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support querying information about available VNF Snapshots (see note 7).
Or-Vnfm.VnfLcm.026	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support deleting information associated to VNF Snapshots.
Or-Vnfm.VnfLcm.027	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support changing the current VNF package.
Or-Vnfm.vnfLcm.028	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support fetching the VNF state snapshot associated to a VNF snapshot.
Or-Vnfm.vnfLcm.029	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing certificate configuration data and modifying information and/or the configuration parameters of certificate management as part of the creating a VNF instance identifier and modifying information and/or the configuration parameters.
Or-Vnfm.vnfLcm.030	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing configuration related to the security group rules associated to the external CPs of a VNF during VNF LCM operations where VNF external CP configuration can be provided.
Or-Vnfm.vnfLcm.031	The VNF Lifecycle Management interface produced by the VNFM on the Or-Vnfm reference point shall support requesting to change the power state of a VNF instance.

Numbering	Requirement
NOTE 1:	Changing the state of a VNF instance refers to starting or stopping a VNF instance. These operations are complementary to instantiating or terminating a VNF.
NOTE 2:	The requirement refers to the information that is writable.
NOTE 3:	This provides information about virtual networks and connections points that are internal to the VNF and whose creation was triggered by the VNFM.
NOTE 4:	Configuration parameters referred to in this clause are declared in the VNFD. They include: those that are set prior to instantiation and that cannot be modified if the VNF is instantiated, those that are set prior to instantiation (are part of initial configuration) and that can be modified later, and those that can be set only after instantiation.
NOTE 5:	It is up to the protocol design stage to design the detailed error handling operations.
NOTE 6:	It depends on the VNF capabilities whether and how the operations are supported by a particular VNF.
NOTE 7:	VNF Snapshot information can include information including creation date, configuration data of included VNF Snapshots, and identifiers of snapshotted VNF instances.

5.3.5 Void

5.3.6 VNF Performance Management interface requirements

Table 5.3.6-1 specifies requirements applicable to the VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point.

Table 5.3.6-1: VNF Performance Management interface requirements

Numbering	Requirement
Or-Vnfm.VnfPm.001	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the NFVO to control the collection and reporting of VNF performance information, resulting from virtualised resources performance information, on the VNF(s) it manages (see note 1).
Or-Vnfm.VnfPm.002	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the capability to notify the availability of VNF performance information.
Or-Vnfm.VnfPm.003	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the NFVO to create a PM job specifying the VNF performance information that the NFVO requires from the VNFM.
Or-Vnfm.VnfPm.004	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the NFVO to delete one or more PM job(s).
Or-Vnfm.VnfPm.005	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall enable the NFVO to receive notifications of data availability for a PM job, and to manage subscriptions to such notifications.
Or-Vnfm.VnfPm.006	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the NFVO to query the details of one or more PM job(s).
Or-Vnfm.VnfPm.007	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the NFVO to manage the thresholds on specified VNF performance information and VNF(s) (see note 2).
Or-Vnfm.VnfPm.008	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall support the capability to notify about a threshold defined for a specified metric of a VNF being crossed.
Or-Vnfm.VnfPm.009	The VNF Performance Management interface produced by the VNFM on the Or-Vnfm reference point shall enable the NFVO to receive notifications related to threshold crossing, and to manage subscriptions to such notifications.
NOTE 1:	Performance information on a given VNF results from collected performance information of the virtualised resources that are mapped to this VNF instance.
NOTE 2:	Management of thresholds include creation, deletion and query the thresholds on specified VNF performance information and VNF(s).

5.3.7 VNF Fault Management interface requirements

Table 5.3.7-1 specifies requirements applicable to the VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point.

Table 5.3.7-1: VNF Fault Management interface requirements

Numbering	Requirement
Or-Vnfm.VnfFm.001	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support collecting VNF fault information (see note).
Or-Vnfm.VnfFm.002	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing alarm notifications related to faults on VNF instances.
Or-Vnfm.VnfFm.003	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing notification when there is a change in alarm information on VNF instances.
Or-Vnfm.VnfFm.004	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support the sending of notification to the NFVO when an alarm on a VNF instance has been created.
Or-Vnfm.VnfFm.005	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support the sending of notification to the NFVO when an alarm on a VNF instance has been cleared.
Or-Vnfm.VnfFm.006	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall allow unambiguous identification of the alarm on a VNF instance sent to the NFVO.
Or-Vnfm.VnfFm.007	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall allow unambiguous identification of the VNF instance causing the alarm.
Or-Vnfm.VnfFm.008	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall allow unambiguous identification of the alarm cause.
Or-Vnfm.VnfFm.009	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the NFVO notifications about alarms on a VNF instance as a consequence of state changes in the virtualised resources used by the VNF.
Or-Vnfm.VnfFm.010	Notifications related to the alarms associated with state changes of virtualised resources of a VNF instance provided on the VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall contain information necessary to identify the VNF and the VNFC(s), the origin (VIM and virtualised resource(s)) of the virtualised resource change notification(s), the type of alarm, and information about the cause of the alarm.
Or-Vnfm.VnfFm.011	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall enable the NFVO to manage subscriptions to notifications related to alarms.
Or-Vnfm.VnfFm.012	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support alarm acknowledgement.
Or-Vnfm.VnfFm.013	The VNF Fault Management interface produced by the VNFM on the Or-Vnfm reference point shall support the sending of notification(s) to the NFVO when the alarm list has been rebuilt.
NOTE: Fault information on a given VNF instance can include the information related to the alarm (e.g. alarm created, alarm cleared, etc.), alarm causes and identification of this VNF instance and fault information concerning the virtualised resources supporting the constituent VNF instance.	

5.3.8 Void

5.3.9 VNF Indicator interface requirements

Table 5.3.9-1 specifies requirements applicable to the VNF Indicator interface produced by the VNFM on the Or-Vnfm reference point.

Table 5.3.9-1: VNF Indicator interface requirements

Numbering	Requirement
Or-Vnfm.VnfInd.001	The VNF Indicator interface produced by the VNFM on the Or-Vnfm reference point shall support providing notifications related to indicator value change, and to manage subscriptions related to such notifications.
Or-Vnfm.VnfInd.002	The VNF Indicator interface produced by the VNFM on the Or-Vnfm reference point shall support retrieving indicator values.

5.3.10 Policy Management interface requirements

Table 5.3.10-1 specifies requirements applicable to the policy management interface produced by the VNFM on the Or-vnfm reference point.

Table 5.3.10-1: Policy management interface requirements

Numbering	Requirement
Or-Vnfm.Plcm.001	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support transferring NFV-MANO policies. See notes 1 and 2.
Or-Vnfm.Plcm.002	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support deleting NFV-MANO policies. See note 1.
Or-Vnfm.Plcm.003	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support querying NFV-MANO policies. See note 1.
Or-Vnfm.Plcm.004	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support activating NFV-MANO policies. See note 1.
Or-Vnfm.Plcm.005	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support deactivating NFV-MANO policies. See note 1.
Or-Vnfm.Plcm.006	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the NFVO notifications about changes of a policy that are related to operations of transferring policy, deleting policy, activating policy, deactivating policy, associating policy and disassociating policy.
Or-Vnfm.Plcm.007	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support providing to the NFVO notifications about any detected policy conflicts.
Or-Vnfm.Plcm.008	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support subscribing to policy management related notifications.
Or-Vnfm.Plcm.009	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support associating NFV-MANO policies to VNF instances.
Or-Vnfm.Plcm.010	The Policy Management interface produced by the VNFM on the Or-Vnfm reference point shall support disassociating NFV-MANO policies from VNF instances.
NOTE 1: For this reference point, NFV-MANO policies include policies applied in VNF lifecycle management (instantiation, scaling, healing and termination). NOTE 2: The case of transferring NFV-MANO policy applies when: - a new policy is imported from the NFVO, which results in the creation of a new policy locally; or - the changes for an existing policy are imported from the NFVO, which results in the update of a policy locally.	

5.3.11 VNF Snapshot Package Management interface requirements

Table 5.3.11-1 specifies requirements applicable to the VNF Snapshot Package Management interface produced by the NFVO on the Or-Vnfm reference point.

Table 5.3.11-1: VNF Snapshot Package Management interface requirements

Numbering	Requirement
Or-Vnfm.VnfSnapPkgm.001	Void.
Or-Vnfm.VnfSnapPkgm.002	Void.
Or-Vnfm.VnfSnapPkgm.003	Void.
Or-Vnfm.VnfSnapPkgm.004	Void.
Or-Vnfm.VnfSnapPkgm.005	The VNF Snapshot Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support querying information about available VNF Snapshot Packages (see note).
Or-Vnfm.VnfSnapPkgm.006	The VNF Snapshot Package Management interface produced by the NFVO on the Or-Vnfm reference point shall support fetching a VNF Snapshot Package, or selected artifacts contained in a VNF Snapshot Package.
Or-Vnfm.VnfSnapPkgm.007	Void.
Or-Vnfm.VnfSnapPkgm.008	Void.
NOTE: VNF Snapshot Package information can include information such as creation date, configuration data of included snapshots, and files contained in the VNF Snapshot Package.	

6 NFVO exposed interfaces

6.1 Introduction

This clause defines the interfaces exposed by the NFVO towards the VNFM over the Or-Vnfm reference point.

NOTE: The fact that information elements and attributes are presented in tabular form does not preclude protocol designs in which these information elements and attributes are encoded in different parts of request and response messages. For example, in a RESTful interface, parts of them can be encoded in the URL, in the message header, in the message body or any combination thereof.

6.2 VNF Package Management interface

6.2.1 Description

This interface allows the VNFM to access VNF Package information.

The interface also includes a notify operation for new VNF Package on-boarding or for VNF Package changes, and operations to manage subscriptions to such notifications.

6.2.2 Query VNF Package Info operation

6.2.2.1 Description

This operation will enable the VNFM to query the NFVO for details of one or more VNF Package information objects. Table 6.2.2.1-1 lists the information flow exchanged between the NFVO and the VNFM.

NOTE: The VNFD is an attribute of the VnfPkgInfo.

Table 6.2.2.1-1: Query VNF Package Info operation

Message	Requirement	Direction
QueryVnfPkgInfoRequest	Mandatory	VNFM → NFVO
QueryVnfPkgInfoResponse	Mandatory	NFVO → VNFM

6.2.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.2.2-1.

Table 6.2.2.2-1: Query VNF Package Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the VNF Packages on which the query applies, based on attributes of the VnfPkgInfo. It can also be used to specify one or more VNF Package information objects to be queried by providing their vnfdId or vnfPkgInfoId. See note.
attributeSelector	M	0..N	String	It provides a list of attribute names of vnfPkgInfo. If present, only these attributes are returned for the vnfPkgInfo matching the filter. If absent, the complete vnfPkgInfo is returned.
NOTE: The vnfdId, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. The vnfPkgInfoId identifies the information related to the onboarding of a VNF package into the NFVO, which implies that it also identifies an onboarded VNF package.				

6.2.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.2.2.3-1.

Table 6.2.2.3-1: Query VNF Package Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	VnfPkgInfo	Details of the VNF Package information objects available to the VNFM matching the input filter. If attributeSelector is present, only the attributes listed in attributeSelector are returned for the selected entities.

6.2.2.4 Operation results

After successful operation, the NFVO has queried the internal VNF Package information objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, the VNF Package information objects that the consumer has access to and that are matching the filter shall be returned.

6.2.3 Subscribe operation

6.2.3.1 Description

This operation enables the VNFM to subscribe with a filter for the notifications related to new VNF Package on-boarded or to changes of VNF Packages sent by the NFVO.

NOTE: Specification of filtering mechanism is part of the protocol design.

Table 6.2.3.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.3.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	VNFM → NFVO
SubscribeResponse	Mandatory	NFVO → VNFM

6.2.3.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.3.2-1.

Table 6.2.3.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for subscribing new VNF Package on-boarded notification or for selecting the VNF Package(s) and the related change notifications to subscribe to. This filter can indicate for subscribing new VNF Package on-boarded, or can contain information about specific types of changes to subscribe to, or attributes of the VNF Package.

6.2.3.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.2.3.3-1.

Table 6.2.3.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription realized.

6.2.3.4 Operation results

After successful subscription, the VNFM is registered to receive notifications related to changes of VNF Packages sent by the NFVO. The result of the operation shall indicate if the subscription has been successful or not with a standard success/error result. For a particular subscription, only notifications matching the filter will be delivered to the VNFM.

6.2.4 Notify operation

6.2.4.1 Description

This operation distributes notifications to subscribers. It is a one-way operation issued by the NFVO that cannot be invoked as an operation by the consumer (VNFM).

In order to receive notifications, the VNFM shall have a subscription.

Table 6.2.4.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.4.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	NFVO → VNFM

The following notifications can be notified/sent by this operation:

- VnfPackageOnBoardingNotification (see clause 8.2.4).
- VnfPackageChangeNotification (see clause 8.2.5).

6.2.5 Void

6.2.6 Fetch VNF Package operation

6.2.6.1 Description

This operation enables the VNFM to fetch a whole on-boarded VNF Package from the NFVO. The package is addressed using an identifier of information held by the NFVO about the specific on-boarded VNF Package. This identifier is contained within the VnfPackageOnBoardingNotification or is returned as a result of Query VNF Package Info operation.

Table 6.2.6.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.6.1-1: Fetch VNF Package operation

Message	Requirement	Direction
FetchVnfPackageRequest	Mandatory	VNFM → NFVO
FetchVnfPackageResponse	Mandatory	NFVO → VNFM

6.2.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.6.2-1.

Table 6.2.6.2-1: Fetch VNF Package operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfPkgInfoId	M	1	Identifier	Identifier of the VNF Package information object associated with the VNF Package to be fetched. This identifier was allocated by the NFVO.

6.2.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.2.6.3-1.

Table 6.2.6.3-1: Fetch VNF Package operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfPackage	M	1	Binary	The VNF Package.

6.2.6.4 Operation results

The result of the operation indicates whether the fetching of the VNF Package has been successful or not in the NFVO with a standard success/error result. After successful operation, the NFVO has provided to the VNFM a copy of the requested VNF Package.

6.2.7 Fetch VNF Package Artifacts operation

6.2.7.1 Description

This operation enables the VNFM to fetch selected artifacts contained in an on-boarded VNF Package. Artifacts are addressed using selector information that can be obtained using the QueryVNF Package Info operation.

NOTE: The VNFD is an attribute of the OnboardedVnfPkgInfo and it is retrieved, if queried individually, with the QueryVnfPackageInfo operation. Fetching the whole VNF Package will also return the VNFD, which is part of the VNF Package.

Table 6.2.7.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.7.1-1: Fetch VNF Package Artifacts operation

Message	Requirement	Direction
FetchVnfPackageArtifactsRequest	Mandatory	VNFM → NFVO
FetchVnfPackageArtifactsResponse	Mandatory	NFVO → VNFM

6.2.7.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.7.2-1.

Table 6.2.7.2-1: Fetch VNF Package Artifacts operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfPkgInfoId	M	1	Identifier	Identifier of the VNF Package information object associated with the VNF Package artifacts to be fetched. This identifier was allocated by the NFVO.
artifactSelector	M	1..N	Not specified	Selector to address an individual VNF package artifact, or list of selectors to address multiple of those. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to obtain multiple artifacts in one go, or as a series of operations that obtain one artifact at a time.				

6.2.7.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.2.7.3-1.

Table 6.2.7.3-1: Fetch VNF Package Artifacts operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfPackageArtifact	M	1..N	Not specified	A VNF package artifact (e.g. files). or multiple thereof. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to obtain multiple artifacts in one go, or as a series of operations that obtain one artifact at a time.				

6.2.7.4 Operation results

The result of the operation indicates whether the fetching of the VNF Package Artifacts has been successful or not in the NFVO with a standard success/error result. After successful operation, the NFVO has provided to the VNFM a copy/copies of the requested artifact(s) contained in the on-boarded VNF Package.

6.2.8 Terminate Subscription operation

6.2.8.1 Description

This operation enables the VNFM to terminate a particular subscription.

Table 6.2.8.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.8.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	VNFM → NFVO
TerminateSubscriptionResponse	Mandatory	NFVO → VNFM

6.2.8.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.8.2-1.

Table 6.2.8.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

6.2.8.3 Output parameters

None.

6.2.8.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the VNFM will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

6.2.9 Query Subscription Info operation

6.2.9.1 Description

This operation enables the VNFM to query information about subscriptions.

Table 6.2.9.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.2.9.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	VNFM → NFVO
QuerySubscriptionInfoResponse	Mandatory	NFVO → VNFM

6.2.9.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.2.9.2-1.

Table 6.2.9.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

6.2.9.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.2.9.3-1.

Table 6.2.9.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

6.2.9.4 Operation results

After successful operation, the NFVO has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF package onboarding or VNF package changes that the VNFM has access to and that are matching the filter shall be returned.

6.3 VNF Lifecycle Operation Granting interface

6.3.1 Description

This interface defines one operation that allows the NFVO to grant lifecycle operations.

6.3.2 Grant VNF Lifecycle Operation operation

6.3.2.1 Description

This operation allows the VNFM to request a grant for authorization of a VNF lifecycle operation. This interface supports multiple use cases, such as:

- The NFVO can approve or reject a request based on policies (e.g. dependencies between VNFs) and available capacity.

- When applicable, the NFVO can reserve resources based on the VNFM's virtualised resources request. Depending on operator policies, the NFVO can decide on whether to reserve virtualised resources or physical compute hosts.

- The NFVO can provide to the VNFM information about the VIM where cloud resources are allocated. This can include additional information such as the resource zone.

- The NFVO can provide to the VNFM container namespaces in which the MCIOs of a VNF with containerized components shall be deployed.

- The NFVO can provide to the VNFM information about the PaaS Services allocated/assigned to the VNF.

When requesting resource creation or modification, the VNFM references the resource definitions that are available to the NFVO in the VNFD. When resources are to be released or modified, the VNFM provides references to the existing resources in the request. When requesting PaaS Services assignment, the VNFM references the PaaS Service request definition from the VNFD, which is also available to the NFVO. When PaaS Services are expected to be released or not used any longer by the VNF, the VNFM references the PaaS Service request that has been granted previously.

Per each VNFM, one of the following operator policies can be selected as a configuration to determine how the NFVO and the VNFM handle resource reservations in a grant request:

1) Policy GRANT_RESERVE: The NFVO guarantees the availability of the VIM resources to be allocated. The NFVO provides to the VNFM reservation identifier(s). Each such identifier identifies the reservation which is applicable to the resource requirements and which the VNFM shall use in the subsequent resource management operation.

2) Policy GRANT_APPROVE: The NFVO approves the VIM resources to be allocated by the VNFM. In general, resource availability is not guaranteed. No explicit reservation identifier is returned to the VNFM. Optionally, to guarantee resource availability, the NFVO may do a reservation and use implicit reservation identification towards the VNFM, i.e. associate the reservation to the VIM access information.

These policies are used to configure the behaviour of both the NFVO and the VNFM identically, also considering the resource reservation capabilities of the VIM:

- resource definitions refer to: either a resource template in the VNFD (VnfVirtualLinkDesc, VirtualComputeDesc, VirtualStorageDesc, OsContainerDesc plus Vdu, if applicable) for the creation of new resources; or

- to information about an existing resource.

In the case of VNF with containerized components, information exchanges associated to resources is performed at two different levels: OS container and MCIO. As specified in clause 5.2.1.2 of ETSI GS NFV-IFA 040 [i.13], the properties of Compute MCIO and Storage MCIO are specified in the VDU of the VNFD, and there is a 1:1 relationship between an instantiated VNFC of a VNF and the Compute MCIO. For storage, the resource requirements are specified as VirtualStorageDesc of the VNFD:

- With regards to the VNF lifecycle operation granting, the NFVO handles the granting of resource requests at MCIO level but with additional exchange of information related to OS containers. Even though there is exposure of information at OS container level, there is no management of OS containers performed neither by the VNFM nor the NFVO. The NFVO can collect detailed information about OS container resource requests to perform the granting of resources as part of the overall resource orchestration. The requests for resources to be allocated to an MCIO is derived from the OsContainerDesc resource templates referenced in the ResourceDefinition. The OsContainerDesc(s) are also referred per VDU in the ResourceDefinition (via the attribute "vduId"). This is determined based on the relationship between VDU and OsContainerDesc as defined in the VNFD (see clause 7.1.6.2 of ETSI GS NFV-IFA 011 [3]). For VDUs which have the attribute isNumOfInstancesClusterBased set to FALSE, the NFVO can derive the number of VNFC instances affected (e.g. to be added, removed, updated) by the VNF LCM operation based on the number of occurrences of ResourceDefinitions or, alternatively, based on the instantiationLevelId or on the targetScaleLevelInfo.

For VDUs which have the attribute isNumOfInstancesClusterBased set to TRUE, the NFVO can derive the number of VNFC instances affected (e.g. to be added, removed, updated) by the VNF LCM operation based on static information about the number of CIS-nodes in the CIS-cluster in a node pool that fulfils the requirements specified in the VDU. The calculated number is, in the general case, a maximum number because not all nodes in the identified pool might have the required resources available at the time. If ResourceDefinition is included in the request, only one occurrence shall be included.

- With regards to the VNF lifecycle management, the NFVO can collect runtime information of the VNF instance at the MCIO level with the QueryVnf operation and VnfLcmOperationOccurrenceNotification specified in clauses 7.2.9 and 7.2.15 of the present document, respectively. The "VnfInfo" provides information about the mapping of the compute, storage and network resources with the MCIO whose declarative descriptor specify corresponding resource requests (refer to clauses 8.5.4, 8.5.5, 8.5.6 and 8.5.7 of the present document). For each VnfcResourceInfo, a mapping to the vduId is also provided, similarly as with the VM-based case for components of a VNF. The NFVO can correlate the number of occurrences of VnfcResourceInfo in the VnfInfo with the information derived from the granting exchanges.

The NFVO is expected to consider the container namespaces, including their resource quota, when evaluating granting requests for lifecycle operations on containerized workloads. Requests for resources to be allocated to MCIOs are derived from the OsContainerDesc resource templates referenced in the grant request. The containerized workloads based on a particular MCIOP are deployed within one container namespace. To determine the container namespace to be returned in the grant response, the NFVO shall resolve the association of the OsContainerDesc resource templates to the corresponding MCIOP. The VNFD contains profiles of all MCIOPs in the VNF deployment flavour information element. An MCIOP profile contains a list of associated VDUs which in turn reference the OsContainerDesc resource templates. By using these associations, the NFVO can return the namespace for the resource definitions related to the MCIOP.

In the GrantVnfLifecycleOperation response, the NFVO can return information that allows to distribute the resources of a VNF over multiple resource zones. This decision is guided by affinity/anti-affinity rules in the VNFD as well as by placement constraints passed in the GrantVnfLifecycleOperation request. The NFVO can also return information that allows to manage the resources of a VNF using multiple VIMs, guided by VIM selection constraints passed in the GrantVnfLifecycleOperation request.

In the present document, as part of the granting mechanism, attributes are defined for signalling the decision to use multiple VIMs per VNF. For the support of VNFs that include resources managed by multiple VIMs, the mechanisms to manage the VNF-internal Virtual Link (VL) requirements across multiple VIMs leverage the concept of "externally managed VL" and the support of "multi-site connectivity services". With these mechanisms, the NFVO can manage the connectivity for the VNF, not only within a given NFVI-PoP (or site), but also across different NFVI-PoPs.

When the VDUs of the VNF are realized by a set of OS containers, the NFVO determines whether the namespace quota associated with the VNF will be impacted by this LCM operation and needs to be changed. If the change is needed, then the NFVO further initiates a request of modification of the namespace quota associated with the VNF to the Container Infrastructure Service Management (CISM) function.

When the VDUs of the VNF are realized by a set of virtual machines, the following applies: some LCM operations, including HealVnf and ChangeCurrentVnfPkg, might try different resource management operations for the same resource to achieve a certain goal. For instance, healing might first try to reboot a virtual compute resource. If this does not fix the problem, it might delete and re-create the virtual compute resource. Because there is a single granting exchange per LCM operation, this granting exchange needs to obtain permission up front, without knowing which of the operations will actually lead to success. Further, temporary resources may not be needed in all cases, depending on the state of the resources and the progress of the operation.

The following provisions therefore apply when the VDUs of the VNF are realized by a set of virtual machines:

1) When requesting a grant, the VNFM need not include information about planned resource management operations that do not change attributes of the VIM-level resource. For example, rebooting a virtual compute resource or rebuilding a virtual compute resource using the image, bootdata and flavour that were used when creating the resource will not lead to observable changes of the virtual compute resource.

2) Even though a grant for a resource management operation has been approved by the NFVO, it is not mandatory for the VNFM to execute the granted resource management operation as part of the LCM operation. The NFVO can learn which resources are affected by an LCM operation e.g. by listening to LcmOpOccNotifications or by reading the related LcmOpOcc object.

3) If, for a particular resource, there are resource modifications and resource deletion/re-creation as alternatives, the VNFM shall include in the grant request information about the more impactful operation i.e. deletion and re-creation. If the deletion/re-creation was granted, a modification of the affected resource may be executed by the VNFM instead.

Table 6.3.2.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.3.2.1-1: Grant VNF Lifecycle Operation operation

Message	Requirement	Direction
GrantVnfLifecycleOperationRequest	Mandatory	VNFM → NFVO
GrantVnfLifecycleOperationResponse	Mandatory	NFVO → VNFM

6.3.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.3.2.2-1.

Table 6.3.2.2-1: Grant VNF Lifecycle Operation operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance which this grant request is related to. Shall also be provided for VNFs that not yet exist but are planned to exist in the future, i.e. if the grant is requested for InstantiateVNF.
vnfdId	M	1	Identifier	Identifier of the VNFD that defines the VNF for which the LCM operation is to be granted. In case the operation changes the current VNF Package, this identifier refers to the VNFD which defines the VNF before the LCM operation to be granted.
dstVnfdId	M	0..1	Identifier	Identifier of the "destination" VNFD which will define the VNF after executing the LCM operation to be granted. Shall be included if the operation changes the current VNF Package and shall be absent otherwise.
flavourId	M	0..1	Identifier	Identifier of the VNF Deployment Flavour (DF) of the VNFD that defines the VNF for which the LCM operation is to be granted. Shall be provided when instantiating the VNF or changing the DF of the VNF instance.
lifecycleOperation	M	1	Enum	The lifecycle management operation for which granting is requested. VALUES: • InstantiateVnf • ScaleVnf • ScaleVnfToLevel • ChangeVnfFlavour • TerminateVnf • HealVnf • OperateVnf • ChangeExtVnfConnectivity • CreateSnapshot • RevertToSnapshot • ChangeCurrentVnfPackage • SelectVnfDeployableModules See note 1.

Parameter	Qualifier	Cardinality	Content	Description
isAutomaticInvocation	M	1	Boolean	Set to true if this VNF LCM operation occurrence has been triggered by an automated procedure inside the VNFM (i.e. ScaleVnf/ScaleVnfToLevel triggered by auto-scale, or HealVnf triggered by auto-heal). Set to false otherwise.
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence associated to the GrantVnfLifecycleOperationRequest.
instantiationLevelId	M	0..1	Identifier	If the granting request is requested for InstantiateVNF, the identifier of the instantiation level may be provided as an alternative way to define the resources to be added. This attribute shall only be used for Instantiate VNF requests. See notes 2, 7 and 10.
targetScaleLevelInfo	M	0..N	ScaleInfo	This attribute shall only be used for Instantiate VNF requests. This is applicable if VNF supports target scale level instantiation. This attribute provides an alternative way to define the resources to be added for the VNFs. For each scaling aspect of the current deployment flavour, the attribute specifies the scale level of VNF constituents (e.g. VDU level) to be instantiated. See notes 2, 7, 8 and 10.
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a selected deployable module. Resources related to VDUs that belong to not selected deployable modules shall not be considered in the granting. This attribute shall only be used for the instantiate VNF requests. See note 10.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor, as received in the VNF LCM operation request. The values indicated in this attribute are applicable to all VNFC instances based on the VDU to which the resourceCapacityDefinition is related.
addResource	M	0..N	ResourceDefinition	List of resource definitions in the VNFD for resources to be added by the LCM operation which is related to this grant request, with one entry per resource. See notes 2, 9 and 10.
tempResource	M	0..N	ResourceDefinition	List of resource definitions in the VNFD for resources to be temporarily instantiated during the runtime of the LCM operation which is related to this grant request, with one entry per resource (see note 3). See note 9.
removeResource	M	0..N	ResourceDefinition	Provides the definitions of resources to be removed by the LCM operation which is related to this grant request, with one entry per resource. See note 9.

Parameter	Qualifier	Cardinality	Content	Description
updateResource	M	0..N	ResourceDefinition	Provides the definitions of resources to be modified by the LCM operation which is related to this grant request, with one entry per resource. See note 9.
placementConstraint	M	0..N	PlacementConstraint	Placement constraints that the VNFM may send to the NFVO in order to influence the resource placement decision. If sent, the NFVO shall take the constraints into consideration when making resource placement decisions, and shall reject the grant if they cannot be honoured (see notes 4, 5 and 6).
vimConstraint	CM	0..N	VimConstraint	Used by the VNFM to require that multiple resources are managed through the same VIM connection. If sent, the NFVO shall take the constraints into consideration when making VIM selection decisions, and shall reject the grant if they cannot be honoured. CONDITION: This parameter shall be supported if VNF-related Resource Management in direct mode is applicable.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the VNFM, specific to the VNF and the LCM operation.

NOTE 1: The VNF LCM operations CreateVnfIdentifier, DeleteVnfIdentifier, QueryVnf, ModifyVnfInformation, ConfirmVnfSnapshot, QuerySnapshotInfo, and DeleteSnapshotInfo can be executed by the VNFM without requesting granting.

NOTE 2: If the granting request is for InstantiateVNF, only one of the three parameters (instantiationLevel or targetScaleLevelInfo or addResource) shall be present.

NOTE 3: The NFVO will assume that the VNFM will be responsible to both allocate and release the temporary resource during the runtime of the LCM operation. This means, the resource can be allocated and consumed after the "start" notification for the LCM operation is sent by the VNFM, and the resource will be released before the "result" notification of the VNF LCM operation is sent by the VNFM. In the case of PaaS Service requests, the handling of the allocation and release of the PaaS Service is a responsibility of the NFVO (supported by corresponding PaaS Service management functions) based on the information about the progress and completion of the VNF LCM operation.

NOTE 4: For the affinity/anti-affinity rules defined in the VNFD using the AffinityOrAntiAffinityGroup and the LocalAffinityOrAntiAffinityRule information elements (see ETSI GS NFV-IFA 011 [3]), and the placement constraints in the GrantVnfLifecycleOperation as defined in this clause, the following applies: assuming unlimited capacity, the combination of all the aforementioned rules shall be satisfiable by at least one possible placement of the new resources, with the exception that some of the rules with fallbackBestEffort may be unsatisfiable due to the actual placement of existing resources. In this case, rules with fallbackBestEffort are allowed to be violated only in relation to the placement of existing resources.

NOTE 5: Passing constraints allows the VNFM or the lifecycle management scripts to influence resource placement decisions by the NFVO to ensure VNF properties such as performance or fault tolerance.

NOTE 6: If fallbackBestEffort is present in placement constraints and set to "true", the NFVO processes the Affinity/AntiAffinity constraint in a best effort manner, in which case, if specified resources cannot be allocated based on specified placement constraint due to limited capacity, the NFVO looks for an alternate best effort placement for the specified resources to be granted.

NOTE 7: The target size for VNF instantiation may be specified in either instantiationLevelId or targetScaleLevelInfo, but not both.

NOTE 8: If targetScaleLevelInfo is specified, information provided in targetScaleLevelInfo shall be used for scalable constituents of the VNF (e.g. VDUs/VLs) in the granting process. For scaling aspects not specified in targetScaleLevelInfo or for the VNF constituents (e.g.VDUs/VLs) that are not scalable, the default instantiation level as declared in the VNFD shall be used in the granting process.

NOTE 9: For resources related to a VDU that has the attribute isNumOfInstancesClusterBased set to TRUE, only one occurrence of the addResource, or tempResource or removeResource, or updateResource shall be included per descriptor indicated in the resourceTemplateId of the ResourceDefinition, not one per VNFC instance.

NOTE 10: If the granting request is for InstantiateVNF and if there are deployable modules defined in the applicable VNF DF in the VNFD, either the selectedDeployableModules attribute or the addResource attribute shall be included but not both. If selectedDeployableModules is included, either the instantiationLevel or targetScaleLevelInfo attribute shall be also included.

6.3.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.3.2.3-1.

Table 6.3.2.3-1: Grant VNF Lifecycle Operation operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vimConnection	CM	0..N	VimConnectionInfo	Provides information regarding VIM and/or CISM connections that are approved to be used by the VNFM to allocate resources, and provides parameters of these VIM and/or CISM connections. Absent in case of rejection or if the VIM or CISM information was configured to the VNFM in another way, present otherwise. This parameter shall be supported when the granted resources are managed by a CISM. CONDITION: This parameter shall be supported when VNF-related Resource Management in direct mode is applicable. See note 1.
cirConnectionInfo	M	0..N	VimConnectionInfo	Provides information regarding a CIR connection that is approved to be used by the VNFM to obtain information about OS container images. Absent in case of rejection or if the CIR information was configured to the VNFM in another way, present otherwise. This parameter shall be supported when the granted resources are managed by a CISM.
mciopRepositoryInfo	M	0..N	VimConnectionInfo	Provides information regarding a MCIOP repository. Absent in case of rejection or if the MCIOP repository information was configured to the VNFM in another way, present otherwise. This parameter shall be supported when the granted resources are managed by a CISM.
zone	M	0..N	ZoneInfo	Identifies resource zones where the resources are approved to be allocated by the VNFM. Absent in case of rejection, present otherwise.
zoneGroup	M	0..N	ZoneGroupInfo	Information about groups of resource zones that are related and that the NFVO has chosen to fulfil a zoneGroup constraint in the GrantVnfLifecycleOperation request. This information confirms that the NFVO has honoured the zoneGroup constraints that were passed as part of "placementConstraints" in the Grant request.
addResource	M	0..N	GrantInfo	List of resources that are approved to be added, with one entry per resource. See note 5.
tempResource	M	0..N	GrantInfo	List of resources that are approved to be temporarily instantiated during the runtime of the lifecycle operation, with one entry per resource. See note 5.
removeResource	M	0..N	GrantInfo	List of resources that are approved to be removed, with one entry per resource. See note 5.

Parameter	Qualifier	Cardinality	Content	Description
updateResource	M	0..N	GrantInfo	List of resources that are approved to be modified, with one entry per resource. See note 5.
vimAssets	M	0..1	VimAssets	Information about assets for the VNF that are managed by the NFVO in the VIM, such as software images and virtualised compute resource flavours.
paasAssets	M	0..N	PaasAsset	Information about PaaS Services assigned to the VNF and that are managed by the NFVO in the PSM.
extVirtualLink	M	0..N	ExtVirtualLinkData	Information about external VLs to connect the VNF to (see note 4).
extManagedVirtualLink	M	0..N	ExtManagedVirtualLinkData	Information about internal VLs that are managed by other entities than the VNFM (see notes 3 and 4).
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO, specific to the VNF and the LCM operation.

NOTE 1: This interface allows to signal the use of multiple VIM connections per VNF. The specification for managing the VNF-internal VL requirements across multiple VIMs is supported via "externally-managed internal VLs" and "multi-site connectivity services".
NOTE 2: Void.
NOTE 3: The indication of externally-managed internal VLs is needed in case networks have been pre-configured for use with certain VNFs, for instance to ensure that these networks have certain properties such as security or acceleration features, or to address particular network topologies. The present document assumes that externally-managed internal VLs are managed by the NFVO and created towards the VIM as supported by the virtualised network resource management interface specified in ETSI GS NFV-IFA 005 [i.4] and the WIM as supported by the multi-site connectivity services management interface specified in ETSI GS NFV-IFA 032 [i.11].
NOTE 4: For any VNF lifecycle management operation request that allows to pass "extVirtualLink" and/or "extManagedVirtualLink" parameters, such as InstantiateVnf, ChangeVnfFlavor, ChangeVnfExtConnectivity or ChangeCurrentVnfPackage, the NFVO may provide the "extVirtualLink" and/or "extManagedVirtualLink" attributes in the grant response to override the values passed in these parameters previously in the associated VNF lifecycle management request, if the lifecycle management request has originated from the NFVO itself. The NFVO shall not provide the "extVirtualLink" and/or "extManagedVirtualLink" attributes in the grant response in case the LCM request did not originate from the NFVO itself or the granted LCM operation request does not allow to pass these parameters.
NOTE 5: For resources related to ResourceDefinition whose VDU that has the attribute isNumOfInstancesClusterBased set to TRUE, only one occurrence of the addResource, or tempResource, or removeResource, or updateResource shall be included.

6.3.2.4 Operation results

In case of permitting the operation, the NFVO returns to the VNFM additional information to be used in the resource management operations during the lifecycle management operation.

Once the NFVO has responded positively with a GrantVnfLifecycleOperationResponse, the VNFM executes the necessary resource management operations either towards the appropriate VIM(s) (a.k.a VNF-related resource management in direct mode) or towards the NFVO which proxies them to the appropriate VIM(s) (a.k.a VNF-related resource management in indirect mode).

In addition to failure situations, the NFVO can reject a GrantVnfLifecycleOperationRequest due to various reasons, such as resource unavailability or operational policy. In case of rejecting the operation or in case of failure, the NFVO returns to the VNFM appropriate error information, describing the reason of rejection or failure.

If placement constraints have been passed with the request, the NFVO shall process the constraints as below:

- If fallbackBestEffort is not present or set to "false" in a set of placement constraint, and if the NFVO cannot find a resource placement that satisfies all these constraints, it shall reject the grant request.

- If fallbackBestEffort is present and set to "true" in one or more placement constraints and the NFVO cannot find a resource placement that satisfies all of these, the NFVO shall process each such affinity/anti-aAffinity constraint in a best effort manner, in which case, if specified resources cannot be allocated based on the specified placement constraint, the NFVO shall look for an alternate best effort placement for the specified resources to be granted. In the best effort anti-affinity case, the resources are spread optimally over all available instances of scope (e.g. zones), and in the best effort affinity case, they are distributed optimally over as few instances of scope as possible. Being able to satisfy a "best-effort" constraint in a best effort manner only, as defined above, shall not lead to rejecting the grant.

6.4 Virtualised Resources Management interfaces in indirect mode

6.4.1 Introduction

In indirect mode of VNF-related resource management, the NFVO produces towards the VNFM the virtualised resource management interfaces defined below.

These interfaces are related to the corresponding interfaces defined in ETSI GS NFV-IFA 006 [1]; however, an additional *resource provider identifier* is introduced. This identifier is used by the NFVO to determine the entity responsible for the management of the virtualised resource, the management of the virtualised resources reservation or the management of the virtualised resources quota (usually one of multiple VIMs with which the NFVO interacts). It is used by the VNFM to uniquely identify resources, resource reservations or resource quotas by means of the pair of the resource provider identifier and the actual identifier of the resource/reservation/quota.

6.4.2 Virtualised Compute interfaces

6.4.2.1 Virtualised Compute Resources Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Compute Resources Management to VNFM. This interface shall comply with the provisions in clause 7.3.1 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualCompute used in output parameters in clause 7.3.1 of ETSI GS NFV-IFA 006 [1] is replaced with ComputeResourceWithRpInfo as defined in clause 8.4.2.2 of the present document.
- For the Terminate Virtualised Compute Resource operation the content of both the input and output parameters is changed from Identifier to ComputeResourceWithRpId as defined in clause 8.4.2.3 of the present document.
- All operations except Query Virtualised Compute Resource and Terminate Virtualised Compute Resource have an additional input parameter, resourceProviderId, defined in table 6.4.2.1-1.

Table 6.4.2.1-1: Definition of the resourceProviderId input parameter for compute resources

Parameter	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, computeId].

6.4.2.2 Virtualised Compute Resources Change Notification interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Compute Resources Change Notifications to be consumed by VNFM. This interface shall comply with the provisions in clause 7.3.2 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The notification VirtualisedResourceChangeNotification sent by means of the Notify operation of clause 7.3.2.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification VirtualisedResourceWithRpChangeNotification defined in clause 8.4.5.2 of the present document.

6.4.2.3 Virtualised Compute Resources Information Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Compute Resources Information Management to VNFM. This interface shall comply with the provisions in clause 7.3.3 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualComputeResourceInformation used in output parameters for the Query Virtualised Compute Resource Information operation in clause 7.3.3.4 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualComputeResourceWithRpInfo as defined in clause 8.4.2.4 of the present document.

- The notification InformationChangeNotification sent by means of the Notify operation of clause 7.3.3.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification InformationWithRpChangeNotification defined in clause 8.4.5.3 of the present document.

6.4.3 Virtualised Network interfaces

6.4.3.1 Virtualised Network Resources Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Network Resources Management to VNFM. This interface shall comply with the provisions in clause 7.4.1 of ETSI GS NFV-IFA 006 [1] with the following change(s):

- The content VirtualNetwork used in output parameters in clause 7.4.1 of ETSI GS NFV-IFA 006 [1] is replaced by NetworkResourceWithRpInfo as defined in clause 8.4.3.2 of the present document.

- For the Terminate Virtualised Network Resource operation the content of both the input and output parameter is changed from Identifier to NetworkResourceWithRpId as defined in clause 8.4.3.3 of the present document.

- All operations except Query Virtualised Network Resource and Terminate Virtualised Network Resource have an additional input parameter, resourceProviderId, defined in table 6.4.3.1-1.

Table 6.4.3.1-1: Definition of the resourceProviderId input parameter for network resources

Parameter	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, networkResourceId].

6.4.3.2 Virtualised Network Resources Change Notification interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Network Resources Change Notifications to be consumed by VNFM. This interface shall comply with the provisions in clause 7.4.2 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The notification VirtualisedResourceChangeNotification sent by means of the Notify operation of clause 7.4.2.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification VirtualisedResourceWithRpChangeNotification defined in clause 8.4.5.2 of the present document.

6.4.3.3 Virtualised Network Resources Information Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Network Resources Information Management to VNFM. This interface shall comply with the provisions in clause 7.4.3 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualNetworkResourceInformation used in output parameters for the Query Virtualised Network Resource Information operation in clause 7.4.3.4 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualNetworkResourceWithRpInfo as defined in clause 8.4.3.4 of the present document.
- The notification InformationChangeNotification sent by means of the Notify operation of clause 7.4.3.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification InformationWithRpChangeNotification defined in clause 8.4.5.3 of the present document.

6.4.4 Virtualised Storage interfaces

6.4.4.1 Virtualised Storage Resources Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Storage Resources Management to VNFM. This interface shall comply with the provisions in clause 7.5.1 of ETSI GS NFV-IFA 006 [1] with the following change(s):

- The content VirtualStorage used in output parameters in clause 7.5.1 of ETSI GS NFV-IFA 006 [1] is replaced by StorageResourceWithRpInfo as defined in clause 8.4.4.2 of the present document.
- For the Terminate Virtualised Storage Resource operation the content of both the input and output parameter is changed from Identifier to StorageResourceWithRpId as defined in clause 8.4.4.3 of the present document.
- All operations except Query Virtualised Storage Resource and Terminate Virtualised Storage Resource have an additional input parameter, resourceProviderId, defined in table 6.4.4.1-1.

Table 6.4.4.1-1: Definition of the resourceProviderId input parameter for storage resources

Parameter	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, storageId].

6.4.4.2 Virtualised Storage Resources Change Notification interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Storage Resources Change Notifications to be consumed by VNFM. This interface shall comply with the provisions in clause 7.5.2 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The notification VirtualisedResourceChangeNotification sent by means of the Notify operation of clause 7.5.2.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification VirtualisedResourceWithRpChangeNotification defined in clause 8.4.5.2 of the present document.

6.4.4.3 Virtualised Storage Resources Information Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Storage Resources Information Management to VNFM. This interface shall comply with the provisions in clause 7.5.3 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualStorageResourceInformation used in output parameters for the Query Virtualised Storage Resources Information operation in clause 7.5.3.4 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualStorageResourceWithRpInfo as defined in clause 8.4.4.4 of the present document.

- The notification InformationChangeNotification sent by means of the Notify operation of clause 7.5.3.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification InformationWithRpChangeNotification defined in clause 8.4.5.3 of the present document.

6.4.5 Virtualised Resource Performance Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Resource Performance Management to be consumed by VNFM. This interface shall comply with the provisions in clause 7.7 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The operations Create PM Job and Create Threshold have an additional input element, resourceProviderId, defined in table 6.4.5-1, with the value received in the response to the GrantVnfLifecycleOperation request.

- The notification PerformanceInformationAvailableNotification notified/sent by means of the Notify operation of clause 7.7.6 of ETSI GS NFV-IFA 006 [1] is replaced with the notification PerformanceInformationWithRpAvailableNotification defined in clause 8.4.6.2 of the present document.

- The notification ThresholdCrossedNotification notified/sent by means of the Notify operation of clause 7.7.6 of ETSI GS NFV-IFA 006 [1] is replaced with the notification ThresholdCrossedWithRpNotification defined in clause 8.4.6.3 of the present document.

Table 6.4.5-1: Definition of the resourceProviderId input parameter for virtual resource performance information

Parameter	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the virtualised resource performance information and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, objectInstanceId].

6.4.6 Virtualised Resource Fault Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Resource Fault Management to be consumed by VNFM. This interface shall comply with the provisions in clause 7.6 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The content Alarm used in the output parameters of the Get Alarm List operation of clause 7.6.4 of ETSI GS NFV-IFA 006 [1] is replaced with AlarmWithRpInfo as defined in clause 8.4.7.2 of the present document in order to distinguish between alarms from different VIM instances managed by the NFVO.

- The notification AlarmNotification published/notified/sent by means of the Notify operation of clause 7.6.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification AlarmWithRpNotification defined in clause 8.4.7.3 of the present document.

- The notification AlarmClearedNotification published/notified/sent by means of the Notify operation of clause 7.6.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification AlarmClearedWithRpNotification defined in clause 8.4.7.4 of the present document.

6.4.7 Virtualised Resources Quota Management interfaces

6.4.7.1 Virtualised Compute Resources Quota Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Compute Resources Quota Management to the VNFM. This interface shall comply with the provisions in clause 7.9.1 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualComputeQuota used in output parameters in clause 7.9.1 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualComputeQuotaWithRpInfo as defined in clause 8.4.8.2 of the present document.

6.4.7.2 Virtualised Network Resources Quota Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Network Resources Quota Management to the VNFM. This interface shall comply with the provisions in clause 7.9.2 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualNetworkQuota used in output parameters in clause 7.9.2 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualNetworkQuotaWithRpInfo as defined in clause 8.4.8.3 of the present document.

6.4.7.3 Virtualised Storage Resources Quota Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Storage Resources Quota Management to the VNFM. This interface shall comply with the provisions in clause 7.9.3 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content VirtualStorageQuota used in output parameters in clause 7.9.3 of ETSI GS NFV-IFA 006 [1] is replaced with VirtualStorageQuotaWithRpInfo as defined in clause 8.4.8.4 of the present document.

6.4.7.4 Virtualised Resources Quota Change Notification interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Resources Quota Change Notification to be consumed by the VNFM. This interface shall comply with the provisions in clause 7.9.4 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The notification VirtualisedResourceQuotaChangeNotification sent by means of the Notify operation of clause 7.9.4.3 of ETSI GS NFV-IFA 006 [1] is replaced with notification VirtualisedResourceQuotaWithRpChangeNotification defined in clause 8.4.8.5 of the present document.

6.4.8 Virtualised Resource Reservation interfaces

6.4.8.1 Virtualised Compute Resources Reservation Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Compute Resources Reservation Management to VNFM. This interface shall comply with the provisions in clause 7.8.1 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content ReservedVirtualCompute used in output parameters in clause 7.8.1 of ETSI GS NFV-IFA 006 [1] is replaced with ReservedVirtualComputeWithRpInfo as defined in clause 8.4.9.2 of the present document.

6.4.8.2 Virtualised Network Resources Reservation Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Network Resources Reservation Management to VNFM. This interface shall comply with the provisions in clause 7.8.2 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content ReservedVirtualNetwork used in output parameters in clause 7.8.2 of ETSI GS NFV-IFA 006 [1] is replaced with ReservedVirtualNetworkWithRpInfo as defined in clause 8.4.9.3 of the present document.

6.4.8.3 Virtualised Storage Resources Reservation Management interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Storage Resources Reservation Management to VNFM. This interface shall comply with the provisions in clause 7.8.3 of ETSI GS NFV-IFA 006 [1] with the following changes:

- The content ReservedVirtualStorage used in output parameters in clause 7.8.3 of ETSI GS NFV-IFA 006 [1] is replaced with ReservedVirtualStorageWithRpInfo as defined in clause 8.4.9.4 of the present document.

6.4.8.4 Virtualised Resources Reservation Change Notification interface

In indirect resource management mode, the NFVO produces an interface for Virtualised Resources Reservation Change Notifications to be consumed by the VNFM. This interface shall comply with the provisions in clause 7.8.4 of ETSI GS NFV-IFA 006 [1] and the related information elements with the following changes:

- The notification VirtualisedResourceReservationChangeNotification sent by means of the Notify operation of clause 7.8.4.3 of ETSI GS NFV-IFA 006 [1] is replaced with the notification VirtualisedResourceReservationWithRpChangeNotification defined in clause 8.4.9.5 of the present document.

6.5 Virtualised Resources Quota Available Notification interface

6.5.1 Description

This interface allows an authorized consumer FB to manage subscriptions regarding information on the availability of the virtualised resources quota(s), and to provide such notification to the subscribed consumer.

Support for this interface is optional.

The VNFM needs to issue a Subscribe request for VirtualisedResourceQuotaAvailable notifications in order to know when a quota applicable to the VNFM is available.

When a quota applicable to the consumer is available, the consumer is notified using the notification VirtualisedResourceQuotaAvailableNotification (see clause 8.11.2).

6.5.2 Subscribe operation

6.5.2.1 Description

This operation enables the VNFM to subscribe with a filter for the notifications related to the availability quota on virtualised resources sent by the NFVO. Specification of filtering mechanism is part of the protocol design.

Table 6.5.2.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.5.2.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	VNFM → NFVO
SubscribeResponse	Mandatory	NFVO → VNFM

6.5.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.5.2.2-1.

Table 6.5.2.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting notifications to subscribe to. This filter can contain information about specific attributes of the virtualised resources quota.

6.5.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.5.2.3-1.

Table 6.5.2.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription realized.

6.5.2.4 Operation results

After successful subscription, the VNFM is registered to receive notifications sent by the NFVO when a virtualised resources quota applicable to the VNFM is available. The result of the operation shall indicate if the subscription has been successful or not with a standard success/error result. For a particular subscription, only notifications matching the filter will be delivered to the VNFM.

6.5.3 Notify operation

6.5.3.1 Description

This operation distributes notifications to subscribers. It is a one-way operation issued by the NFVO that cannot be invoked as an operation by the consumer (VNFM).

In order to receive notifications, the VNFM shall have a subscription.

Table 6.5.3.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.5.3.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	NFVO → VNFM

The following notification is sent by this operation:

- VirtualisedResourceQuotaAvailableNotification. See clause 8.11.2.

6.5.4 Terminate Subscription operation

6.5.4.1 Description

This operation enables the VNFM to terminate a particular subscription.

Table 6.5.4.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.5.4.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	VNFM → NFVO
TerminateSubscriptionResponse	Mandatory	NFVO → VNFM

6.5.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.5.4.2-1.

Table 6.5.4.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

6.5.4.3 Output parameters

None.

6.5.4.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the VNFM will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

6.5.5 Query Subscription Info operation

6.5.5.1 Description

This operation enables the VNFM to query information about subscriptions.

Table 6.5.5.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 6.5.5.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	VNFM → NFVO
QuerySubscriptionInfoResponse	Mandatory	NFVO → VNFM

6.5.5.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.5.5.2-1.

Table 6.5.5.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

6.5.5.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.5.5.3-1.

Table 6.5.5.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

6.5.5.4 Operation results

After successful operation, the NFVO has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF quota availability that the VNFM has access to and that are matching the filter shall be returned.

6.6 VNF Snapshot Package Management interface

6.6.1 Description

This interface allows the VNFM to access VNF Snapshot Package information and to fetch the content of an available VNF Snapshot package from the NFVO.

6.6.2 Fetch VNF Snapshot Package operation

6.6.2.1 Description

This operation enables the VNFM to fetch the content of a VNF Snapshot Package. The package is addressed using an identifier of information held by the NFVO about the specific VNF Snapshot Package.
Table 6.6.2.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 6.6.2.1-1: Fetch VNF Snapshot Package operation

Message	Requirement	Direction
FetchVnfSnapshotPackageRequest	Mandatory	VNFM → NFVO
FetchVnfSnapshotPackageResponse	Mandatory	NFVO → VNFM

6.6.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.6.2.2-1.

Table 6.6.2.2-1: Fetch VNF Snapshot Package operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotPkgInfoId	M	1	Identifier (Reference to VnfSnapshotPkgInfo)	References the information held by the NFVO about the specific VNF Snapshot Package. This identifier was allocated by the NFVO.

6.6.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.6.2.3-1.

Table 6.6.2.3-1: Fetch VNF Snapshot Package operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotPackage	M	1	Binary	The VNF Snapshot Package.

6.6.2.4 Operation results

After successful operation, the NFVO has provided to the VNFM a copy of the requested VNF Snapshot Package.

6.6.3 Fetch VNF Snapshot Package Artifacts operation

6.6.3.1 Description

This operation enables the VNFM to fetch selected artifacts contained in a VNF Snapshot Package. Artifacts are addressed using selector information that can be obtained using the QueryVnfSnapshotPkgInfo operation. Table 6.6.3.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 6.6.3.1-1: Fetch VNF Snapshot Package Artifacts operation

Message	Requirement	Direction
FetchVnfSnapshotPackageArtifactsRequest	Mandatory	VNFM → NFVO
FetchVnfSnapshotPackageArtifactsResponse	Mandatory	NFVO → VNFM

6.6.3.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.6.3.2-1.

Table 6.6.3.2-1: Fetch VNF Snapshot Package Artifacts operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotPkgInfoId	M	1	Identifier (Reference to VnfSnapshotPkgInfo)	References the information held by the NFVO about the specific VNF Snapshot Package.
artifactSelector	M	1..N	Not specified	Selector to address an individual VNF Snapshot Package artifact, or list of selectors to address multiple of those. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to obtain multiple artifacts in one go, or as a series of operations that obtain one artifact at a time.				

6.6.3.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.6.3.3-1.

Table 6.6.3.3-1: Fetch VNF Snapshot Package Artifacts operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotPackage Artifact	M	1..N	Not specified	A VNF Snapshot Package artifact (e.g. file), or multiple thereof. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to obtain multiple artifacts in one go, or as a series of operations that obtain one artifact at a time.				

6.6.3.4 Operation results

After successful operation, the NFVO has provided to the VNFM a copy/copies of the requested artifact(s) contained in the VNF Snapshot Package.

6.6.4 Query VNF Snapshot Package Information operation

6.6.4.1 Description

This operation enables the VNFM to query the NFVO for information about one or more VNF Snapshot Packages. Table 6.6.4.1-1 lists the information flow exchanged between the VNFM and the NFVO.

The operation allows querying specific components of the information about a VNF Snapshot Package, for instance, retrieving the vnfSnapshotInfoId.

NOTE: The vnfSnapshotInfoId is an attribute of the VnfSnapshotPkgInfo.

Table 6.6.4.1-1: Query VNF Snapshot Package Information operation

Message	Requirement	Direction
QueryVnfSnapshotPkgInfoRequest	Mandatory	VNFM → NFVO
QueryVnfSnapshotPkgInfoResponse	Mandatory	NFVO → VNFM

6.6.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 6.6.4.2-1.

Table 6.6.4.2-1: Query VNF Snapshot Package Information operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the VNF Snapshot Packages on which the query applies, based on attributes of the VnfSnapshotPkgInfo. It can also be used to specify one or more VNF Snapshot Packages to be queried by providing their vnfSnapshotInfoId, vnfcSnapshotInfoId or vnfSnapshotPkgInfoId. See note.
attributeSelector	M	0..N	String	It provides a list of attribute names of VnfSnapshotPkgInfo. If present, only these attributes are returned for the VnfSnapshotPkgInfo matching the filter. If absent, the complete VnfSnapshotPkgInfo is returned.
NOTE: The vnfSnapshotInfoId, assigned by the VNFM at VNF Snapshot creation or at VNF Snapshot Package extraction, identifies the information related to a VNF Snapshot. The vnfSnapshotPkgInfoId identifies the information related to the creation or storage of a VNF Snapshot Package in the NFVO, which implies that it also identifies an available VNF Snapshot Package.				

6.6.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 6.6.4.3-1.

Table 6.6.4.3-1: Query VNF Snapshot Package Information operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	VnfSnapshotPkgInfo	Details of the VNF Snapshot Packages matching the input filter. If attributeSelector is present, only the attributes listed in attributeSelector are returned for the selected entities. Cardinality is 0 if no data is matching the input filter.

6.6.4.4 Operation results

After successful operation, the NFVO has queried the internal VNF Snapshot Package information objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the VNF Snapshot Package that the consumer has access to and that are matching the filter shall be returned.

7 VNFM exposed interfaces

7.1 Introduction

This clause defines the interfaces exposed by the VNFM towards the NFVO over the Or-Vnfm reference point.

NOTE: The fact that information elements and attributes are presented in tabular form does not preclude protocol designs in which these information elements and attributes are encoded in different parts of request and response messages. For example, in a RESTful interface, parts of them can be encoded in the URL, in the message header, in the message body or any combination thereof.

7.2 VNF Lifecycle Management interface

7.2.1 Description

This interface allows the NFVO to invoke VNF lifecycle management operations towards the VNFM.

The following operations are defined:

- Create VNF Identifier.
- Instantiate VNF.
- Scale VNF.
- Scale VNF to Level.
- Change VNF Flavour.
- Terminate VNF.
- Delete VNF Identifier.
- Query VNF.
- Heal VNF.
- Operate VNF.
- Modify VNF Information.
- Get Operation Status.
- Change External VNF connectivity.
- Query VNF Snapshot information.
- Create VNF Snapshot.
- Revert to VNF Snapshot.
- Delete VNF Snapshot information.
- Fetch VNF state snapshot.
- Change current VNF Package.
- Select VNF Deployable Modules.

An identifier (i.e. lifecycleOperationOccurrenceId) is generated for each VNF lifecycle operation occurrence, except for Query VNF, Create VNF Identifier, Delete VNF Identifier and Get Operation Status.

Furthermore, this interface allows the NFVO to manage subscriptions to notifications sent by the VNFM which inform about changes of a VNF instance that are related to VNF lifecycle management operation occurrences, as well as related to the creation/deletion of a VNF instance identifier and the associated instance of a VnfInfo information element. It further allows the VNFM to provide such notifications to the subscriber.

The use of direct-mode of certificate management specified in clause 5.2.3.2 of ETSI GS NFV-IFA 026 [i.15] does not require any certificate specific attributes in the interfaces described in the present document.

7.2.2 Create VNF Identifier operation

7.2.2.1 Description

This operation creates a VNF instance identifier, and an associated instance of a VnfInfo information element, identified by that identifier, in the NOT_INSTANTIATED instantiation state without instantiating the VNF or doing any additional lifecycle operation(s). It allows returning right away a VNF instance identifier that can be used in subsequent lifecycle operations, like the Instantiate VNF operation.

This operation shall be supported for all VNFs.

Table 7.2.2.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.2.1-1: Create VNF Identifier operation

Message	Requirement	Direction
CreateVnfIdentifierRequest	Mandatory	NFVO → VNFM
CreateVnfIdentifierResponse	Mandatory	VNFM → NFVO

7.2.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.2.2-1.

Table 7.2.2.2-1: Create VNF Identifier operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfdId	M	1	Identifier	Identifier that identifies the VNFD which defines the VNF instance to be created. See note.
vnfInstanceName	M	0..1	String	Human-readable name of the VNF instance to be created.
vnfInstanceDescription	M	0..1	String	Human-readable description of the VNF instance to be created.
metadata	M	0..N	KeyValuePair	This parameter provides values for the "metadata" attribute in "VnfInfo".
certificateConfigurationData	CM	0..1	CertificateConfigurationData	Configuration for certificate management such as certificate profile, information of CMF and security policy in this VNF. It shall be present when using delegation-mode, otherwise it shall be absent. CONDITION: This attribute shall be supported when delegation-mode of certificate management is applicable.
NOTE: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2.				

7.2.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.2.3-1.

Table 7.2.2.3-1: Create VNF Identifier operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	VNF instance identifier just created.

7.2.2.4 Operation results

In case of success, an instance of a VnfInfo information element, in the NOT_INSTANTIATED instantiation state, has been created and can be used in subsequent lifecycle operations and the corresponding VnfIdentifierCreationNotification has been sent. In case of failure, appropriate error information is returned.

7.2.3 Instantiate VNF operation

7.2.3.1 Description

This operation instantiates a particular DF of a VNF that has been in the NOT_INSTANTIATED instantiation state, based on the definition in the VNFD.

This operation shall be supported for all VNFs.

Table 7.2.3.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.3.1-1: Instantiate VNF operation

Message	Requirement	Direction
InstantiateVnfRequest	Mandatory	NFVO → VNFM
InstantiateVnfResponse	Mandatory	VNFM → NFVO

7.2.3.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.3.2-1.

Table 7.2.3.2-1: Instantiate VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance.
flavourId	M	1	Identifier	Identifier of the VNF DF to be instantiated.
instantiationLevelId	M	0..1	Identifier	Identifier of the instantiation level of the DF to be instantiated. See notes 3 and 5.
targetScaleLevelInfo	M	0..N	ScaleInfo	This attribute is applicable if VNF supports target scale level instantiation. For each scaling aspect of the current deployment flavour, the attribute specifies the scale level of VNF constituents (e.g. VDU level) to be instantiated. See notes 3, 4 and 5.
extVirtualLink	M	0..N	ExtVirtualLinkData	Information about external VLs to connect the VNF to.
extManagedVirtualLink	M	0..N	ExtManagedVirtualLinkData	Information about internal VLs that are managed by other entities than the VNFM (see notes 1 and 2).

Parameter	Qualifier	Cardinality	Content	Description
vimConnectionInfo	CM	0..N	VimConnectionInfo	Information about VIM or CISM connection(s) for managing resources for the VNF instance, or external/externally-managed virtual links. This attribute shall be supported when the VNF is realized by a set of OS containers. CONDITION: This attribute shall be supported if VNF-related resource management in direct mode is applicable. In that case, this attribute shall be present if there is the need to communicate VIM connection information for external or externally-managed virtual links.
localizationLanguage	M	0..1	Not specified	Localization language of the VNF to be instantiated. The localization languages supported by a VNF can be declared in the VNFD. If this parameter is not provided and the "defaultLocalizationLanguage" attribute is declared in the VNFD, the "defaultLocalizationLanguage" shall be used to determine the localization language VNF to be instantiated.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the instantiation process, specific to the VNF being instantiated as declared in the VNFD (see clause 7.1.5.3 in ETSI GS NFV-IFA 011 [3]).
extension	M	0..N	KeyValuePair	This parameter provides values for the "extension" attribute in "VnfInfo" (see clause 8.5.2).
vnfConfigurableProperty	M	0..N	KeyValuePair	This parameter provides values for the VNF configurable properties attribute in the "VnfInfo" (see clause 8.5.2).
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a selected deployable module. Only VNFCs based on VDUs that belong to deployable modules listed in this attribute are requested to be instantiated.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor. Values can only be provided for resource capacity related attributes that have been defined in the VNFD as being configurable. Furthermore, provided values shall be within the allowed values indicated in the VNFD.

Parameter	Qualifier	Cardinality	Content	Description
NOTE 1: The indication of externally-managed internal VLs is needed in case networks have been pre-configured for use with certain VNFs, for instance to ensure that these networks have certain properties such as security or acceleration features, or to address particular network topologies. The present document assumes that externally-managed internal VLs are managed by the NFVO and created towards the VIM as supported by the virtualised network resource management interface specified in ETSI GS NFV-IFA 005 [i.4]. NOTE 2: It is possible to have several ExtManagedVirtualLinkData for the same VNF internal VL in case of a multi-site VNF spanning several VIMs. The set of ExtManagedVirtualLinkData corresponding to the same VNF internal VL shall indicate so by referencing to the same VnfVirtualLinkDesc and externally-managed multi-site VL instance (refer to clause 8.12.4.2). NOTE 3: The target size for VNF instantiation may be specified in either instantiationLevelId or targetScaleLevelInfo, but not both. If none of the two attributes (instantiationLevelId or targetScaleLevelInfo) are present, the default instantiation level as declared in the VNFD shall be used. NOTE 4: If targetScaleLevelInfo is specified, information provided in targetScaleLevelInfo shall be used for instantiating scalable constituents of the VNF (e.g. VDUs/VLs). For scaling aspects not specified in targetScaleLevelInfo or for the VNF constituents (e.g. VDUs/VLs) that are not scalable, the default instantiation level as declared in the VNFD shall be used for instantiation. NOTE 5: If the referenced instantiationLevel or the targetScaleLevelInfo contain information related to VNFCs that are not going to be instantiated due to the selection of deployable modules, the information is stored in the VNFM for later use and included in the instantiatedVnfInfo. If none of the attributes is present, the information from the defaultInstantiationLevel related to those VNFCs is stored and included in the instantiatedVnfInfo. If, during the lifecycle of the VNF, as a result of a change of the selected deployable modules any of those VNFCs is going to be instantiated, the stored information determines the number of instances, unless the request that triggered the change also contains information about the number of instances.				

7.2.3.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.3.3-1.

Table 7.2.3.3-1: Instantiate VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.3.4 Operation results

In case of success, the VNF has been instantiated and initially configured, and the associated instance of a VnfInfo information element has been updated. The VNF instance is in the INSTANTIATED instantiation state. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.4 Scale VNF operation

7.2.4.1 Description

This operation provides methods to request scaling a VNF in multiple ways:

- horizontal scaling:
  - scale out: adding additional VNFC instances to the VNF to increase capacity;
  - scale in: removing VNFC instances from the VNF, in order to release unused capacity;

- vertical scaling:
  - scale up adding further resources to existing VNFC instances, e.g. increase memory, Central Processing Unit (CPU) capacity or storage size of the virtualisation container hosting a VNFC instance, in order to increase VNF capacity;
  - scale down: removing resources from existing VNFC instances, e.g. decrease memory, CPU capacity or storage size of the virtualisation container hosting a VNFC instance, in order to release unused capacity.

Potentially, different aspects of a VNF can be scaled independently. For example, a VNF could be designed to provide static capacity such as database nodes and dynamic capacity such as query processing nodes. Such a VNF might be scaled with regards to two separate aspects: the 'static capacity' aspect can be scaled by adding VNFCs from VNF Deployment Units (VDUs) defining database nodes, and the 'dynamic capacity' aspect can be scaled by adding VNFCs from VDUs defining query processing nodes.

In complex VNF designs, scaling a VNF often requires adding/removing a number of related VNFC instances of several different types, possibly based on multiple VDUs. For example, in a high availability configuration, it might be required to add in each scaling step a pair of VNFC instances, one in active and one in standby configuration.

The ScaleVnfRequest in the interface allows the consumer to specify the scaling aspect. The scaling aspects valid for a particular VNF are defined in the VNFD. After receiving a scale request, the VNFM will figure out the necessary set of VNFCs and the related set of resources based on VNF-specific rules, for instance using the lifecycle management script associated to the Scale VNF event.

When scaling a VNF for a particular aspect, the number of scaling steps to apply to that aspect can be provided as a parameter. A scaling step is the smallest unit by which a particular aspect of a VNF can be scaled, and is mapped by the VNFM to the addition (or removal) of a certain number of resources, based on one or more VDUs. For each scaling aspect, the maximum scale level is defined in the VNFD. The minimum scale level is assumed as zero; the maximum scale level corresponds to the maximum number of steps that can be performed within this aspect, starting at the minimum scale level (i.e. zero). At each point in time between the completed VNF instantiation and the VNF termination, the "size" of the VNF with regards to a particular aspect can be expressed by the current scale level with regards to that aspect, and can be obtained, among other information, by invoking the "QueryVNF" operation. When the VNF is instantiated, the current scale level is initialized with values that are defined as part of the instantiation level in the VNFD for the associated aspect. Figure 7.2.4.1-1 illustrates the concepts described above.

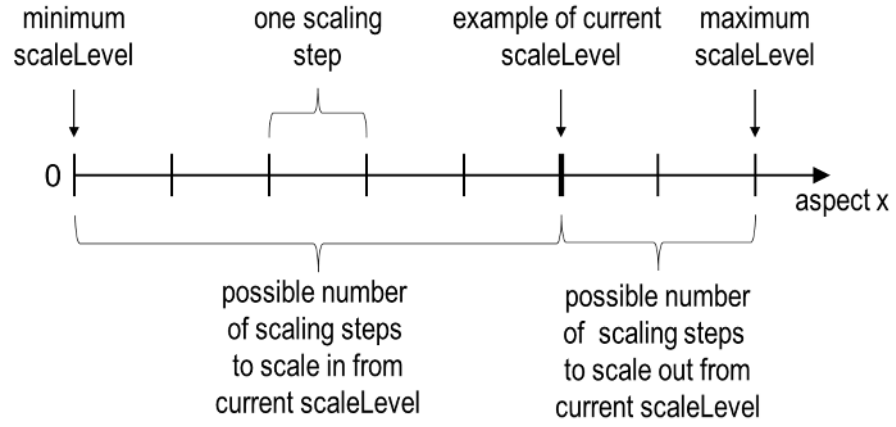


Figure 7.2.4.1-1: Illustrating the concepts of scaleLevel and scaling steps for a particular scaling aspect

The VNFM will then communicate information about the necessary resource changes via the GrantVnfLifecycleOperationRequest to the NFVO.

It depends on the VNF capabilities, and is declared in the VNFD, whether and how this operation is supported for a particular VNF.

Table 7.2.4.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.4.1-1: Scale VNF operation

Message	Requirement	Direction
ScaleVnfRequest	Mandatory	NFVO → VNFM
ScaleVnfResponse	Mandatory	VNFM → NFVO

7.2.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.4.2-1.

Table 7.2.4.2-1: Scale VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to which this scaling request is related.
type	M	1	Enum	Defines the type of the scale operation requested. The set of types actually supported depends on the capabilities of the VNF being managed, as declared in the VNFD. VALUES: • SCALE_OUT • SCALE_IN • SCALE_VERTICAL See note 1.
aspectId	M	0..1	Identifier	Identifies the aspect of the VNF that is requested to be scaled, as declared in the VNFD. It shall be present when 'type' indicates SCALE_OUT or SCALE_IN and absent otherwise.
numberOfSteps	M	0..1	Integer	Number of scaling steps to be executed as part of this ScaleVnf operation. It shall be a positive number. Defaults to 1. The VNF Provider defines in the VNFD whether or not a particular VNF supports performing more than one step at a time. Such a property in the VNFD applies for all instances of a particular VNF. See note 2.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor. The indicated values are absolute (target) values, as opposed to relative (delta) values. Values can only be provided for resource capacity related attributes that have been defined in the VNFD as being configurable. Furthermore, provided values shall be within the allowed values indicated in the VNFD. It shall be present when 'type' indicates SCALE_VERTICAL and absent otherwise.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the scaling process, specific to the VNF being scaled as declared in the VNFD (see clause 7.1.5.4 in ETSI GS NFV-IFA 011 [3]).
NOTE 1: ETSI GS NFV-IFA 010 [2] specifies that the lifecycle management operations that expand or contract a VNF instance include scale in, scale out, scale up and scale down. NOTE 2: A scaling step is the smallest unit by which a VNF can be scaled w.r.t a particular scaling aspect.				

7.2.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.4.3-1.

Table 7.2.4.3-1: Scale VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.4.4 Operation results

In case of success, the VNF has been scaled according to the request, and the associated instance of a VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.5 Scale VNF to Level operation

7.2.5.1 Description

This operation scales an instantiated VNF of a particular DF to a target size. The target size is either expressed as an instantiation level of that DF as defined in the VNFD, or given as a list of scale levels, one per scaling aspect of that DF. Instantiation levels and scaling aspects are declared in the VNFD. Typically, the result of this operation is adding and/or removing Network Functions Virtualisation Infrastructure (NFVI) resources to/from the VNF.

This operation also enables a consumer to request scaling an instantiated VNF of a particular DF to a power profile for the VNF instance. This can be done either by referring to a defined instantiation level or list of scale levels defined in the VNFD (see above) that have an associated power profile or to a specified power profile in the VNFD. With this operation, the power management component of the VNFM selects the power state for the VNF and controls the VNF to switch to the selected power state.

It depends on the VNF capabilities, and is declared in the VNFD, whether this operation is supported for a particular VNF.

Table 7.2.5.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.5.1-1: Scale VNF to Level operation

Message	Requirement	Direction
ScaleVnfToLevelRequest	Mandatory	NFVO → VNFM
ScaleVnfToLevelResponse	Mandatory	VNFM → NFVO

7.2.5.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.5.2-1.

Table 7.2.5.2-1: Scale VNF to Level operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to which this scaling request is related.
instantiationLevelId	M	0..1	Identifier	Identifier of the target instantiation level of the current DF to which the VNF is requested to be scaled. See note.
scaleInfo	M	0..N	ScaleInfo	For each scaling aspect of the current DF, defines the target scale level to which the VNF is to be scaled. The VNF Provider defines in the VNFD whether or not a particular VNF supports scaling according to this parameter. Such a property in the VNFD applies for all instances of a particular VNF. See note.
powerProfileId	M	0..1	Identifier	Identifier of the target power profile in the current DF, as defined in the VNFD, from which the target power state can be selected and be switched to. See note.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the scaling process, specific to the VNF being scaled as declared in the VNFD (see clause 7.1.5.5 in ETSI GS NFV-IFA 011 [3]).
NOTE: Either instantiationLevelId or scaleInfo or powerProfileId shall be present, but not multiple of them. If the referenced instantiationLevel, the scaleInfo or powerProfileId attribute contain information related to VNFCs that are not going to be instantiated due to the selection of deployable modules, the information is stored in the VNFM for later use and included in the instantiatedVnfInfo. If, during the lifecycle of the VNF, as a result of a change of the selected deployable modules any of those VNFCs is going to be instantiated, the stored information determines the number of instances, unless the request that triggered the change also contains information about the number of instances.				

7.2.5.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.5.3-1.

Table 7.2.5.3-1: Scale VNF to Level operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.5.4 Operation results

In case of success, the VNF has been scaled according to the request, and the associated instance of a VnfInfo information element has been updated. Furthermore, in case of success, if the operation input requests to change the power state of the VNF (either directly through corresponding input parameter, or indirectly through instantiation level or list of scale levels), the VNF has been configured and is operating according to the power profile corresponding to the selected power state. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.6 Change VNF Flavour operation

7.2.6.1 Description

This operation changes the DF of a VNF instance.

It depends on the VNF capabilities, and is declared in the VNFD, whether this operation is supported for a particular VNF. This operation may be service-disruptive.

Table 7.2.6.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.6.1-1: Change VNF Flavour operation

Message	Requirement	Direction
ChangeVnfFlavourRequest	Mandatory	NFVO → VNFM
ChangeVnfFlavourResponse	Mandatory	VNFM → NFVO

7.2.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.6.2-1.

Table 7.2.6.2-1: Change VNF Flavour operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to be modified.
newFlavourId	M	1	Identifier	Identifier of the new VNF DF to apply to this VNF instance.
instantiationLevelId	M	0..1	Identifier	Identifier of the instantiation level of the DF to be used. See note 3.
targetScaleLevelInfo	M	0..N	ScaleInfo	This attribute is applicable if VNF supports target scale level instantiation. For each scaling aspect of the current deployment flavour, the attribute specifies the scale level of VNF constituents (e.g. VDU level) to be instantiated. See notes 3 and 4.
extVirtualLink	M	0..N	ExtVirtualLinkData	Information about external VLs to connect the VNF to.
extManagedVirtualLink	M	0..N	ExtManagedVirtualLink Data	Information about internal VLs that are managed by other entities than the VNFM (see notes 1 and 2).
vimConnectionInfo	CM	0..N	VimConnectionInfo	Information about VIM or CISM connection(s) for managing resources for the VNF instance, or external/externally-managed virtual links. This attribute shall be supported when the VNF is realized by a set of OS containers. CONDITION: This attribute shall be supported and present if VNF-related resource management in direct mode is applicable. In that case, this attribute shall be present if there is the need to communicate VIM connection information for external or externally-managed virtual links.

Parameter	Qualifier	Cardinality	Content	Description
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the flavour change process, specific to the VNF being modified as declared in the VNFD (see clause 7.1.5.9 in ETSI GS NFV-IFA 011 [3]).
extension	M	0..N	KeyValuePair	This parameter provides changes to the values for the "extension" attribute in "VnfInfo".
vnfConfigurableProperty	M	0..N	KeyValuePair	This parameter provides changes to the values for the "vnfConfigurableProperty" attribute in "VnfInfo".
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a selected deployable module. Only VNFCs based on VDUs that belong to deployable modules listed in this attribute are requested to be instantiated or preserved if they were already instantiated.
certificateConfigurationData	CM	0..1	CertificateConfigurationData	Configuration for certificate management such as certificate profile and security policy in this VNF. Can be present when setting a new certificate to the VNF instance in delegation-mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode of certificate management is applicable.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor. Values can only be provided for resource capacity related attributes that have been defined in the VNFD as being configurable. Furthermore, provided values shall be within the allowed values indicated in the VNFD.

NOTE 1: The indication of externally-managed internal VLs is needed in case networks have been pre-configured for use with certain VNFs, for instance to ensure that these networks have certain properties such as security or acceleration features, or to address particular network topologies. The present document assumes that externally-managed internal VLs are managed by the NFVO and created towards the VIM as supported by the virtualised network resource management interface specified in ETSI GS NFV-IFA 005 [i.4].

NOTE 2: It is possible to have several ExtManagedVirtualLinkData for the same VNF internal VL in case of a multi-site VNF spanning several VIMs. The set of ExtManagedVirtualLinkData corresponding to the same VNF internal VL shall indicate so by referencing to the same VnfVirtualLinkDesc and externally-managed multi-site VL instance (refer to clause 8.12.4.2).

NOTE 3: The target size for VNF instantiation may be specified in either instantiationLevelId or targetScaleLevelInfo, but not both. If none of the two attributes (instantiationLevelId or targetScaleLevelInfo) are present, the default instantiation level as declared in the VNFD shall be used.

NOTE 4: If targetScaleLevelInfo is specified, information provided in targetScaleLevelInfo shall be used for instantiating scalable constituents of the VNF (e.g. VDUs/VLs). For scaling aspects not specified in targetScaleLevelInfo or for the VNF constituents (e.g.VDUs/VLs) that are not scalable, the default instantiation level as declared in the VNFD shall be used for instantiation.

7.2.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.6.3-1.

Table 7.2.6.3-1: Change VNF Flavour operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.6.4 Operation results

In case of success, the VNF has been modified to use the new DF and initially configured, and the associated instance of a VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.7 Terminate VNF operation

7.2.7.1 Description

This operation terminates a VNF instance that has been in the INSTANTIATED instantiation state.

A VNF can be terminated gracefully or forcefully. Graceful termination means that the VNFM arranges to take the VNF out of service, e.g. by asking the VNF's EM to take the VNF out of service, and only after that shuts down the VNF and releases the resources. Forceful termination means that the VNFM immediately shuts down the VNF and releases the resources. A time interval can be specified for taking the VNF out of service, after which the VNF is shut down if taking it out of service has not completed.

Terminating a VNF instance does not delete the instance of the VnfInfo information element. This operation shall be supported for all VNFs.

Table 7.2.7.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.7.1-1: Terminate VNF operation

Message	Requirement	Direction
TerminateVnfRequest	Mandatory	NFVO → VNFM
TerminateVnfResponse	Mandatory	VNFM → NFVO

7.2.7.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.7.2-1.

Table 7.2.7.2-1: Terminate VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to be terminated.

Parameter	Qualifier	Cardinality	Content	Description
terminationType	M	1	Enum	Signals whether forceful or graceful termination is requested. VALUES: • FORCEFUL • GRACEFUL In case of forceful termination, the VNF is shut down immediately, and resources are released (see note 1). In case of graceful termination, the VNFM first arranges to take the VNF out of service (e.g. involving interaction with EM, using the LCM coordination interface defined in clause 6.4 or 8.3 of ETSI GS NFV-IFA 008 [i.5] or using means out of scope of the present document). Once this was successful, or after a timeout, the VNFM shuts down the VNF and releases the resources.
gracefulTerminationTimeout	M	0..1	TimeDuration	The time interval to wait for the VNF to be taken out of service during graceful termination, before shutting down the VNF and releasing the resources. If not given, it is expected that the VNFM waits for the successful taking out of service of the VNF, no matter how long it takes, before shutting down the VNF and releasing the resources (see note 2). Minimum timeout or timeout range are specified by the VNF Provider (e.g. defined in the VNFD or communicated by other means). Not relevant in case of forceful termination.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the Terminate VNF operation, specific to the VNF being terminated as declared in the VNFD (see clause 7.1.5.7 of ETSI GS NFV-IFA 011 [3]).
NOTE 1: If the VNF is still in service, this can adversely impact network service, and therefore, operator policies apply to determine if forceful termination is allowed in the particular situation. NOTE 2: This implies that no VNF shutdown and resource release will be attempted if taking the VNF out of service fails or hangs.				

7.2.7.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.7.3-1.

Table 7.2.7.3-1: Terminate VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.7.4 Operation results

In case of success, the VNF instance has been terminated and resources used by the VNF have been released, and the associated instance of a VnfInfo information element has been updated. The VNF instance is in the NOT_INSTANTIATED instantiation state. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.8 Delete VNF Identifier operation

7.2.8.1 Description

This operation deletes a VNF instance identifier and the associated instance of a VnfInfo information element in the NOT_INSTANTIATED instantiation state.

This operation shall be supported for all VNFs.

Table 7.2.8.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.8.1-1: Delete VNF Identifier operation

Message	Requirement	Direction
DeleteVnfIdentifierRequest	Mandatory	NFVO → VNFM
DeleteVnfIdentifierResponse	Mandatory	VNFM → NFVO

7.2.8.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.8.2-1.

Table 7.2.8.2-1: Delete VNF Identifier operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	VNF instance identifier to be deleted

7.2.8.3 Output parameters

No output parameter.

7.2.8.4 Operation results

In case of success, the VNF instance identifier and the associated instance of the VnfInfo information element has been deleted and can no longer be used; and the corresponding VnfIdentifierDeletionNotification has been sent. If the VNF instance was not terminated (i.e. the VNF is in INSTANTIATED instantiation state), the operation shall be rejected.

In case of failure, appropriate error information is returned.

7.2.9 Query VNF operation

7.2.9.1 Description

This operation provides information about VNF instances. The applicable VNF instances can be chosen based on filtering criteria, and the information can be restricted to selected attributes.

This operation shall be supported for all VNFs.

Table 7.2.9.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.9.1-1: Query VNF operation

Message	Requirement	Direction
QueryVnfRequest	Mandatory	NFVO → VNFM
QueryVnfResponse	Mandatory	VNFM → NFVO

7.2.9.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.9.2-1.

Table 7.2.9.2-1: Query VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter to select the VNF instance(s) about which information is queried.
attributeSelector	M	0..N	String	Provides a list of attribute names. If present, only these attributes are returned for the VNF instance(s) matching the filter. If absent, the complete information is returned for the VNF instance(s) matching the filter.

7.2.9.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.9.3-1.

Table 7.2.9.3-1: Query VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInfo	M	0..N	VnfInfo	The information items about the selected VNF instance(s) that are returned. If attributeSelector is present, only the attributes listed in attributeSelector are returned for the selected VNF instance(s). See note.
NOTE: The lower cardinality is 0 since there may be no matches to the provided filter.				

7.2.9.4 Operation results

In case of success, information related to the VNF instances that match the filter is returned. In case of failure, appropriate error information is returned.

7.2.10 Heal VNF operation

7.2.10.1 Description

This operation enables the NFVO to request a VNFM to perform a VNF healing procedure.

It depends on the VNF capabilities, and is declared in the VNFD, whether this operation is supported for a particular VNF.

Table 7.2.10.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.10.1-1: Heal VNF operation

Message	Requirement	Direction
HealVnfRequest	Mandatory	NFVO → VNFM
HealVnfResponse	Mandatory	VNFM → NFVO

7.2.10.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.10.2-1.

Table 7.2.10.2-1: Heal VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifies the VNF instance requiring a healing action.
cause	M	0..1	String	Indicates the reason why a healing procedure is required.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the healing process, specific to the VNF being healed as declared in the VNFD (see clause 7.1.5.6 in ETSI GS NFV-IFA 011 [3]). EXAMPLE: Input parameters to VNF-specific healing procedures.
healingResource	M	0..N	Enum	Indicate kinds of the virtual resource to be healed. Values: • VL • LINKPORT • STORAGE • VIRTUALCP • COMPUTE • OSCONTAINER Default value is COMPUTE when the VNF is realized by a virtual resource and OSCONTAINER when the VNF is realized by a set of OS containers.

7.2.10.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.10.3-1.

Table 7.2.10.3-1: Heal VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.10.4 Operation results

In case of success, the VNF has been healed, and the associated instance of a VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.11 Operate VNF operation

7.2.11.1 Description

This operation enables requesting to change the state of a VNF instance, including starting and stopping the VNF instance.

NOTE 1: These operations are complementary to instantiating and terminating a VNF.

NOTE 2: In the present document, only starting and stopping the VNF instance(s) are supported. Extension of this operation to support other VNF state changes is part of the protocol design.

A VNF instance can be in the following states:

- STARTED: the VNF instance is up and running.
- STOPPED: the VNF instance has been shut down. A VNF instance is stopped if all its VNFC instances are also stopped.

In the state STOPPED, the virtualised container(s), where the VNFC instance(s) of the VNF run, are shut down but not terminated. In addition, if the workflow requires a graceful stop, as part of this process the VNFM (producer of the interface) will interact with VNF/EM to gracefully stop the VNF application. Once a VNF is instantiated, i.e. all instantiation steps have been completed, the VNF instance is in the state STARTED.

Figure 7.2.11.1-1 illustrates the VNF operate state diagram. The desired change of state is indicated as an input in the OperateVnfRequest operation.

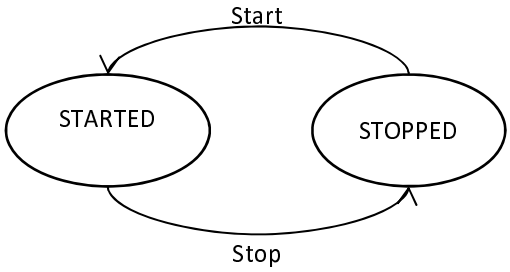


Figure 7.2.11.1-1: Operate VNF state diagram

It depends on the VNF capabilities, and is declared in the VNFD, whether this operation is supported for a particular VNF.

Table 7.2.11.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.11.1-1: Operate VNF operation

Message	Requirement	Direction
OperateVnfRequest	Mandatory	NFVO → VNFM
OperateVnfResponse	Mandatory	VNFM → NFVO

7.2.11.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.11.2-1.

Table 7.2.11.2-1: Operate VNF operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance.
changeStateTo	M	1	Enum	The desired state to change the VNF to. VALUES: • START • STOP
stopType	M	0..1	Enum	It signals whether forceful or graceful stop is requested. VALUES: • FORCEFUL • GRACEFUL In case of forceful stop, the VNF is stopped immediately. Note that if the VNF is still in service, this may adversely impact network service, and therefore, operator policies apply to determine if forceful stop is allowed in the particular situation. In case of graceful stop, the VNFM first arranges to take the VNF out of service (e.g. involving interaction with EM, using the LCM coordination interface defined in clauses 6.4 or 8.3 of ETSI GS NFV-IFA 008 [i.5] or using means out of scope of the present document). Once this is successful, or after a timeout, the VNFM stops the VNF. Only applicable when changing state to stop.
gracefulStopTimeout	M	0..1	TimeDuration	The time interval to wait for the VNF to be taken out of service during graceful stop, before stopping the VNF. If not given, it is expected that the VNFM waits for the successful taking out of service of the VNF, no matter how long it takes, before stopping the VNF (see note). Minimum timeout or timeout range are specified by the VNF vendor (e.g. defined in the VNFD or communicated by other means). The parameter is not relevant in case of forceful stop.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the Operate VNF operation, specific to the VNF being operated as declared in the VNFD (see clause 7.1.5.8 in ETSI GS NFV-IFA 011 [3]).
NOTE: This implies that no VNF stop will be attempted if taking the VNF out of service fails or hangs.				

7.2.11.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.11.3-1.

Table 7.2.11.3-1: Operate VNF operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.11.4 Operation results

In case of success, the state of the VNF has been changed, and the associated instance of a VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.12 Modify VNF Information operation

7.2.12.1 Description

This operation allows updating information about a VNF instance.

This operation shall be supported for all VNFs.

Table 7.2.12.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.12.1-1: Modify VNF Information operation

Message	Requirement	Direction
ModifyVnfInfoRequest	Mandatory	NFVO → VNFM
ModifyVnfInfoResponse	Mandatory	VNFM → NFVO

7.2.12.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.12.2-1.

Table 7.2.12.2-1: Modify VNF Information operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance for which the writeable attributes of VnfInfo are requested to be modified.
newValues	M	1..N	KeyValuePair	Contains the set of attributes to update. The key in the KeyValuePair indicates the name of an attribute that is writable through the interface whose value is to be updated. The value in the KeyValuePair indicates the new attribute value.

7.2.12.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.12.3-1.

Table 7.2.12.3-1: Modify VNF Information operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.12.4 Operation results

In case of success:

- if the operation handles changes to the VNF configurable properties, the configuration in the VNF has been modified according to the input parameters specified in the operation;
- if the operation handles other changes to the VNF instance information, the VNF information has been changed according to the input parameters specified in the operation.

In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification. In particular, error information shall indicate the reason why the requested attribute has not been updated, e.g. changing the value of the attribute is not supported, input attribute name is not recognized, etc.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.13 Get Operation Status operation

7.2.13.1 Description

This operation provides the status of a VNF lifecycle management operation. This means, it is not a VNF lifecycle management operation itself, but an operation on VNF lifecycle management operations. Therefore, this operation shall be supported for all VNFs.

Table 7.2.13.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.13.1-1: Get Operation Status operation

Message	Requirement	Direction
GetOperationStatusRequest	Mandatory	NFVO → VNFM
GetOperationStatusResponse	Mandatory	VNFM → NFVO

7.2.13.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.13.2-1.

Table 7.2.13.2-1: Get Operation Status operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	Identifier of the VNF lifecycle operation occurrence.

7.2.13.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.13.3-1.

Table 7.2.13.3-1: Get Operation Status operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
operationStatus	M	1	Not specified	Indicates the operation status (which includes, for example: Processing, Successfully done, Failed, but can also include operation-specific states).

7.2.13.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.2.14 Subscribe operation

7.2.14.1 Description

This operation enables the NFVO to subscribe with a filter for the notifications sent by the VNFM which are related to VNF lifecycle management operation occurrences, as well as creation/deletion of VNF instance identifiers and the associated VnfInfo information element instances.

NOTE: Specification of filtering mechanism is part of the protocol design.

Table 7.2.14.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.14.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	NFVO → VNFM
SubscribeResponse	Mandatory	VNFM → NFVO

7.2.14.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.14.2-1.

Table 7.2.14.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting e.g. the VNF instances of interest and the specific types of changes. See note.
NOTE: When subscribing for notifications regarding the creation of VNF identifiers and the associated VNF information object instances, selecting the VNF instances in the filter is not possible.				

7.2.14.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.14.3-1.

Table 7.2.14.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription realized.

7.2.14.4 Operation results

After successful subscription, the consumer (NFVO) is registered to receive notifications related to VNF lifecycle management operation occurrences, as well as creation/deletion of VNF instance identifiers and the associated VnfInfo information element instances.

The result of the operation shall indicate if the subscription has been successful or not with a standard success/error result. For a particular subscription, only notifications matching the filter will be delivered to the consumer.

7.2.15 Notify operation

7.2.15.1 Description

This operation notifies a subscriber about events related to VNF lifecycle operation occurrences, as well as creation/deletion of VNF instance identifiers and the associated VnfInfo information element instances.

This operation distributes notifications to subscribers. It is a one-way operation issued by the producer (VNFM) that cannot be invoked as an operation by the consumer (NFVO). In order to receive notifications, the consumer (NFVO) has to perform an explicit Subscribe operation beforehand.

Table 7.2.15.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.15.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	VNFM → NFVO

The following notifications can be notified/sent by this operation:

- VnfLcmOperationOccurrenceNotification (see clause 8.6.2).
- VnfIdentifierCreationNotification (see clause 8.6.7).
- VnfIdentifierDeletionNotification (see clause 8.6.8).

7.2.16 Terminate Subscription operation

7.2.16.1 Description

This operation enables the NFVO to terminate a particular subscription.

Table 7.2.16.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.2.16.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	NFVO → VNFM
TerminateSubscriptionResponse	Mandatory	VNFM → NFVO

7.2.16.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.16.2-1.

Table 7.2.16.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

7.2.16.3 Output parameters

None.

7.2.16.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the NFVO will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

7.2.17 Query Subscription Info operation

7.2.17.1 Description

This operation enables the NFVO to query information about subscriptions.

Table 7.2.17.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.2.17.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	NFVO → VNFM
QuerySubscriptionInfoResponse	Mandatory	VNFM → NFVO

7.2.17.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.17.2-1.

Table 7.2.17.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

7.2.17.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.17.3-1.

Table 7.2.17.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

7.2.17.4 Operation results

After successful operation, the VNFM has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF lifecycle management that the NFVO has access to and that are matching the filter shall be returned.

7.2.18 Change External VNF Connectivity operation

7.2.18.1 Description

This operation enables changing the external connectivity of a VNF instance. The types of changes that this operation supports are:

- Disconnect external CPs that are connected to a particular external VL, and connect them to a different external VL.
- Disconnect external CPs that are connected to a particular external VL.
- Disconnect and delete external CPs that are connected to a particular external VL and that represent sub-ports of a trunk port, i.e. CP instances that are created from external CPDs that have trunk mode configured according to clause 7.1.6.3 in ETSI GS NFV-IFA 011 [3]. If the parent port is exposed as an extCp, the VNFM shall ensure that the parent port is not deleted. If the parent port is exposed as an extCp and there are other subports connected, the VNFM shall ensure that the parent port is not disconnected.

NOTE 1: Subports need not be used for containerized VNFCs. The application container can send and receive IP packets with any VLAN tag as long as the network interface to connect to the secondary container cluster network has been configured appropriately.

- Change the connectivity parameters of the existing external CPs, including changing addresses.

NOTE 2: Depending on the capabilities of the underlying VIM resources, certain changes (e.g. modifying the IP address assignment) might not be supported without deleting the resource and creating another one with the modified configuration.

- Connect CPs to a particular external VL.
- Create new CPs that represent sub-ports of a trunk port, i.e. CP instances that are created from external CPDs that have trunk mode configured according to clause 7.1.6.3 in ETSI GS NFV-IFA 011 [3], and connect them to a particular external VL. Creation of the parent port with this operation is not supported.

NOTE 3: Subports need not be used for containerized VNFCs. The application container can send and receive IP packets with any VLAN tag as long as the network interface to connect to the secondary container cluster network has been configured appropriately.

- Create new external CPs, which are not sub-ports of a trunk port, and connect them to an existing external VL.
- Disconnect and delete external CPs, which are not sub-ports of a trunk port, that are connected to a particular external VL.

VNFs shall support this operation. This operation may be service-disruptive.

Table 7.2.18.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.2.18.1-1: Change External VNF Connectivity

Message	Requirement	Direction
ChangeExtVnfConnectivityRequest	Mandatory	NFVO → VNFM
ChangefExtVnfConnectivityResponse	Mandatory	VNFM → NFVO

7.2.18.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.18.2-1. The parameters passed for this operation override those passed at instantiation time.

Table 7.2.18.2-1: Change External VNF Connectivity operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance.
extVirtualLink	M	1..N	ExtVirtualLinkData	Information about external VLs to change (e.g. connect the VNF to).
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the Change External VNF Connectivity operation, specific to the VNF of which the external VLs are being changed as declared in the VNFD (see clause 7.1.5.10 in ETSI GS NFV-IFA 011 [3]).

Parameter	Qualifier	Cardinality	Content	Description
vimConnectionInfo	CM	0..N	VimConnectionInfo	Information about VIM or CISM connection(s) for managing resources for the VNF instance, or external virtual links. This attribute shall be supported when the VNF is realized by a set of OS containers. CONDITION: This attribute shall be supported if VNF-related resource management in direct mode is applicable. In that case, this attribute shall be present if there is the need to communicate VIM connection information for external virtual links.
certificateConfigurationData	CM	0..1	CertificateConfigurationData	Configuration for certificate management such as certificate profile and security policy in this VNF. Can be present when setting a new certificate to the VNF instance in delegation mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode of certificate management is applicable.

The following behaviour applies for the changes that can be performed with this operation:

- To change the connection of external CP instances based on certain external CPDs from a "source" external VL to a different "target" external VL, the identifier of the "target" external VL shall be sent in the "extVirtualLinkId" attribute of the "extVirtualLink" parameter, and the "extCp" attributes of that parameter shall refer via the "cpdId" attribute to the external CPDs of the corresponding external connection point instances that are to be reconnected to the target external VL.

NOTE 1: For CP instances that are not part of a trunk, this means that all CP instances based on a given external CPD will be reconnected. See clause A.3 in annex A for an illustration. For CP instances that are part of a trunk the change of connectivity can be requested individually per CP instance.

- To change the connectivity parameters of the external CPs connected to a particular external VL, including changing addresses, the identifier of that external VL shall be sent in the "extVirtualLinkId" attribute of the "extVirtualLink" parameter, and the "extCp" attribute of that parameter shall contain at least those entries with modified parameters.

- To create a new external CP instance, this shall be based on a certain external CPD that is referenced by at least a VDU from which the VNF instance has at least one VNFC instantiated. The newly external CP instance connects to the same external VL as other CP instances created based on the same CPD. For creating a new external CP instance:

  - The "extVirtualLinkId" attribute of the "extVirtualLink" parameter indicates the identifier of the external VL instance to which the new external CP instance is requested to be connected (refer to above behavior).

  - The "extCp" attribute shall refer via the "cpdId" attribute to the external CPDs from which a new instance is to be created.

NOTE 2: For the capability to connect to different external VLs, refer to the use of subports.

- To delete an existing external CP instance:

  - The "extVirtualLinkId" attribute of the "extVirtualLink" parameter indicates the identifier of the external VL instance from which the existing external CP instance is requested to be disconnected.

  - The "extCpDeletion" attribute shall refer via the "cpdId" attribute to the external CPD from which a corresponding CP instance will be deleted.

  - The "cpInstanceId" of the contained "cpConfig" in the "extCp" shall refer to the particular external CP instance to be deleted.

NOTE 3: If external CPs are requested to be created and connected, and disconnected and deleted in a same operation request, the consumer can provide respective "extVirtualLink", e.g., one for the external CPs to be created and connected, and one for the external CPs to disconnect and delete.

7.2.18.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.18.3-1.

Table 7.2.18.3-1: Change External VNF Connectivity operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.18.4 Operation results

In the case of success, the connectivity of the VNF has been changed according to the input parameters, and the associated instance of a VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.19 Query Snapshot Information operation

7.2.19.1 Description

This operation enables the NFVO to query the VNFM for information it has stored about one or more VNF Snapshots. Table 7.2.19.1-1 lists the information flow exchanged between the VNFM and the NFVO.

The operation allows querying specific components of the information stored in the VNFM about a VNF Snapshot, for instance, retrieving the vnfSnapshotInfoId.

This operation shall be supported by the VNF if the Create Snapshot Operation is supported.

Table 7.2.19.1-1: Query Snapshot Information operation

Message	Requirement	Direction
QuerySnapshotInfoRequest	Mandatory	NFVO → VNFM
QuerySnapshotInfoResponse	Mandatory	VNFM → NFVO

7.2.19.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.19.2-1.

Table 7.2.19.2-1: Query Snapshot Information operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the VNF Snapshot on which the query applies, based on attributes of the VnfSnapshotInfo. It can also be used to specify one or more VNF Snapshots to be queried by providing their vnfSnapshotInfoId or vnfInstanceId. See note.
attributeSelector	M	0..N	String	It provides a list of attribute names of VnfSnapshotInfo. If present, only these attributes are returned for the VnfSnapshotInfo matching the filter. If absent, the complete VnfSnapshotInfo is returned.
NOTE: The vnfSnapshotInfoId, assigned at VNF Snapshot creation or at VNF Snapshot Package extraction, identifies the information related to a VNF Snapshot.				

7.2.19.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.19.3-1.

Table 7.2.19.3-1: Query Snapshot Information operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotInfo	M	0..N	VnfSnapshotInfo	Details of the VNF Snapshots available to the NFVO matching the input filter. If attributeSelector is present, only the attributes listed in attributeSelector are returned for the selected entities. The lower cardinality is 0 since there may be no matches to the provided filter.

7.2.19.4 Operation results

After successful operation, the VNFM has queried the internal VNF Snapshot information objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the VNF Snapshot that the consumer has access to and that are matching the filter shall be returned.

7.2.20 Create Snapshot operation

7.2.20.1 Description

This operation enables the NFVO to request the creation of a VNF Snapshot. The VNF instance to be snapshotted is addressed using an identifier held by the VNFM about a specific VNF instance.

It depends on the VNF capabilities, and is declared in the VNFD (refer to the "supportedOperation" attribute in the VnfDf information element; see clause 7.1.8.2 in ETSI GS NFV-IFA 011 [3]), whether this operation is supported for a particular VNF.

Table 7.2.20.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.20.1-1: Create Snapshot operation

Message	Requirement	Direction
CreateSnapshotRequest	Mandatory	NFVO → VNFM
CreateSnapshotResponse	Mandatory	VNFM → NFVO

7.2.20.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.20.2-1.

Table 7.2.20.2-1: Create Snapshot operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to be snapshotted.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the snapshot creation process, specific for the VNF being "snapshotted" as declared in the VNFD (see clause 7.1.5.11 in ETSI GS NFV-IFA 011 [3]).
userDefinedData	O	0..N	KeyValuePair	User defined data for the VNF Snapshot.

7.2.20.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.20.3-1.

Table 7.2.20.3-1: Create Snapshot operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotInfoId	M	1	Identifier (Reference to VnfSnapshotInfo)	Identifier of information held by the VNFM about the specific VNF Snapshot. This identifier was allocated by the VNFM if the create operation was successful.

7.2.20.4 Operation results

The result of the operation indicates if the creation of the VNF Snapshot has been successful or not with a standard success/error result.

Before the VNF Snapshot creation starts and ends, the VNF/EM have been notified about the operation via VnfLcmOperationOccurrenceNotification in case they have subscribed for this type of notifications.

After successful operation, the VNFM has created a VNF Snapshot from the specified VNF instance, including information associated with this VNF Snapshot. Once created, the VNF Snapshot is known to the VNFM. It is enabled to be queried for its associated information, it is enabled to be reverted to, and it is enabled to create a VNF Snapshot Package from it.

7.2.21 Revert-to Snapshot operation

7.2.21.1 Description

This operation enables the NFVO to request the reversion of a VNF instance to a VNF Snapshot. The VNF Snapshot to be reverted to is addressed using an identifier held by the VNFM about a specific VNF Snapshot.

It depends on the VNF capabilities, and is declared in the VNFD (refer to the "supportedOperation" attribute in the VnfDf information element; see clause 7.1.8.2 in ETSI GS NFV-IFA 011 [3]), whether this operation is supported for a particular VNF. This operation may be service-disruptive.

Table 7.2.21.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.21.1-1: Revert-to Snapshot operation

Message	Requirement	Direction
RevertToSnapshotRequest	Mandatory	NFVO → VNFM
RevertToSnapshotResponse	Mandatory	VNFM → NFVO

7.2.21.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.21.2-1.

Table 7.2.21.2-1: Revert-to Snapshot operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to be reverted.
vnfSnapshotInfoId	M	1	Identifier (Reference to VnfSnapshotInfo)	Identifier of information held by the VNFM about the VNF Snapshot to be reverted to. This identifier was allocated by the VNFM.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the snapshot reversion process, specific for the VNF instance being "reverted" to snapshot as declared in the VNFD (see clause 7.1.5.12 in ETSI GS NFV-IFA 011 [3]).

7.2.21.3 Output parameters

No output parameter.

7.2.21.4 Operation results

The result of the operation indicates if the reversion to the VNF Snapshot has been successful or not with a standard success/error result.

Before the VNF Snapshot reversion starts and ends, the VNF/EM have been notified about the operation via VnfLcmOperationOccurrenceNotification in case they have subscribed for this type of notifications.

7.2.22 Delete Snapshot Information operation

7.2.22.1 Description

This operation enables the NFVO to request the deletion of the held information associated to a VNF Snapshot. The VNF Snapshot information to be deleted is addressed using an identifier of information held by the VNFM about a specific VNF Snapshot.

It depends on the VNF capabilities, and is declared in the VNFD (refer to the "supportedOperation" attribute in the VnfDf information element; see clause 7.1.8.2 in ETSI GS NFV-IFA 011 [3]), whether this operation is supported for a particular VNF.

Table 7.2.22.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.22.1-1: Delete Snapshot Information operation

Message	Requirement	Direction
DeleteSnapshotInfoRequest	Mandatory	NFVO → VNFM
DeleteSnapshotInfoResponse	Mandatory	VNFM → NFVO

7.2.22.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.22.2-1.

Table 7.2.22.2-1: Delete Snapshot Information operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotInfoId	M	1	Identifier (Reference to VnfSnapshotInfo)	Identifier of information held by the VNFM about a specific VNF Snapshot. This identifier was allocated by the VNFM.

7.2.22.3 Output parameters

No output parameter.

7.2.22.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

After successful operation, the VNFM has deleted the held information associated to the specified VNF Snapshot. After deletion of the held information associated to a VNF Snapshot, this VNF Snapshot is not known any longer to the VNFM and it is not possible to revert to this VNF Snapshot or to create a VNF Snapshot Package from it.

7.2.23 Change current VNF package operation

7.2.23.1 Description

This operation enables the NFVO to request the VNFM to change the current VNF Package, i.e. the VNF package on which a VNF instance is based.

Refer to ETSI GS NFV-REL 006 [i.10] for more information related to the types of changes to the current VNF Package (a.k.a "VNF software modification"). Clause B.3 of the present document illustrates the variants of changes to the current VNF Package and information flow procedures.

This operation encompasses the following scenarios:

- Changes of the VNF virtualised resources, such as requirements, composition and structure between the VNF versions, without changing the VNF software version.
- Changes of both the VNF software version and the VNF virtualised resources. This case includes replacing the VNF software version by means of virtualised resources management, such as terminating the current virtualised resource instances running the current software version and instantiating new virtualised resource instances with the destination VNF software version. The new virtualised resource instances may have the same characteristics as the current virtualised resource instances.
- Changes related to the VNFD, such as correction of bugs in the VNFD, changes in the naming scheme of VNFD components (e.g. name of the VDU, vduId), and adding/removing descriptors of VNF Package changes (VnfPackageChangeInfo).

NOTE: For software updates that are executed by functional entities outside NFV-MANO and that require synchronization of the information held by the NFV-MANO entities with a new VNF package that reflects the same changes, a separate procedure using the Modify VNF Information operation has been defined, as illustrated in clause B.2. This procedure assumes certain restrictions on the characteristics of the new VNF package, as defined in note 4 in table 8.5.2.2-1.

As part of changing the current VNF Package, the VNFM shall be capable to add temporary virtualised resources used in the modification process, e.g. virtualised resources for a VNFC which will be responsible for handling or supporting the change of the current VNF Package process. The need for temporary virtualised resources shall be indicated as "tempResource" to the NFVO during the VNF LCM operation granting exchange. In addition, the VNFM shall be capable to add and remove virtualised resources as required for the "change of current VNF Package" process. The need for addition and removal of existing virtualised resources shall be indicated as "addResource" and "removeResource" in the VNF LCM operation granting exchange.

The following applies to the existing resources of the VNF instance: in the course of the successful execution of this operation, the VNFM shall replace or update those resources of the VNF instance that are based on descriptors (e.g. VDUs, VLDs, CPDs) that have changed between source and destination VNFD to align them with the updated descriptors, with the only allowed exception that the references to software images need not be updated if the resources are not replaced. Further, the VNFM shall remove resources that relate to descriptors in the source VNFD that have no corresponding descriptor in the destination VNFD. For newly-created resources, the VNFM shall use the descriptors of the destination VNFD.

All VNFs shall support this operation. This operation may be service-disruptive. It is declared in the VNFD (refer to the "selector" attribute in the " VnfPackageChangeInfo" information element; see clause 7.1.15.2.2 in ETSI GS NFV-IFA 011 [3]), whether a change from a particular "source" VNF package to a particular "destination" VNF package is possible.

Table 7.2.23.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.2.23.1-1: Change current VNF Package operation

Message	Requirement	Direction
ChangeCurrentVnfPackageRequest	Mandatory	NFVO → VNFM
ChangeCurrentVnfPackageResponse	Mandatory	VNFM → NFVO

7.2.23.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.23.2-1.

Table 7.2.23.2-1: Change current VNF Package operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance to which this change request is related.
vnfdId	M	1	Identifier	Identifier of the VNFD which defines the destination VNF Package for the change.
extVirtualLink	M	0..N	ExtVirtualLinkData	Information about external VLs.
extManagedVirtualLink	M	0..N	ExtManagedVirtualLinkData	Information about internal VLs that are managed by other entities than the VNFM (see notes 1 and 2).
vimConnectionInfo	CM	0..N	VimConnectionInfo	Information about VIM or CISM connection(s) for managing resources for the VNF instance, or external/externally-managed virtual links. This attribute shall be supported when the VNF is realized by a set of OS containers. CONDITION: This attribute shall be supported if VNF-related resource management in direct mode is applicable. In that case, this attribute shall be present if there is the need to communicate modified VIM connection information for external or externally-managed virtual links.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the modification process, specific to the VNF, whose VNF Package is requested to be changed, as declared in the VNFD (see clause 7.1.5.13 in ETSI GS NFV-IFA 011 [3]).
extension	M	0..N	KeyValuePair	This parameter provides changes to the values for the "extension" attribute in "VnfInfo", including new values for extensions that are declared in the VNFDs of both the source and the destination VNF Packages (see clause 7.1.14.2 in ETSI GS NFV-IFA 011 [3]) and values for new extensions declared in the VNFD of the destination VNF Package.

Parameter	Qualifier	Cardinality	Content	Description
vnfConfigurableProperties	M	0..N	KeyValuePair	This parameter provides changes to the values for the "vnfConfigurableProperties" attribute in "VnfInfo", including new values for configurable properties that are declared in the VNFDs of both the source and the destination VNF Packages (see clause 7.1.12.2 in ETSI GS NFV-IFA 011 [3]) and values for new configurable properties declared in the VNFD of the destination VNF Package.
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a selected deployable module. If this attribute is present only VNFCs based on VDUs that belong to deployable modules listed in this attribute are requested to be instantiated or preserved if they were already instantiated. If this attribute is not present the deployable modules that were selected before the operation, and that still are defined in the VNFD in the destination package, or the corresponding ones according to the component mappings, remain valid. See note 3.
certificateConfigurationData	CM	0..1	CertificateConfigurationData	Configuration for certificate management such as certificate profile and security policy in this VNF. Can be present when setting a new certificate to the VNF instance in delegation mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode of certificate management is applicable.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor. Values can only be provided for resource capacity related attributes that have been defined in the VNFD as being configurable. Furthermore, provided values shall be within the allowed values indicated in the VNFD.
upgradeType	M	0..1	Enum	Indicate upgrade type when change the current VNF Package on which a VNF instance is based. Values: • ROLLING_UPGRADE • BLUE_GREEN • etc.
NOTE 1: The indication of externally-managed internal VLs is needed in case networks have been pre-configured for use with certain VNFs, for instance to ensure that these networks have certain properties such as security or acceleration features, or to address particular network topologies. The present document assumes that externally-managed internal VLs are managed by the NFVO and created towards the VIM as supported by the virtualised network resource management interface specified in ETSI GS NFV-IFA 005 [i.4].				

Parameter	Qualifier	Cardinality	Content	Description
NOTE 2: It is possible to have several ExtManagedVirtualLinkData for the same VNF internal VL in case of a multi-site VNF spanning several VIMs. The set of ExtManagedVirtualLinkData corresponding to the same VNF internal VL shall indicate so by referencing to the same VnfVirtualLinkDesc and externally-managed multi-site VL instance (refer to clause 8.12.4.2). NOTE 3: Component mappings are defined in the VNFD in the source or destination package for the relevant change path. See clause 7.1.15.2 in ETSI GS NFV-IFA 011 [3].				

7.2.23.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.23.3-1.

Table 7.2.23.3-1: Change current VNF Package operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.23.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

After successful operation, the requested change of the current VNF Package has been completed on the VNF instance, and the associated instance of the VnfInfo information element has been updated. In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.2.24 Fetch VNF state snapshot

7.2.24.1 Description

As part of a VNF snapshot creation, VNF-specific state data associated to the VNF snapshot might be created by the VNFM. Such data can be used during VNF snapshot reversions, root cause analysis, etc. and might need to be also compiled by the NFVO into a VNF snapshot package.

This operation enables the NFVO to fetch the content of a VNF state snapshot from the VNFM.

Table 7.2.24.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.2.24.1-1: Fetch VNF state snapshot operation

Message	Requirement	Direction
FetchVnfStateSnapshotRequest	Mandatory	NFVO → VNFM
FetchVnfStateSnapshotResponse	Mandatory	VNFM → NFVO

7.2.24.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.24.2-1.

Table 7.2.24.2-1: Fetch VNF state snapshot operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfSnapshotInfoId	M	1	Identifier (Reference to VnfSnapshotInfo)	References information held by the VNFM about a specific VNF Snapshot.

7.2.24.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.24.3-1.

Table 7.2.24.3-1: Fetch VNF state snapshot operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfStateSnapshot	M	1	Not specified	VNF state snapshot.

7.2.24.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

After successful operation, the NFVO has fetched the requested VNF state snapshot from the VNFM. In case of failure, appropriate error information is returned.

7.2.25 Select VNF deployable modules operation

7.2.25.1 Description

This operation enables the NFVO to perform a selection of deployable modules for a running VNF instance and, by that, modify a previously indicated selection. Table 7.2.25.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.2.25.1-1: Select VNF deployable modules operation

Message	Requirement	Direction
SelectVnfDeployableModulesRequest	Mandatory	NFVO → VNFM
SelectVnfDeployableModulesResponse	Mandatory	VNFM → NFVO

7.2.25.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.2.25.2-1.

Table 7.2.25.2-1: Select VNF deployable modules operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance on which the selection is requested.
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a selected deployable module, as defined in the VNFD. VNFCs based on VDUs that belong to deployable modules listed in this attribute will be instantiated if not already instantiated. VNFCs based on VDUs that belong to deployable modules not listed in this attribute and that were already instantiated will be terminated.
targetScaleLevelInfo	M	0..N	ScaleInfo	Defines the target scale levels of scaling aspects of the VDUs that belong to selected deployable modules.

Parameter	Qualifier	Cardinality	Content	Description
				If this attribute is not present or if there are VDUs that belong to selected deployable modules that take no part in any of the scaling aspects indicated in this attribute, the VNFCs based on those VDUs shall be instantiated according to the currently valid VNF scale level or instantiation level. This attribute should only contain scale level information of scaling aspects associated to VDUs that will be used to instantiate VNFCs as a result of this operation. If it contains other scale level information it shall be ignored. See note 1. The VNF Provider defines in the VNFD whether or not a particular VNF supports scaling according to this parameter. Such a property in the VNFD applies for all instances of a particular VNF.
resourceCapacity Definition	M	0..N	ResourceCapa cityDefinition	Indicates values for resource capacity related attributes pertaining to a descriptor. Values can only be provided for resource capacity related attributes that have been defined in the VNFD as being configurable. Furthermore, provided values shall be within the allowed values indicated in the VNFD. This attribute should only contain information about resource capacity related attributes of VDUs that will be used to instantiate VNFCs as a result of this operation. If it contains information about attributes of other VDUs it shall be ignored. See note 2.
additionalParam	M	0..N	KeyValuePair	Additional parameters passed by the NFVO as input to the selection of deployable modules process, specific to the VNF being affected by the operation as declared in the VNFD (see clause 7.1.5.14 in ETSI GS NFV-IFA 011 [3]).
NOTE 1: Thus, the select VNF deployable modules operation cannot be used as a scale VNF operation to horizontally scale VNFCs that were already instantiated. NOTE 2: Thus, the select VNF deployable modules operation cannot be used as a scale VNF operation to vertically scale VNFCs that were already instantiated.				

7.2.25.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.2.25.3-1.

Table 7.2.25.3-1: Select VNF deployable modules operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle operation occurrence.

7.2.25.4 Operation results

In case of success, the VNFCs based on VDUs that belong to the selected deployable modules have been instantiated, if they were not already. VNFCs based on VDUs that belong to deployable modules that have not been selected in the operation and that were already instantiated have been terminated.

In case of failure, appropriate error information is provided in the "result" LCM Operation Occurrence Notification.

The VNFM shall return a lifecycleOperationOccurrenceId that identifies the LCM operation. The LCM operation shall trigger the sending of the "start" LCM Operation Occurrence Notification before additional notifications as part of this operation are triggered, or operations towards the NFVO or VIM are invoked.

On successful as well as unsuccessful completion of the operation, the VNFM shall send the "result" LCM Operation Occurrence Notification.

7.3 Void

7.4 VNF Performance Management interface

7.4.1 Description

This interface allows providing performance management (measurement results collection and notifications) related to VNFs. Performance information on a given VNF related measured object instance (see note 1) results from performance information of the virtualised resources that is collected from the VIM and mapped to this VNF related measured object instance.

NOTE 1: The VNF related measured object instance is the instance of one of the measured object type(s) for which the performance measurements applicable to Or-Vnfm reference point are defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].

Collection and reporting of performance information is controlled by a PM job that groups details of performance collection and reporting information.

When new performance information is available, the consumer is notified using the notification PerformanceInformationAvailableNotification (see clause 8.7.8). The details of the performance measurements are provided using the PerformanceReport information element (see clause 8.7.5).

NOTE 2: Delivery mechanism for the performance reports is not specified in the present document.

The following operations are defined for this interface which will be consumed by the NFVO:

- Create PM Job operation.
- Delete PM Jobs operation.
- Subscribe operation.
- Notify operation.
- Query PM Job operation.
- Create Threshold operation.
- Delete Thresholds operation.
- Query Threshold operation.
- Terminate Subscription operation.
- Query Subscription Info operation.

7.4.2 Create PM Job operation

7.4.2.1 Description

This operation will create a PM job, enabling an NFVO to specify a one or more measured object(s) related to VNF, that the VNFM is managing, for which it wants to receive performance information. This will allow the requesting NFVO to specify its performance information requirements with the VNFM.

The VNFM needs to be subscribed to receive PerformanceInformationAvailable notifications in order to know when new collected performance information is available.

Table 7.4.2.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.2.1-1: Create PM Job operation

Message	Requirement	Direction
CreatePmJobRequest	Mandatory	NFVO → VNFM
CreatePmJobResponse	Mandatory	VNFM → NFVO

7.4.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.2.2-1.

Table 7.4.2.2-1: Create PM Job operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
objectSelector	M	1	ObjectSelection	Defines the VNF related measured object(s) for which performance information is requested to be collected.
performanceMetric	M	0..N	String	This defines the type of performance metric(s) for the specified measured object(s). At least one of the two attributes (performance metric or group) shall be present.
performanceMetricGroup	M	0..N	String	Group of performance metrics. A metric group is a pre-defined list of metrics, known to the producer that it can decompose to individual metrics. At least one of the two attributes (performance metric or group) shall be present.
collectionPeriod	M	1	Not specified	Specifies the periodicity at which the VNFM will collect performance information (see note).
reportingPeriod	M	1	Not specified	Specifies the periodicity at which the VNFM will report to the NFVO about performance information (see note).
reportingBoundary	O	0..1	Not specified	Identifies a boundary after which the reporting will stop. The boundary shall allow a single reporting as well as periodic reporting up to the boundary.
NOTE: At the end of each reportingPeriod, the VNFM will inform NFVO about availability of the performance data collected for each completed collection period during this reportingPeriod. While the exact definition of the types for collectionPeriod and reportingPeriod is part of the protocol design, it is recommended that the reportingPeriod be equal or a multiple of the collectionPeriod. In the latter case, the performance data for the collection periods within one reporting period would be reported together.				

7.4.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.2.3-1.

Table 7.4.2.3-1: Create PM Job operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
pmJobId	M	1	Identifier	Identifier of the created PM job.

7.4.2.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

The pmJobId is returned when the operations has been successful.

7.4.3 Delete PM Jobs operation

7.4.3.1 Description

This operation will delete one or more PM job(s).

Table 7.4.3.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.3.1-1: Delete PM Jobs operation

Message	Requirement	Direction
DeletePmJobsRequest	Mandatory	NFVO → VNFM
DeletePmJobsResponse	Mandatory	VNFM → NFVO

7.4.3.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.3.2-1.

Table 7.4.3.2-1: Delete PM Jobs operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
pmJobId	M	1..N	Identifier	Identifiers of the PM jobs to be deleted.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to delete multiple PM Jobs in one request, or as a series of requests that delete one PM Job at a time.				

7.4.3.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.3.3-1.

Table 7.4.3.3-1: Delete PM Jobs operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
deletedPmJobId	M	1..N	Identifier	Identifiers of the PM jobs successfully deleted.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to delete multiple PM Jobs in one request, or as a series of requests that delete one PM Job at a time.				

7.4.3.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.4.4 Subscribe operation

7.4.4.1 Description

This operation enables the NFVOs to subscribe with a filter for the notifications related to performance information with the VNFM.

NOTE 1: Specification of filtering mechanism is part of the protocol design.

NOTE 2: It is part of the protocol design whether subscribing is represented as a separate "Subscribe" operation or whether subscription-related information is managed as part of managing PM jobs and Thresholds.

Table 7.4.4.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.4.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	NFVO → VNFM
SubscribeResponse	Mandatory	VNFM → NFVO

7.4.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.4.2-1.

Table 7.4.4.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting notifications. The filter can be on VNF, type of notification or attribute of the notification.

7.4.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.4.3-1.

Table 7.4.4.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription returned.

7.4.4.4 Operation results

As a result of this operation, the VNFM shall indicate to the NFVO in the subscribeResponse message whether the subscription was successful or not.

For a particular subscription, only notifications matching the filter will be delivered to the consumer.

7.4.5 Notify operation

7.4.5.1 Description

This operation distributes notifications to subscribers. It is a one-way operation issued by the VNFM that cannot be invoked as an operation by the consumer (NFVO). In order to receive notifications, the NFVO shall have a subscription.

Table 7.4.5.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.5.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	VNFM → NFVO

The following notifications can be notified/sent by this operation:

- PerformanceInformationAvailableNotification (see clause 8.7.8).
- ThresholdCrossedNotification (see clause 8.7.9).

7.4.6 Query PM Job operation

7.4.6.1 Description

This operation will enable the NFVO to solicit from the VNFM the details of one or more PM job(s).

This operation is not returning performance reports.

Table 7.4.6.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.6.1-1: Query PM Job operation

Message	Requirement	Direction
QueryPmJobRequest	Mandatory	NFVO → VNFM
QueryPmJobResponse	Mandatory	VNFM → NFVO

7.4.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.6.2-1.

Table 7.4.6.2-1: Query PM Job operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the PM Jobs on which the query applies. It can be a single identifier, multiple identifiers or a wildcard.

7.4.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.6.3-1.

Table 7.4.6.3-1: Query PM Job operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
pmJob	M	0..N	PmJob	Details of PM jobs matching the input filter.

7.4.6.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.4.7 Create Threshold operation

7.4.7.1 Description

This operation will allow the NFVO to create a threshold to specify threshold levels on specified performance metric and VNF related measured object(s) for which notifications will be generated when crossed.

Creating a threshold does not trigger collection of metrics. In order for the threshold to be active, there needs to be a PM job collecting the needed metric for the selected entities.

Table 7.4.7.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.7.1-1: Create Threshold operation

Message	Requirement	Direction
CreateThresholdRequest	Mandatory	NFVO → VNFM
CreateThresholdResponse	Mandatory	VNFM → NFVO

7.4.7.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.7.2-1.

Table 7.4.7.2-1: Create Threshold operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
objectSelector	M	1	ObjectSelection	Defines the VNF related measured object(s) for which the threshold will be defined.
performanceMetric	M	1	String	Defines the performance metric on which the threshold will be defined.
thresholdType	M	1	Enum	Defines the type of threshold. The list of possible values is part of the protocol design and might include: single/multi valued threshold, static/dynamic threshold, template based threshold, etc. VALUES: • SIMPLE: Single-valued static threshold • Etc.
thresholdDetails	M	1	Not specified	Details of the threshold: value to be crossed, and direction in which it is crossed, details on the notification to be generated, etc.

7.4.7.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.7.3-1.

Table 7.4.7.3-1: Create Threshold operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
thresholdId	M	1	Identifier	Identifier of created threshold.

7.4.7.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

The thresholdId is returned when the operations has been successful.

7.4.8 Delete Thresholds operation

7.4.8.1 Description

This operation will allow the NFVO to delete one or more existing threshold(s).

Table 7.4.8.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.8.1-1: Delete Thresholds operation

Message	Requirement	Direction
DeleteThresholdsRequest	Mandatory	NFVO → VNFM
DeleteThresholdsResponse	Mandatory	VNFM → NFVO

7.4.8.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.8.2-1.

Table 7.4.8.2-1: Delete Thresholds operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
thresholdId	M	1..N	Identifier	Identifiers of the thresholds to be deleted.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to delete multiple thresholds in one request, or as a series of requests that delete one threshold at a time.				

7.4.8.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.8.3-1.

Table 7.4.8.3-1: Delete Thresholds operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
deletedThresholdId	M	1..N	Identifier	Identifiers of the thresholds that have been deleted successfully.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to delete multiple thresholds in one request, or as a series of requests that delete one threshold at a time.				

7.4.8.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.4.9 Query Threshold operation

7.4.9.1 Description

This operation will allow the NFVO to query the details of an existing threshold.

Table 7.4.9.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.4.9.1-1: Query Threshold operation

Message	Requirement	Direction
QueryThresholdRequest	Mandatory	NFVO → VNFM
QueyThresholdResponse	Mandatory	VNFM → NFVO

7.4.9.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.9.2-1.

Table 7.4.9.2-1: Query Threshold operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the thresholds on which the query applies. It can be a single identifier, multiple identifiers or a wildcard.

7.4.9.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.9.3-1.

Table 7.4.9.3-1: Query Threshold operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
threshold	M	0..N	Threshold	List of threshold details matching the input filter.

7.4.9.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.4.10 Terminate Subscription operation

7.4.10.1 Description

This operation enables the NFVO to terminate a particular subscription.

NOTE: It is part of the protocol design whether terminating a subscription is represented as a separate "Terminate Subscription" operation or whether subscription-related information is managed as part of managing PM jobs and Thresholds.

Table 7.4.10.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.4.10.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	NFVO → VNFM
TerminateSubscriptionResponse	Mandatory	VNFM → NFVO

7.4.10.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.10.2-1.

Table 7.4.10.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

7.4.10.3 Output parameters

None.

7.4.10.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the NFVO will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

7.4.11 Query Subscription Info operation

7.4.11.1 Description

This operation enables the NFVO to query information about subscriptions.

NOTE: It is part of the protocol design whether querying information about subscriptions is represented as a separate "Query Subscription Info" operation or whether subscription-related information is managed as part of managing PM jobs and Thresholds.

Table 7.4.11.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.4.11.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	NFVO → VNFM
QuerySubscriptionInfoResponse	Mandatory	VNFM → NFVO

7.4.11.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.4.11.2-1.

Table 7.4.11.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

7.4.11.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.4.11.3-1.

Table 7.4.11.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

7.4.11.4 Operation results

After successful operation, the VNFM has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF fault management that the NFVO has access to and that are matching the filter shall be returned.

7.5 VNF Fault Management interface

7.5.1 Description

This interface shall allow the VNFM to provide alarms related to the VNFs visible to the consumer.

Virtualised resource alarms collected by the VNFM will be filtered, correlated and modified by the VNFM and mapped to the corresponding VNF instance, resulting in alarms on the corresponding VNF.

NOTE: The NFVO is enabled in the alarms to observe information on changes in the state of the virtualised resources due to upcoming NFVI operation and maintenance.

The fault management interface shall support the following operations:

- Subscribe operation (Subscription of NFVOs with the VNFM for the notifications related to the alarms).
- Notify operation (Notifications of alarms or alarm state change from VNFM to NFVO).
- Get alarm list operation (Accessing active alarms by the NFVO).

- Acknowledge alarms operation (Acknowledging alarms by the NFVO).
- Terminate Subscription operation.
- Query Subscription Info operation.

7.5.2 Subscribe operation

7.5.2.1 Description

This operation enables the NFVO to subscribe with a filter for the notifications related to VNF alarms sent by the VNFM.

NOTE: Specification of filtering mechanism is part of the protocol design.

Table 7.5.2.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.5.2.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	NFVO → VNFM
SubscribeResponse	Mandatory	VNFM → NFVO

7.5.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.5.2.2-1.

Table 7.5.2.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting VNFs and related alarm notifications. This filter can contain information to select VNF instances, information to select notification types as defined in clause 7.5.3.1, and additional filter criteria on further attributes of the Alarm information element defined in clause 8.8.4 as chosen in the protocol design stage.

7.5.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.5.2.3-1.

Table 7.5.2.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription returned.

7.5.2.4 Operation results

As a result of this operation, the VNFM shall indicate to the NFVO in the SubscribeResponse message whether the subscription was successful or not.

For a particular subscription, only notifications matching the filter will be delivered to the consumer.

7.5.3 Notify operation

7.5.3.1 Description

This operation distributes notifications to subscribers. It is a one-way operation issued by the VNFM towards the NFVO that cannot be invoked as an operation by the consumer (NFVO).

In order to receive notifications, the NFVO shall have a subscription.

Table 7.5.3.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.5.3.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	VNFM → NFVO

The following notifications can be notified/sent by this operation:

- AlarmNotification (see clause 8.8.2).
- AlarmClearedNotification (see clause 8.8.3).
- AlarmListRebuiltNotification (see clause 8.8.6).

7.5.4 Get Alarm List operation

7.5.4.1 Description

This operation enables the NFVOs to query the active alarms from the VNFM.

Table 7.5.4.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.5.4.1-1: Get Alarm List operation

Message	Requirement	Direction
GetAlarmListRequest	Mandatory	NFVO → VNFM
GetAlarmListResponse	Mandatory	VNFM → NFVO

7.5.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.5.4.2-1.

Table 7.5.4.2-1: Get Alarm List operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting alarms. This can contain the list of the VNF Identifiers, fault type, severity and cause.

7.5.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.5.4.3-1.

Table 7.5.4.3-1: Get Alarm List operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
alarm	M	0..N	Alarm	Information about alarms including alarmId, affected VNF identifier, and FaultDetails. The cardinality can be "0" to indicate that no Alarm could be retrieved based on the input Filter information (e.g. no matching alarm).

7.5.4.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular request, only alarms matching the filter are delivered to the NFVO.

7.5.5 Terminate Subscription operation

7.5.5.1 Description

This operation enables the NFVO to terminate a particular subscription.

Table 7.5.5.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.5.5.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	NFVO → VNFM
TerminateSubscriptionResponse	Mandatory	VNFM → NFVO

7.5.5.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.5.5.2-1.

Table 7.5.5.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

7.5.5.3 Output parameters

None.

7.5.5.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the NFVO will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

7.5.6 Query Subscription Info operation

7.5.6.1 Description

This operation enables the NFVO to query information about subscriptions.

Table 7.5.6.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.5.6.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	NFVO → VNFM
QuerySubscriptionInfoResponse	Mandatory	VNFM → NFVO

7.5.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.5.6.2-1.

Table 7.5.6.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

7.5.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.5.6.3-1.

Table 7.5.6.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

7.5.6.4 Operation results

After successful operation, the VNFM has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF fault management that the NFVO has access to and that are matching the filter shall be returned.

7.5.7 Acknowledge alarms operation

7.5.7.1 Description

This operation enables the NFVO to acknowledge alarms at VNFM.

Table 7.5.7.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.5.7.1-1: Acknowledge alarms operation

Message	Requirement	Direction
AcknowledgeAlarmsRequest	Mandatory	NFVO → VNFM
AcknowledgeAlarmsResponse	Mandatory	VNFM → NFVO

7.5.7.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.5.7.2-1.

Table 7.5.7.2-1: Acknowledge alarms operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
alarmId	M	1..N	Identifier (Reference to Alarm)	Identifier of an individual alarm to be acknowledged, or multiple identifiers of the alarms to be acknowledged. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to acknowledge multiple alarms in one request, or as a series of requests that acknowledge one alarm at a time.				

7.5.7.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.5.7.3-1.

Table 7.5.7.3-1: Acknowledge alarms operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
acknowledgedAlarmId	M	1..N	Identifier (Reference to Alarm)	Identifier of an individual alarm that is acknowledged, or multiple identifiers of the alarms that are acknowledged. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to acknowledge multiple alarms in one request, or as a series of requests that acknowledge one alarm at a time.				

7.5.7.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result.

7.6 Void

7.7 VNF Indicator interface

7.7.1 Description

This interface allows the VNFM to provide information on value changes of VNF related indicators. VNF related indicators are declared in the VNFD. This interface is originally produced by the EM and/or VNF on the Ve-Vnfm-em and/or Ve-Vnfm-vnf reference point respectively (see ETSI GS NFV-IFA 008 [i.5]) and is re-exposed by the VNFM.

The following operations are defined for this interface:

- Subscribe.
- Notify.
- Get Indicator Value.
- Terminate Subscription.
- Query Subscription Info.

7.7.2 Subscribe operation

7.7.2.1 Description

This operation enables the NFVO to subscribe with a filter for the notifications related to VNF indicator value changes sent by the VNFM.

NOTE: Specification of filtering mechanism is part of the protocol design.

Table 7.7.2.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.7.2.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	NFVO → VNFM
SubscribeResponse	Mandatory	VNFM →NFVO

7.7.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.7.2.2-1.

Table 7.7.2.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting VNFs and related indicators.

7.7.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.7.2.3-1.

Table 7.7.2.3-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription returned.

7.7.2.4 Operation results

As a result of this operation, the VNFM shall indicate to the NFVO in the SubscribeResponse message whether the subscription was successful or not. For a particular subscription, only notifications matching the filter will be delivered to the consumer.

7.7.3 Notify operation

7.7.3.1 Description

This operation distributes notifications to subscribers. It is a one-way operation issued by the VNFM towards the NFVO that cannot be invoked as an operation by the consumer (NFVO). In order to receive notifications, the NFVO shall have a subscription.

Table 7.7.3.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.7.3.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	VNFM → NFVO

The following notification can be notified/sent by this operation:

- IndicatorValueChangeNotification (see clause 8.10.2).
- SupportedIndicatorsChangeNotification (see clause 8.10.4).

7.7.4 Get Indicator Value operation

7.7.4.1 Description

This operation enables NFVO to request from the VNFM information about available indicators and their actual values.

Table 7.7.4.1-1 lists the information flow exchanged between the VNFM and the NFVO.

Table 7.7.4.1-1: Get Indicator Value operation

Message	Requirement	Direction
GetIndicatorValueRequest	Mandatory	NFVO → VNFM
GetIndicatorValueResponse	Mandatory	VNFM → NFVO

7.7.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.7.4.2-1.

Table 7.7.4.2-1: Get Indicator Value operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting VNFs and related indicators.

7.7.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.7.4.3-1.

Table 7.7.4.3-1: Get Indicator Value operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
indicatorInformation	M	0..N	IndicatorInformation	The requested indicator values as complex structures having the VNF Instance ID, Indicator and the value of the Indicator.

7.7.4.4 Operation results

The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular request, only indicators matching the filter will be delivered to the NFVO.

7.7.5 Terminate Subscription operation

7.7.5.1 Description

This operation enables the NFVO to terminate a particular subscription.

Table 7.7.5.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.7.5.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	NFVO → VNFM
TerminateSubscriptionResponse	Mandatory	VNFM → NFVO

7.7.5.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.7.5.2-1.

Table 7.7.5.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

7.7.5.3 Output parameters

None.

7.7.5.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the NFVO will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

7.7.6 Query Subscription Info operation

7.7.6.1 Description

This operation enables the NFVO to query information about subscriptions.

Table 7.7.6.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.7.6.1-1: Query Subscription operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	NFVO → VNFM
QuerySubscriptionInfoResponse	Mandatory	VNFM → NFVO

7.7.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.7.6.2-1.

Table 7.7.6.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

7.7.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.7.6.3-1.

Table 7.7.6.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

7.7.6.4 Operation results

After successful operation, the VNFM has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to VNF indicator value changes that the NFVO has access to and that are matching the filter shall be returned.

7.8 Policy Management interface

7.8.1 Description

This interface allows the NFVO to invoke policy management operations towards the VNFM.

The following policy management operations are defined for this interface:

- Transfer Policy
- Delete Policy
- Query Policy
- Activate Policy
- Deactivate Policy
- Associate Policy
- Disassociate Policy

This interface allows the NFVO to manage subscriptions to notifications sent by the VNFM which inform about changes of a policy and about any detected policy conflicts. It allows the VNFM to provide such notifications to the subscriber (e.g. NFVO).

7.8.2 Transfer Policy operation

7.8.2.1 Description

This operation enables the NFVO to transfer a NFV-MANO policy to the VNFM. Table 7.8.2.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.2.1-1: Transfer Policy operation

Message	Requirement	Direction
TransferPolicyRequest	Mandatory	NFVO → VNFM
TransferPolicyResponse	Mandatory	VNFM → NFVO

7.8.2.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.2.2-1.

Table 7.8.2.2-1: Transfer Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
designer	M	1	String	Human readable name of designer of the policy.
name	M	1	String	Human readable name of the policy.
version	M	1	Version	Version of the policy. Its value shall be the same as the one within the policy being transferred, i.e. the "policyVersion" attribute in the "Policy" information element specified in ETSI GS NFV-IFA 048 [6].
policy	M	1	Policy	Specifies the policy. See notes 1 and 2. "Policy" is specified in clause 5.2 of ETSI GS NFV-IFA 048 [6].
NOTE 1: An identifier for uniquely identifying the policy is included in the policy. NOTE 2: The NFVO may use this operation to update an existing policy with a new version. Different policy versions share the same internal identifier of the policy but having different PolicyInfo instances. The design of different policy versions and their business logic is out of the scope of the present document.				

7.8.2.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.2.3-1.

Table 7.8.2.3-1: Transfer Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1	Identifier	Identifier of the policy information created by the VNFM.

7.8.2.4 Operation results

In case of success, the NFV-MANO policy is transferred to the VNFM and corresponding policy information is created by the VNFM. In case of failure, appropriate error information is returned.

7.8.3 Delete Policy operation

7.8.3.1 Description

This operation enables the NFVO to delete one or multiple NFV-MANO policy(ies) from the VNFM. Table 7.8.3.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.3.1-1: Delete Policy operation

Message	Requirement	Direction
DeletePolicyRequest	Mandatory	NFVO → VNFM
DeletePolicyResponse	Mandatory	VNFM→ NFVO

7.8.3.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.3.2-1.

Table 7.8.3.2-1: Delete Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1..N	Identifier(Reference to PolicyInfo)	Identifier(s) of policy information.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to delete multiple policies in one request, or as a series of requests that delete one policy at a time.				

7.8.3.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.3.3-1.

Table 7.8.3.3-1: Delete Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
deletedPolicyInfoId	M	0..N	Identifier (Reference to PolicyInfo)	Identifier(s) of the deleted NFV-MANO policy information.

7.8.3.4 Operation results

In case of success, the NFV-MANO policy(ies) are deleted from the VNFM, and a success indicator is returned to the NFVO. In case of failure, appropriate error information is returned.

7.8.4 Query Policy operation

7.8.4.1 Description

This operation enables the NFVO to query the information from the VNFM on one or multiple NFV-MANO policy(ies). Table 7.8.4.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.4.1-1: Query Policy operation

Message	Requirement	Direction
QueryPolicyRequest	Mandatory	NFVO → VNFM
QueryPolicyResponse	Mandatory	VNFM → NFVO

7.8.4.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.4.2-1.

Table 7.8.4.2-1: Query Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filter defining the NFV-MANO policy information on which the query applies, based on attributes of NFV-MANO policy information. It can also be used to specify one or more NFV-MANO policy(ies) information to be queried by providing their identifiers.
attributeSelector	M	0..N	String	Provides a list of attribute names of NFV-MANO policy information. If present, only these attributes are returned for the policy information matching the filter. If absent, the complete policy information is returned.

7.8.4.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.4.3-1.

Table 7.8.4.3-1: Query Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryNsPolicyInfoResult	M	0..N	PolicyInfo	NFV-MANO policy information matching the input filter. If attributeSelector is present, only the attributes listed in attributeSelector are returned for the selected policy information. See note.
NOTE: The lower cardinality is 0 since there may be no matches to the provided filter.				

7.8.4.4 Operation results

After success operation, the VNFM has queried the internal NFV-MANO policy information. The result of the operation indicates whether it has been successful or not with a standard success/error result. For a particular query, policy information that is matching the filter shall be returned.

7.8.5 Activate Policy operation

7.8.5.1 Description

This operation enables the NFVO to activate one or multiple NFV-MANO policy(ies) in the VNFM. Table 7.8.5.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.5.1-1: Activate Policy operation

Message	Requirement	Direction
ActivatePolicyRequest	Mandatory	NFVO → VNFM
ActivatePolicyResponse	Mandatory	VNFM → NFVO

7.8.5.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.5.2-1.

Table 7.8.5.2-1: Activate Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1..N	Identifier(Reference to PolicyInfo)	Identifier(s) of policy information. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to activate multiple policies in one request, or as a series of requests that activate one policy at a time.				

7.8.5.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.5.3-1.

Table 7.8.5.3-1: Activate Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
activatedPolicyInfoId	M	0..N	Identifier (Reference to PolicyInfo)	Identifier(s) of the activated NFV-MANO policy(ies).

7.8.5.4 Operation results

In case of success, the NFV-MANO policy(ies) are activated in the VNFM, and a success indicator is returned to the NFVO. In case of failure, appropriate error information is returned.

7.8.6 Deactivate Policy operation

7.8.6.1 Description

This operation enables the NFVO to deactivate one or multiple NFV-MANO policy(ies) in the VNFM. Table 7.8.6.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.6.1-1: Deactivate Policy operation

Message	Requirement	Direction
DeactivatePolicyRequest	Mandatory	NFVO → VNFM
DeactivatePolicyResponse	Mandatory	VNFM → NFVO

7.8.6.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.6.2-1.

Table 7.8.6.2-1: Deactivate Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1..N	Identifier(Reference to PolicyInfo)	Identifier(s) of policy information. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to deactivate multiple policies in one request, or as a series of requests that deactivate one policy at a time.				

7.8.6.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.6.3-1.

Table 7.8.6.3-1: Deactivate Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
deactivatedPolicyInfoId	M	0..N	Identifier (Reference to PolicyInfo)	Identifier(s) of the deactivated NFV-MANO policy(ies).

7.8.6.4 Operation results

In case of success, the NFV-MANO policy(ies) are deactivated in the VNFM, and a success indicator is returned to the NFVO. In case of failure, appropriate error information is returned.

7.8.7 Subscribe operation

7.8.7.1 Description

This operation enables the NFVO to subscribe with a filter for the notifications sent by the VNFM which are related to changes of a policy and any detected policy conflicts. Changes of a policy are related to operations of transferring policy, deleting policy, activating policy, deactivating policy, associating policy and disassociating policy.

Table 7.8.7.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.7.1-1: Subscribe operation

Message	Requirement	Direction
SubscribeRequest	Mandatory	NFVO → VNFM
SubscribeResponse	Mandatory	VNFM → NFVO

7.8.7.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.7.2-1.

Table 7.8.7.2-1: Subscribe operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Input filter for selecting the notifications. This filter can contain information about specific types of notifications to subscribe to, or attributes of the PolicyInfo. Details are part of the protocol design.

7.8.7.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.7.3-1.

Table 7.8.7.3-1: Subscribe operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription realized.

7.8.7.4 Operation results

After successful subscription, the consumer (NFVO) is registered to receive notifications about events related to changes of a policy and any detected policy conflicts.

The result of the operation shall indicate if the subscription has been successful or not with a standard success/error result. For a particular subscription, only notifications matching the filter will be delivered to the consumer.

7.8.8 Notify operation

7.8.8.1 Description

This operation notifies a subscriber about events related to notifications about changes of a policy and any detected policy conflicts.

This operation distributes notifications to subscribers. It is a one-way operation issued by the producer (VNFM) that cannot be invoked as an operation by the consumer (NFVO). In order to receive notifications, the consumer (NFVO) has to perform an explicit Subscribe operation beforehand.

Table 7.8.8.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.8.1-1: Notify operation

Message	Requirement	Direction
Notify	Mandatory	VNFM → NFVO

The following notifications can be notified/sent by this operation:

- PolicyChangeNotification. See clause 8.13.3.
- PolicyConflictNotification. See clause 8.13.4.

7.8.9 Terminate Subscription operation

7.8.9.1 Description

This operation enables the NFVO to terminate a particular subscription.

Table 7.8.9.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.9.1-1: Terminate Subscription operation

Message	Requirement	Direction
TerminateSubscriptionRequest	Mandatory	NFVO → VNFM
TerminateSubscriptionResponse	Mandatory	VNFM → NFVO

7.8.9.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.9.2-1.

Table 7.8.9.2-1: Terminate Subscription operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
subscriptionId	M	1	Identifier	Identifier of the subscription to be terminated.

7.8.9.3 Output parameters

None.

7.8.9.4 Operation results

After successful termination of a subscription, the identified subscription does not exist anymore, and the NFVO will not receive notifications related that subscription any longer. The result of the operation shall indicate if the subscription termination has been successful or not with a standard success/error result.

7.8.10 Query Subscription Info operation

7.8.10.1 Description

This operation enables the NFVO to query information about subscriptions.

Table 7.8.10.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.10.1-1: Query Subscription Info operation

Message	Requirement	Direction
QuerySubscriptionInfoRequest	Mandatory	NFVO → VNFM
QuerySubscriptionInfoResponse	Mandatory	VNFM → NFVO

7.8.10.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.10.2-1.

Table 7.8.10.2-1: Query Subscription Info operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
filter	M	1	Filter	Filtering criteria to select one or a set of subscriptions. Details are part of the protocol design.

7.8.10.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.10.3-1.

Table 7.8.10.3-1: Query Subscription Info operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
queryResult	M	0..N	Not specified	Information about the subscription(s) matching the query.

7.8.10.4 Operation results

After successful operation, the VNFM has queried the internal subscription objects. The result of the operation indicates if it has been successful or not with a standard success/error result. For a particular query, information about the subscriptions to notifications related to changes of a policy and any detected policy conflicts that the NFVO has access to and that are matching the filter shall be returned.

7.8.11 Associate Policy operation

7.8.11.1 Description

This operation enables the NFVO to associate a NFV-MANO policy to one or multiple VNF instances in the VNFM.

Table 7.8.11.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.11.1-1: Associate Policy operation

Message	Requirement	Direction
AssociatePolicyRequest	Mandatory	NFVO → VNFM
AssociatePolicyResponse	Mandatory	VNFM → NFVO

7.8.11.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.11.2-1.

Table 7.8.11.2-1: Associate Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1	Identifier (Reference to PolicyInfo	Identifier of policy information.
vnfInstanceId	M	1..N	Identifier	Identifier(s) of the VNF instance(s) to associate policy to. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to associate a policy to multiple VNF instances in one request, or as a series of requests that associate the policy to one VNF instance at a time.				

7.8.11.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.11.3-1.

Table 7.8.11.3-1: Associate Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	0..N	Identifier	Identifier(s) of the VNF instance(s) to which the policy has been associated.

7.8.11.4 Operation results

After successful operation, the VNFM has associated the NFV-MANO policy to the VNF instance(s), and a success indicator is returned to the NFVO. In case of failure, appropriate error information is returned. The associations performed via the present interface operation take precedence and override any of the associations defined by "targetObjectId", if present, within the policy itself as defined by the "Policy" information element specified in ETSI GS NFV-IFA 048 [**Error! Reference source not found.**].

7.8.12 Disassociate Policy operation

7.8.12.1 Description

This operation enables the NFVO to disassociate a NFV-MANO policy from one or multiple VNF instances in the VNFM.

Table 7.8.12.1-1 lists the information flow exchanged between the NFVO and the VNFM.

Table 7.8.12.1-1: Disassociate Policy operation

Message	Requirement	Direction
DisassociatePolicyRequest	Mandatory	NFVO → VNFM
DisassociatePolicyResponse	Mandatory	VNFM → NFVO

7.8.12.2 Input parameters

The input parameters sent when invoking the operation shall follow the indications provided in table 7.8.12.2-1.

Table 7.8.12.2-1: Disassociate Policy operation input parameters

Parameter	Qualifier	Cardinality	Content	Description
policyInfoId	M	1	Identifier (Reference to PolicyInfo	Identifier of policy information.
vnfInstanceId	M	1..N	Identifier	Identifier(s) of the VNF instance(s) to disassociate policy from. See note.
NOTE: It is up to the protocol design stage to determine whether this operation will be modelled as a "bulk" operation that allows to disassociate a policy from multiple VNF instances in one request, or as a series of requests that disassociate the policy from one VNF instance at a time.				

7.8.12.3 Output parameters

The output parameters returned by the operation shall follow the indications provided in table 7.8.12.3-1.

Table 7.8.12.3-1: Disassociate Policy operation output parameters

Parameter	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	0..N	Identifier	Identifier(s) of the VNF instance(s) from which the policy has been disassociated.

7.8.12.4 Operation results

After successful operation, the VNFM has disassociated the NFV-MANO policy from the VNF instance(s), and a success indicator is returned to the NFVO. In case of failure, appropriate error information is returned. The disassociations performed via the present interface operation take precedence and override any of the associations defined by "targetObjectId", if present, within the policy itself as defined by the "Policy" information element specified in ETSI GS NFV-IFA 048 [6].

7.9 Void

8 Information elements exchanged

8.1 Introduction

This clause defines, or references, definitions of information elements used in the interfaces defined in the present document.

The specification of the following information elements is part of the protocol design:

- String.
- Integer.
- Identifier.
- Filter.
- DateTime.
- Value.
- Version.
- KeyValuePair.

8.2 Information elements and notifications related to VNF Package Management

8.2.1 Introduction

This clause defines information elements related to VNF Package Management.

8.2.2 VnfPkgInfo information element

8.2.2.1 Description

This information element provides the details of a VNF Package.

NOTE: The definition below is aligned with the definition of the VnfPkgInfo information element in ETSI GS NFV-IFA 013 [8].

8.2.2.2 Attributes

The VnfPkgInfo information element shall follow the indications provided in table 8.2.2.2-1.

Table 8.2.2.2-1: Attributes of the VnfPkgInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfPkgInfoId	M	1	Identifier	Identifier of the VNF Package information object. This identifier was allocated by the NFVO.
vnfdId	M	0..1	Identifier	Identifier of the onboarded VNF Package. See notes 1, 2 and 3.

Attribute	Qualifier	Cardinality	Content	Description
vnfdExtInvariantId	M	0..1	Identifier	Identifies a VNFD in a version independent manner. This attribute is invariant across versions of the VNFD that fulfil certain conditions related to the external connectivity and management of the VNF. See notes 2 and 5.
vnfProvider	M	0..1	String	Provider of the on-boarded VNF package. See notes 2 and 3.
vnfProductName	M	0..1	String	Product name of the on-boarded VNF package. See notes 2 and 3.
vnfSoftwareVersion	M	0..1	Version	Software version of the on-boarded VNF package. See notes 2 and 3.
vnfdVersion	M	0..1	Version	VNFD version of the on-boarded VNF package. See notes 2 and 3.
checksum	M	0..1	Not specified	Checksum of the on-boarded VNF Package. See notes 2 and 3.
vnfd	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD contained in the on-boarded VNF Package, e.g. URL to the on-boarded VNFD. See note 3.
softwareImage	M	0..N	VnfPackageSoftwareImageInfo	Information about VNF Package artifacts that are software images. See note 3.
additionalArtifact	M	0..N	VnfPackageArtifactInformation	Information about VNF Package artifacts contained in the VNF Package that are not software images. See note 4.
onboardingState	M	1	Enum	On-boarding state of the VNF Package. VALUES: • CREATED • UPLOADING • PROCESSING • ONBOARDED
operationalState	M	1	Enum	Operational state of the VNF Package. VALUES: • ENABLED • DISABLED
usageState	M	1	Enum	Usage state of the VNF Package. VALUES: • IN_USE • NOT_IN_USE
userDefinedData	O	0..N	KeyValuePair	User defined data for the VNF Package.

NOTE 1: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. This information is copied from the VNFD of the on-boarded VNF Package.
NOTE 2: This information is copied from the VNFD of the on-boarded VNF Package.
NOTE 3: These attributes shall be present after the VNF Package is on-boarded.
NOTE 4: It may be present after the VNF Package is on-boarded and shall be absent otherwise.
NOTE 5: This attribute may be present after the VNF Package is on-boarded.

8.2.3 Vnfd information element

8.2.3.1 Description

This information element provides the details of the VNFD.

8.2.3.2 Attributes

The structure of the Vnfd information element shall comply with the provisions for the Vnfd information element as defined in ETSI GS NFV-IFA 011 [3], clause 7.1.2.

8.2.4 VnfPackageOnBoardingNotification

8.2.4.1 Description

This notification indicates that a VNF Package is on-boarded, after all the on-boarding steps (e.g. uploading and processing) are done. A change in on-boarding state before the VNF Package is on-boarded is not reported. Support of this notification is mandatory.

8.2.4.2 Trigger Conditions

- New VNF Package on-boarded.

8.2.4.3 Attributes

The VnfPackageOnBoardingNotification shall follow the indications provided in table 8.2.4.3-1.

Table 8.2.4.3-1: Attributes of the VnfPackageOnBoardingNotification

Attribute	Qualifier	Cardinality	Content	Description
onboardedVnfPkgInfoId	M	1	Identifier	Identifier of the VNF Package information object.
vnfdId	M	1	Identifier	Identifier of the on-boarded VNF Package (see note).
NOTE: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. This information is copied from the VNFD of the on-boarded VNF Package.				

8.2.5 VnfPackageChangeNotification

8.2.5.1 Description

This notification indicates a change of status in an on-boarded VNF Package. Only changes in operational state and the deletion of the VNF package will be reported. Change in usage state is not reported.

Support of this notification is mandatory.

8.2.5.2 Trigger Conditions

- Change of the operational state of an on-boarded VNF Package.
- Deletion of an on-boarded VNF Package.

8.2.5.3 Attributes

The VnfPackageChangeNotification shall follow the indications provided in table 8.2.5.3-1.

Table 8.2.5.3-1: Attributes of the VnfPackageChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
onboardedVnfPkgInfoId	M	1	Identifier	Identifier of the VNF Package information object.
vnfdId	M	1	Identifier	Identifier of the on-boarded VNF Package (see note).
changeType	M	1	Enum	It categorizes the type of change. VALUES: • OP_STATE_CHANGE: change of operational state of an on-boarded VNF Package • PKG_DELETE: deletion of a VNF Package
operationalState	M	0..1	Enum	New operational state of the VNF Package. VALUES: • ENABLED • DISABLED
NOTE: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. This information is copied from the VNFD of the on-boarded VNF Package.				

8.2.6 Void

8.2.7 VnfPackageSoftwareImageInfo information element

8.2.7.1 Description

This information element represents Software Image Information.

8.2.7.2 Attributes

The VnfPackageSoftwareImageInfo information element shall follow the indications provided in table 8.2.7.2-1.

Table 8.2.7.2-1: Attributes of the VnfPackageSoftwareImageInfo information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this software image.
name	M	1	Not specified	The name of this software image.
provider	M	1	Not specified	The provider of this software image.
version	M	1	Not specified	The version of this software image.
checksum	M	1	Not specified	The checksum of the software image file.
containerFormat	M	1	Not specified	The container format indicates whether the software image is in a file format that also contains metadata about the actual software.
diskFormat	M	0..1	Not specified	The disk format of a software image is the format of the underlying disk image. See note 1.
createdAt	M	1	Not specified	The time when this software image was created.
minDisk	M	0..1	Not specified	The minimal Disk for this software image. See note 2.
minRam	M	0..1	Not specified	The minimal RAM for this software image. See note 1.
size	M	1	Not specified	The size of this software image.
userMetadata	M	0..N	KeyValuePair	User-defined metadata.
accessInformation	M	1	Not specified	Information (such as a URL, a path in the VNF Package, or an identifier) that allows to access a copy of this software image artifact.
NOTE 1: The attribute shall be present for VM-based software images referenced from a Vdu, and shall be absent otherwise. NOTE 2: The attribute shall be present for software images referenced from a VirtualStorageDesc, and shall be absent otherwise.				

8.2.8 VnfPackageArtifactInformation information element

8.2.8.1 Description

This information element represents an artifact other than a Software Image which is contained in the VNF Package.

8.2.8.2 Attributes

The VnfPackageArtifactInformation information element shall follow the indications provided in table 8.2.8.2-1.

Table 8.2.8.2-1: Attributes of the VnfPackageArtifactInformation information element

Attribute	Qualifier	Cardinality	Content	Description
selector	M	1	Not specified	Information (such as a URL, a path in the VNF Package, or an identifier) that allows to access a copy of this artifact.
metadata	M	1	Not specified	The metadata of the artifact that are available in the VNF Package, such as Content type, size, creation date, etc.

8.2.9 Void

8.3 Information elements related to VNF Lifecycle Operation Granting

8.3.1 Introduction

This clause defines information elements related to VNF Lifecycle Operation Granting.

8.3.2 ResourceDefinition information element

8.3.2.1 Description

This information element provides information of an existing or proposed resource or PaaS Service used by the VNF.

8.3.2.2 Attributes

The ResourceDefinition information element shall follow the indications provided in table 8.3.2.2-1.

Table 8.3.2.2-1: Attributes of the ResourceDefinition information element

Attribute	Qualifier	Cardinality	Content	Description
resourceDefinitionId	M	1	Identifier	Identifier of this ResourceDefinition information element, unique at least within the scope of the grant request.
type	M	1	Enum	Type of the resource definition referenced. VALUES: • COMPUTE • VL • LINKPORT • STORAGE • OSCONTAINER • VIRTUALCP • PAASSERVICE • Etc.
vduId	M	0..1	Identifier (Reference to Vdu)	Reference to the related Vdu applicable to this resource in the VNFD. Shall only be present if a VDU is applicable to this resource in the VNFD.
vnfdId	M	0..1	Identifier	Identifier of the VNFD to which resourceTemplateId and vduId refer. Shall be present if at least one of resourceTemplateId and vduId is present and the operation to be granted changes the current VNF Package. May be absent otherwise.
resourceTemplateId	M	0..N	Identifier (Reference to VnfVirtualLinkDesc, VirtualComputeDesc, VnfExtCpd, VirtualStorageDesc, OsContainerDesc, or PaasServiceRequest)	Reference to a resource template (VnfVirtualLinkDesc, VirtualComputeDesc, VnfExtCpd, VirtualStorageDesc, OsContainerDesc or PaaS Service request) in the VNFD. Cardinality may be greater than "1" when type=OSCONTAINER and multiple references to OsContainerDesc are present in the VDU indicated by the "vduId". See note 2.

Attribute	Qualifier	Cardinality	Content	Description
osContainerDesc	M	0..N	OsContainerDesc	Resource requirements for the OS Containers. The content of OsContainerDesc is defined in clause 7.1.6.13 of ETSI GS NFV-IFA 011 [3]. See note 2 and 3.
secondaryResourceTemplateId	M	0..1	Identifier (Reference to VnfExtCpd)	Reference to a secondary resource template (VnfExtCpd) in the VNFD. Shall be present if type=LINKPORT and the linkport is shared by two external CP instances, one exposing a VNFC CP instance (based on a VnfExtCpd referenced by "resourceTemplateId") and another one exposing a VIP CP instance (based on a VnfExtCpd referenced by this attribute). Shall be absent otherwise. See note 1.
resourceHandle	M	0..1	ResourceHandle	Resource information for an existing resource. Shall be present for resources that are planned to be deleted or modified. Shall be absent otherwise.
snapshotResDef	M	0..1	SnapshotResourceDefinition	Information to identify a snapshot resource. Shall only be present if the operation to be granted concerns to creating a VNF snapshot from the VNF or to reverting the VNF to a VNF snapshot.
resourceUsageRequirement	M	0..1	Enum	The type of resource usage requirement. A resource can be shared among VNF instances or dedicated to a single VNF instance. VALUES: • DEDICATED • SHARED Default value is "DEDICATED".
NOTE 1: The use cases UC#4 and UC#5 in clause A.4 provide examples for such a configuration. NOTE 2: In case of the type of the ResourceDefinition is OSCONTAINER, either resourceTemplateId or osContainerDesc shall be present. NOTE 3: In case osContainerDesc is not present in the VNFD, while MCIOP(s) is present, osContainerDesc shall be included in the GrantVnfLifecycleOperationRequest based on the processing result of the MCIOP from CISM.				

8.3.3 GrantInfo information element

8.3.3.1 Description

This information element contains information about a Compute, storage or network resource whose addition/update/deletion was granted in a GrantVnfLifecycleOperationResponse. It enables also referencing to a PaaS Service request definition which has been granted in a GrantVnfLifecycleOperationResponse.

8.3.3.2 Attributes

The GrantInfo information element shall follow the indications provided in table 8.3.3.2-1.

Table 8.3.3.2-1: Attributes of the GrantInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceDefinitionId	M	1	Identifier (Reference to ResourceDefinition)	Identifier of the related ResourceDefinition information element from the grant request. See note.

Attribute	Qualifier	Cardinality	Content	Description
reservationId	M	0..1	Identifier (Reference to ReservedVirtualCompute, ReservedVirtualNetwork, ReservedVirtualStorage, or ReservedComputeHosts)	The reservation identifier applicable to the VNFC/VirtualLink/VirtualStorage/ compute hosts. It shall be present for new and unallocated resources when policy is GRANT_RESERVE and an applicable reservation exists; shall not be present otherwise. See note.
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Reference to the identifier of the VimConnectionInfo information element defining the VIM or CISM connection to be used to manage this resource. Shall be present for new resources, and shall be absent for resources that have already been allocated. This parameter shall be supported when the granted resources are managed by a CISM. CONDITION: This attribute shall be supported when VNF-related Resource Management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the virtualised resource. Shall be present for new resources, and shall be absent for resources that have already been allocated. See note. CONDITION: This attribute shall be supported when VNF-related Resource Management in indirect mode is applicable.
zoneId	M	0..1	Identifier (Reference to ZoneInfo)	Reference to the identifier of the ZoneInfo information element defining the resource zone into which this resource is to be placed. Shall be present for new resources, and shall be absent for resources that have already been allocated.
resourceGroupId	M	0..1	Identifier	Identifier of the "infrastructure resource group", logical grouping of virtual resources assigned to a tenant within an Infrastructure Domain, to be provided when allocating the resource. If the VIM connection referenced by "vimConnectionId" applies to multiple infrastructure resource groups, this attribute shall be present for new resources. If the VIM connection referenced by "vimConnectionId" applies to a single infrastructure resource group, this attribute may be present for new resources. This attribute shall be absent for resources that have already been allocated.

Attribute	Qualifier	Cardinality	Content	Description
containerNamespace	M	0..1	String	The value of the namespace in which the MCIOs of a VNF with containerized components shall be deployed. This attribute shall be present if the granted resources are managed by a CISM. The attribute shall be ignored if the granted resources are not managed by a CISM.
mcioConstraints	M	0..N	KeyValuePair	The constraint values to be assigned to MCIOs of a VNF with containerized components. The key in the key-value pair indicates the parameter name of the MCIO constraint in the MCIO declarative descriptor and shall be one of the possible enumeration values of the "mcioConstraintsParams" attribute as specified in clause 7.1.6.2.2 of ETSI GS NFV-IFA 011 [3]. The value in the key-value pair indicates the value to be assigned to the MCIO constraint. This attribute shall be present if the granted resources are managed by a CISM. The attribute shall be ignored if the granted resources are not managed by a CISM.
sharedResource	M	0..1	ResourceHandle	Resource information for an existing resource. Shall be set when an existing allocated resource is approved to be shared and reused for the resource defined by ResourceDefinition. See note.
NOTE: If a "sharedResource" is provided, the "reservationId" and "resourceProviderId" shall not be set.				

8.3.4 ZoneInfo information element

8.3.4.1 Description

This information element provides information regarding a resource zone.

8.3.4.2 Attributes

The ZoneInfo information element shall follow the indications provided in table 8.3.4.2-1.

Table 8.3.4.2-1: Attributes of the ZoneInfo information element

Attribute	Qualifier	Cardinality	Content	Description
zoneInfoId	M	1	Identifier	The identifier of this ZoneInfo instance, for the purpose of referencing it from other information elements.
zoneId	M	1	Identifier	The identifier of the resource zone, as managed by the resource management layer (typically, the VIM).
vimConnectionId	CM	1	Identifier (Reference to VimConnectionInfo)	The identifier of the connection to the VIM that manages the resource zone. CONDITION: This attribute shall be supported when VNF-related Resource Management in direct mode is applicable.

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	CM	1	Identifier	Identifies the entity responsible for the management the resource zone. CONDITION: This attribute shall be supported when VNF-related Resource Management in indirect mode is applicable.

8.3.5 ZoneGroupInfo information element

8.3.5.1 Description

This information element provides information regarding a resource zone group. A resource zone group is a group of one or more related resource zones which can be used in resource placement constraints. To fulfil such constraint, the NFVO may decide to place a resource into any zone that belongs to a particular group.

NOTE: A resource zone group can be used to support overflow from one resource zone into another, in case a particular deployment supports only non-elastic resource zones.

8.3.5.2 Attributes

The ZoneGroupInfo information element shall follow the indications provided in table 8.3.5.2-1.

Table 8.3.5.2-1: Attributes of the ZoneGroupInfo information element

Attribute	Qualifier	Cardinality	Content	Description
zoneId	M	1..N	Identifier (Reference to ZoneInfo)	References of identifiers of ZoneInfo instances, each of which provides information about a resource zone that belongs to this group.

8.3.6 PlacementConstraint information element

8.3.6.1 Description

This information element provides information regarding a resource placement constraint. A set of such constraints may be sent by the VNFM to the NFVO to influence the resource placement decisions made by the NFVO as part of the granting process. A placement constraint defines a condition to the placement of new resources, considering other new resources as well as existing resources.

EXAMPLE: The following rules influence the placement of a set of resources such that they are placed in the same Network Function Virtualisation Infrastructure Point of Presence (NFVI-PoP) but in different resource zones:

```
{type="affinity"; scope="NFVI-PoP"; {resource1,resource2}}
{type="anti-affinity"; scope="Zone"; {resource1,resource2}}
```

Annex B in ETSI GS NFV-IFA 011 [3] provides additional description and examples about the usage of the affinity/anti-affinity rules.

8.3.6.2 Attributes

The PlacementConstraint information element shall follow the indications provided in table 8.3.6.2-1.

Table 8.3.6.2-1: Attributes of the PlacementConstraint information element

Attribute	Qualifier	Cardinality	Content	Description
affinityOrAntiAffinity	M	1	Enum	The type of the constraint. VALUES: • AFFINITY • ANTI_AFFINITY
scope	M	1	Enum	The scope of the placement constraint indicating the category of the "place" where the constraint applies. VALUES: • NFVI_POP • ZONE • ZONE_GROUP • NFVI_NODE • CIS_NODE • CONTAINER_NAMESPACE See note.
resource	M	2..N	ConstraintResourceRef	References to resources in the constraint rule.
fallbackBestEffort	M	0..1	Boolean	Indication if the constraint is handled with fall back best effort. Default value is "false". If set to true, the Affinity/Anti_Affinity placement constraint need not be fully satisfied due to capacity constraints and/or due to the actual placement of existing resources, i.e. if resource placement cannot honour the placement constraint, the request is processed in a best effort manner.
NOTE: The "CIS_NODE" and "CONTAINER_NAMESPACE" scopes shall only be applicable to express affinity or anti-affinity relationship between containerized workloads.				

8.3.7 VimConstraint information element

8.3.7.1 Description

This information element provides information regarding a VIM selection constraint. A set of such constraints may be sent by the VNFM to the NFVO to influence the VIM selection decisions made by the NFVO as part of the granting process.

8.3.7.2 Attributes

The VimConstraint information element shall follow the indications provided in table 8.3.7.2-1.

Table 8.3.7.2-1: Attributes of the VimConstraint information element

Attribute	Qualifier	Cardinality	Content	Description
sameResourceGroup	M	0..1	Boolean	If present and set to true, this signals that the constraint applies not only to the same VIM connection, but also to the same infrastructure resource group.
resource	M	2..N	ConstraintResourceRef	References to resources in the constraint rule. The NFVO shall ensure that all resources in this list are managed through the same VIM connection. If "sameResourceGroup" is set to true, the NFVO shall further ensure that all resources in this list are part of the same infrastructure resource group in that VIM connection.

8.3.8 ConstraintResourceRef information element

8.3.8.1 Description

This information element references a resource either by its VIM-level identifier for existing resources, or by the identifier of a resourceDefinition information element in the grant request for new resources.

8.3.8.2 Attributes

The ConstraintResourceRef information element shall follow the indications provided in table 8.3.8.2-1.

Table 8.3.8.2-1: Attributes of the ConstraintResourceRef information element

Attribute	Qualifier	Cardinality	Content	Description
idType	M	1	Enum	The type of the identifier. VALUES: • RES_MGMT: Resource-management-level identifier; this identifier is managed by the VIM in direct mode and is managed by the NFVO in indirect mode • GRANT: Reference to identifier in the ResourceDefinition in the grant request
resourceId	M	1	Identifier	An actual resource-management-level identifier (idType=RES_MGMT), or an identifier that references the ResourceDefinition in the related grant request (idType=GRANT).
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM Connection. It shall only be present when idType = RES_MGMT. CONDITION: It shall be supported when VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifier of the resource provider. It shall only be present when idType = RES_MGMT. CONDITION: It shall be supported when VNF-related resource management in indirect mode is applicable.

8.3.9 VimAssets information element

8.3.9.1 Description

This information element contains references to the asset which are defined in VNFD and managed in the VIM by the NFVO, such as compute resource flavours and/or software images.

8.3.9.2 Attributes

The VimAssets information element shall follow the indications provided in table 8.3.9.2-1.

Table 8.3.9.2-1: Attributes of the VimAssets information element

Attribute	Qualifier	Cardinality	Content	Description
computeResourceFlavour	M	0..N	VimComputeResourceFlavour	Mappings between virtual compute descriptors defined in the VNFD and compute resource flavours managed in the VIM.
softwareImage	M	0..N	VimSoftwareImage	Mappings between software images defined in the VNFD and software images managed in the VIM.
snapshotResource	M	0..N	VimSnapshotResource	Mappings between snapshot resources defined in the VNF snapshot package and resources managed in the VIM.
storageAsset	M	0..N	StorageAsset	Mappings between virtual storages defined in the VNFD and virtual storages managed in the NFVI.

8.3.10 VimComputeResourceFlavour information element

8.3.10.1 Description

If the VIM requires the use of virtual compute resource flavours during compute resource instantiation, it is assumed that such flavours are selected or created by the NFVO based on the information in the VirtualComputeDesc information elements defined in the VNFD.

This information element defines the mapping between a VirtualComputeDesc in the VNFD and the corresponding compute resource flavour managed by the NFVO in the VIM.

8.3.10.2 Attributes

The VimComputeResourceFlavour information element shall follow the indications provided in table 8.3.10.2-1.

Table 8.3.10.2-1: Attributes of the VimComputeResourceFlavour information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM connection to access the flavour referenced in this information element. CONDITION: Shall be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the virtualised resource. CONDITION: Shall be supported and present if VNF-related resource management in indirect mode is applicable.
vnfdVirtualComputeDescId	M	1	Identifier (Reference to VirtualComputeDesc)	Identifier which references the VirtualComputeDesc in the VNFD that maps to this flavour.
vimFlavourId	M	1	Identifier	Identifier of the compute resource flavour in the resource management layer (i.e. VIM).

8.3.11 VimSoftwareImage information element

8.3.11.1 Description

This information element contains a mapping between a software image definition the VNFD and the corresponding software image managed by the NFVO in the VIM which is needed during compute resource instantiation.

8.3.11.2 Attributes

The VimSoftwareImage information element shall follow the indications provided in table 8.3.11.2-1.

Table 8.3.11.2-1: Attributes of the VimSoftwareImage information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM or CIR connection to access the software image referenced in this information element. Shall be supported and present when the VNF is realized by a set of OS containers. CONDITION: Shall be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifier used by NFVO to determine the entity responsible for the management of the VIM asset. CONDITION: Shall be supported and present if VNF-related resource management in indirect mode is applicable.
vnfdSoftwareImageId	M	1	Identifier (Reference to SwImageDesc)	Identifier of the software image descriptor in the VNFD.
vimSoftwareImageId	M	1	Identifier	Identifier of the software image in the resource management layer (i.e. VIM).

8.3.12 VimSnapshotResource information element

8.3.12.1 Description

This information element contains a mapping between a snapshot resource definition related to a VNF snapshot and the corresponding resource managed by the NFVO in the VIM which is needed during the revert to VNF snapshot operation.

8.3.12.2 Attributes

The VimSnapshotResource information element shall follow the indications provided in table 8.3.12.2-1.

Table 8.3.12.2-1: Attributes of the VimSnapshotResource information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	References the VIM connection to access the software image referenced in this structure. The applicable "VimConnectionInfo" structure, which is referenced by vimConnectionId, can be obtained from the "vimConnectionInfo" attribute of the "VnfInfo" structure. CONDITION: This attribute shall only be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the virtualised resource. CONDITION: This attribute shall only be supported and present if VNF-related resource management in indirect mode is applicable.
vnfSnapshotId	M	1	Identifier	Identifier of the VNF snapshot related to the snapshot resource.
vnfcSnapshotId	M	1	Identifier	Identifier of the information about a specific VNFC snapshot (refer to "VnfcSnapshotInfo") of the VNF snapshot.
storageSnapshotId	M	0..1	Identifier	Identifier of the virtual storage resource that has been snapshotted as referred in the VNFC snapshot information. Shall only be present if the snapshot resource in the VIM is a storage resource (as indicated by the "type=STORAGE" in the parent resource definition).
vimSnapshotResourceId	M	1	Identifier	Identifier of the snapshot resource in the resource management layer (i.e. VIM).

8.3.13 SnapshotResourceDefinition information element

8.3.13.1 Description

This information element provides information related to a snapshot resource.

8.3.13.2 Attributes

The SnapshotResourceDefinition information element shall follow the indications provided in table 8.3.13.2-1.

Table 8.3.13.2-1: Attributes of the SnapshotResourceDefinition information element

Attribute	Qualifier	Cardinality	Content	Description
vnfSnapshotId	M	1	Identifier	Identifier of the VNF snapshot related to the resource change for the VNF instance. Shall only be present if the operation to be granted concerns to creating a VNF snapshot from the VNF or to reverting the VNF to a VNF snapshot.
vnfcSnapshotId	M	0..1	Identifier	Identifier of the information about a specific VNFC snapshot (refer to "VnfcSnapshotInfo") of the VNF snapshot. Shall only be present if the operation to be granted concerns to reverting the VNF to a

Attribute	Qualifier	Cardinality	Content	Description
				VNF snapshot, and the resource is planned to be added based on the VNFC snapshot and the type of resource is "COMPUTE" or "STORAGE". See note 1 and note 2.
storageSnapshotId	M	0..1	Identifier	Identifier of a snapshotted storage resource associated to the VNFC snapshot. Shall only be present if the operation to be granted concerns to reverting the VNF to a VNF snapshot, and the storage resource is planned to be added based on the VNFC snapshot and the type of resource is "STORAGE". See note 2.
snapshotResource	M	0..1	ResourceHandle	Resource information for an existing snapshot resource. Shall only be present if the operation to be granted concerns to reverting the VNF to a VNF snapshot and the resource is planned to be added based on an existing VNF snapshot that has been created by the VNFM. Shall be absent otherwise. See note 2.
NOTE 1: If present, the value of the "vduId" (for a related VDU) in the "VnfcResourceInfo" referred by the "vnfcInfoId" of the "VnfcSnapshotInfo" shall match the value of the "vduId" in the resource definition that is signalled in the granting request. NOTE 2: For snapshot resource definitions from a VNF snapshot package, only the "vnfcSnapshotId" and "storageSnapshotId" (in case of a storage type of resource) are applicable. If the snapshot resource definition generated as part of a VNF snapshot created by the VNFM (that is, not extracted from a VNF snapshot package), the "snapshotResource" is applicable. This is a similar specification as the one defined with the "vduId", "resourceTemplateId" and "resourceHandle" attributes provided in the ResourceDefinition, but in this case applicable to resources that are defined from VNF snapshots instead of VNFD.				

8.3.14 StorageAsset information element

8.3.14.1 Description

This information element contains a mapping between a VirtualStorageDesc in the VNFD and the corresponding virtual storage managed by the NFVO in the NFVI.

8.3.14.2 Attributes

The StorageAsset information element shall follow the indications provided in table 8.3.14.2-1.

Table 8.3.14.2-1: Attributes of the StorageAsset information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM or CISM connection to access the virtual storage referenced in this information element. Shall be supported and present when the VNF is realized by a set of OS containers. CONDITION: Shall be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifier used by NFVO to determine the entity responsible for the management of the storage asset. CONDITION: Shall be supported and present if VNF-related resource management in indirect mode is applicable.

Attribute	Qualifier	Cardinality	Content	Description
vnfdVirtualStorageDescId	M	1	Identifier (Reference to VirtualStorageDesc)	Identifier of the virtual storage descriptor in the VNFD.
storageClassName	M	1	String	Name of storage class, which represents features and policies concerning a virtual storage.

8.3.15 PaasAsset information element

8.3.15.1 Description

This information element contains references to PaaS Services, which are requested in a VNFD, assigned to a VNF, and managed in the PSM by the NFVO.

8.3.15.2 Attributes

The PaasAsset information element shall follow the indications provided in table 8.3.15.2-1.

Table 8.3.15.2-1: Attributes of the PaasAsset information element

Attribute	Qualifier	Cardinality	Content	Description
paasServiceType	M	1	String	The type of PaaS Service. The value of this attribute is expected to be matched against values of the registered PaaS Services in the PSR.
paasServiceId	M	1	Identfier	Identifier of the assigned PaaS Service as managed by the PSM.
paasServiceVersion	M	0..1	Version	Version of the PaaS Service. It shall be present if the PaaS Service is versioned.
paasServiceRequestId	M	1	Identifier (Reference to PaasServiceRequest)	Identifier of the PaaS Service request in the VNFD that maps to the assigned PaaS Service.
paasServiceHandle	M	1	Not specified	A handle enabling the access and use of the PaaS Service by the VNF. This can include for instance certain interface endpoint URI together with necessary credentials to access it. The type and format of the handle depends on the form that the PaaS Service.

8.4 Information elements and notifications related to Virtualised Resources Management in indirect mode

8.4.1 Introduction

This clause defines information elements related to Virtualised Resources Management. These information elements shall be supported when VNF-related resource management in indirect mode is applicable.

8.4.2 Information elements related to Virtualised Compute

8.4.2.1 Introduction

The clauses below define information elements related to the management of virtualised compute resources and virtualised compute resources information.

8.4.2.2 ComputeResourceWithRpInfo information element

8.4.2.2.1 Description

The ComputeResourceWithRpInfo information element encapsulates data of an instantiated virtualised compute resource in indirect mode.

8.4.2.2.2 Attributes

The ComputeResourceWithRpInfo information element shall comply with the provisions in clause 8.4.3.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.2.2.2-1. All attributes of the VirtualCompute are also attributes of the ComputeResourceWithRpInfo.

Table 8.4.2.2.2-1: Attributes of the ComputeResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, computeId].
(inherited attributes)				All attributes inherited from VirtualCompute.

8.4.2.3 ComputeResourceWithRpId information element

8.4.2.3.1 Description

This information element defines the identity of a virtualised compute resource in indirect mode.

8.4.2.3.2 Attributes

The ComputeResourceWithRpId information element shall follow the indications provided in table 8.4.2.3.2-1.

Table 8.4.2.3.2-1: Attributes of the ComputeResourceWithRpId information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, computeId].
computeId	M	1	Identifier	Identifier of the compute resource within the VIM.

8.4.2.4 VirtualComputeResourceWithRpInfo information element

8.4.2.4.1 Description

The VirtualComputeResourceWithRpInfo information element defines the characteristics of a consumable virtualised compute resources in indirect mode.

8.4.2.4.2 Attributes

The VirtualComputeResourceWithRpInfo information element shall comply with the provisions in clause 8.3.3.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.2.4.2-1. All attributes of the VirtualComputeResourceInformation are also attributes of the VirtualComputeResourceWithRpInfo.

Table 8.4.2.4.2-1: Attributes of the VirtualComputeResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the consumable virtualised resource and is used by the VNFM to uniquely identify consumable compute type resources by means of the tuple [resourceProviderId, computeResourceTypeId].
(inherited attributes)				All attributes inherited from VirtualComputeResourceInformation.

8.4.3 Information elements related to Virtualised Network

8.4.3.1 Introduction

The clauses below define information elements related to the management of virtualised network resources and virtualised network resources information.

8.4.3.2 NetworkResourceWithRpInfo information element

8.4.3.2.1 Description

The NetworkResourceWithRpInfo information element encapsulates data of an instantiated virtualised network resource in indirect mode.

8.4.3.2.2 Attributes

The NetworkResourceWithRpInfo information element shall comply with the provisions in clause 8.4.5.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.3.2.2-1. All attributes of the VirtualNetwork are also attributes of the NetworkResourceWithRpInfo.

Table 8.4.3.2.2-1: Attributes of the NetworkResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, networkResourceId].
(inherited attributes)				All attributes inherited from VirtualNetwork.

8.4.3.3 NetworkResourceWithRpId information element

8.4.3.3.1 Description

This information element defines the identity of a virtualised network resource in indirect mode.

8.4.3.3.2 Attributes

The NetworkResourceWithRpId information element shall follow the indications provided in table 8.4.3.3.2-1.

Table 8.4.3.3.2-1: Attributes of the NetworkResourceWithRpId information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, networkResourceId].
networkResourceId	M	1	Identifier	Identifier of the network resource within the VIM.

8.4.3.4 VirtualNetworkResourceWithRpInfo information element

8.4.3.4.1 Description

The VirtualNetworkResourceWithRpInfo information element defines the characteristics of a consumable virtualised network resource in indirect mode.

8.4.3.4.2 Attributes

The VirtualNetworkResourceWithRpInfo information element shall comply with the provisions in clause 8.3.5 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.3.4.2-1. All attributes of the VirtualNetworkResourceInformation are also attributes of the VirtualNetworkResourceWithRpInfo.

Table 8.4.3.4.2-1: Attributes of the VirtualNetworkResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the consumable virtualised resource and is used by the VNFM to uniquely identify consumable network type resources by means of the tuple [resourceProviderId, networkResourceTypeId].
(inherited attributes)				All attributes inherited from VirtualNetworkResourceInformation.

8.4.4 Information elements related to Virtualised Storage

8.4.4.1 Introduction

The clauses below define information elements related to the management of virtualised storage resources and virtualised storage resources information.

8.4.4.2 StorageResourceWithRpInfo information element

8.4.4.2.1 Description

The StorageResourceWithRpInfo information element encapsulates data of an instantiated virtualised storage resource.

8.4.4.2.2 Attributes

The StorageResourceWithRpInfo information element shall comply with the provisions in clause 8.4.7.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.4.2.2-1. All attributes of the VirtualStorage are also attributes of the StorageResourceWithRpInfo.

Table 8.4.4.2.2-1: Attributes of the StorageResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, storageId].
(inherited attributes)				All attributes inherited from StorageResourceWithRpInfo.

8.4.4.3 StorageResourceWithRpId information element

8.4.4.3.1 Description

This information element defines the identity of a virtualised storage resource in indirect mode.

8.4.4.3.2 Attributes

The StorageResourceWithRpId information element shall follow the indications provided in table 8.4.4.3.2-1.

Table 8.4.4.3.2-1: Attributes of the StorageResourceWithRpId information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, storageId].
storageId	M	1	Identifier	Identifier of the storage resource within the VIM.

8.4.4.4 VirtualStorageResourceWithRpInfo information element

8.4.4.4.1 Description

The VirtualStorageResourceWithRpInfo information element defines the characteristics of a consumable virtualised storage resource in indirect mode.

8.4.4.4.2 Attributes

The VirtualStorageResourceWithRpInfo information element shall comply with the provisions in clause 8.3.4 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.4.4.2-1. All attributes of the VirtualStorageResourceInformation are also attributes of the VirtualStorageResourceWithRpInfo.

Table 8.4.4.4.2-1: Attributes of the VirtualStorageResourceWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the consumable virtualised resource and is used by the VNFM to uniquely identify consumable storage type resources by means of the tuple [resourceProviderId, storageResourceTypeId].
(inherited attributes)				All attributes inherited from VirtualStorageResourceInformation.

8.4.5 Notifications related to changes of virtualised resources

8.4.5.1 Introduction

The clauses below define notifications related to changes of virtualised resources.

8.4.5.2 VirtualisedResourceWithRpChangeNotification

8.4.5.2.1 Description

This notification informs the receiver of changes in the virtualised resources that are allocated and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.5.2.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the VirtualisedResourceChangeNotification in clause 8.4.9 of ETSI GS NFV-IFA 006 [1].

8.4.5.2.3 Attributes

The VirtualisedResourceWithRpChangeNotification shall comply with the indications in clause 8.4.9 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.5.2.3-1. All attributes of the VirtualisedResourceChangeNotification are also attributes of the VirtualisedResourceWithRpChangeNotification.

Table 8.4.5.2.3-1: Attributes of the VirtualisedResourceWithRpChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the change notification and is used by the VNFM to uniquely identify the resource by means of the tuple [resourceProviderId, resourceTypeId].
(inherited attributes)				All attributes inherited from VirtualisedResourceChangeNotification.

8.4.5.3 InformationWithRpChangeNotification

8.4.5.3.1 Description

This notification informs the receiver that information related to consumable virtualised resources is changed and is applicable in the indirect mode.

Support of this notification is mandatory.

8.4.5.3.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the InformationChangeNotification in clause 8.3.2 of ETSI GS NFV-IFA 006 [1].

8.4.5.3.3 Attributes

The InformationWithRpChangeNotification shall comply with the indications in clause 8.3.2 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.5.3.3-1. All attributes of the InformationChangeNotification are also attributes of the InformationWithRpChangeNotification.

Table 8.4.5.3.3-1: Attributes of the InformationWithRpChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the change notification and is used by the VNFM to uniquely identify the consumable resource by means of the tuple [resourceProviderId, resourceTypeId].
(inherited attributes)				All attributes inherited from InformationChangeNotification.

8.4.6 Notifications related to Virtualised Resource Performance Management

8.4.6.1 Introduction

The clauses below define notifications related of virtualised resource performance management.

8.4.6.2 PerformanceInformationWithRpAvailableNotification

8.4.6.2.1 Description

This notification informs the receiver that performance information is available and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.6.2.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the PerformanceInformationAvailableNotification in clause 8.5.8 of ETSI GS NFV-IFA 006 [1].

8.4.6.2.3 Attributes

The PerformanceInformationWithRpAvailableNotification shall comply with the indications in clause 8.5.8 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.6.2.3-1. All attributes of the PerformanceInformationAvailableNotification are also attributes of the PerformanceInformationWithRpAvailableNotification.

Table 8.4.6.2.3-1: Attributes of the PerformanceInformationWithRpAvailableNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the resources and is used by the VNFM to uniquely identify the resources for which information is available by means of the tuple [resourceProviderId, objectInstanceId].
(inherited attributes)				All attributes inherited from PerformanceInformationAvailableNotification.

8.4.6.3 ThresholdCrossedWithRpNotification

8.4.6.3.1 Description

This notification informs the receiver that a threshold value has been crossed and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.6.3.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the PerformanceInformationAvailableNotification in clause 8.5.9 of ETSI GS NFV-IFA 006 [1].

8.4.6.3.3 Attributes

The ThresholdCrossedWithRpNotification shall comply with the indications in clause 8.5.9 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.6.3.3-1. All attributes of the ThresholdCrossedNotification are also attributes of the ThresholdCrossedWithRpNotification.

Table 8.4.6.3.3-1: Attributes of the ThresholdCrossedWithRpNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the resources and is used by the VNFM to uniquely identify the resources for which the threshold is crossed by means of the tuple [resourceProviderId, objectInstanceId].
(inherited attributes)				All attributes inherited from ThresholdCrossedNotification.

8.4.7 Information elements and notifications related to Virtualised Resource Fault Management

8.4.7.1 Introduction

The clauses below define notifications related to virtualised resources fault management.

8.4.7.2 AlarmWithRpInfo information element

8.4.7.2.1 Description

The AlarmWithRpInfo information element encapsulates data of a virtualised resource alarm in indirect mode.

8.4.7.2.2 Attributes

The AlarmWithRpInfo information element shall comply with the provisions in clause 8.6.4 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.7.2.2-1. All attributes of the Alarm are also attributes of the AlarmWithRpInfo.

Table 8.4.7.2.2-1: Attributes of the AlarmWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for issuing the alarm, and is used by the VNFM to uniquely identify resources by means of the tuple [resourceProviderId, managedObjectId].
(inherited attributes)				All attributes inherited from Alarm.

8.4.7.3 AlarmWithRpNotification

8.4.7.3.1 Description

This notification encapsulates information on an alarm and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.7.3.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the AlarmNotification in clause 8.6.2 of ETSI GS NFV-IFA 006 [1].

8.4.7.3.3 Attributes

The AlarmWithRpNotification shall comply with the indications in clause 8.6.2 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.7.3.3-1. All attributes of the AlarmNotification are also attributes of the AlarmWithRpNotification.

Table 8.4.7.3.3-1: Attributes of the AlarmWithRpNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the alarm and is used by the VNFM to uniquely identify the alarm by means of the tuple [resourceProviderId, alarmId].
(inherited attributes)				All attributes inherited from AlarmNotification.

8.4.7.4 AlarmClearedWithRpNotification

8.4.7.4.1 Description

This notification encapsulates information on a cleared alarm and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.7.4.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the AlarmClearedNotification in clause 8.6.3 of ETSI GS NFV-IFA 006 [1].

8.4.7.4.3 Attributes

The AlarmClearedWithRpNotification shall comply with the indications in clause 8.6.3 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.7.4.3-1. All attributes of the AlarmClearedNotification are also attributes of the AlarmClearedWithRpNotification.

Table 8.4.7.4.3-1: Attributes of the AlarmClearedWithRpNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the alarm and is used by the VNFM to uniquely identify the alarm by means of the tuple [resourceProviderId, alarmId].
(inherited attributes)				All attributes inherited from AlarmClearedNotification.

8.4.8 Information elements and notifications related to Virtualised Resources Quota

8.4.8.1 Introduction

The clauses below define information elements and notifications related to the management of virtualised resources quota.

8.4.8.2 VirtualComputeQuotaWithRpInfo information element

8.4.8.2.1 Description

The VirtualComputeQuotaWithRpInfo information element encapsulates information about a quota for virtualised compute resources.

8.4.8.2.2 Attributes

The VirtualComputeQuotaWithRpInfo information element shall comply with the provisions in clause 8.8.2.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.8.2.2-1. All attributes of the VirtualComputeQuota are also attributes of the VirtualComputeQuotaWithRpInfo.

Table 8.4.8.2.2-1: Attributes of the VirtualComputeQuotaWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by the NFVO to determine the entity responsible for the management of the virtualised resources quota and is used by the VNFM to uniquely identify resources quota by means of the tuple [resourceProviderId, resourceGroupId].
(inherited attributes)				All attributes inherited from VirtualComputeQuota.

8.4.8.3 VirtualNetworkQuotaWithRpInfo information element

8.4.8.3.1 Description

The VirtualNetworkQuotaWithRpInfo information element encapsulates information about a quota for virtualised network resources.

8.4.8.3.2 Attributes

The VirtualNetworkQuotaWithRpInfo information element shall comply with the provisions in clause 8.8.3.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.8.3.2-1. All attributes of the VirtualNetworkQuota are also attributes of the VirtualNetworkQuotaWithRpInfo.

Table 8.4.8.3.2-1: Attributes of the VirtualNetworkQuotaWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by the NFVO to determine the entity responsible for the management of the virtualised resources quota and is used by the VNFM to uniquely identify resources quota by means of the tuple [resourceProviderId, resourceGroupId].
(inherited attributes)				All attributes inherited from VirtualNetworkQuota.

8.4.8.4 VirtualStorageQuotaWithRpInfo information element

8.4.8.4.1 Description

The VirtualStorageQuotaWithRpInfo information element encapsulates information about a quota for virtualised storage resources.

8.4.8.4.2 Attributes

The VirtualStorageQuotaWithRpInfo information element shall comply with the provisions in clause 8.8.4.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.8.4.2-1. All attributes of the VirtualStorageQuota are also attributes of the VirtualStorageQuotaWithRpInfo.

Table 8.4.8.4.2-1: Attributes of the VirtualStorageQuotaWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by the NFVO to determine the entity responsible for the management of the virtualised resources quota and is used by the VNFM to uniquely identify resources quota by means of the tuple [resourceProviderId, resourceGroupId].
(inherited attributes)				All attributes inherited from VirtualStorageQuota.

8.4.8.5 VirtualisedResourceQuotaWithRpChangeNotification

8.4.8.5.1 Description

This notification indicates a change in a virtualised resource quota and is applicable in the indirect mode of resource quota management. Support of this notification is mandatory.

8.4.8.5.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the VirtualisedResourceQuotaChangeNotification in clause 8.8.5.2 of ETSI GS NFV-IFA 006 [1].

8.4.8.5.3 Attributes

The VirtualisedResourceQuotaWithRpChangeNotification shall comply with the provisions in clause 8.8.5 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.8.5.3-1. All attributes of the VirtualisedResourceQuotaChangeNotification are also attributes of the VirtualisedResourceQuotaWithRpChangeNotification.

Table 8.4.8.5.3-1: Attributes of the VirtualisedResourceQuotaWithRpChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the change notification and is used by the VNFM to uniquely identify the resource quota by means of the tuple [resourceProviderId, resourceTypeId].
(inherited attributes)				All attributes inherited from VirtualisedResourceQuotaChangeNotification.

8.4.9 Information elements and notifications related to Virtualised Resources Reservation

8.4.9.1 Introduction

The clauses below define information elements and notifications related to the management of virtualised resources reservations.

8.4.9.2 ReservedVirtualComputeWithRpInfo information element

8.4.9.2.1 Description

The ReservedVirtualComputeWithRpInfo information element encapsulates information about a reservation for virtualised compute resources.

8.4.9.2.2 Attributes

The ReservedVirtualComputeWithRpInfo information element shall comply with the provisions in clause 8.7.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.9.2.2-1. All attributes of the ReservedVirtualCompute are also attributes of the ReservedVirtualComputeWithRpInfo.

Table 8.4.9.2.2-1: Attributes of the ReservedVirtualComputeWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources reservation by means of the tuple [resourceProviderId, reservationId].
(inherited attributes)				All attributes inherited from ReservedVirtualCompute.

8.4.9.3 ReservedVirtualNetworkWithRpInfo information element

8.4.9.3.1 Description

The ReservedVirtualNetworkWithRpInfo information element encapsulates information about a reservation for virtualised network resources.

8.4.9.3.2 Attributes

The ReservedVirtualNetworkWithRpInfo information element shall comply with the provisions in clause 8.7.4.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.9.3.2-1. All attributes of the ReservedVirtualNetwork are also attributes of the ReservedVirtualNetworkWithRpInfo.

Table 8.4.9.3.2-1: Attributes of the ReservedVirtualNetworkWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources reservation by means of the tuple [resourceProviderId, reservationId].
(inherited attributes)				All attributes inherited from ReservedVirtualNetwork.

8.4.9.4 ReservedVirtualStorageWithRpInfo information element

8.4.9.4.1 Description

The ReservedVirtualStorageWithRpInfo information element encapsulates information about a reservation for virtualised storage resources.

8.4.9.4.2 Attributes

The ReservedVirtualStorageWithRpInfo information element shall comply with the provisions in clause 8.7.6.2 of ETSI GS NFV-IFA 006 [1] with additional attributes provided in table 8.4.9.4.2-1. All attributes of the ReservedVirtualStorage are also attributes of the ReservedVirtualStorageWithRpInfo.

Table 8.4.9.4.2-1: Attributes of the ReservedVirtualStorageWithRpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the management of the Virtualised resource and is used by the VNFM to uniquely identify resources reservation by means of the tuple [resourceProviderId, reservationId].
(inherited attributes				All attributes inherited from ReservedVirtualStorage.

8.4.9.5 VirtualisedResourceReservationWithRpChangeNotification

8.4.9.5.1 Description

This notification indicates a change in a virtualised resource reservation and is applicable in the indirect mode of VNF-related resource reservation management.

Support of this notification is mandatory.

8.4.9.5.2 Trigger conditions

This notification is triggered with the same trigger conditions applicable to the VirtualisedResourceReservationChangeNotification in clause 8.7.7.2 of ETSI GS NFV-IFA 006 [1].

8.4.9.5.3 Attributes

The VirtualisedResourceReservationWithRpChangeNotification shall comply with the provisions in clause 8.7.7 of ETSI GS NFV-IFA 006 [1] with additional attributes of the notification according to table 8.4.9.5.3-1. All attributes of the VirtualisedResourceReservationChangeNotification are also attributes of the VirtualisedResourceReservationWithRpChangeNotification.

Table 8.4.9.5.3-1: Attributes of the VirtualisedResourceReservationWithRpChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceProviderId	M	1	Identifier	It is used by NFVO to determine the entity responsible for the change notification and is used by the VNFM to uniquely identify the resource reservation by means of the tuple [resourceProviderId, reservationId].
(inherited attributes)				All attributes inherited from VirtualisedResourceReservationChangeNotification.

8.5 Information elements related to VNF Lifecycle Management

8.5.1 Introduction

This clause defines information elements related to VNF Lifecycle Management.

8.5.2 VnfInfo information element

8.5.2.1 Description

The VnfInfo information element provides run-time information about a VNF instance.

8.5.2.2 Attributes

The VnfInfo information element shall follow the indications provided in table 8.5.2.2-1.

Table 8.5.2.2-1: Attributes of the VnfInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifier of the VNF instance that is represented by this VnfInfo information element.
vnfInstanceName	M	0..1	String	VNF instance name. See note 1.
vnfInstanceDescription	M	0..1	String	Human-readable description of the VNF instance. See note 1.
vnfdId	M	1	Identifier (Reference to Vnfd)	Identifier of the VNFD on which the VNF instance is based. See notes 1, 2 and 4.

Attribute	Qualifier	Cardinality	Content	Description
vnfProvider	M	1	String	See note 3.
vnfProductName	M	1	String	See note 3.
vnfSoftwareVersion	M	1	Version	See note 3.
vnfdVersion	M	1	Version	See note 3.
vnfConfigurableProperty	M	0..N	KeyValuePair	Additional VNF-specific attributes that provide the current values of the configurable properties of the VNF instance. These attributes represent values that are stored persistently in the VnfInfo information element and that correspond to configuration parameters of the VNF instance. Modifying the values of these attributes directly affects the configuration of the VNF instance if it exists. Configurable properties referred in this attribute shall be declared in the VNFD (see clause 7.1.12 in ETSI GS NFV-IFA 011 [3]). See notes 1 and 5.
vimConnectionInfo	CM	0..N	VimConnectionInfo	Information about VIM or CISM connection(s) for managing resources for the VNF instance. CONDITION: Shall be supported and present if VNF-related resource management in direct mode is applicable. If VIM connection information is provisioned to the VNFM by means outside the scope of the present document, the information in the "vimConnectionInfo" attribute provides necessary information for binding the VnfInfo to the applicable VIM connection information used to perform resource management for the VNF instance. See also the definition of the "VimConnectionInfo" in clause 8.12.5. See note 1.
cirConnectionInfo	M	0..N	VimConnectionInfo	Information about the CIR connection for managing OS container images for the VNF instance. Shall be present when the VNF is realized by a set of OS containers. See note 1.
mciopRepositoryInfo	M	0..N	VimConnectionInfo	Information about the MCIOP repository for the VNF instance. Shall be present when the VNF is realized by a set of OS containers. See note 1.
certificateInfo	M	0..1	CertificateInfo	Information about certificate and certificate management in this VNF. Shall be present when using delegation-mode, otherwise shall be absent. Content of this attribute shall not be either added (if previously not set) or removed (if previously set) by using the Modify VnfInfo operation. See notes 1 and 5.
instantiationState	M	1	Enum	The instantiation state of the VNF instance. VALUES: • NOT_INSTANTIATED: VNF instance is terminated or not instantiated, and the identifier of the VNF instance exists) • INSTANTIATED: VNF instance is instantiated

Attribute	Qualifier	Cardinality	Content	Description
instantiatedVnfInfo	M	0..1	InstantiatedVnfInfo	Information specific to an instantiated VNF instance. Shall be present if the VNF is in INSTANTIATED instantiation state.
metadata	M	0..N	KeyValuePair	Additional VNF-specific attributes that provide metadata describing the VNF instance. These attributes represent values that are stored persistently in the VnfInfo information element for consumption by functional blocks that invoke the VNF lifecycle management interface. They are not consumed by the VNFM or the lifecycle management scripts. Modifying the values of these attributes has no effect on the VNF instance, it only affects the information represented in VnfInfo. Metadata that the VNF provider foresees shall be declared in the VNFD (see clause 7.1.14.2 in ETSI GS NFV-IFA 011 [3]), like information about supported protocols and data models for configuring the VNF. The VNFM shall accept requests to write metadata that are not declared in the VNFD. See note 1.
extension	M	0..N	KeyValuePair	Additional VNF-specific attributes that affect the lifecycle management of this VNF instance. These attributes represent values that are stored persistently in the VnfInfo information element for consumption by the VNFM or the lifecycle management scripts during the execution of VF lifecycle management operations. Modifying the values of these attributes has no direct effect on the VNF instance; however, the modified attribute values can be considered during subsequent VNF lifecycle management operations, which means that the modified values can indirectly affect the configuration of the VNF instance. All extensions that are allowed for the VNF shall be declared in the VNFD (see clause 7.1.14.2 in ETSI GS NFV-IFA 011 [3]). See note 1.

NOTE 1: This attribute in the VnfInfo shall be writable through the Modify VNF information operation (refer to clause 7.2.12).
NOTE 2: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way.
NOTE 3: See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. This information is copied from the VNFD of the on-boarded VNF Package which was used to instantiate the VNF instance.
NOTE 4: Modifying the value of this attribute can be performed when no conflicts exist between the previous and the newly referred VNF Package, e.g. when the new VNFD is not changed with respect to the previous VNFD apart from referencing to other VNF software image(s). In order to avoid misalignment of the VnfInfo with the current VNF's on-boarded VNF Package, the values copied from the VNFD of the on-boarded VNF Package (see note 3) need to be kept in sync.
NOTE 5: These are sometimes also referred to as configuration parameters applicable to a VNF. Some of these are set prior to instantiation and cannot be modified if the VNF is instantiated, some are set prior to instantiation (are part of initial configuration) and can be modified later, and others can be set only after instantiation. The applicability of certain configuration may depend on the VNF and the required operation of the VNF at a certain point in time.

8.5.3 InstantiatedVnfInfo information element

8.5.3.1 Description

This information element provides run-time information specific to an instantiated VNF instance.

Annex A provides examples illustrating the relationship among the different run-time information elements (CP, VL and link ports) used to represent the connectivity of a VNF.

8.5.3.2 Attributes

The InstantiatedVnfInfo information element shall follow the indications provided in table 8.5.3.2-1.

Table 8.5.3.2-1: Attributes of the InstantiatedVnfInfo information element

Attribute	Qualifier	Cardinality	Content	Description
flavourId	M	1	Identifier (Reference to VnfDf)	Identifier of the VNF DF applied to this VNF instance. See note 1.
vnfState	M	1	Enum	The state of the VNF instance. VALUES: • STARTED • STOPPED
vnfPowerState	M	0..1	PowerStateInfo	The power state of the VNF instance. Only present if the VNF instance is in a power state or a scale status (refer to attribute "scaleStatus") that has an associated power profile, as defined in the VNFD.
scaleStatus	M	0..N	ScaleInfo	Scale status of the VNF, one entry per aspect. Shall be present if the VNF supports scaling. Represents for every scaling aspect how "big" the VNF has been scaled with regards to that aspect. For an aspect that has not been deployed because it consists of VDUs that belong to a deployable module that has not been selected (or to multiple deployable modules, none of which have been selected), it indicates the scale level that has been requested in the instantiation or in a scaling operation, or, if none has been requested in any of them, the scale level applicable to the aspect based on the default instantiation level. See notes 2 and 5.
maxScaleLevel	M	0..N	ScaleInfo	Maximum allowed scale levels of the VNF, one entry per aspect, as defined in the VNFD. This attribute shall be present if the VNF supports scaling. Represents for every scaling aspect how "big" the VNF can be scaled with regards to that aspect. See note 2.

Attribute	Qualifier	Cardinality	Content	Description
selectedDeployableModule	M	0..N	Identifier (Reference to DeployableModule)	References a currently selected deployable module, as defined in the VNFD, that has already completed the instantiation of its VNFCs.
resourceCapacityDefinition	M	0..N	ResourceCapacityDefinition	Shows current values of VDU attributes related to resource capacity, if different to the default values from the VNFD, as indicated in the request(s) of the last completed VNF LCM operation(s) that contain this attribute. If an attribute value has been modified multiple times, only the last value is shown. The values indicated in this attribute are applicable to all VNFC instances based on the VDU to which the resourceCapacityDefinition is related.
extCpInfo	M	1..N	VnfExtCpInfo	External CPs exposed by the VNF instance. See note 6.
vipCpInfo	M	0..N	VipCpInfo	VIP CPs that are part of the VNF instance. Shall be present when that particular VIP CP of the VNFC instance is associated to an external CP of the VNF instance. May be present otherwise.
virtualCpInfo	M	0..N	VirtualCpInfo	Virtual CPs that are part of the VNF instance. Shall be present when a particular Virtual CP is associated to an external CP of the VNF instance. May be present otherwise.
extVirtualLinkInfo	M	0..N	ExtVirtualLinkInfo	External VLs the VNF instance is connected to.
extManagedVirtualLinkInfo	M	0..N	ExtManagedVirtualLinkInfo	Externally-managed internal VLs of the VNF instance. See note 4.
monitoringParameter	M	0..N	Not specified	Performance metrics tracked by VNFM (e.g. for auto-scaling purposes). See note 3.
localizationLanguage	M	0..1	Not specified	Information about localization language of the VNF (includes e.g. strings in the VNFD). The localization languages supported by a VNF can be declared in the VNFD, and localization language selection can take place at instantiation time.
vnfcResourceInfo	M	0..N	VnfcResourceInfo	Information on the virtualised compute and storage resource(s) used by the VNFCs of the VNF instance.
vnfVirtualLinkResourceInfo	M	0..N	VnfVirtualLinkResourceInfo	Information on the virtualised network resource(s) used by the VLs of the VNF instance.
virtualStorageResourceInfo	M	0..N	VirtualStorageResourceInfo	Information on the virtualised storage resource(s) used as storage for the VNF instance.
mcioInfo	M	0..N	McioInfo	Information on the MCIO(s) representing VNFC instance(s) realized by one or a set of OS containers and created from the same VDU for the VNF instance.

Attribute	Qualifier	Cardinality	Content	Description
vnfPaasServiceInfo	M	0..N	PaasServiceInfo	Information on the PaaS Services assigned and used by the VNF instance.

NOTE 1: The VnfDf information element is defined in ETSI GS NFV-IFA 011 [3], clause 7.1.8.2.
NOTE 2: For every scaling aspect, the information provided by the "scaleStatus" and "maxScaleLevel" attributes allows an external entity to derive how many scaling steps are possible for scaling in or scaling out a VNF instance. Per aspect, the number of steps possible to scale in corresponds to the "scaleLevel" attribute for that aspect in the "scaleStatus" information element, and the possible number of steps to scale out corresponds to the difference between "maxScaleLevel" for that aspect, and the "scaleLevel" attribute for that aspect in the "scaleStatus" information element.
NOTE 3: The monitoring parameters to be tracked by VNFM are identified by VNF provider in the VNFD. The VNFM collects the values of identified performance metrics using one or more locally initiated PM Jobs.
NOTE 4: It is possible to have several ExtManagedVirtualLinkInfo for the same VNF internal VL in case of a multi-site VNF spanning several VIMs. The set of ExtManagedVirtualLinkInfo corresponding to the same VNF internal VL shall indicate so by referencing to the same VnfVirtualLinkDesc and externally-managed multi-site VL instance (refer to clause 8.5.10).
NOTE 5: For a scaling aspect whose related VNFCs have not been instantiated due to the selection of deployable modules, the "scaleStatus" indicates the scale level that would be applicable to the aspect if a VNF LCM operation changes the selected deployable modules and the related VNFCs are instantiated, unless the VNF LCM operation explicitly indicates the scale level for the aspect.
NOTE 6: Subports need not be used for containerized VNFCs. The application container can send and receive IP packets with any VLAN tag as long as the network interface to connect to the secondary container cluster network has been configured appropriately. Thus, no individual extCpInfo, except the one representing the trunk, need be modelled to allow traffic tagged with a particular VLAN through the connection point.

8.5.4 VnfcResourceInfo information element

8.5.4.1 Description

This information element provides information on virtualised compute and storage resources used by a VNFC in a VNF instance.

Depending on the form of virtualisation container of the VNFC:

- For a VNFC based on VM, a reference to the corresponding VirtualCompute shall be provided; and
- For a VNFC based on OS container(s), a reference to the Compute MCIO shall be provided. Hence, exposure of information by the VNFM to the NFVO is at the MCIO level.

In addition, the references to the storage resources depend on the form of the VNFC:

- For a VNFC based on VM, storage resource identifiers shall refer to VirtualStorage resources; and
- For a VNFC based on OS container(s), storage resource identifiers shall refer to Storage MCIOs.

8.5.4.2 Attributes

The VnfcResourceInfo information element shall follow the indications provided in table 8.5.4.2-1.

Table 8.5.4.2-1: Attributes of the VnfcResourceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfcInstanceId	M	1	Identifier	Identifier of this VNFC instance.
vduId	M	1	Identifier (Reference to Vdu)	Reference to the applicable Vdu information element in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure). See note 1.

Attribute	Qualifier	Cardinality	Content	Description
computeResource	M	1	ResourceHandle	Reference to the VirtualCompute resource or reference to a Compute MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.
storageResourceId	M	0..N	Identifier (Reference to VirtualStorageResourceInfo)	Reference(s) to the VirtualStorage resource(s) or references to Storage MCIO(s). Information about the resource(s) is available from the VIM or the CISM.
reservationId	M	0..1	Identifier	The reservation identifier applicable to the resource. It shall be present when an applicable reservation exists.
vnfcCpInfo	M	0..N	VnfcCpInfo	CP(s) of the VNFC instance. Shall be present when that particular CP of the VNFC instance is associated to an external CP of the VNF instance. May be present otherwise. See note 2.
metadata	M	0..N	KeyValuePair	Metadata about this resource.
trunkPortsInfo	M	0..N	TrunkPortsInfo	Collections of CPs of the VNFC instance in trunk(s). Shall be present when the VNFC has CPs working in trunk mode, as parent port of a trunk, and other CPs working as subports of the same trunk, and the referred CP instances are also present in the vnfcCpInfo attribute.
certificateContentId	CM	0..N	Identifier (Reference to CertificateContent)	The identifier(s) of certificate(s) that this VNFC instance uses. Shall be present when using in delegation-mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode in certificate management is applicable.
NOTE 1: If only the value or the presence of this attribute is changed in the "VnfcResourceInfo" information element by an LCM operation occurrence, this does not represent a change that requires including a related "AffectedVnfc" information element in the VNF LCM operation occurrence notifications related to this LCM operation occurrence. NOTE 2: Subports need not be used for containerized VNFCs. The application container can send and receive IP packets with any VLAN tag as long as the network interface to connect to the secondary container cluster network has been configured appropriately. Thus, no individual vnfcCpInfo, except the one representing the trunk, need be modelled to allow traffic tagged with a particular VLAN through the connection point.				

8.5.5 VnfVirtualLinkResourceInfo information element

8.5.5.1 Description

This information element provides information on virtualised network resources used by an internal VL instance in a VNF.

8.5.5.2 Attributes

The VnfVirtualLinkResourceInfo information element shall follow the indications provided in table 8.5.5.2-1.

Table 8.5.5.2-1: Attributes of the VnfVirtualLinkResourceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
virtualLinkInstanceId	M	1	Identifier	Identifier of this VL instance.
vnfVirtualLinkDescId	M	1	Identifier (Reference to VnfVirtualLinkDesc)	Identifier of the VNF Virtual Link Descriptor (VLD) in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure). See note.
networkResource	M	1	ResourceHandle	Reference to the VirtualNetwork resource or reference to a Network MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.
reservationId	M	0..1	Identifier	The reservation identifier applicable to the resource. It shall be present when an applicable reservation exists.
vnfLinkPort	M	0..N	VnfLinkPortInfo	Links ports of this VL. Shall be present when the linkPort is used for external connectivity by the VNF (refer to VnfLinkPortInfo in clause 8.5.11). May be present otherwise.
metadata	M	0..N	KeyValuePair	Metadata about this resource.
NOTE: If only the value or the presence of this attribute is changed in the "VnfVirtualLinkResourceInfo" information element by an LCM operation occurrence, this does not represent a change that requires including a related "AffectedVirtualLink" information element in the VNF LCM operation occurrence notifications related to this LCM operation occurrence.				

8.5.6 VirtualStorageResourceInfo information element

8.5.6.1 Description

This information element provides information on virtualised storage resources used by a storage instance in a VNF.

8.5.6.2 Attributes

The VirtualStorageResourceInfo information element shall follow the indications provided in table 8.5.6.2-1.

Table 8.5.6.2-1: Attributes of the VirtualStorageResourceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
virtualStorageInstanceId	M	1	Identifier	Identifier of this virtual storage resource instance.
virtualStorageDescId	M	1	Identifier (Reference to VirtualStorageDesc)	Identifier of the VirtualStorageDesc in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure). See note.

Attribute	Qualifier	Cardinality	Content	Description
storageResource	M	1	ResourceHandle	Reference to the VirtualStorage resource or reference to a Storage MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.
reservationId	M	0..1	Identifier	The reservation identifier applicable to the resource. It shall be present when an applicable reservation exists.
metadata	M	0..N	KeyValuePair	Metadata about this resource.
NOTE: If only the value or the presence of this attribute is changed in the "VirtualStorageResourceInfo" information element by an LCM operation occurrence, this does not represent a change that requires including a related "AffectedVirtualStorage" information element in the VNF LCM operation occurrence notifications related to this LCM operation occurrence.				

8.5.7 ResourceHandle information element

8.5.7.1 Description

This information element provides information that allows addressing a resource that is used by a VNF instance.

Information about the resource is available from the corresponding Virtualised Compute/Storage/Network Resource Management interfaces or the OS container compute/storage/network management service interfaces. Table 8.5.7.1-1 shows the relationship between the resourceId attribute of ResourceHandle specified in the present document and the resource identifiers used in the aforementioned interfaces specified in ETSI GS NFV-IFA 005 [i.4] and ETSI GS NFV-IFA 006 [1].

Table 8.5.7.1-1: Relationship between resource identifiers managed by a VIM

Attribute in Or-Vnfm ref. point	Type, Interface, information element and attribute in ETSI GS NFV-IFA 005 [i.4] and ETSI GS NFV-IFA 006 [1]		
	Type	Interface	Information element and attribute
ResourceHandle:resourceId	Compute	Virtualised Compute Resource Management	VirtualCompute:computeId
	Storage	Virtualised Storage Resource Management	VirtualStorage:storageId
	Network	Virtualised Network Resource Management	VirtualNetwork:networkResourceId

Table 8.5.7.1-2 shows the relationship between the resourceId attribute of ResourceHandle specified in the present document and the managed object used in the interface requirements specified in ETSI GS NFV-IFA 040 [i.13].

Table 8.5.7.1-2: Relationship between resource identifiers managed by a CISM

Attribute in Or-Vnfm ref. point	Type, Interface, managed object in ETSI GS NFV-IFA 040 [i.13]		
	Type	Interface	Identifier
ResourceHandle:resourceId	Compute	OS container compute management service	ID of Compute MCIOs
	Storage	OS container storage management service	ID of Storage MCIOs
	Network	OS container network management service	ID of Network MCIOs

8.5.7.2 Attributes

The ResourceHandle information element shall follow the indications provided in table 8.5.7.2-1.

Table 8.5.7.2-1: Attributes of the ResourceHandle information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Reference to the identifier of the VimConnectionInfo information element defining the VIM or CISM Connection to manage this resource. This parameter shall be supported when the resources are managed by a CISM. CONDITION: This attribute shall be supported when VNF-related Resource Management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the virtualised resource. CONDITION: This attribute shall be supported when VNF-related Resource Management in indirect mode is applicable.
resourceId	M	1	Identifier	Identifier of the resource in the scope of the VIM or the CISM or the resource provider.
vimLevelResourceType	M	0..1	Not specified	Type of the resource in the scope of the VIM or the CISM or the resource provider. See note 1.
vimLevelAdditionalResourceInfo	M	0..1	Not specified	Additional resource information which is specific to this resource and its type, and which is available from the VIM or the CISM or the resource provider. See note 2.
containerNamespace	M	0..1	String	The value of the namespace in which the MCIO corresponding to the resource is deployed. This attribute shall be present if the resource is managed by a CISM and it shall be absent otherwise.

NOTE 1: The value set of the "vimLevelResourceType" attribute is within the scope of the VIM or the CISM or the resource provider and can be used as information that complements the ResourceHandle. This value set is different from the value set of the "type" attribute in the ResourceDefinition (refer to clause 8.3.2).

NOTE 2: Which structure and content of the resource information to be expected depends on the type of resource and its provider. The information shall be limited to properties directly owned by the resource referenced in this ResourceHandle.

8.5.8 ScaleInfo information element

8.5.8.1 Description

This information element provides information about the scale level of a VNF instance with regards to one scaling aspect.

8.5.8.2 Attributes

The ScaleInfo information element shall follow the indications provided in table 8.5.8.2-1.

Table 8.5.8.2-1: Attributes of the ScaleInfo information element

Attribute	Qualifier	Cardinality	Content	Description
aspectId	M	1	Identifier (Reference to ScalingAspect)	Reference to the scaling aspect.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure).
scaleLevel	M	1	Integer	The scale level for that aspect. Minimum value 0, maximum value maxScaleLevel as declared in the VNFD (see ETSI GS NFV-IFA 011 [3], clause 7.1.10.2.2).

8.5.9 ExtVirtualLinkInfo information element

8.5.9.1 Description

This information element provides a reference to an external VL.

8.5.9.2 Attributes

The ExtVirtualLinkInfo information element shall follow the indications provided in table 8.5.9.2-1.

Table 8.5.9.2-1: Attributes of the ExtVirtualLinkInfo information element

Attribute	Qualifier	Cardinality	Content	Description
extVirtualLinkId	M	1	Identifier	Identifier of this external VL. The identifier is assigned by the NFV-MANO entity that manages this VL instance.
resourceHandle	M	1	ResourceHandle	Reference to the resource realizing this VL.
extLinkPort	M	0..N	ExtLinkPortInfo	Link ports of this VL.
extNetAttDefResource	M	0..N	NetAttDefResourceInfo	Network attachment definition resources that provide the specification of the interface to attach connection points to this VL.
routingResource	M	0..N	ResourceHandle	Network resources that provide the forwarding/routing capabilities associated to the network resource realizing this VL.

8.5.10 ExtManagedVirtualLinkInfo information element

8.5.10.1 Description

This information element provides a reference to an externally-managed internal VL.

8.5.10.2 Attributes

The ExtManagedVirtualLinkInfo information element shall follow the indications provided in table 8.5.10.2-1.

Table 8.5.10.2-1: Attributes of the ExtManagedVirtualLinkInfo information element

Attribute	Qualifier	Cardinality	Content	Description
extManagedVirtualLinkId	M	1	Identifier	Identifier of this externally-managed internal VL. The identifier is assigned by the NFV-MANO entity that manages this VL instance.

Attribute	Qualifier	Cardinality	Content	Description
vnfVirtualLinkDescId	M	1	Identifier (Reference to VnfVirtualLinkDesc)	Identifier of the VNF Virtual Link Descriptor (VLD) in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure).
networkResource	M	1	ResourceHandle	Reference to the VirtualNetwork resource providing this VL.
vnfLinkPort	M	0..N	VnfLinkPortInfo	Link ports of this VL. See note.
vnfNetAttDefResource	M	0..N	NetAttDefResourceInfo	Network attachment definition resources that provide the specification of the interface to attach connection points to this VL. See note.
extManagedMultisiteVirtualLinkId	M	0..1	Identifier	Identifier of the externally-managed multi-site VL instance. The identifier is assigned by the NFV-MANO entity that manages the externally managed multi-site VL instance. It shall be present when the externally-managed internal VL is part of a multi-site VL, e.g. in support of multi-site VNF spanning several VIMs. All externally-managed internal VL instances corresponding to a an internal VL created based on the same virtualLinkDescId shall refer to the same extManagedMultisiteVirtualLinkId.
routingResource	M	0..N	ResourceHandle	Network resources that provide the forwarding/routing capabilities associated to the network resource realizing this VL.
NOTE: Both vnfLinkPort and vnfNetAttDefResource can be present in an ExtManagedVirtualLinkInfo to indicate that a single internal virtual link is providing connectivity for both VM-based and container-based VNFCs.				

8.5.11 VnfLinkPortInfo information element

8.5.11.1 Description

This information element provides information about a port of a VNF's internal VL. See also VnfVirtualLinkResourceInfo in clause 8.5.5.

8.5.11.2 Attributes

The attributes of the VnfLinkPortInfo information element shall follow the indications provided in table 8.5.11.2-1.

Table 8.5.11.2-1: Attributes of the VnfLinkPortInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfLinkPortId	M	1	Identifier	Identifier of this link port as provided by the entity that has created the link port.
resourceHandle	M	1	ResourceHandle	Reference to the virtualised resource realizing this link port.

Attribute	Qualifier	Cardinality	Content	Description
associatedExtCpId	M	0..1	Identifier (Reference to VnfExtCpInfo)	External CP of the VNF associated to this link port. When an external CP is associated to an internal VL, this attribute reflects this association. Shall be present when the link port is used for external connectivity by the VNF. See notes 1, 2, 3 and 4.
vnfcCpInstanceId	M	0..1	Identifier (Reference to VnfcCpInfo)	VNFC CP of the VNF connected to this link port. May be present. See notes 1, 3 and 4.
vipCpInstanceId	M	0..1	Identifier (Reference to VipCpInfo)	VIP CP instance of the VNF connected to this link port. May be present. See notes 1, 3, 4 and 5.
NOTE 1: There shall be at most one link port associated with any external connection point instance or internal connection point (i.e. VNFC CP) instance or VIP CP instance. NOTE 2: A VnfLinkPort does not terminate on an external CP, as external CPs are connected to external VLs. NOTE 3: Either associatedExtCpId or any combination of vnfcCpInstanceId and vipCpInstanceId (i.e. one or both of them) shall be present for a VnfLinkPortInfo. In case both vnfcCpInstanceId and vipCpInstanceId are present, the two different CP instances share the linkport. NOTE 4: The attributes "associatedExtCpId" and "vnfcCpInstanceId" model two separate associations in the information model. To represent these in the data model during the protocol design stage, an alternative representation of these associations could be chosen as well. NOTE 5: Clause A.4 provides examples for configurations where both vipCpInstanceId and vnfcCpInstanceId are present (UC#5 and UC#5-b), only vnfcCpInstanceId is present (UC#2), or only vipCpInstanceId is present (UC6 and UC#6-b).				

8.5.12 VnfExtCpInfo information element

8.5.12.1 Description

This information element provides information related to an external CP.

8.5.12.2 Attributes

The VnfExtCpInfo information element shall follow the indications provided in table 8.5.12.2-1.

Table 8.5.12.2-1: Attributes of the VnfExtCpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier	Identifier of this external CP instance and of this VnfExtCpInfo information element.
cpdId	M	1	Identifier (Reference to VnfExtCpd)	Identifier of the external Connection Point Descriptor (CPD), VnfExtCpd, in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure).
cpProtocolInfo	M	0..N	CpProtocolInfo	Protocol information for this CP. There shall be one cpProtocolInfo for each layer protocol supported.
associatedVnfcCpId	M	0..1	Identifier (Reference to VnfcCpInfo)	Identifier of the VnfcCp that is exposed as this VnfExtCp, either directly or via a floating IP address. Shall be present if the cpdId of this VnfExtCp has an intCpd attribute. See note 1.
associatedVipCpId	M	0..1	Identifier (Reference to VipCpInfo)	Identifier of the VIP CP that is exposed as this VnfExtCp instance, either directly or via a floating IP address. Shall be present if the cpdId of this VnfExtCp has a vipCpd attribute. See note 1.

Attribute	Qualifier	Cardinality	Content	Description
associatedVirtualCpId	M	0..1	Identifier (Reference to VirtualCpInfo)	Identifier of the VirtualCp that is exposed as this VnfExtCp. Shall be present if the cpdId of this VnfExtCp has a virtualCpd attribute. See note 1.
associatedVnfVirtualLinkId	M	0..1	Identifier (reference to VnfVirtualLinkResourceInfo)	Identifier of the Vnf VL that this VnfExtCP maps to. Shall be present if the cpdId of this VnfExtCp has an intVirtualLinkDesc attribute. See note 1.
extLinkPortId	M	0..1	Identifier (Reference to ExtLinkPortInfo)	Identifier of the "ExtLinkPortInfo" information element in the "ExtVirtualLinkInfo" information element. Shall be present if the CP is associated to a link port. See note 2.
netAttDefResourceId	M	0..N	Identifier (Reference to NetAttDefResourceInfo)	Identifier of the network attachment definition resource(s) that provides the specification of the interface to attach the connection point to a secondary container cluster network. See notes 3 and 4. It shall be present if the external CP is associated to a VNFC realized by one or a set of OS containers and is connected to a secondary container cluster network. It shall not be present otherwise.
metadata	M	0..N	KeyValuePair	Metadata about this external CP.
certificateContentId	CM	0..N	Identifier (Reference to CertificateContent)	The identifier(s) of certificate(s) that this VNF CP instance uses. Shall be present when using in delegation-mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode in certificate management is applicable.
cpSecurityGroupInfo	M	0..1	Not specified	List of active security group rules applied to this external CP instance. The attribute identifies which "SecurityGroupRule" specified in the VNFD are activated and the values of its attributes.

NOTE 1: The attributes associatedVnfcCpId, associatedVipCpId, associatedVirtualCpId and associatedVnfVirtualLinkId are mutually exclusive. Exactly one shall be present.
NOTE 2: An external CP is not associated to a link port in the cases indicated for the "extLinkPorts" attribute in clause 8.12.2.2.
NOTE 3: Cardinality greater than 1 is only applicable for specific cases where more than one network attachment definition resource is needed to fulfil the connectivity requirements of the extCP, e.g. to build a link redundant mated pair in SR-IOV cases.
NOTE 4: When more than one netAttDefResourceId is indicated, all shall belong to the same namespace.

8.5.13 ExtLinkPortInfo information element

8.5.13.1 Description

This information element provides information about a port of an external VL, i.e. a port providing connectivity for the VNF to an NS VL.

8.5.13.2 Attributes

The attributes of the ExtLinkPortInfo information element shall follow the indications provided in table 8.5.13.2-1.

Table 8.5.13.2-1: Attributes of the ExtLinkPortInfo information element

Attribute	Qualifier	Cardinality	Content	Description
extLinkPortId	M	1	Identifier	Identifier of this link port as provided by the entity that has created the link port.
resourceHandle	M	1	ResourceHandle	Reference to the virtualised resource realizing this link port.
cpInstanceId	M	0..1	Identifier (Reference to VnfExtCpInfo)	External CP of the VNF connected to this link port. See note 1.
secondaryCpInstanceId	M	0..1	Identifier (Reference to VnfExtCpInfo)	Additional external CP of the VNF connected to this link port. If present, this attribute shall refer to a "secondary" ExtCpInfo item in the VNF instance that exposes a virtual IP CP instance which shares this linkport with the external CP instance referenced by the "cpInstanceId" attribute. See note 1 and note 2.
NOTE 1: There shall be at most one link port associated with any external connection point instance. NOTE 2: The use cases UC#4 and UC#5 in clause A.4 provide examples for such a configuration.				

8.5.14 VnfcCpInfo information element

8.5.14.1 Description

This information element provides information related to a CP of a VNFC.

8.5.14.2 Attributes

The VnfcCpInfo information element shall follow the indications provided in table 8.5.14.2-1.

Table 8.5.14.2-1: Attributes of the VnfcCpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier	Identifier of this VnfcCpInfo information element.
cpdId	M	1	Identifier (Reference to VduCpd)	Identifier of the VDU CPD, cpdId, in the VNFD.
vnfExtCpId	M	0..1	Identifier (Reference to VnfExtCpInfo)	When the VNFC CP is exposed as external CP of the VNF, the identifier of this external VNF CP.
cpProtocolInfo	M	0..N	CpProtocolInfo	Protocol information for this CP. There shall be one cpProtocolInfo for each layer protocol supported.
vnfLinkPortId	M	0..1	Identifier (Reference to VnfLinkPortInfo)	Identifier of the "VnfLinkPortInfo" information element in the "VnfVirtualLinkResourceInfo" information element. Shall be present if the CP is associated to a link port.
netAttDefResourceId	M	0..N	Identifier (Reference to NetAttDefResourceInfo)	Identifier of the network attachment definition resource(s) that provides the specification of the interface to attach the connection point to a secondary container cluster network. See notes 1 and 2. It shall be present if the internal CP is associated to a VNFC realized by one or a set of OS containers and is connected to a secondary container cluster network. It shall not be present otherwise.
certificateContentId	CM	0..N	Identifier (Reference to CertificateContent)	The identifier(s) of certificate(s) that this VNFC CP instance uses. Shall be present when using in delegation-mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode in certificate management is applicable.
metadata	M	0..N	KeyValuePair	Metadata about this VNFC CP.
NOTE 1: Cardinality greater than 1 is only applicable for specific cases where more than one network attachment definition resource is needed to fulfil the connectivity requirements of the internal CP, e.g. to build a link redundant mated pair in SR-IOV cases. NOTE 2: When more than one netAttDefResourceId is indicated, all shall belong to the same namespace.				

8.5.15 CpProtocolInfo information element

8.5.15.1 Description

This information element describes and associates the protocol layer that a CP uses together with other protocol-related information, like addresses.

8.5.15.2 Attributes

The CpProtocolInfo information element shall follow the indications provided in table 8.5.15.2-1.

Table 8.5.15.2-1: Attributes of the CpProtocolInfo information element

Attribute	Qualifier	Cardinality	Content	Description
layerProtocol	M	1	Enum	Identifies which protocol the CP uses for connectivity purposes. See note 1. VALUES: • IP_OVER_ETHERNET • IP_FOR_VIRTUAL_CP • Etc.
address	M	1..N	Not specified.	List of network addresses that have been configured (statically or dynamically) on the link port that connects the CP to a VL. See note 2.
domainNames	M	0..N	Not specified	Domain names that have been configured (statically or dynamically) and been associated to the CP.
NOTE 1: The layerProtocol values shall be compatible with the ones defined in the CPD. NOTE 2: The address information shall be compatible with the layerProtocol attribute.				

8.5.16 VnfSnapshotInfo information element

8.5.16.1 Description

This information element provides the details of a VNF Snapshot, which the VNFM creates and stores as part of the ongoing VNF Lifecycle Management operations related to VNF Snapshots.

8.5.16.2 Attributes

The VnfSnapshotInfo information element shall follow the indications provided in table 8.5.16.2-1.

Table 8.5.16.2-1: Attributes of the VnfSnapshotInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfSnapshotInfoId	M	1	Identifier	Identifier of information held by the VNFM about a specific VNF Snapshot. This identifier was allocated by the VNFM.
triggeredAt	M	1	DateTime	Timestamp indicating when the VNF Snapshot creation has been started.
createdAt	M	0..1	DateTime	Timestamp indicating when the VNF Snapshot creation has been completed. Cardinality is 0 when the VNF Snapshot creation has not yet completed and shall be 1 afterwards.
vnfInstanceId	M	1	Identifier	Identifier of the snapshotted VNF instance.
vnfdId	M	1	Identifier (Reference to Vnfd)	References the VNFD in use at the time the snapshot of the VNF instance has been created. See notes 1 and 2.
vnfInfo	M	1	VnfInfo	VnfInfo of the snapshotted VNF instance.
vnfcSnapshotInfo	M	1..N	VnfcSnapshotInfo	Information about VNFC Snapshots constituting this VNF Snapshot.
vnfStateSnapshotInfo	M	0..1	VnfStateSnapshotInfo	Information about VNF-specific state snapshot data. This attribute shall not be present before the VNF snapshot has been completed. Otherwise, this attribute shall be present if the VNF snapshot has associated additional VNF-specific state data.
userDefinedData	O	0..N	KeyValuePair	User defined data for the VNF Snapshot.

Attribute	Qualifier	Cardinality	Content	Description
NOTE 1: This identifier, which is managed by the VNF provider, identifies the VNF Package and the VNFD in a globally unique way. NOTE 2: See ETSI GS NFV-IFA 011 [3], clause 7.1.2.2. This information is copied from the VNFD of the on-boarded VNF Package which was used to instantiate the VNF instance.				

8.5.17 VnfcSnapshotInfo information element

8.5.17.1 Description

This information element provides the details of a VNFC Snapshot, which the VNFM creates and stores as part of the ongoing VNF Lifecycle Management operations related to VNF Snapshots.

8.5.17.2 Attributes

The VnfcSnapshotInfo information element shall follow the indications provided in table 8.5.17.2-1.

Table 8.5.17.2-1: Attributes of the VnfcSnapshotInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfcSnapshotInfoId	M	1	Identifier	Identifier of information held by the VNFM about a specific VNFC Snapshot. This identifier was allocated by the VNFM.
triggeredAt	M	1	DateTime	Timestamp indicating when the VNFC Snapshot creation has been started.
createdAt	M	0..1	DateTime	Timestamp indicating when the VNFC Snapshot creation has been completed. Cardinality is 0 when the VNF Snapshot creation has not yet completed and shall be 1 afterwards.
vnfcInstanceId	M	1	Identifier	Identifier of the snapshotted VNFC instance.
vnfcInfoId	M	1	Identifier (Reference to VnfcResourceInfo)	Reference to the information about the snapshotted VNFC instance.
computeSnapshotResource	M	0..1	ResourceHandle	Reference to a compute snapshot resource. See note.
storageSnapshotResource	M	0..N	StorageSnapshotResource	Mapping of the storage resources associated to the VNFC with the storage snapshot resources.
userDefinedData	O	0..N	KeyValuePair	User defined data for the VNFC Snapshot.
NOTE: The identifier of the compute snapshot resource is assigned during creation of a VNFC Snapshot being returned from the VIM as output data in the response message of the individual resource operations. This attribute shall only be present for a VNFC snapshot that has been newly created by the VNFM as a result of the "Create Snapshot operation".				

8.5.18 StorageSnapshotResource information element

8.5.18.1 Description

This information element provides a mapping of the storage resources associated to the VNFC with the storage snapshot resources.

8.5.18.2 Attributes

The StorageSnapshotResource information element shall follow the indications provided in table 8.5.18.2-1.

Table 8.5.18.2-1: Attributes of the StorageSnapshotResource information element

Attribute	Qualifier	Cardinality	Content	Description
storageResourceId	M	1	Identifier (Reference to VirtualStorageResourceInfo)	Reference to a virtual storage resource.
storageSnapshotResource	M	0..1	ResourceHandle	Reference to a storage snapshot resource. See note.
NOTE: The identifier of the storage snapshot resource is assigned during creation of a VNFC Snapshot being returned from the VIM as output data in the response message of the individual resource operations. This attribute shall only be present for a VNFC snapshot with associated storage resources and that has been newly created by the VNFM as a result of the "Create Snapshot operation".				

8.5.19 TrunkPortsInfo information element

8.5.19.1 Description

The information element provides runtime information of a collection of CP(s) of the VNFC instance which has one CP working in trunk mode, as parent port of a trunk, and other CPs working as subports of the same trunk.

8.5.19.2 Attributes

The attributes of the TrunkPortsInfo information element shall follow the indications provided in table 8.5.19.2-1.

Table 8.5.19.2-1: Attributes of the TrunkPortsInfo information element

Attribute	Qualifier	Cardinality	Content	Description
parentPort	M	1	Identifier (Reference to VnfcCpInfo)	Reference to the CP instance which is used as parent port in the trunk.
subportList	M	0..N	Identifier (Reference to VnfcCpInfo)	Reference to the CP instance(s) working as subport(s) in the trunk.

8.5.20 VipCpInfo information element

8.5.20.1 Description

This information element provides information related to a VIP CP.

8.5.20.2 Attributes

The VipCpInfo information element shall follow the indications provided in table 8.5.20.2-1.

Table 8.5.20.2-1: Attributes of the VipCpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier	Identifier of this VIP CP instance and of this VipCpInfo information element.
cpdId	M	1	Identifier (Reference to VipCpd)	Identifier of the VIP Connection Point Descriptor, VipCpd, in the VNFD.

Attribute	Qualifier	Cardinality	Content	Description
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case the value differs from the vnfdId attribute of the VNF instance (e.g. during a "Change current VNF package" operation or due to its final failure). See note 2.
vnfExtCpId	M	0..1	Identifier (Reference to VnfExtCpInfo)	When the VIP CP is exposed as external CP of the VNF, the identifier of this external VNF CP instance.
cpProtocolInfo	M	0..N	CpProtocolInfo	Protocol information for this CP. There shall be one cpProtocolInfo for layer 3. There may be one cpProtocolInfo for layer 2.
associatedVnfcCpId	M	0..N	Identifier (Reference to VnfcCpInfo)	Identifiers of the VnfcCps that share the virtual IP addresse allocated to the VIP CP instance. See note 1.
vnfLinkPortId	M	0..1	Identifier (Reference to VnfLinkPortInfo)	Identifier of the "VnfLinkPortInfo" information element in the "VnfVirtualLinkResourceInfo" information element. Shall be present if the CP is associated to a link port in an internal VL.
metadata	M	0..N	KeyValuePair	Metadata about this VIP CP.
NOTE 1: It is possible that there is no associated VnfcCp because the VIP CP is available but not associated yet. NOTE 2: If only the value or the presence of this attribute is changed in the "VipCpInfo" information element by an LCM operation occurrence, this does not represent a change that requires including a related "AffectedVipCp" information element in the VNF LCM operation occurrence notifications related to this LCM operation occurrence.				

8.5.21 VnfStateSnapshotInfo information element

8.5.21.1 Description

This information element represents information about VNF-specific state snapshot data and where to retrieve it.

8.5.21.2 Attributes

The VnfStateSnapshotInfo information element shall follow the indications provided in table 8.5.21.2-1.

Table 8.5.21.2-1: Attributes of the VnfStateSnapshotnfo information element

Attribute	Qualifier	Cardinality	Content	Description
accessInformation	M	1	Not specified	Information (such as a path) that identifies/addresses this VNF state snapshot.
metadata	M	1	Not specified	The metadata of the VNF state snapshot, such as content type, size, creation date, etc.

8.5.22 McioInfo information element

8.5.22.1 Description

This information element provides information about an MCIO representing the set of VNFC instances realized by one or a set of OS containers which have been created based on the same VDU.

Within the CISM, an MCIO controller monitors the actual state of an MCIO representing the set of VNFC instances realized by one or a set of OS containers and compare it to the desired state. For an MCIO related to a VDU that has the attribute isNumOfInstancesClusterBased set to FALSE the desired state is specified in the respective declarative descriptor. For an MCIO related to a VDU that has the attribute isNumOfInstancesClusterBased set to TRUE, the desired state is determined by the number of CIS-nodes in the cluster that fulfil the VDU requirements. It triggers actions toward the CIS to align the actual to the desired state. Monitoring the actual state includes monitoring the number of MCIO instances available at any specific point in time. In addition, an MCIO controller maintains properties and runtime information on the MCIO instances which have been created based on the same VDU. The McioInfo information element provides the runtime information on the MCIOs obtained from the respective MCIO controllers.

NOTE: There are different types of MCIOs. The set of VNFC instances based on the same VDU is represented by one MCIO. Each individual VNFC instance is represented by another type of MCIO.

Runtime information of the set of OS containers realizing an individual VNFC instance is not part of the McioInfo information element; such runtime information is provided in the ResourceHandle information element referenced from the VnfcResourceInfo. The McioInfo does not provide runtime information of a constituent VNFC instance created based on a specific VDU.

8.5.22.2 Attributes

The McioInfo information element shall follow the indications provided in table 8.5.22.2-1.

Table 8.5.22.2-1: Attributes of the McioInfo information element

Attribute	Qualifier	Cardinality	Content	Description
mcioId	M	1	Identifier	Identifier of this MCIO, created by the CISM.
mcioName	M	1	String	Human readable name of this MCIO.
mcioNamespace	M	1	String	Namespace of this MCIO.
vduId	M	1	Identifier (Reference to Vdu)	Reference to the applicable Vdu information element in the VNFD.
cismId	M	1	Identifier	Identifier of the CISM managing this MCIO.
mcioType	M	1	Not Specified	The type of MCIO. See note 1.
desiredInstances	M	1	Integer	Number of desired MCIO instances.
availableInstances	M	1	Integer	Number of available MCIO instances.
additionalInfo	M	0..1	Not Specified	Additional information which is specific to the MCIO, its type, and which is available from the CISM. See note 2.
certificateContentId	CM	0..N	Identifier (Reference to CertificateContent)	The identifier(s) of certificate(s) that this MCIO instance uses. Shall be present when using in delegation-mode. Otherwise shall not be present. CONDITION: This attribute shall be supported when delegation-mode in certificate management is applicable.

NOTE 1: The type of MCIO as specified in the declarative descriptor of the MCIO, and that can be read from the CISM.
EXAMPLE: In case of MCIOs managed by Kubernetes®, the type of MCIO corresponds to the "kind" property of the declarative descriptor.
NOTE 2: If the attribute additionalInfo is present, it may contain runtime information on the actual and desired state of the MCIO(s).

8.5.23 VirtualCpInfo information element

8.5.23.1 Description

This information element provides information related to a Virtual CP of a VNF.

8.5.23.2 Attributes

The VirtualCpInfo information element shall follow the indications provided in table 8.5.23.2-1.

Table 8.5.23.2-1: Attributes of the VirtualCpInfo information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier	Identifier of this VirtualCpInfo information element.
cpdId	M	1	Identifier (Reference to VirtualCpd)	Identifier of the VirtualCpd, cpdId, in the VNFD.
resourceHandle	M	1	ResourceHandle	Reference to the virtualised resource realizing this Virtual CP.
vnfExtCpId	M	0..1	Identifier (Reference to VnfExtCpInfo)	When the Virtual CP is exposed as external CP of the VNF, the identifier of this external VNF CP.
cpProtocolInfo	M	0..N	CpProtocolInfo	Protocol information for this CP. There shall be one cpProtocolInfo for each layer protocol supported.
vduId	M	1..N	Identifier (Reference to Vdu)	Reference to the VDU(s) which implement the service accessible via the Virtual CP. See note.
additionalServiceInfo	M	0..N	AdditionalServiceInfo	Additional service identification information of the Virtual CP.
metadata	M	0..N	KeyValuePair	Metadata about this Virtual CP.
NOTE: A consumer of the VNF LCM interface can learn the actual VNFC instances implementing the service accessible via the Virtual CP by querying the "vnfcResourceInfo" from the "InstantiatedVnfInfo" and filtering by corresponding "vduId" values.				

8.5.24 AdditionalServiceInfo information element

8.5.24.1 Description

This information element describes the additional service information of the Virtual CP used to expose properties of the Virtual CP to NFV-MANO.

See also description in clause 7.1.18.3 of ETSI GS NFV-IFA 011 [3].

8.5.24.2 Attributes

The attributes of the AdditionalServiceInfo information element shall follow the indications provided in table 8.5.24.2-1.

Table 8.5.24.2-1: Attributes of the AdditionalServiceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
portInfo	M	1..N	ServicePortInfo	Service port numbers exposed by the Virtual CP.
serviceInfo	M	0..1	Not specified	Service matching information exposed by the Virtual CP. See note.
NOTE: This attribute shall only be present if additional information is needed to identify the service termination within the VNF, such as for example a url path information in an HTTP request required to allow a single Virtual CP IP address to be used for several HTTP based services that use the same port number.				

8.5.25 ServicePortInfo information element

8.5.25.1 Description

This information element describes the service identifying port properties exposed by the Virtual CP.

8.5.25.2 Attributes

The attributes of the ServicePortInfo information element shall follow the indications provided in table 8.5.25.2-1.

Table 8.5.25.2-1: Attributes of the ServicePortInfo information element

Attribute	Qualifier	Cardinality	Content	Description
name	M	1	String	The name of the port exposed by the Virtual CP.
protocol	M	1	Enum	The L4 protocol for this port exposed by the Virtual CP. VALUES: • TCP • UDP • SCTP
port	M	1	Integer	The L4 port number exposed by the Virtual CP.
portConfigurable	M	1	Boolean	Specifies whether the port attribute value is allowed to be configurable.

8.5.26 NetAttDefResourceInfo information element

8.5.26.1 Description

This information element contains information related to a network attachment definition resource that provides the specification of the interface used to connect one or multiple connection points to a secondary container cluster network.

8.5.26.2 Attributes

The NetAttDefResourceInfo information element shall follow the indications provided in table 8.5.26.2-1.

Table 8.5.26.2-1: Attributes of the NetAttDefResourceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
netAttDefResourceInfoId	M	1	Identifier	Identifier of this network attachment definition resource as provided by the entity that has created it.
netAttDefResource	M	1	ResourceHandle	Resource handle of the resource in the scope of the CISM.
associatedExtCpId	M	0..N	Identifier (Reference to VnfExtCpInfo)	External CP of the VNF associated to this network attachment definition resource. Shall be present when the network attachment definition resource is used for external connectivity by the VNF.
associatedVnfcCpId	M	0..N	Identifier (Reference to VnfcCpInfo)	VNFC CP of the VNF associated to this network attachment definition resource. May be present when the network attachment definition resource is used for internal connectivity by the VNF.

8.5.27 PaasServiceInfo information element

8.5.27.1 Description

This information element provides information about a PaaS Service that is used by the VNF instance.

The PaasServiceInfo is comprised of various sets of information. Some information comes from the VNFD, other information comes from the PaaS Service assets provided by the NFVO to the VNFM, and other information is provided as runtime information about the usage of the PaaS Service.

8.5.27.2 Attributes

The PaasServiceInfo information element shall follow the indications provided in table 8.5.27.2-1.

Table 8.5.27.2-1: Attributes of the PaasServiceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
paasServiceId	M	1	Identifier	Identifier of the assigned PaaS Service as managed by the PSM.
paasServiceType	M	1	String	The type of PaaS Service. The value of this attribute is expected to be matched against values of the registered PaaS Services in the PSR.
paasServiceVersion	M	0..1	Version	Version of the PaaS Service. It shall be present if the PaaS Service is versioned.
paasServiceRequestId	M	1	Identifier (Reference to PaasServiceRequest)	Identifier of the PaaS Service request in the descriptor that maps to the assigned PaaS Service.
paasServiceHandle	M	1	Not specified	A handle enabling the access and use of the PaaS Service. This can include for instance certain interface endpoint URI together with necessary credentials to access it. The type and format of the handle depends on the form that the PaaS Service is formed.
additionalInfo	M	0..1	Not specified	Additional information which is specific to the PaaS Service, its type, and which is available from the PaaS Service instance.

8.5.28 CertificateConfigurationData information element

8.5.28.1 Description

This information element provides input information related to certificate management.

ETSI GS NFV-IFA 031 [i.14] specifies the means to discover the applicable certificate management mode of VNFM and configure into the NFVO applicable certificate management mode via the "NFV-MANO Configuration and Information Management" interface.

8.5.28.2 Attributes

The CertificateConfigurationData information element shall follow the indications provided in table 8.5.28.2-1.

Table 8.5.28.2-1: Attributes of the CertificateConfigurationData information element

Attribute	Qualifier	Cardinality	Content	Description
overridingCertificateProfile	M	0..N	CertificateBaseProfile	Overriding certificate profile. This overrides the certificateBaseProfile provided in the VNFD, and the CA and CMF can additionally override aspects of this certificateBaseProfile at later point in the VNF lifecycle if necessary to meet operator security policy.
securityPolicy	M	1..N	SecurityPolicy	Security policy to be satisfied for certificate.
cmfData	M	0..1	CmfData	Information for CMF.

8.5.29 SecurityPolicy information element

8.5.29.1 Description

This information element provides input information related to security policy for certificate management.

8.5.29.2 Attributes

The SecurityPolicy information element shall follow the indications provided in table 8.5.29.2-1.

Table 8.5.29.2-1: Attributes of the SecurityPolicy information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this security policy.
maxValidityPeriod	M	0..1	Number	Allowed max validity period for certificates.
allowedAlgorithm	M	0..1	String	Allowed signature algorithm.
minimumKeyLength	M	0..1	Number	Minimum key length for certificates.

8.5.30 CertificateBaseProfile information element

8.5.30.1 Description

This information element provides input information to override certificate base profile for certificate management.

8.5.30.2 Attributes

The CertficateBaseProfile information element shall follow the indications provided in table 8.5.30.2-1.

Table 8.5.30.2-1: Attributes of the CertificateBaseProfile information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this certificate profile.
issuer	M	0..1	String	Issuer of certificates. See note.
isssuerUniqueIdentifier	M	0..1	Identifier	Identifier of this issuer of certificates. See note.
subject	M	0..1	CertSubjectData	Subject of certificates. See note.
subjectUniqueIdentifier	M	0..1	Identifier	Identifier of this subject of certificates. See note.
basicConstraints	M	0..1	String	Basic constraints of certificates. See note.
issuerAltName	M	0..N	String	Alternative issuer of certificates in this NS. See note.
subjectAltName	M	0..N	String	Alternative subject of certificates. Shall be present when this certificate is used for encrypted communication using IP address and subjectAltName attribute of CertificateBaseProfile in CertificateDesc of VNFD is empty (see ETSI GS NFV-IFA 011[3], clause 7.1.19). See note.
nameConstraints	M	0..N	Not specified	Name constraints of certificates. See note.
NOTE: At least one overriding attributes shall be present, otherwise shall be absent.				

8.5.31 CmfData information element

8.5.31.1 Description

This information element provides input information related to CMF for certificate management.

8.5.31.2 Attributes

The CmfData information element shall follow the indications provided in table 8.5.31.2-1.

Table 8.5.31.2-1: Attributes of the CmfData information element

Attribute	Qualifier	Cardinality	Content	Description
endPoint	M	1	Not specified	End point of CMF instance.
supportedProtocol	M	1..N	Not specified	Supported protocol by CMF instance.

8.5.32 CertSubjectData information element

8.5.32.1 Description

This information element provides input information related to subject of certificate.

8.5.32.2 Attributes

The CertSubjectDatainformation element shall follow the indications provided in table 8.5.32.2-1.

Table 8.5.32.2-1: Attributes of the CertSubjectData information element

Attribute	Qualifier	Cardinality	Content	Description
commonName	M	0..1	Not specified	Information of certification target subject FQDN. Can be set empty when this certificate is used for encrypted communication using IP address. See note.
organization	M	0..1	Not specified	Information of certification target subject Organization. See note.
country	M	0..1	Not specified	Information of certification target subject Country. See note.
state	M	0..1	Not specified	Information of certification target subject State. See note.
locality	M	0..1	Not specified	Information of certification target subject Locality. See note.
emailAddress	O	0..1	Not specified	Information of certification contact email address. See note.
NOTE: At least one overriding attributes shall be present, otherwise shall be absent.				

8.5.33 CertificateInfo information element

8.5.33.1 Description

This information element provides information related to certificate and certificate management.

8.5.33.2 Attributes

The CertificateInfo information element shall follow the indications provided in table 8.5.33.2-1.

Table 8.5.33.2-1: Attributes of the CertificateInfo information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this certificate information.
certificateConfigurationInfo	M	0..1	CertificateConfigurationInfo	Configuration for certificate management such as certificate profile and security policy.
certificateContents	M	0..N	CertificateContent	Information for contents of issued certificates. The information contained in this attribute may be updated over time during the VNF LCM, e.g. certificate(s) renewal.

8.5.34 CertificateConfigurationInfo information element

8.5.34.1 Description

This information element provides information related to certificate management.

8.5.34.2 Attributes

The CertificateConfigurationInfo information element shall follow the indications provided in table 8.5.34.2-1.

Table 8.5.34.2-1: Attributes of the CertificateConfigurationInfo information element

Attribute	Qualifier	Cardinality	Content	Description
certificateBaseProfile	M	0..N	CertificateBaseProfile	Information for certificate profile.
securityPolicy	M	1..N	SecurityPolicy	Information for security policy to be satisfied for certificate.
delegationSupportedCertificateManagements	M	0..1	Not specified	Describes supported certificate management information.
cmfInfo	M	0..1	CmfInfo	Information for CMF.

8.5.35 CmfInfo information element

8.5.35.1 Description

This information element provides information related to CMF for certificate management.

8.5.35.2 Attributes

The CmfInfo information element shall follow the indications provided in table 8.5.35.2-1.

Table 8.5.35.2-1: Attributes of the CmfInfo information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this CMF information.
endPoint	M	1	Not specified	End point of CMF instance.
supportedProtocol	M	1..N	Not specified	Supported protocol by CMF instance.
certificateChain	M	0..N	Not specified	Certificate chain that this CMF provides.

8.5.36 CertificateContent information element

8.5.36.1 Description

This information element provides information related to certificate content.

ETSI GS NFV-IFA 031 [i.14] specifies the means to discover the applicable certificate management mode of VNFM and configure into the NFVO applicable certificate management mode via the "NFV-MANO Configuration and Information Management" interface.

8.5.36.2 Attributes

The CertificateContent information element shall follow the indications provided in table 8.5.36.2-1.

Table 8.5.36.2-1: Attributes of the CertificateContent information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	The identifier of this certificate.
certficateDescId	M	1	Identifier (Reference to CertificateDesc)	The identifier of certificate description in VNFD to be used to issue this certificate. See note.
certificateType	M	1	Enum	The type of this certificate. Values: • VNFCI_CERT • VNFOAM_CERT
delegationSupportedCertificateManagements	M	0..1	Not specified	Describes supported certificate management information.
version	M	0..1	Not specified	The version of this certificate.
serialNumber	M	0..1	Not specified	The serial number of this certificate.
signatureAlgorithm	M	0..1	Not specified	The algorithm of this certificate's signature.
issuer	M	0..1	Not specified	The issuer of this certificate.
notBefore	M	0..1	Not specified	The start date of valid period for this certificate.
notAfter	M	0..1	Not specified	The end date of valid period for this certificate.
subject	M	0..1	Not specified	The subject of this certificate.
publicKeyAlgorithm	M	0..1	Not specified	The algorithm of this certificate's public key.
publicKey	M	0..1	Not specified	The public key of this certificate.
certificateExtensions	O	0..N	Not specified	The extension of this certificate.
NOTE: The CertificateDesc information element is defined in ETSI GS NFV-IFA 011 [3], clause 7.1.19.				

8.5.37 ResourceCapacityDefinition information element

8.5.37.1 Description

This information element provides values for the capacity related attributes of a resource used by the VNF.

8.5.37.2 Attributes

The ResourceCapacityDefinition information element shall follow the indications provided in table 8.5.37.2-1.

Table 8.5.37.2-1: Attributes of the ResourceCapacityDefinition information element

Attribute	Qualifier	Cardinality	Content	Description
tag	M	0..N	Not specified	Tag assigned by the issuer of a VNF LCM operation request that contains this information element with values to be applied to a VDU. It is used for tracking purposes. The tag is preserved in the run time record as long as at least one value of the capacity related attributes associated with that tag is still valid, i.e. it has not been modified by a later VNF LCM operation requests. Details are left for protocol design phase.
type	M	1	Enum	Type of the resource definition referenced. VALUES: COMPUTE STORAGE OSCONTAINER

Attribute	Qualifier	Cardinality	Content	Description
vduId	M	0..1	Identifier (Reference to Vdu)	Reference to the related Vdu applicable to this resource in the VNFD. It shall be present when the referenced resource definition is related to a VDU. See note.
osContainerDescData	M	0..N	OsContainerDescData	Indicates values for resource capacity related attributes in an OsContainerDesc. It shall be present when the attribute 'type' indicates OSCONTAINER and absent otherwise.
virtualComputeDescData	M	0..1	VirtualComputeDescData	Indicates values for resource capacity related attributes in an VirtualComputeDesc. It shall be present when the attribute 'type' indicates COMPUTE and absent otherwise.
virtualStorageDescData	M	0..N	VirtualStorageDescData	Indicates the value for the storage size related attribute in an VirtualStorageDesc. It shall be present when the attribute 'type' indicates STORAGE and absent otherwise.
NOTE: Resource definitions not related to a VDU are not considered in this version of the present document.				

8.5.38 OsContainerDescData information element

8.5.38.1 Description

This information element provides values for the capacity related attributes of an OsContainer resource used by the VNF.

8.5.38.2 Attributes

The OsContainerDescData information element shall follow the indications provided in table 8.5.38.2-1.

Table 8.5.38.2-1: Attributes of the OsContainerDescData information element

Attribute	Qualifier	Cardinality	Content	Description
resourceTemplateId	M	1	Identifier	Identifier of an osContainerDesc in the VNFD.
requestedCpuResources	M	0..1	Integer	Number of CPU resources requested for the container (e.g. in milli-CPU-s). See note.
requestedMemoryResources	M	0..1	Number	Amount of memory resources requested for the container (e.g. in MB). See note.
requestedEphemeralStorageResources	M	0..1	Number	Size of ephemeral storage resources requested for the container (e.g. in GB). See note.
extendedResourceRequests	M	0..N	Not specified	An array of key-value pairs of extended resources required by the container. Only the amount can be modified, the extendedResource is as specified in the VNFD. See note.
cpuResourceLimit	M	0..1	Integer	Number of CPU resources the container can maximally use (e.g. in milli-CPU). See note.
memoryResourceLimit	M	0..1	Number	Amount of memory resources the container can maximally use (e.g. in MB). See note.
ephemeralStorageResourceLimit	M	0..1	Number	Size of ephemeral storage resources the container can maximally use (e.g. in GB). See note.

Attribute	Qualifier	Cardinality	Content	Description
hugePageResources	M	0..1	Not specified	Specifies HugePages resources requested for the container, which the container can maximally use (e.g. "hugepages-2Mi: 100Mi"). Only the total size can be modified, the hugePage size is as specified in the VNFD. See note.
cpuPowerStateRequirements	M	0..1	Not specified	The virtual CPU power state configuration data for the OS container. Refer to clause 7.1.9.2.5 of ETSI GS NFV-IFA 011 [3] for an indicative definition of parameterization of CPU power states management.
NOTE: At least one of the attributes shall be present.				

8.5.39 VirtualComputeDescData information element

8.5.39.1 Description

This information element provides values for the capacity related attributes of the virtual compute resource of a VM used by the VNF.

8.5.39.2 Attributes

The VirtualComputeDescData information element shall follow the indications provided in table 8.5.39.2-1.

Table 8.5.39.2-1: Attributes of the VirtualComputeDescData information element

Attribute	Qualifier	Cardinality	Content	Description
resourceTemplateId	M	1	Identifier	Identifier of an VirtualComputeDesc in the VNFD.
numVirtualCpu	M	0..1	Integer	Number of virtual CPUs. See note 1.
virtualMemSize	M	0..1	Number	Amount of virtual Memory (e.g. in MB). See note 1.
sizeOfVirtualDisk	M	0..1	Number	Size of virtualised storage resource in GB. See notes 1 and 2.
hugePagesRequirements	M	0..1	Not specified	Specifies required total size for hugePages of a particular size. Only the total size can be modified, the hugePage size is as specified in the VNFD.
cpuPowerStateRequirements	M	0..1	Not specified	The virtual CPU power state configuration data for the virtualised compute resource. Refer to clause 7.1.9.2.5 of ETSI GS NFV-IFA 011 [3] for an indicative definition of parameterization of CPU power states management.
NOTE 1: At least one of the attributes shall be present. NOTE 2: The attribute corresponds to the sizeOfStorage attribute in the BlockStorageData IE when used for a local disk in the virtual compute descriptor.				

8.5.40 VirtualStorageDescData information element

8.5.40.1 Description

This information element provides values for the capacity related attributes of the virtual storage resource used by the VNF.

8.5.40.2 Attributes

The VirtualStorageDescData information element shall follow the indications provided in table 8.5.40.2-1.

Table 8.5.40.2-1: Attributes of the VirtualStorageDescData information element

Attribute	Qualifier	Cardinality	Content	Description
resourceTemplateId	M	1	Identifier	Identifier of an VirtualStorageDesc in the VNFD.
sizeOfStorage	M	1	Number	If the 'typeOfStorage' attribute in the VirtualStorageDesc (see clause 7.1.9.4.2.2 in ETSI GS NFV-IFA 011 [3]) referenced by the resourceTemplateId indicates BLOCK or FILE it is the size of the virtualised storage resource. If the 'typeOfStorage' attribute in the VirtualStorageDesc (see clause 7.1.9.4.2.2 in ETSI GS NFV-IFA 011 [3]) referenced by the resourceTemplateId indicates OBJECT it is the maximum size of the virtualised storage resource.

8.5.41 PowerStateInfo information element

8.5.41.1 Description

This information element provides information about the power state of a VNF instance.

8.5.41.2 Attributes

The PowerStateInfo information element shall follow the indications provided in table 8.5.41.2-1.

Table 8.5.41.2-1: Attributes of the PowerStateInfo information element

Attribute	Qualifier	Cardinality	Content	Description
powerProfileId	M	1	Identifier (Reference to PowerProfile)	Identifier of the power profile, as defined in the VNFD, applied to the VNF instance.
name	M	1	String	Name of the power profile as provided in the VNFD.
powerConsumptionInfo	M	0..1	Not specified	Estimated power consumption information of the resources associated with this power profile, as provided in the VNFD. VNFM exposes and can use this information to control the VNF instance to operate according to the power profile and selected power state, e.g. monitoring actual energy or power consumption of the VNF instance and compare against the estimated one.

8.6 Information elements and notifications related to VNF Lifecycle Changes

8.6.1 Introduction

This clause defines notifications related to VNF lifecycle changes and update of VNF information.

8.6.2 VnfLcmOperationOccurrenceNotification

8.6.2.1 Description

This notification informs the receiver of changes in the VNF lifecycle caused by a VNF lifecycle management operation occurrence. The support of the notification is mandatory.

8.6.2.2 Trigger conditions

This notification is produced when there is a change in the VNF lifecycle caused by a VNF lifecycle management operation occurrence, including:

- Instantiation of the VNF.
- Scaling of the VNF instance (including auto-scaling).
- Healing of the VNF instance (including auto-healing).
- Change of the state of the VNF instance (i.e. Operate VNF).
- Change of the DF of the VNF instance.
- Changing the external connectivity of the VNF instance.
- Termination of the VNF instance.
- Modification of VNF instance information and/or VNF configurable properties explicitly through Modify VNF Information operation.
- Create a VNF Snapshot.
- Revert to a VNF Snapshot.
- Change of current VNF Package.
- Selection of VNF Deployable Modules.

If this is a notification about the start of an LCM operation occurrence, the notification shall be sent before any action (including sending the grant request) is taken, however, after acknowledging the LCM operation request to the consumer.

If this is a notification about the result of an LCM operation occurrence, the notification shall be sent after all other actions of the LCM operation have been executed.

8.6.2.3 Attributes

The VnfLcmOperationOccurrenceNotification shall follow the indications provided in table 8.6.2.3-1.

Table 8.6.2.3-1: Attributes of the VnfLcmOperationOccurrenceNotification

Attribute	Qualifier	Cardinality	Content	Description
notificationStatus	M	1	Enum	Indicates whether this notification reports about the start of a lifecycle management operation occurrence or the result of a lifecycle management operation occurrence. VALUES: • START: Informs about the start of the VNF LCM operation occurrence • RESULT: Informs about the final or intermediate result of the VNF LCM operation occurrence
operationStatus	M	1	Not specified	Indicates the operation status. See note.
vnfInstanceId	M	1	Identifier	The identifier of the VNF instance affected.
operation	M	1	String	The lifecycle management operation.
isAutomaticInvocation	M	1	Boolean	Set to true if this VNF LCM operation occurrence has been triggered by an automated procedure inside the VNFM (i.e. ScaleVnf / ScaleVnfToLevel triggered by auto-scale, or HealVnf triggered by auto-heal). Set to false otherwise.
lifecycleOperationOccurrenceId	M	1	Identifier	The identifier of the VNF lifecycle management operation occurrence associated to the notification.
affectedVnfc	M	0..N	AffectedVnfc	Information about VNFC instances that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence.
affectedVirtualLink	M	0..N	AffectedVirtualLink	Information about VL instances that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence.
affectedExtLinkPort	M	0..N	AffectedExtLinkPort	Information about external VNF link ports that were affected during the lifecycle operation.
affectedVirtualStorage	M	0..N	AffectedVirtualStorage	Information about virtualised storage instances that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence.

Attribute	Qualifier	Cardinality	Content	Description
affectedVipCp	M	0..N	AffectedVipCp	Information about virtual IP CP instances that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence.
changedInfo	M	0..1	Not specified	Information about the changed VNF information, including changed VNF configurable properties, if this notification represents the result of a lifecycle management operation occurrence.
changedExtConnectivity	M	0..N	ExtVirtualLinkInfo	Information about changed external connectivity, if this notification represents the result of a lifecycle management operation occurrence.
affectedVirtualCp	M	0..N	AffectedVirtualCp	Information about virtual CP instances that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence.
affectedCertificate	CM	0..N	AffectedCertificate	Information about certificate content that were affected during the execution of the lifecycle management operation, if this notification represents the result of a lifecycle management operation occurrence. Shall be present when delegation-mode, otherwise shall be absent. CONDITION: This attribute shall be supported when delegation-mode in certificate management is applicable.
NOTE: If this notification represents the result of a lifecycle management operation occurrence that was not successful, the notification shall contain appropriate error information.				

8.6.3 AffectedVnfc information element

8.6.3.1 Description

This information element provides information about added, deleted, modified and temporary VNFCs.

8.6.3.2 Attributes

The AffectedVnfc information element shall follow the indications provided in table 8.6.3.2-1.

Table 8.6.3.2-1: Attributes of the AffectedVnfc information element

Attribute	Qualifier	Cardinality	Content	Description
vnfcInstanceId	M	1	Identifier (Reference to VnfcResourceInfo)	Identifier of the VNFC instance.

Attribute	Qualifier	Cardinality	Content	Description
vduId	M	1	Identifier (Reference to Vdu)	Identifier of the VDU in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case of a "change current VNF Package" to identify whether the affected VNFC instance is associated to a VDU which is referred from the source or destination VNFD.
changeType	M	1	Enum	Signals the type of change. VALUES: • ADDED • REMOVED • MODIFIED • TEMPORARY For a temporary resource, an AffectedVnfc IE exists as long as the temporary resource exists.
computeResource	M	1	ResourceHandle	Reference to the VirtualCompute resource or reference to a Compute MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.
metadata	M	0..N	KeyValuePair	Metadata about this resource. The content of this attribute shall be a copy of the content of the "metadata" attribute of the VnfcResourceInfo information element.
affectedVnfcCpInstances	M	0..N	Identifier (Reference to VnfcCpInfo)	Identifiers of CP(s) of the VNFC instance that were affected by the change. Shall be present for those affected CPs of the VNFC instance that are associated to an external CP of the VNF instance. May be present for further affected CPs of the VNFC instance.
addedStorageResourceIds	M	0..N	Identifier (Reference to VirtualStorageResourceInfo)	Reference(s) to VirtualStorage resource(s) that were added. Each value refers to a VirtualStorageResourceInfo item in the VnfInfo that was added to the VNFC. It shall be provided if at least one storage resource was added to the VNFC.
removedStorageResourceIds	M	0..N	Identifier (Reference to VirtualStorageResourceInfo)	Reference(s) to VirtualStorage resource(s) that were removed. The value contains the identifier of a VirtualStorageResourceInfo item that has been removed from the VNFC, and might no longer exist in the VnfInfo. It shall be provided if at least one storage resource was removed from the VNFC.

8.6.4 AffectedVirtualLink information element

8.6.4.1 Description

This information element provides information about added, deleted, modified and temporary VLs, as well as about link port changes.

8.6.4.2 Attributes

The AffectedVirtualLink information element shall follow the indications provided in table 8.6.4.2-1.

Table 8.6.4.2-1: Attributes of the AffectedVirtualLink information element

Attribute	Qualifier	Cardinality	Content	Description
virtualLinkInstanceId	M	1	Identifier (Reference to VnfVirtualLinkResourceInfo)	Identifier of the VL instance.
vnfVirtualLinkDescId	M	1	Identifier (Reference to VnfVirtualLinkDesc)	Identifier of the VLD in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case of a "change current VNF Package" to identify whether the affected VL instance is associated to a VLD which is referred from the source or destination VNFD.
changeType	M	1	Enum	Signals the type of change including, not limited to, changes made to the characteristics of the existing VL, new VL added, existing VL removed, temporary VL exists, link port added, link port removed. VALUES: • ADDED • REMOVED • MODIFIED • TEMPORARY • LINK_PORT_ADDED • LINK_PORT_REMOVED • etc. For a temporary resource, an AffectedVirtualLink IE exists as long as the temporary resource exists.
networkResource	M	1	ResourceHandle	Reference to the VirtualNetwork resource or reference to a Network MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.

Attribute	Qualifier	Cardinality	Content	Description
vnfLinkPortId	M	0..N	Identifier (Reference to VnfLinkPortInfo)	Identifiers of the link ports of the affected VL related to the change. Shall be set when changeType is equal to "LINK_PORT_ADDED" or "LINK_PORT_REMOVED", and the related links ports are present (case "added") or have been present (case "removed") in the VNF internal VL (represented by "vnfVirtualLinkResourceInfo" attribute in the "InstantiatedVnfInfo") or externally-managed VL resources of the VNF (represented by the "extManagedVirtualLinkInfo" attribute in the "InstantiatedVnfInfo").
metadata	M	0..N	KeyValuePair	Metadata about this resource. The content of this attribute shall be a copy of the content of the "metadata" attribute of the VnfVirtualLinkResourceInfo information element.

8.6.4a AffectedExtLinkPort information element

8.6.4a.1 Description

This information element provides information about added and deleted external link ports (link ports attached to external virtual links).

8.6.4a.2 Attributes

The AffectedExtLinkPort information element shall follow the indications provided in table 8.6.4a.2-1.

Table 8.6.4a.2-1: Attributes of the AffectedExtLinkPort information element

Attribute	Qualifier	Cardinality	Content	Description
extLinkPortId	M	1	Identifier (Reference to ExtLinkPortInfo)	Identifier of the link port.
changeType	M	1	Enum	Signals the type of change VALUES: • ADDED • MODIFIED • REMOVED
extCpInstanceId	M	1	Identifier (Reference to VnfExtCpInfo)	Identifier of the related external CP.
resourceHandle	M	1	ResourceHandle	Resource handle of the virtualised resource that realizes the external link port. Detailed information is (for added resources) or has been (for removed resources) available from the VIM.

8.6.5 AffectedVirtualStorage information element

8.6.5.1 Description

This information element provides information about added, deleted, modified and temporary virtual storage resources.

8.6.5.2 Attributes

The AffectedVirtualStorage information element shall follow the indications provided in table 8.6.5.2-1.

Table 8.6.5.2-1: Attributes of the AffectedVirtualStorage information element

Attribute	Qualifier	Cardinality	Content	Description
virtualStorageInstanceId	M	1	Identifier (Reference to VirtualStorageResourceInfo)	Identifier of the virtual storage instance.
virtualStorageDescId	M	1	Identifier (Reference to VirtualStorageDesc)	Identifier of the VirtualStorageDesc in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case of a "change current VNF Package" to identify whether the affected virtual storage instance is associated to a VirtualStorageDesc which is referred from the source or destination VNFD.
changeType	M	1	Enum	Signals the type of change. VALUES: • ADDED • REMOVED • MODIFIED • TEMPORARY For a temporary resource, an AffectedVirtualStorage IE exists as long as the temporary resource exists.
storageResource	M	1	ResourceHandle	Reference to the VirtualStorage resource or reference to a Storage MCIO. Detailed information is (for new and modified resources) or has been (for removed resources) available from the VIM or the CISM.
metadata	M	0..N	KeyValuePair	Metadata about this resource. The content of this attribute shall be a copy of the content of the "metadata" attribute of the VirtualStorageResourceInfo information element.

8.6.6 AffectedVipCp information element

8.6.6.1 Description

This information element provides information about added, deleted and modified virtual IP CP instances.

8.6.6.2 Attributes

The AffectedVipCp information element shall follow the indications provided in table 8.6.6.2-1.

Table 8.6.6.2-1: Attributes of the AffectedVipCp information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier (Reference to VipCpInfo)	Identifier of the virtual IP CP instance.
cpdId	M	1	Identifier (Reference to VipCpd)	Identifier of the VipCpd in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case of a "change current VNF Package" to identify whether the affected virtual CP instance is associated to a VipCpd which is referred from the source or destination VNFD.
changeType	M	1	Enum	Signals the type of change. VALUES: • ADDED • REMOVED • MODIFIED

8.6.6a AffectedVirtualCp information element

8.6.6a.1 Description

This information element provides information about added, deleted and modified virtual CP instances.

8.6.6a.2 Attributes

The AffectedVirtualCp information element shall follow the indications provided in table 8.6.6a.2-1.

Table 8.6.6a.2-1: Attributes of the AffectedVirtualCp information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	1	Identifier (Reference to VirtualCpInfo)	Identifier of the virtual CP instance.
cpdId	M	1	Identifier (Reference to VirtualCpd)	Identifier of the VirtualCpd in the VNFD.
vnfdId	M	0..1	Identifier (Reference to Vnfd)	Reference to the VNFD. Shall be present in case of a "change current VNF Package" to identify whether the affected virtual CP instance is associated to a VirtualCpd which is referred from the source or destination VNFD.
changeType	M	1	Enum	Signals the type of change. VALUES: • ADDED • REMOVED • MODIFIED

8.6.7 VnfIdentifierCreationNotification

8.6.7.1 Description

This notification informs the receiver of the creation of a new VNF instance identifier and the associated instance of a VnfInfo information element, identified by that identifier. The support of the notification is mandatory.

8.6.7.2 Trigger conditions

- Creation of a VNF instance identifier and the associated instance of a VnfInfo information element.

8.6.7.3 Attributes

The VnfIdentifierCreationNotification shall follow the indications provided in table 8.6.7.3-1.

Table 8.6.7.3-1: Attributes of the VnfIdentifierCreationNotification

Attribute	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	The newly created VNF instance identifier.

8.6.8 VnfIdentifierDeletionNotification

8.6.8.1 Description

This notification informs the receiver of the deletion of a VNF instance identifier and the associated instance of a VnfInfo information element identified by that identifier. The support of the notification is mandatory.

8.6.8.2 Trigger conditions

- Deletion of a VNF instance identifier and the associated instance of a VnfInfo information element.

8.6.8.3 Attributes

The VnfIdentifierDeletionNotification shall follow the indications provided in table 8.6.8.3-1.

Table 8.6.8.3-1: Attributes of the VnfIdentifierDeletionNotification

Attribute	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	The VNF instance identifier that has been deleted.

8.6.9 AffectedCertificate information element

8.6.9.1 Description

This information element provides information about added, deleted and modified certificate contents.

ETSI GS NFV-IFA 031 [i.14] specifies the means to discover the applicable certificate management mode of VNFM and configure into the NFVO applicable certificate management mode via the "NFV-MANO Configuration and Information Management" interface.

8.6.9.2 Attributes

The AffectedCertificate information element shall follow the indications provided in table 8.6.9.2-1.

Table 8.6.9.2-1: Attributes of the AffectedCertificate information element

Attribute	Qualifier	Cardinality	Content	Description
certificateInfoId	M	1	Identifier (Reference to CertificateInfo)	The identifier of certificate information.
certificateBaseProfileId	M	0..1	Identifier (Reference to CertificateBaseProfile)	The identifier of certificate base profile.
securityPolicyId	M	0..1	Identifier (Reference to SecurityPolicy)	The identifier of security policy.
cmfInfoId	M	0..1	Identifier (Reference to CmfInfo)	The identifier of CMF information.
certificateContentId	M	0..1	Identifier (Reference to certificateContent)	The identifier of certificate content.
changeType	M	1	Enum	Signals the type of change. VALUES: ADD REMOVE MODIFY

8.7 Information elements and notifications related to VNF Performance Management

8.7.1 Introduction

This clause defines information elements and notifications related to VNF Performance Management.

8.7.2 ObjectSelection information element

8.7.2.1 Description

This information element allows to specify VNF related measured object instances on which performance information will be provided.

The ObjectSelection is a pattern to select object instances. The pattern is used in multiple interfaces. In the present interface, the ObjectSelection pattern is used to select VNF related measured object instances.

The pattern proposes 2 exclusive options:

1) Provide a list of object types and a filter to specify object properties.

2) Provide a list of object instances.

In the present interface, the object type will be the VNF related measured object types (see note).

NOTE: The VNF related measured object types are the measured object type(s) for which the performance measurements applicable to Or-Vnfm reference point are defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].

8.7.2.2 Attributes

The ObjectSelection information element shall follow the indications provided in table 8.7.2.2-1.

Table 8.7.2.2-1: Attributes of the ObjectSelection information element

Attribute	Qualifier	Cardinality	Content	Description
objectType	M	0..N	String	Defines the measured object types. The object types for this information element will be the VNF related measured object types. One of the two attributes (objectType + objectFilter or objectInstanceId) shall be present.
objectFilter	M	0..1	Filter	The filter will apply on the object types to specify on which object instances the performance information is requested to be collected. One of the two attributes (objectType + objectFilter or objectInstanceId) shall be present.
objectInstanceId	M	0..N	Identifier	Identifies the object instances for which performance information is requested to be collected. The object instances for this information element will be instances corresponding to the VNF related measured object types. One of the two attributes (objectType+ objectFilter or objectInstanceId) shall be present.

8.7.3 PmJob information element

8.7.3.1 Description

This information element provides the details of the PM Job. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.3.2 Attributes

The PmJob information element shall follow the indications provided in table 8.7.3.2-1.

Table 8.7.3.2-1: Attributes of the PmJob information element

Attribute	Qualifier	Cardinality	Content	Description
pmJobId	M	1	Identifier	Identifier of this PM job.
objectSelector	M	1	ObjectSelection	Defines the object instances for which performance information is requested to be collected. The object instances for this information element will be instances corresponding to the VNF related measured object types.
performanceMetric	M	0..N	String	This defines the type(s) of performance metric(s) for the specified object instances. At least one of the two attributes (performance metric or group) shall be present.
performanceMetricGroup	M	0..N	String	Group of performance metrics. A metric group is a pre-defined list of metrics, known to the producer that it can decompose to individual metrics. Valid values are specified as "Measurement Name" values of the performance measurements applicable to Or-Vnfm reference point, as defined in clause 7.2 of ETSI GS NFV-IFA 027 [5]. At least one of the two attributes (performance metric or group) shall be present.
collectionPeriod	M	1	Not specified	Specifies the periodicity at which the producer will collect performance information (see note).

Attribute	Qualifier	Cardinality	Content	Description
reportingPeriod	M	1	Not specified	Specifies the periodicity at which the producer will report to the consumer about performance information (see note).
reportingBoundary	O	0..1	Not specified	Identifies a boundary after which the reporting will stop. The boundary shall allow a single reporting as well as periodic reporting up to the boundary.
pmJobConnection	M	0..N	MonitoringConnection	An access information and interface information of PM job to monitor the PM of VNF instance by the VNFM. This can include for instance certain interface endpoint URI together with necessary credentials to access it.
NOTE: At the end of each reportingPeriod, the producer will inform the consumer about availability of the performance data collected for each completed collection period during this reportingPeriod. While the exact definition of the types for collectionPeriod and reportingPeriod is part of the protocol design, it is recommended that the reportingPeriod be equal or a multiple of the collectionPeriod. In the latter case, the performance data for the collection periods within one reporting period would be reported together.				

8.7.4 Threshold information element

8.7.4.1 Description

This information element provides the details of a threshold. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.4.2 Attributes

The Threshold information element shall follow the indications provided in table 8.7.4.2-1.

Table 8.7.4.2-1: Attributes of the Threshold information element

Attribute	Qualifier	Cardinality	Content	Description
thresholdId	M	1	Identifier	Identifier of this Threshold information element.
objectSelector	M	1	ObjectSelection	Defines the object instances associated with the threshold. The object instances for this information element will be instances corresponding to the VNF related measured object types.
performanceMetric	M	1	String	Defines the performance metric associated with the threshold. Valid values are specified as "Measurement Name" values of the performance measurements applicable to Or-Vnfm reference point, as defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].
thresholdType	M	1	Enum	Type of threshold. The list of possible values is part of the protocol design and might include: single/ multi valued threshold, static/dynamic threshold, template based threshold, etc. VALUES: • SIMPLE: Single-valued static threshold • Etc.
thresholdDetails	M	1	Not specified	Details of the threshold: value to be crossed, details on the notification to be generated, etc.

8.7.5 PerformanceReport information element

8.7.5.1 Description

This information element defines the format of a performance report provided by the producer to the consumer on a specified object instance or a set of them. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.5.2 Attributes

The PerformanceReport information element shall follow the indications provided in table 8.7.5.2-1.

Table 8.7.5.2-1: Attributes of the PerformanceReport information element

Attribute	Qualifier	Cardinality	Content	Description
performanceReport	M	1..N	PerformanceReportEntry	List of performance information entries.

8.7.6 PerformanceReportEntry information element

8.7.6.1 Description

This information element defines a single performance report entry. This performance report entry is for a given metric of a given object instance, but can include multiple collected values. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.6.2 Attributes

The PerformanceReportEntry information element shall follow the indications provided in table 8.7.6.2-1.

Table 8.7.6.2-1: Attributes of the PerformanceReportEntry information element

Attribute	Qualifier	Cardinality	Content	Description
objectType	M	1	String	Defines the object type. The object types for this information element will be the VNF related measured object types.
objectInstanceId	M	1	Identifier	The object instance for which the performance metric is reported. The object instances for this information element will be the instances corresponding to the VNF related measured object types.
performanceMetric	M	1	String	Name of the metric collected. This attribute's value contains the related "Measurement Name" values of the performance measurements applicable to Or-Vnfm reference point, as defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].
performanceValue	M	1..N	PerformanceValueEntry	List of performance values with associated timestamp and measurement context (see ETSI GS NFV-IFA 027 [5]).

8.7.7 PerformanceValueEntry information element

8.7.7.1 Description

This information element defines a single performance value with its associated time stamp and measurement context (see ETSI GS NFV-IFA 027 [5]).

8.7.7.2 Attributes

The PerformanceValueEntry information element shall follow the indications provided in table 8.7.7.2-1.

Table 8.7.7.2-1: Attributes of the PerformanceValueEntry information element

Attribute	Qualifier	Cardinality	Content	Description
timeStamp	M	1	DateTime	Timestamp indicating when the data was collected.
performanceValue	M	1	Value	Value of the metric collected. The type of this attribute corresponds to the related "Measurement Unit" for the performance measurements applicable to Or-Vnfm reference point, as defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].
measurementContext	M	0..1	Not specified	Measurement context of the metric collected. The specific measurement context for each kind of performance metrics is defined in ETSI GS NFV-IFA 027 [5].

8.7.8 PerformanceInformationAvailableNotification

8.7.8.1 Description

This notification informs the receiver that performance information is available. Delivery mechanism for the performance reports is not specified in the present document. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.8.2 Trigger Conditions

- New performance information is available.

8.7.8.3 Attributes

The PerformanceInformationAvailableNotification shall follow the indications provided in table 8.7.8.3-1.

Table 8.7.8.3-1: Attributes of the PerformanceInformationAvailableNotification

Attribute	Qualifier	Cardinality	Content	Description
objectInstanceId	M	1..N	Identifier	Object instance(s) for which performance information is available. The object instances for this information element will be instances corresponding to the VNF related measured object types.

8.7.9 ThresholdCrossedNotification

8.7.9.1 Description

This notification informs the receiver that a threshold value has been crossed. The object instances for this information element will be the instances corresponding to the VNF related measured object types.

8.7.9.2 Trigger Condition

A Threshold has been crossed. Depending on threshold type, there might be a single or multiple crossing values.

8.7.9.3 Attributes

The ThresholdCrossedNotification shall follow the indications provided in table 8.7.9.3-1.

Table 8.7.9.3-1: Attributes of the ThresholdCrossedNotification

Attribute	Qualifier	Cardinality	Content	Description
thresholdId	M	1	Identifier (Reference to Threshold)	Threshold which has been crossed.
crossingDirection	M	1	Enum	An indication of whether the threshold was crossed in upward or downward direction.VALUES: • UP • DOWN
objectInstanceId	M	1	Identifier	Object instance for which the threshold has been crossed. The object instances for this information element will be instances corresponding to the VNF related measured object types.
performanceMetric	M	1	String	Performance metric associated with the threshold. This attribute's value contains the related "Measurement Name" values of the performance measurements applicable to Or-Vnfm reference point, as defined in clause 7.2 of ETSI GS NFV-IFA 027 [5].
performanceValue	M	1	Value	Value of the metric that resulted in threshold crossing.
measurementContext	M	0..1	Not specified	Measurement context of the metric collected. The specific measurement context for each kind of performance metrics is defined in ETSI GS NFV-IFA 027 [5].

8.7.10 MonitoringConnection

8.7.10.1 Description

This information element provides detail of monitoring connection data. The object instances for this information element provide the way to access to monitoring tool.

8.7.10.2 Attributes

The MonitoringConnection shall follow the indications provided in table 8.7.10.2-1.

Table 8.7.10.2-1: Attributes of the MonitoringConnection

Attribute	Qualifier	Cardinality	Content	Description
monitoringConnectionId	M	1	Identifier	Identifier of the monitoring connection. See note.
monitoringType	M	1	Enum	Type of monitoring way. VALUES: VIM_CISM EXTERNAL PAAS
vimId	M	0..1	Identifier (Reference to vimId)	Information about VIM or CISM connection(s) for monitoring resources for the VNF instance. Can be set when MonitoringType is equal to "VIM_CISM".
paasServiceId	M	0..1	Identifier (Reference to paasServiceId)	Information about PaasAsset to be used. Can be set when MonitoringType is equal to "PAAS".
interfaceInfo	M	0..1	Not specified	Information about the interface(s) to the external monitoring tool, if available, including interface endpoint e.g. URL API version, and protocol type. Can be set when MonitoringType is equal to "EXTERNAL".
accessInfo	M	0..1	Not specified	Authentication credentials for accessing the external monitoring tool. Examples can include those to support different authentication schemes, e.g. OAuth, Token, Username/password, etc. Can be set when MonitoringType is equal to "EXTERNAL".
Extra	M	0..1	String	Type specific additional information, if applicable.
NOTE: The VNFM can be made aware of monitoring connection information, including their identifiers to be used by configuration means outside the scope of the present document (e.g. using relevant NFV-MANO management APIs as defined in ETSI GS NFV-IFA 031 [i.14]).				

8.8 Information elements and notifications related to VNF Fault Management

8.8.1 Introduction

This clause defines information elements and notifications related to VNF Fault Management.

8.8.2 AlarmNotification

8.8.2.1 Description

This notification informs the receiver of alarms related to the VNFs managed by the VNFM. Alarms are created in response to:

- faults detected by the VNFM; and
- faults generated due to changes in the state of virtualised resources used by the VNF instances managed by the VNFM, including changes in the state of the virtualised resources due to upcoming NFVI operation and maintenance; and
- faults generated by the VIM on virtualised resources used by the VNFs and their constituent VNFC instances managed by the VNFM.

The notification is mandatory.

8.8.2.2 Trigger conditions

- An alarm has been created.
- An alarm has been updated, e.g. if the severity of the alarm has changed.

8.8.2.3 Attributes

The AlarmNotification shall follow the indications provided in table 8.8.2.3-1.

Table 8.8.2.3-1: Attributes of the AlarmNotification

Attribute	Qualifier	Cardinality	Content	Description
alarm	M	1	Alarm	Information about an alarm including AlarmId, affected VNF identifier, and FaultDetails. For notifications related to changes in the state of virtualised resources (indicated using the attribute faultType), the alarm shall indicate: • The cause for the state change of the virtualised resource using the attribute probableCause, with possible values such as: maintenance of NFVI component, evacuation of NFVI component, etc. • The identifier of the origin (VIM) responsible for the management of the virtualised resource with state change using the attribute faultDetails.

8.8.3 AlarmClearedNotification

8.8.3.1 Description

This notification informs the receiver of the clearing of an alarm related to the VNFs managed by the VNFM, e.g. the alarm's perceived severity is set to "cleared" since the corresponding fault has been solved. The notification is mandatory.

8.8.3.2 Trigger conditions

- An alarm has been cleared.

8.8.3.3 Attributes

The AlarmClearedNotification shall follow the indications provided in table 8.8.3.3-1.

Table 8.8.3.3-1: Attributes of the AlarmClearedNotification

Attribute	Qualifier	Cardinality	Content	Description
alarmId	M	1	Identifier (Reference to Alarm)	Alarm identifier.
alarmClearedTime	M	1	DateTime	The timestamp indicating when the alarm was cleared.

8.8.4 Alarm information element

8.8.4.1 Description

The Alarm information element encapsulates information about an alarm.

The Managed Objects for this information element will be VNF instances.

NOTE: The NFVO is enabled in the alarms to observe information on changes in the state of the virtualised resources due to upcoming NFVI operation and maintenance.

8.8.4.2 Attributes

The Alarm information element shall follow the indications provided in table 8.8.4.2-1.

Table 8.8.4.2-1: Attributes of the Alarm information element

Attribute	Qualifier	Cardinality	Content	Description
alarmId	M	1	Identifier	Identifier of this Alarm information element.
managedObjectId	M	1	Identifier	Identifier of the affected managed object. The managed objects for this information element will be VNF instances. A VNF instance can have fault monitored sub-object types and identification information is carried as defined in the respective Alarm definition as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7], e.g. using the "faultDetails" attribute.
rootCauseFaultyResource	M	0..1	FaultyResourceInfo	The virtualised resources that are causing the VNF fault. Shall be present if the alarm affects virtualised resources. See note 1.
alarmRaisedTime	M	1	DateTime	Timestamp indicating when the alarm is raised by the managed object.
alarmChangedTime	M	0..1	DateTime	Timestamp indicating when the alarm was last changed. It shall be present if the alarm has been updated.
alarmClearedTime	M	0..1	DateTime	Timestamp indicating when the alarm was cleared. It shall be present if the alarm has been cleared.
ackState	M	1	Enum	State of the alarm. VALUES: • ACKNOWLEDGED • UNACKNOWLEDGED
perceivedSeverity	M	1	Enum	Perceived severity of the managed object failure. VALUES: • CRITICAL • MAJOR • MINOR • WARNING • INDETERMINATE • CLEARED Valid values applicable to specific Alarms are specified as "Perceived severity" values of the Alarm applicable to Or-Vnfm reference point, as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7].
eventTime	M	1	DateTime	Timestamp indicating when the fault was observed. See note 2.

Attribute	Qualifier	Cardinality	Content	Description
eventType	M	1	Enum	Type of the event. The values for the eventType attribute use the event type defined in Recommendation ITU-T X.733 [4]. VALUES: • COMMUNICATIONS_ALARM • PROCESSING_ERROR_ALARM • ENVIRONMENTAL_ALARM • QOS_ALARM • EQUIPMENT_ALARM Valid values applicable to specific Alarms are specified as "Event type" values of the Alarm applicable to Or-Vnfm reference point, as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7].
faultType	M	0..1	String	Additional information related to the type of the fault. Valid values applicable to specific Alarms are specified as "Alarm definition identifier" values of the Alarm applicable to Or-Vnfm reference point, as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7].
probableCause	M	1	String	Information about the probable cause of the fault. Valid values applicable to specific Alarms are specified as "Probable cause" values of the Alarm applicable to Or-Vnfm reference point, as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7].
isRootCause	M	1	Boolean	Attribute indicating if this fault is the root for other correlated alarms. If TRUE, then the alarms listed in the attribute CorrelatedAlarmId are caused by this fault.
correlatedAlarmId	M	0..N	Identifier (Reference to Alarm)	List of identifiers of other alarms correlated to this fault.
faultDetails	M	0..N	Not specified	Provides additional information about the fault. See notes 1 and 2. Valid values applicable to specific Alarms are specified as "Fault details" values of the Alarm applicable to Or-Vnfm reference point, as defined in clause 7.3 of ETSI GS NFV-IFA 045 [7].
NOTE 1: For an alarm about upcoming impact due to NFVI operation and maintenance, the rootCauseFaultyResource indicates a resource to be impacted. Further information on the upcoming impact (e.g. group of impacted resources, time of impact) is provided in the faultDetails attribute. NOTE 2: When alarms are due to upcoming NFVI operation and maintenance, the faultDetails shall include information about the anticipated time of the maintenance.				

8.8.5 FaultyResourceInfo information element

8.8.5.1 Description

The FaultyResourceInfo information element encapsulates information about faulty resource that has a negative impact on a VNF.

8.8.5.2 Attributes

The FaultyResourceInfo information element shall follow the indications provided in table 8.8.5.2-1.

Table 8.8.5.2-1: Attributes of the FaultyResourceInfo information element

Attribute	Qualifier	Cardinality	Content	Description
faultyResource	M	1	ResourceHandle	Information that identifies the faulty resource instance and its managing entity. See clause 8.5.7.
faultyResourceType	M	1	Enum	Type of the faulty resource. VALUES: • COMPUTE • STORAGE • NETWORK

8.8.6 AlarmListRebuiltNotification

8.8.6.1 Description

This notification informs the receiver that the active alarm list has been rebuilt by the VNFM. Upon receipt of this notification, the receiver needs to use the "Get Alarm List" operation to synchronize its view on current active alarms with that of the VNFM.

The notification is mandatory.

8.8.6.2 Trigger conditions

- Active alarm list has been rebuilt by the VNFM, e.g. if the VNFM detects its storage holding the alarm list is corrupted.

8.8.6.3 Attributes

The AlarmListRebuiltNotification does not contain any attributes.

8.9 Void

8.10 Information elements and notifications related to VNF Indicators

8.10.1 Introduction

The clauses below define information elements which represent indicator values, and notifications about changes of these.

8.10.2 IndicatorValueChangeNotification

8.10.2.1 Description

This notification informs the receiver of a value change of an indicator related to the VNF. The notification is mandatory.

8.10.2.2 Trigger conditions

- The value of an indicator has changed.

8.10.2.3 Attributes

The IndicatorValueChangeNotification information element shall follow the indications provided in table 8.10.2.3-1.

Table 8.10.2.3-1: Attributes of the IndicatorValueChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
indicatorInformation	M	1	IndicatorInformation	This is to provide the indicator, the value of the indicator and the VNF instance the indicator is related to.

8.10.3 IndicatorInformation information element

8.10.3.1 Description

This information element provides the indicator values of a VNF instance.

8.10.3.2 Attributes

The IndicatorInformation information element shall follow the indications provided in table 8.10.3.2-1.

Table 8.10.3.2-1: Attributes of the IndicatorInformation information element

Attribute	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier	Identifies the VNF instance which provides the indicator value(s).
vnfcInstanceId	O	0..N	Identifier	List of identifiers of VNFC instances which provides the indicator value(s).
indicatorId	M	1	Identifier (Reference to VnfIndicator)	Identifies the indicator.
indicatorValue	M	1	Value	Provides the value of the indicator. The value format is defined in the VNFD (see ETSI GS NFV-IFA 011 [3]).
indicatorName	M	0..1	String	Human readable name of the indicator. Shall be present if defined in the VNFD according to clause 7.1.2 of ETSI GS NFV-IFA 011 [3].

8.10.4 SupportedIndicatorsChangeNotification

8.10.4.1 Description

This notification informs the receiver that the set of indicators supported by a VNF instance has changed. Such change can occur as a side effect of the "Change current VNF package" operation.

8.10.4.2 Trigger conditions

- The set of indicators supported by a VNF instance has changed.

8.10.4.3 Attributes

The SupportedIndicatorsChangeNotification information element shall follow the indications provided in table 8.10.4.3-1.

Table 8.10.4.3-1: Attributes of the SupportedIndicatorsChangeNotification

Attribute	Qualifier	Cardinality	Content	Description
vnfInstanceId	M	1	Identifier (Reference to VnfInfo)	Identifies the VNF instance which provides the indicators.
supportedIndicator	M	0..N	SupportedIndicatorInformation	Set of VNF indicators supported by the VNF instance.

8.10.5 SupportedIndicatorInformation information element

8.10.5.1 Description

This information element provides information about a supported VNF indicator.

8.10.5.2 Attributes

The SupportedIndicatorInformation information element shall follow the indications provided in table 8.10.5.2-1.

Table 8.10.5.2-1: Attributes of the SupportedIndicatorInformation information element

Attribute	Qualifier	Cardinality	Content	Description
indicatorId	M	1	Identifier (Reference to VnfIndicator)	Identifies the indicator.
indicatorName	M	0..1	String	Human readable name of the indicator. Shall be present if defined in the VNFD according to clause 7.1.2 of ETSI GS NFV-IFA 011 [3].

8.11 Notifications related to Virtualised Resources Quota

8.11.1 Introduction

This clause defines notifications related to virtualised resources quota.

8.11.2 VirtualisedResourceQuotaAvailableNotification

8.11.2.1 Description

This notification indicates the availability of a quota applicable to the consumer. Support of this notification is mandatory if the Virtualised Resources Quota Available Notification interface is supported.

8.11.2.2 Trigger Conditions

- A virtualised resources quota applicable to the consumer has been set.

8.11.2.3 Attributes

The VirtualisedResourceQuotaAvailableNotification shall follow the indications provided in table 8.11.2.3-1.

Table 8.11.2.3-1: Attributes of the VirtualisedResourceQuotaAvailableNotification

Attribute	Qualifier	Cardinality	Content	Description
resourceGroupId	M	1	Identifier	Unique identifier of the "infrastructure resource group", logical grouping of virtual resources assigned to a tenant within an Infrastructure Domain.
vimConnectionInfo	CM	0..1	VimConnectionInfo	Information about the VIM connection to manage the virtualised resources quota. CONDITION: This attribute shall be supported when VNF-related Resource Management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the virtualised resources quota. CONDITION: This attribute shall be supported when VNF-related Resource Management in indirect mode is applicable.

8.12 Information elements and notifications related to multiple interfaces

8.12.1 Introduction

This clause defines information elements that are referenced by other information elements related to multiple interfaces.

8.12.2 ExtVirtualLinkData information element

8.12.2.1 Description

This information element provides the information of an external VL to be used as a parameter passed to multiple interfaces.

8.12.2.2 Attributes

The ExtVirtualLinkData information element shall follow the indications provided in table 8.12.2.2-1.

Table 8.12.2.2-1: Attributes of the ExtVirtualLinkData information element

Attribute	Qualifier	Cardinality	Content	Description
extVirtualLinkId	M	1	Identifier	Identifier of this external VL instance. The identifier is assigned by the NFV-MANO entity that manages this VL instance.
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM connection to manage this resource. CONDITION: This attribute shall be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the resource. CONDITION: This attribute shall be supported and present when VNF-related Resource Management in indirect mode is applicable.
resourceId	M	1	Identifier	Identifier of the resource in the scope of the VIM or the resource provider.
extCp	M	0..N	VnfExtCpData	External CPs of the VNF to be connected to this external VL. See note 3.
extLinkPorts	M	0..N	ExtLinkPortData	Externally provided link ports to be used to connect external connection points to this external VL. If this attribute is not present, the VNFM shall create the link ports on the external VL except in the cases defined below. See note 1.
extNetAttDefResourceData	M	0..N	NetAttDefResourceData	Externally provided network attachment definition resource(s) that provide the specification of the interface to attach external CPs to this external VL. See note 2. It is only applicable if the external VL is realized by a secondary container cluster network. It shall not be present otherwise.
extCpDeletion	M	0..N	VnfExtCpData	External CPs of the VNF to be disconnected from this external VL and to be deleted. See note 3.

Attribute	Qualifier	Cardinality	Content	Description
NOTE 1: A link port is not needed for an external CP instance that exposes a CP in the following cases: 1) For a VIP CP directly exposed as extCP: 1.1) No dedicated IP address is allocated as VIP address, as indicated in the VNFD. 1.2) A dedicated IP address is allocated as VIP address, but the NFVO indicates that no port is needed (createExtLinkPort = false). 2) For a VIP CP exposed as extCP via a floating IP address: 2.1) No dedicated IP address is allocated as VIP address, as indicated in the VNFD, and the VNFC CP associated to the VIP CP is also exposed via a floating IP address. 3) For a VIRTUAL CP exposed as extCp. 4) For a VNFC CP exposed as extCp in a secondary container cluster external network or a secondary container cluster internal network. NOTE 2: An example of the network attachment definition resource when the container infrastructure service is a Kubernetes® instance is a Network Attachment Definition (NAD). NOTE 3: Either "extCp" or "extCpDeletion" shall be present, but not both.				

8.12.2a ExtLinkPortData information element

8.12.2a.1 Description

This information element represents an externally provided link port to be used to connect an external connection point to an external VL.

8.12.2a.2 Attributes

The ExtLinkPortData information element shall follow the indications provided in table 8.12.2a.2-1.

Table 8.12.2a.2-1: Attributes of the ExtLinkPortData information element

Attribute	Qualifier	Cardinality	Content	Description
extLinkPortId	M	1	Identifier	Identifier of this link port as provided by the entity that has created the link port.
resourceHandle	M	1	ResourceHandle	Resource handle of the virtualised resource that realizes the external link port.

8.12.3 VnfExtCpData information element

8.12.3.1 Description

This information element provides input information related to one or more external CP instances created based on the same CPD.

8.12.3.2 Attributes

The VnfExtCpData information element shall follow the indications provided in table 8.12.3.2-1.

Table 8.12.3.2-1: Attributes of the VnfExtCpData information element

Attribute	Qualifier	Cardinality	Content	Description
cpdId	M	1	Identifier	Identifier of the CPD in the VNFD.
cpConfig	M	1..N	VnfExtCpConfig	List of instance data that need to be configured on the CP instances created from the respective CPD. See notes 1 and 2. In case of deleting an external CP, the list of instances to be deleted.
NOTE 1: Subports need not be used for containerized VNFCs. The application container can send and receive IP packets with any VLAN tag as long as the network interface to connect to the secondary container cluster network has been configured appropriately. Thus, no individual cpConfig, except the one representing the trunk, need be modelled to allow traffic tagged with a particular VLAN through the connection point. NOTE 2: In the case that the cloud native template included in the MCIOP describes the set of VNFC instances, for containerized VNFCs, individual connection points need not be configured for each VNFC instance. It is only required to configure one cpConfig per cpdId, not per VNFC instance. The case of using, for a scalable VDU, a cloud native template in the MCIOP that describes one single VNFC instance is not specified in the present document version.				

8.12.3a VnfExtCpConfig information element

8.12.3a.1 Description

This information element represents an externally provided link port, or a network attachment definition resource of secondary container cluster network, or network address information per instance of an external connection point.

In the case of VM-based deployment of the VNFC exposing the external CP:

- in case a link port is provided, the VNFM shall use that link port when connecting the external CP to the external VL;
- in case no link port is provided, the VNFM shall create a link port on the external VL, and use that link port to connect the external CP to the external VL.

In the case of container-based deployment of the VNFC exposing the external CP, the VNFM shall use the network attachment definition resource of secondary container cluster network when connecting the CP to the external VL.

8.12.3a.2 Attributes

The VnfExtCpConfig information element shall follow the indications provided in table 8.12.3a.2-1.

Table 8.12.3a.2-1: Attributes of the VnfExtCpConfig information element

Attribute	Qualifier	Cardinality	Content	Description
cpInstanceId	M	0..1	Identifier	Identifier of the external CP instance to which this set of configuration parameters is requested to be applied. In case of deletion of an external CP, the identifier of the external CP instance to delete. Shall be present if this instance has already been created.
linkPortId	M	0..1	Identifier (Reference to ExtLinkPortData)	Identifier of a pre-configured link port to which the external CP will be associated. See notes 1 and 4.
createExtLinkPort	M	0..1	Boolean	Indicates to the VNFM the need to create a dedicated link port for the external CP. If set to True, the VNFM shall create a link port. If set to False, the VNFM shall not create a link port. This attribute is only applicable for external CP instances without a floating IP address that expose a VIP CP instance for which a dedicated IP address is allocated.

Attribute	Qualifier	Cardinality	Content	Description
netAttDefResourceId	M	0..N	Identifier (Reference to NetAttDefResourceData)	Identifiers of network attachment definition resources that provide the specification of the interface to attach the external CP to a secondary container cluster network. It is only applicable if the external CP is connected or to be connected to a secondary container cluster network. It shall not be present if the external CP is related to a virtual network not categorized as secondary container cluster network. See notes 2, 3 and 4.
cpProtocolData	M	0..N	Not specified	Parameters for configuring fixed and dynamic network addresses for the CP, including the information on applicable layer protocol(s). For dynamic addresses, it should be possible to define per parameter set the number of network addresses to be assigned dynamically. Other parameters could be, e.g. valid address ranges or subnets, domain name information, etc. See notes 1 and 2.
cpSecurityGroupData	M	0..1	Not specified	Parameters for modifying and overriding security groups information bound to this CP. The parameters identify which "SecurityGroupRule" specified in the VNFD is activated/deactivated and for an activated security group rule the values of attributes defined in the VNFD that can be overridden such as "portRangeMin" and "portRangeMax". If no security group configuration is provided for a given "SecurityGroupRule" specified in the VNFD as part of this attribute, the VNFM shall use the configuration as specified in the VNFD.

NOTE 1: The following conditions apply to the attributes "linkPortId" and " cpProtocolData" for an external CP instance connected or to be connected to a virtual network not categorized as secondary container cluster network:
1) The "linkPortId" and "cpProtocolData" attributes shall both be absent for the deletion of an existing external CP instance addressed by cpInstanceId.
2) At least one of these attributes shall be present for a to-be-created external CP instance or an existing external CP instance.
3) If the "linkPortId" attribute is absent, the VNFM shall create a link port.
4) If the "cpProtocolData" attribute is absent, the "linkPortId" attribute shall be provided referencing a pre-created link port, and the VNFM can use means outside the scope of the present document to obtain the pre-configured address information for the connection point from the resource representing the link port.
5) If both "cpProtocolData" and "linkportId" are provided, the NFVO shall ensure that the cpProtocolData can be used with the pre-created link port referenced by "linkPortId".

NOTE 2: The following conditions apply to the attributes "netAttDefResourceId" and "cpProtocolData" for an external CP instance connected or to be connected to a secondary container cluster network:
1) The "netAttDefResourceId" and "cpProtocolData" attributes shall both be absent for the deletion of an existing external CP instance addressed by cpInstanceId.
2) The "netAttDefResourceId" attribute shall be present and the "cpProtocolData" attribute may be present for a to-be-created external CP instance or an existing external CP instance.

NOTE 3: Cardinality greater than 1 is only applicable for specific cases where more than one network attachment definition resource is needed to fulfil the connectivity requirements of the external CP, e.g. to build a link redundant mated pair in SR-IOV cases. When more than one netAttDefResourceId is indicated, all shall belong to the same namespace as defined by the corresponding "containerNamespace" attribute in the "resourceHandle" attribute in the "NetAttDefResourceData".

NOTE 4: Either linkPortId or netAttDefResourceId may be included, but not both.

8.12.4 ExtManagedVirtualLinkData information element

8.12.4.1 Description

This information element provides the information of an externally-managed internal VL to be used as a parameter passed to multiple interfaces.

8.12.4.2 Attributes

The ExtManagedVirtualLinkData information element shall follow the indications provided in table 8.12.4.2-1.

Table 8.12.4.2-1: Attributes of the ExtManagedVirtualLinkData information element

Attribute	Qualifier	Cardinality	Content	Description
extManagedVirtualLinkId	M	1	Identifier	Identifier of this externally-managed internal VL instance. The identifier is assigned by the NFV-MANO entity that manages this VL instance.
vnfVirtualLinkDescId	M	1	Identifier (Reference to VnfVirtualLinkDesc)	Identifier of the VLD in the VNFD for this VL.
vimConnectionId	CM	0..1	Identifier (Reference to VimConnectionInfo)	Identifier of the VIM connection to manage this resource. CONDITION: This attribute shall be supported and present if VNF-related resource management in direct mode is applicable.
resourceProviderId	CM	0..1	Identifier	Identifies the entity responsible for the management of the resource. CONDITION: This attribute shall be supported and present when VNF-related Resource Management in indirect mode is applicable.
resourceId	M	1	Identifier	Identifier of the resource in the scope of the VIM or the resource provider.
netAttDefResourceData	M	0..N	NetAttDefResourceData	Externally provided network attachment definition resource(s) that provide the specification of the interface to attach VNFC connection points to this externally-managed VL. See notes 1,3, and 5.
intCp	M	0..N	IntVnfCpData	Internal CPs of the VNF to be connected to this externally-managed VL. See notes 1 and 4.
vnfLinkPort	M	0..N	VnfLinkPortData	Externally provided link ports to be used to connect VNFC connection points to this externally-managed VL on this network resource. If this attribute is not present, the VNFM shall create the link ports on the externally-managed VL. See notes 2 and 5.
extManagedMultisiteVirtualLinkId	M	0..1	Identifier	Identifier of the externally-managed multi-site VL instance. The identifier is assigned by the NFV-MANO entity that manages the externally managed multi-site VL instance. It shall be present when the present externally-managed internal VL (indicated by extManagedVirtualLinkId) is part of a multi-site VL, e.g. in support of multi-site VNF spanning several VIMs. All externally-managed internal VL instances corresponding to a an internal VL created based on the same

Attribute	Qualifier	Cardinality	Content	Description
				virtualLinkDescId shall refer to the same extManagedMultisiteVirtualLinkId.

NOTE 1: It is only applicable if the externally-managed VL is realized by a secondary container cluster network. It shall not be present otherwise.
NOTE 2: A link port is not needed for a VNFC internal connection point connected to a secondary container cluster network.
NOTE 3: An example of the network attachment definition resource when the container infrastructure service is a Kubernetes® instance is a Network Attachment Definition (NAD).
NOTE 4: In the case that the cloud native template included in the MCIOP describes the set of VNFC instances, an instance of intCp need not be included for each VNFC instance as all instances would contain the same information. It is sufficient to include one intCp for the related CPD. The case of using, for a scalable VDU, a cloud native template in the MCIOP that describes one single VNFC instance is not specified in the present document version.
NOTE 5: Both vnfLinkPort and vnfNetAttDefResource can be provided in a ExtManagedVirtualLinkData to indicate that a single internal virtual link is providing connectivity for both VM-based and container-based VNFCs.

8.12.5 VimConnectionInfo information element

8.12.5.1 Description

This information element provides information regarding a VIM, a CISM, a CIR or a MCIOP repository connection.

It is assumed that during the protocol design stage, VimConnectionInfo will be specified such that it allows interfacing to different VIM, CISM, CIR or MCIOP repository types. VIM, a CISM, a CIR or a MCIOP repository may be configured into the VNFM by means outside the scope of the present document and bound to the identifier of that VIM. ETSI GS NFV-IFA 031 [i.14] specifies the means to configure into the VNFM applicable VIM connection information via the "NFV-MANO Configuration and Information Management" interface.

8.12.5.2 Attributes

The VimConnectionInfo information element shall follow the indications provided in table 8.12.5.2-1.

Table 8.12.5.2-1: Attributes of the VimConnectionInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vimConnectionInfoId	M	1	Identifier	The identifier of this VimConnectionInfo information element, for the purpose of referencing it from other information elements. This identifier is managed by the NFVO.
vimId	M	0..1	Identifier	The identifier of the VIM, CISM, CIR or MCIOP repository. This identifier is managed by the NFVO. See note 2. Shall be present to address additional information about the VIM, CISM, CIR or MCIOP repository if such information has been configured into the VNFM by means outside the scope of the present document, and should be absent otherwise.
interfaceInfo	M	0..N	Not specified	Information about the interface(s) to the VIM, CISM, CIR or MCIOP repository, if available, including interface endpoint e.g. URL API version, and protocol type. Alternatively, such information may have been configured into the VNFM out-of-band and bound to the vimId by means outside the scope of the present document. If present and VimConnectionInfo bound to the vimId has already been configured into the VNFM out-of-band, the information values provided by the present attribute shall be used for the resources management of the VNF by the VNFM. See note 3.

Attribute	Qualifier	Cardinality	Content	Description
accessInfo	M	0..N	Not specified	Authentication credentials for accessing the VIM, CISM, CIR or MCIOP repository. Examples can include those to support different authentication schemes, e.g. OAuth, Token, Username/password, etc. See note 1. Alternatively, such information may have been configured into the VNFM out-of-band and bound to the vimId by means outside the scope of the present document. If present and VimConnectionInfo bound to the vimId has already been configured into the VNFM out-of-band, the information values provided by the present attribute shall be used for the resources management of the VNF by the VNFM. See note 3.
extra	M	0..N	Not specified	VIM, CISM, CIR or MCIOP repository type specific additional information, if applicable. Alternatively, such information may have been configured into the VNFM out-of-band and bound to the vimId by means outside the scope of the present document. If present and VimConnectionInfo bound to the vimId has already been configured into the VNFM out-of-band, the information values provided by the present attribute shall be used for the resources management of the VNF by the VNFM. See note 3.

NOTE 1: If needed, this attribute also provides information about the resourceGroupIds that are accessible using a particular set of credentials.
NOTE 2: The NFVO can be made aware of VIM instances information, including their identifiers to be used by configuration means outside the scope of the present document (e.g. using relevant NFV-MANO management APIs as defined in ETSI GS NFV-IFA 031 [i.14]). Likewise, the NFVO can be made aware of which VIM instances information has been configured into a specific VNFM by means outside the scope of the present document (e.g. using relevant NFV-MANO management APIs as defined in ETSI GS NFV-IFA 031 [i.14]).
NOTE 3: Due to the possibility of configuring such information into the VNFM out-of-band, by means outside the scope of the present document, as well as in-band, by means specified in the present document, care should be taken to avoid unintended conflicts in the VNFM when managing such information,

8.12.6 VnfLinkPortData information element

8.12.6.1 Description

This information element represents an externally provided link port to be used to connect a VNFC connection point to an externally-managed VL.

8.12.6.2 Attributes

The VnfLinkPortData information element shall follow the indications provided in table 8.12.6.2-1.

Table 8.12.6.2-1: Attributes of the VnfLinkPortData information element

Attribute	Qualifier	Cardinality	Content	Description
vnfLinkPortId	M	1	Identifier	Identifier of this link port as provided by the entity that has created the link port.
resourceHandle	M	1	ResourceHandle	Resource handle of the virtualised resource that realizes the link port.

8.12.7 NetAttDefResourceData information element

8.12.7.1 Description

This information element represents a network attachment definition resource that provides the specification of the interface to be used to connect one or multiple connection points to a secondary container cluster network realizing a VL.

8.12.7.2 Attributes

The NetAttDefResourceData information element shall follow the indications provided in table 8.12.7.2-1.

Table 8.12.7.2-1: Attributes of the NetAttDefResourceData information element

Attribute	Qualifier	Cardinality	Content	Description
netAttDefResourceId	M	1	Identifier	Identifier of this network attachment definition resource as provided by the entity that has created it.
resourceHandle	M	1	ResourceHandle	Resource handle of the resource identifying the network attachment definition resource that provides the specification of the interface to attach the connection points to a secondary container cluster network.

8.12.8 IntVnfCpData information element

8.12.8.1 Description

This information element provides input information related to one or more VNF internal CP instances created based on the same CPD.

8.12.8.2 Attributes

The IntVnfCpData information element shall follow the indications provided in table 8.12.8.2-1.

Table 8.12.8.2-1: Attributes of the IntVnfCpData information element

Attribute	Qualifier	Cardinality	Content	Description
cpdId	M	1	Identifier	Identifier of the CPD in the VNFD.
netAttDefResourceId	M	1..N	Identifier (Reference to NetAttDefResourceData)	Identifiers of network attachment definition resources that provide the specification of the interface to attach the VNF internal CP created from the CPD identified by cpdId to a secondary container cluster network. See note.
NOTE: Cardinality greater than 1 is only applicable for specific cases where more than one network attachment definition resource is needed to fulfil the connectivity requirements of the VNF internal CP, e.g. to build a link redundant mated pair in SR-IOV cases. When more than one netAttDefResourceId is indicated, all shall belong to the same namespace as defined by the corresponding "containerNamespace" attribute in the "resourceHandle" attribute in the "NetAttDefResourceData".				

8.13 Information elements and notifications related to Policy Management

8.13.1 Introduction

The clauses below define information elements and notifications related to policy management.

8.13.2 Information elements related to Policy Management Operations

8.13.2.1 Introduction

The clauses below define information elements related to policy management operations.

8.13.2.2 PolicyInfo information element

8.13.2.2.1 Description

This information element provides policy related information. It contains the policy itself and additional information related to the policy.

8.13.2.2.2 Attributes

The structure of the PolicyInfo information element shall comply with the provisions for the PolicyInfo information element as defined in ETSI GS NFV-IFA 013 [8], clause 8.8.2.2.2.

8.13.3 PolicyChangeNotification

8.13.3.1 Description

This notification indicates a change of a NFV-MANO policy related to operations of transferring policy, deleting policy, activating policy, deactivating policy, associating policy and disassociating policy.

Support of this notification is mandatory.

8.13.3.2 Trigger Conditions

The notification is produced when a policy has been changed as a result of an operation of TransferPolicy, DeletePolicy, ActivatePolicy, DeactivatePolicy, AssociatePolicy or DisassociatePolicy.

8.13.3.3 Attributes

The PolicyChangeNotification shall comply with the provisions in clause 8.8.3.3 of ETSI GS NFV-IFA 013 [8].

8.13.4 PolicyConflictNotification

8.13.4.1 Description

This notification indicates a policy conflict is detected by the VNFM. A policy conflict can include any conflicted monitored events, conditions or actions among two or more polices enforced by the VNFM.

Support of this notification is mandatory.

8.13.4.2 Trigger Conditions

The notification is produced when a policy conflict is detected by the VNFM.

8.13.4.3 Attributes

The PolicyConflictNotification shall comply with the provisions in clause 8.8.4.3 of ETSI GS NFV-IFA 013 [8].

8.14 Information elements related to VNF Snapshot Package Management

8.14.1 Introduction

This clause defines information elements related to VNF Snapshot Package Management.

8.14.2 VnfSnapshotPkgInfo information element

8.14.2.1 Description

This information element provides the details of a VNF Snapshot Package, which the NFVO creates and stores as part of the ongoing operational VNF Snapshot Package management process.

8.14.2.2 Attributes

The VnfSnapshotPkgInfo information element shall follow the indications provided in table 8.14.2.2-1.

Table 8.14.2.2-1: Attributes of the VnfSnapshotPkgInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfSnapshotPkgInfoId	M	1	Identifier	Identifier of information held by the VNFM about a specific VNF Snapshot Package. This identifier was allocated by the NFVO.
vnfSnapshotPkgId	M	0..1	Identifier	Identifier that identifies the VNF Snapshot Package. See notes 1 and 2.
name	M	1	String	Human-readable name of the VNF Snapshot Package.
checksum	M	0..1	Not specified	Checksum of the stored VNF Snapshot Package. See note 2.
createdAt	M	0..1	DateTime	Timestamp indicating when the VNF Snapshot Package creation has been completed. See note 2.
vnfSnapshotInfoId	M	0..1	Identifier (Reference to VnfSnapshotInfo)	References information about a specific VNF Snapshot. This identifier was allocated by the VNFM. See note 2.
isFullSnapshot	M	1	Boolean	Value is 1 (true) in case of a "full" VNF Snapshot Package, i.e. containing all snapshotted VNFC instances; otherwise the value is 0 (false).
vnfd	M	0..1	Vnfd	VNFD of the snapshotted VNF instance that is contained in the stored VNF Snapshot Package. See note 2.
vnfInfo	M	0..1	VnfInfo	VnfInfo of the snapshotted VNF instance that is contained in the stored VNF Snapshot Package. See note 2.
vnfcSnapshotInfoId	M	0..N	Identifier (Reference to VnfcSnapshotInfo)	References information about specific VNFC Snapshot(s). These identifiers were allocated by the VNFM. See note 2.
vnfcSnapshotImage	M	0..N	VnfcSnapshotImageInfo	Information about VNFC Snapshot artifact(s) that are VNFC Snapshot Images. See note 2.
additionalArtifact	M	0..N	SnapshotPkgArtifactInformation	Information about Snapshot artifact(s) that are not VNFC Snapshot Images.

Attribute	Qualifier	Cardinality	Content	Description
state	M	1	Enum	State of the VNF Snapshot Package. VALUES: • CREATED • BUILDING • UPLOADING • AVAILABLE • EXTRACTING • PROCESSING • ERROR
userDefinedData	O	0..N	KeyValuePair	User defined data for the VNF Snapshot Package.
accessInformation	M	0..1	Not specified	Information (such as a URL, or an identifier) that allows to access a copy of this VNF Snapshot Package. See note 2.
NOTE 1: This identifier identifies the VNF Snapshot Package in a globally unique way. It is created during the Build VNF Snapshot Package operation. Multiple instances of the same VNF Snapshot Package share the same vnfSnapshotPkgId. NOTE 2: Cardinality is 0 when the VnfSnapshotPkgInfo was created but the VNF Snapshot Package was not yet built or uploaded.				

8.14.3 SnapshotPkgArtifactInformation information element

8.14.3.1 Description

This information element represents an artifact other than a VNFC Snapshot Image which is contained in the VNF Snapshot Package.

8.14.3.2 Attributes

The SnapshotPkgArtifactInformation information element shall follow the indications provided in table 8.14.3.2-1.

Table 8.14.3.2-1: Attributes of the SnapshotPkgArtifactInformation information element

Attribute	Qualifier	Cardinality	Content	Description
selector	M	1	Not specified	Information (such as a path) that identifies/addresses this artifact in the VNF Snapshot Package.
metadata	M	1	Not specified	The metadata of the artifact that are available in the VNF Snapshot Package, such as content type, size, creation date, etc.

8.14.4 VnfcSnapshotImageInfo information element

8.14.4.1 Description

This information element represents VNFC Snapshot Image Information.

8.14.4.2 Attributes

The VnfcSnapshotImageInfo information element shall follow the indications provided in table 8.14.4.2-1.

Table 8.14.4.2-1: Attributes of the VnfcSnapshotImageInfo information element

Attribute	Qualifier	Cardinality	Content	Description
vnfcSnapshotImageId	M	1	Identifier	The identifier of this VNFC Snapshot image.
name	M	1	Not specified	The name of this VNFC Snapshot image.
checksum	M	1	Not specified	The checksum of the VNFC Snapshot image file.
vnfcInstanceId	M	1	Identifier	Identifier of the snapshotted VNFC instance that this VNFC Snapshot image belongs to.

Attribute	Qualifier	Cardinality	Content	Description
containerFormat	M	1	Not specified	The container format indicates whether the VNFC Snapshot image is in a file format that also contains metadata about the actual snapshot.
diskFormat	M	1	Not specified	The disk format of a VNFC Snapshot image is the format of the underlying disk image.
createdAt	M	1	DateTime	The time when this VNFC Snapshot image creation has been completed.
minDisk	M	1	Not specified	The minimal Disk for this VNFC Snapshot image.
minRam	M	1	Not specified	The minimal RAM for this VNFC Snapshot image.
size	M	1	Not specified	The size of this VNFC Snapshot image.
userMetadata	M	0..N	KeyValuePair	User-defined metadata.
accessInformation	M	1	Not specified	Information such as a path (if the image is included in the VNF Snapshot Package) or an URL or identifier (if the image is not included in the VNF Snapshot Package) that allows to access a copy of this VNFC Snapshot Image.

8.14.5 Void

Annex A (informative): Examples of VNF connectivity patterns

A.1 Introduction

This annex illustrates examples of possible connectivity patterns for a VNF. The purpose is to illustrate the relationship among the different information elements specified in clause 8.5 that are used to describe the connectivity of and within a VNF instance.

The present annex A also illustrates the use of the "Change External VNF Connectivity" operation to re-connect external CPs of a VNF instance to a different external VL.

NOTE: The information related to connectivity as shown in the annex A is to be understood in the context of the present document, i.e. availability of certain information on the Or-Vnfm reference point follows the conditions that are detailed in the respective attribute descriptions and notes in the present document.

A.2 Example of a VNF with two different types of external connections points

The present example shows a regular connectivity pattern of a VNF where the two external CPs of the VNF use different connectivity patterns. Figure A.2-1 illustrates the example, from which it is highlighted the following:

- An external CP of the VNF instance (see VnfExtCp #1) that maps to an internal CP, i.e. a CP of a specific VNFC.
- An external CP of the VNF instance (see VnfExtCp #2) that refers to a link port of an internal VL of the VNF, typically a port in a router function (see VnfLinkPort #2.2).
- An internal VL of the VNF instance (see VnfVirtualLink #1) that is only used for connectivity of VNFCs within the VNF.
- An internal VL of the VNF instance (see VnfVirtualLink #2) that is used as provider of a link port for connectivity of external CPs of the VNF.
- Link ports of internal VL(s) of the VNF instance (see VnfLinkPort #1.1 to #1.3 and VnfLinkPort #2.1) that are optionally exposed on Or-Vnfm reference point.
- Internal CPs, i.e. CPs of specific VNFCs (see grey VNFC CPs) that are optionally exposed on the Or-Vnfm reference point.

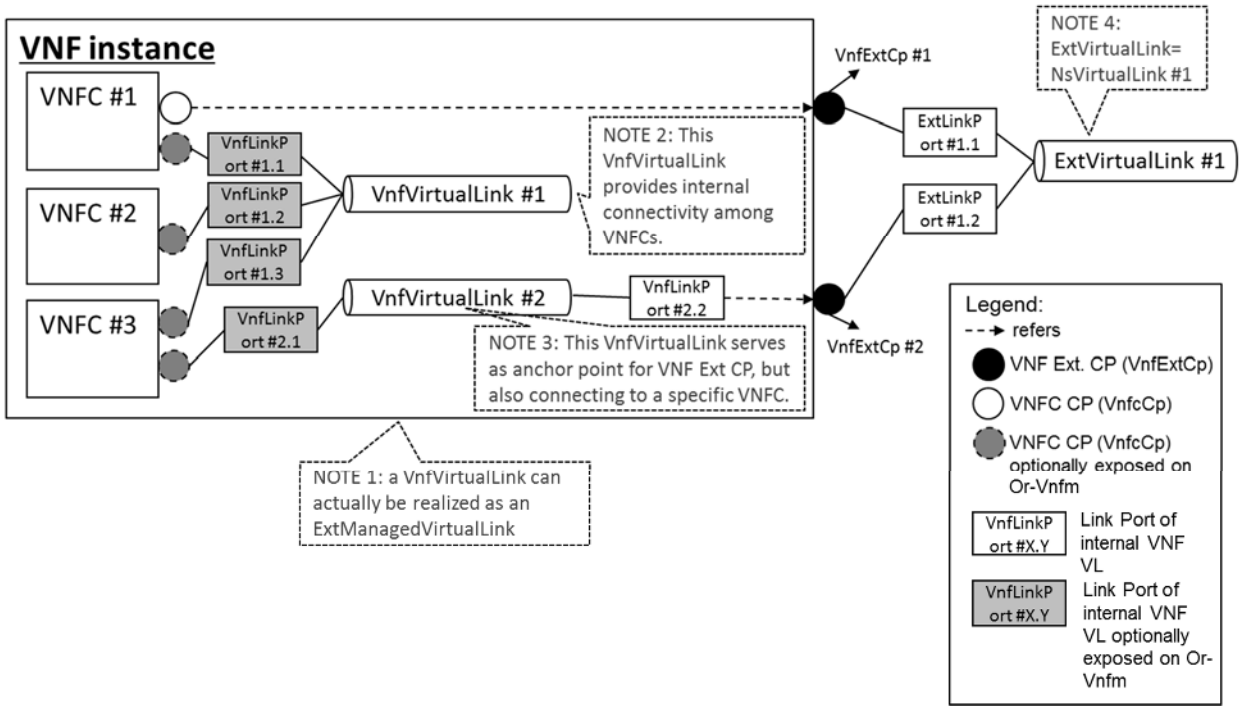


Figure A.2-1: Example of a VNF with two different types of external connection points

The example cases above only depict an initial, very basic set. Clause A.4 provides a more detailed set of use cases and related examples.

A.3 Example of changing VNF connectivity

This example illustrates the operation "Change external VNF connectivity" (clause 7.2.18). The scenario depicted disconnects all external CP instances that were created based on a particular CPD from a "source" external VL and connects them to a "target" external VL.

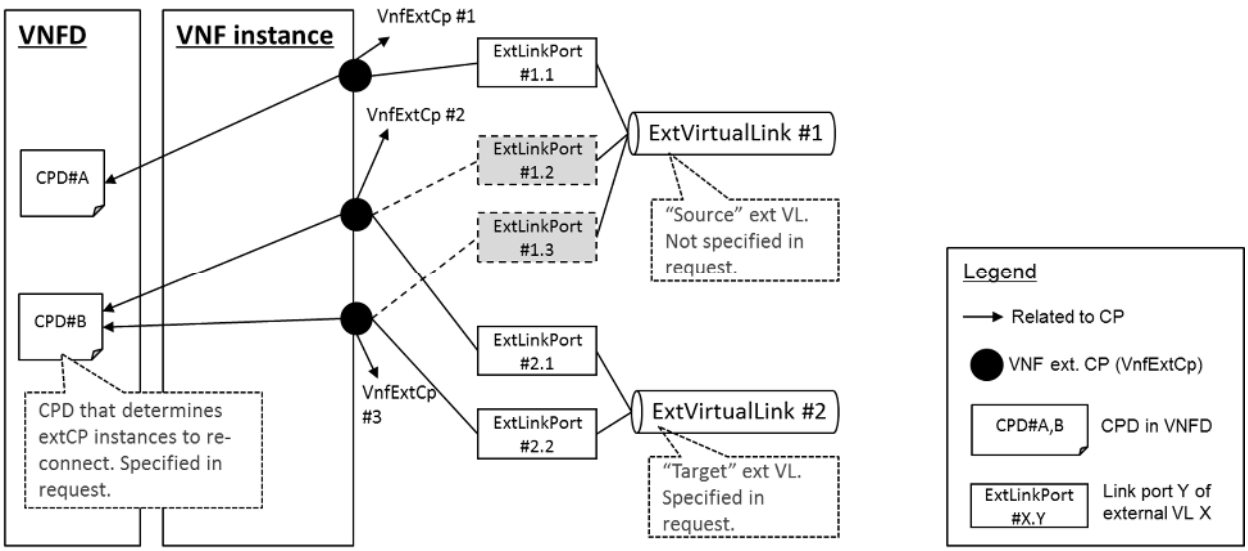


Figure A.3-1: Illustration of disconnecting external CPs from one external VL and connecting them to another external VL

A.4 VNF external connectivity use cases

A.4.1 Introduction

This clause illustrates the different use cases that expose VNF external connection points.

For each of the use cases the following aspects are shown:

- Networking topology
- VNFD representation of related descriptors
- Related parameters sent by NFVO to VNFM (as part of extVirtualLinkData)
- Related run time information sent by VNFM to NFVO

A.4.2 UC 1: Directly exposed VnfcCps

A.4.2.1 Network topology

VnfcCps are directly exposed as VnfExtCps, i.e. they are connected to an external link.

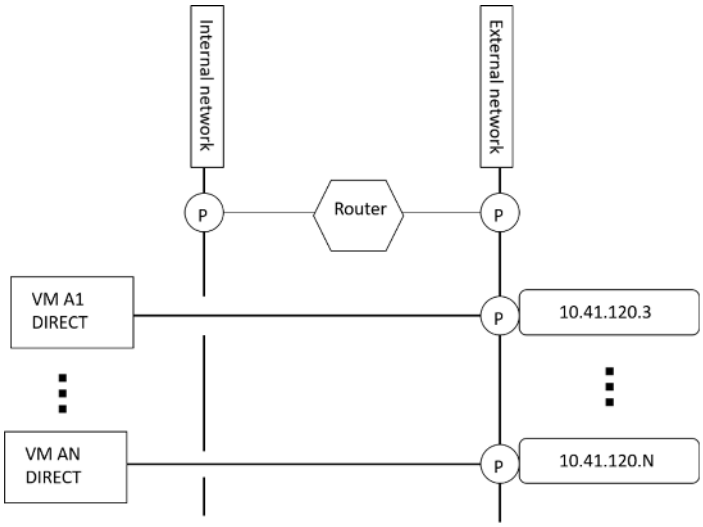


Figure A.4.2.1-1: VduCps directly exposed as VnfExtCps

Figure A.4.2.1-1 shows A1, ..., AN instances of a VNFC with a VnfcCp directly connected to an external virtual link.

A.4.2.2 VNFD representation

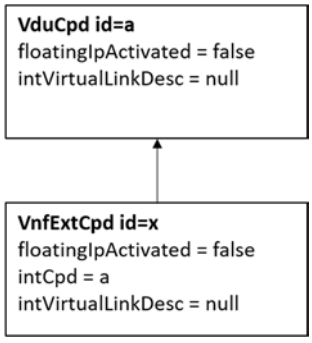


Figure A.4.2.2-1: CPDs of VduCps directly exposed as VnfExtCps

Figure A.4.2.2-1 shows the related CPDs in the VNFD, a VduCpd and a VnfExtCpd exposing it, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VduCpd id=a; no reference to an intVirtualLinkDesc: It indicates that the VduCp is not connected to an internal VL.
- In the VnfExtCpd id=x; floatingIpActivated = false: It indicates that the ExtCp is represented by a port on the external VL which connects to the VM (direct connection of the VM to the external VL).

A.4.2.3 Interface parameters

NFVO provides one VnfExtCpData structure as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response.

The VnfExtCpData structure contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.3)
     …
    cpConfigN: cpProtocolData (10.41.120.N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM A1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (10.41.120.3)
…
VnfcResourceInfo: VM AN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (10.41.120.N)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.3)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.N)
```

A.4.3 UC 2: VnfcCps exposed via a floating IP as VnfExtCp

A.4.3.1 Network topology

VnfcCps are connected to an internal link but are externally exposed via a floating IP address.

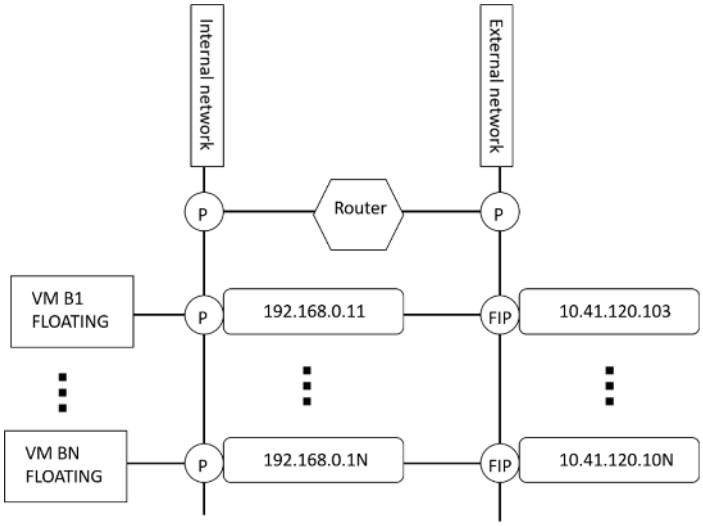


Figure A.4.3.1-1: VduCps connected to an internal virtual link and exposed via a floating IP address

Figure A.4.3.1-1 shows B1, ..., BN instances of a VNFC connected to an internal virtual link where the VnfcCps are assigned floating IP addresses from the external VL in addition to their default addresses from the internal VL.

A.4.3.2 VNFD representation

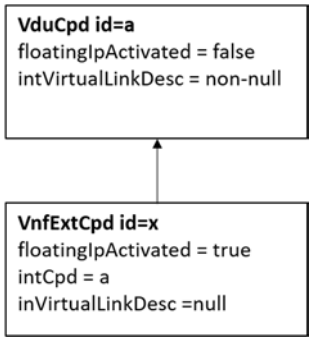


Figure A.4.3.2-1: CPDs of VduCps connected to an internal virtual link and exposed externally via a floating IP

Figure A.4.3.2-1 shows the related CPDs in the VNFD, a VduCpd and a VnfExtCpd exposing it via a floating IP address, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VduCpd id=a; reference to an intVirtualLinkDesc: It indicates that the VduCp is connected to an internal VL.
- In the VnfExtCpd id=x; floatingIpActivated = true: It indicates that the ExtCp is represented by a floating IP address that exposes the internal VNFC CP.

A.4.3.3 Interface parameters

NFVO provides one VnfExtCpData structure as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response.

The VnfExtCpData structure contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData record:

```
VnfExtCpData:
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.103)
     …
    cpConfigN: cpProtocolData (10.41.120.10N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM B1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (192.168.0.11)
…
VnfcResourceInfo: VM BN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (192.168.0.1N)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.103)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.10N)
```

A.4.4 UC 3: Directly exposed VipCp re-uses IP address of one of the exposed VnfcCps

A.4.4.1 Network topology

VnfcCps are directly exposed as VnfExtCps, i.e. they are connected to an external link. The IP address of one of the VnfcCp instances is re-used as VIP address, shared by all the VnfcCp instances.

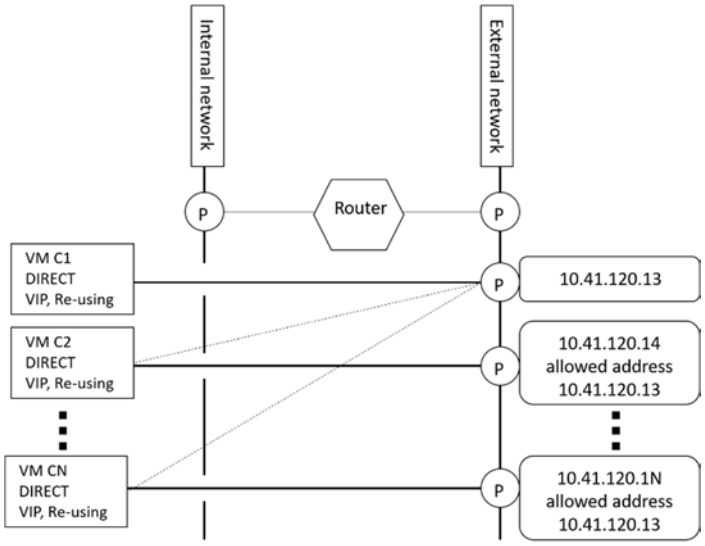


Figure A.4.4.1-1: Directly exposed VduCps with one of the VnfcCp addresses used as VIP

Figure A.4.4.1-1 shows C1, C2, ..., CN instances of a VNFC directly connected to an external virtual link. A VIP address is allocated and is also exposed in the external VL. The IP address of one of the VnfcCp instances is re-used as VIP.

A.4.4.2 VNFD representation

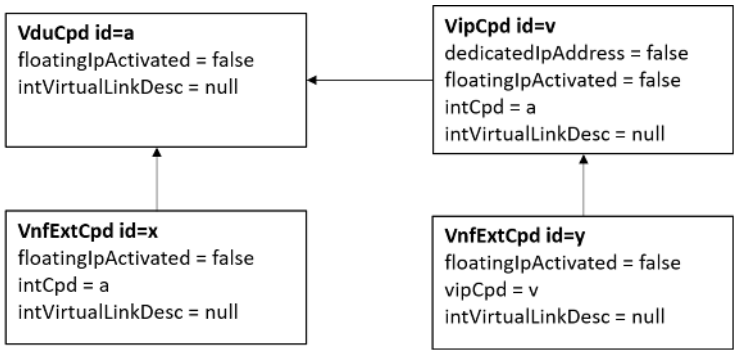


Figure A.4.4.2-1: CPDs of VduCps directly exposed as VnfExtCps with one of the addresses re-used as VIP

Figure A.4.4.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and two VnfExtCpds exposing them, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = false: It indicates that that the VIP does not have a dedicated IP address but that it re-uses one.

A.4.4.3 Interface parameters

NFVO provides two VnfExtCpData structures as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response. One corresponds to the VipCpd and the other one to the VduCpd.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has one cpConfig entry in this example.

NOTE 1: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

NOTE 2: Based on the information from the VNFD (dedicatedIpAddress = false) the VNFM will not create a port to allocate the VIP address since one of the addresses allocated to the VduCp instances is re-used as VIP address.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VduCpd contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.13)
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.13)
    cpConfig1: cpProtocolData (10.41.120.14)
     …
    cpConfigN: cpProtocolData (10.41.120.1N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

VnfcResourceInfo: VM C1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (10.41.120.13)
VnfcResourceInfo: VM C2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, vnfExtCp = 22, cpProtocolInfo (10.41.120.14)
…
VnfcResourceInfo: VM CN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (10.41.120.1N)

VipCpInfo records:

vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (10.41.120.13)

VnfExtCpInfo records:

vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.13)
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.13)
vnfExtCpInfo: cpInstanceId = 22, cpdId = x, associatedVnfcCpId = 2, cpProtocolInfo (10.41.120.14)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.1N)

A.4.5 UC 4: Directly exposed VipCp with dedicated IP address and port

A.4.5.1 Network topology

VnfcCps are directly exposed as VnfExtCps, i.e. they are connected to an external link. A dedicated VIP address is allocated to a dedicated port and shared by the VnfcCp instances.

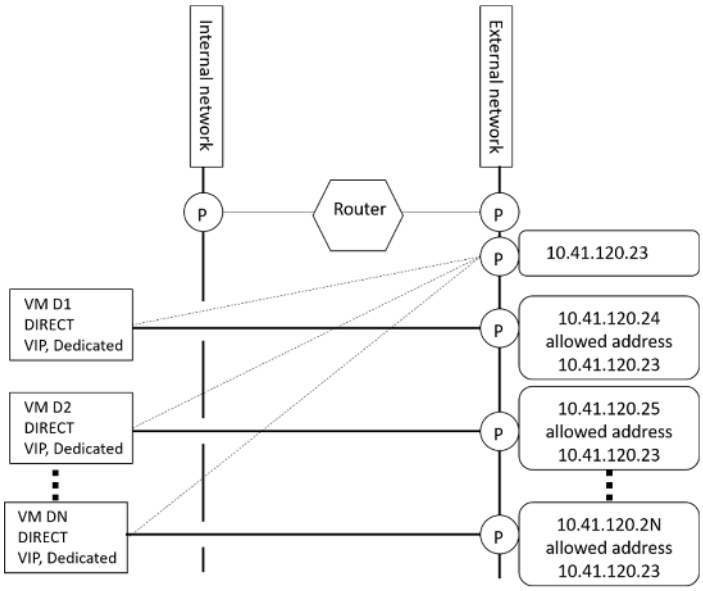


Figure A.4.5.1-1: VduCps directly and dedicated VIP port connected to an external virtual link

Figure A.4.5.1-1 shows D1, D2, ..., DN instances of a VNFC directly connected to an external virtual link. A dedicated port is created on the external virtual link and allocated a VIP address. The VIP address is shared by the VnfcCp instances.

A.4.5.2 VNFD representation

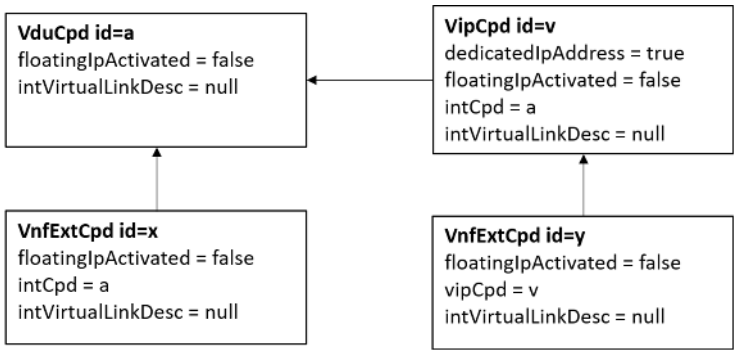


Figure A.4.5.2-1: CPDs of VduCps directly exposed as VnfExtCps with a dedicated VIP address

Figure A.4.5.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and two VnfExtCpds exposing them, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = true: It indicates that that the VIP has its own dedicated IP address.

A.4.5.3 Interface parameters

NFVO provides two VnfExtCpData structures as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response. One corresponds to the VipCpd and the other one to the VduCpd.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has one cpConfig entry in this example.

NOTE 1: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

The "createExtLinkPort" flag set to true indicates to the VNFM the need to create a port, to allocate the VIP address.

NOTE 2: If the NFVO provides an already created port in the external virtual link, the "createExtLinkPort" flag is not provided.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VduCpd contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.23), createExtLinkPort = true
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.24)
    cpConfig1: cpProtocolData (10.41.120.25)
     …
    cpConfigN: cpProtocolData (10.41.120.2N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM D1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (10.41.120.24)
VnfcResourceInfo: VM D2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, vnfExtCp = 22, cpProtocolInfo (10.41.120.25)
…
VnfcResourceInfo: VM DN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (10.41.120.2N)
```

VipCpInfo records:

```
vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (10.41.120.23)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.23)
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.24)
vnfExtCpInfo: cpInstanceId = 22, cpdId = x, associatedVnfcCpId = 2, cpProtocolInfo (10.41.120.25)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.2N)
```

A.4.6 UC 4-a: Directly exposed VipCp with dedicated IP address without dedicated port

A.4.6.1 Network topology

VnfcCps are directly exposed as VnfExtCps, i.e. they are connected to an external link. The VnfcCp instances share a dedicated VIP address allocated without a dedicated port.

It is an NFVO's decision whether to create a port or not in order to allocate the VIP address.

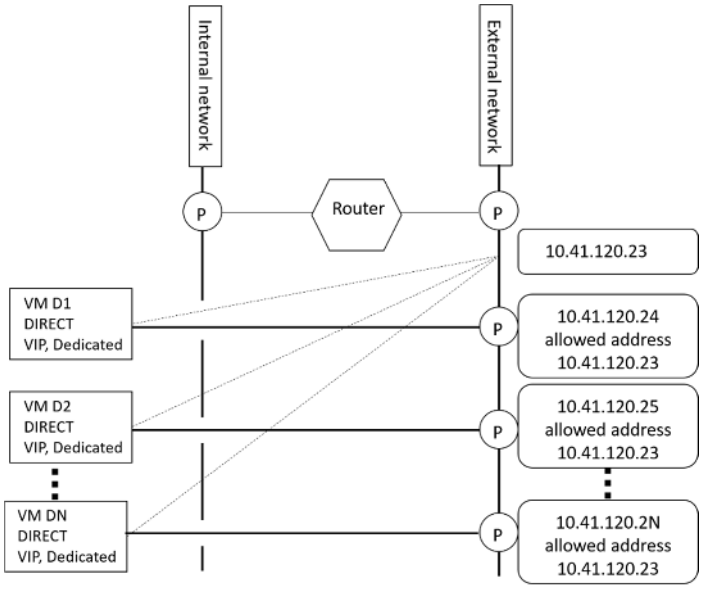


Figure A.4.6.1-1: VduCps directly and dedicated VIP port connected to an external virtual link

Figure A.4.6.1-1 shows D1, D2, ..., DN instances of a VNFC directly connected to an external virtual link. The VIP address shared by the VnfcCp instances is allocated without a dedicated port.

A.4.6.2 VNFD representation

The VNFD representation is the same as in UC 4 (see clause A.4.5.2).

A.4.6.3 Interface parameters

NFVO provides two VnfExtCpData structures as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response. One corresponds to the VipCpd and the other one to the VduCpd.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has one cpConfig entry in this example.

NOTE: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

The createExtLinkPort flag set to false indicates the VNFM not to create a port to allocate the VIP address.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VduCpd contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.23), createExtLinkPort = false
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.24)
    cpConfig1: cpProtocolData (10.41.120.25)
     …
    cpConfigN: cpProtocolData (10.41.120.2N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM D1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (10.41.120.24)
VnfcResourceInfo: VM D2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, vnfExtCp = 22, cpProtocolInfo (10.41.120.25)
…
VnfcResourceInfo: VM DN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (10.41.120.2N)
```

VipCpInfo records:

```
vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (10.41.120.23)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.23)
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.24)
vnfExtCpInfo: cpInstanceId = 22, cpdId = x, associatedVnfcCpId = 2, cpProtocolInfo (10.41.120.25)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.2N)
```

The values in the records shown above are the same as in UC 4 (see clause A.4.5.3). However, although not shown, the vnfExtCpInfo record that represents the VIP CP does not contain a reference to an extLinkPortId, as there is no link port attached in use case 4-a. Note that in use case 4, there is such reference as the VnfExtCp is associated to a port. This is a difference in the vnfExtCpInfo record exposing the VipCp between use cases 4 and 4-a.

A.4.7 UC 5: VipCp exposed as floating IP re-uses IP address of one of the exposed VnfcCps

A.4.7.1 Network topology

VnfcCps are connected to an internal link and exposed via floating IP addresses. They share a VIP address that is exposed via a floating IP address. The VIP address re-uses one of the VnfcCp addresses.

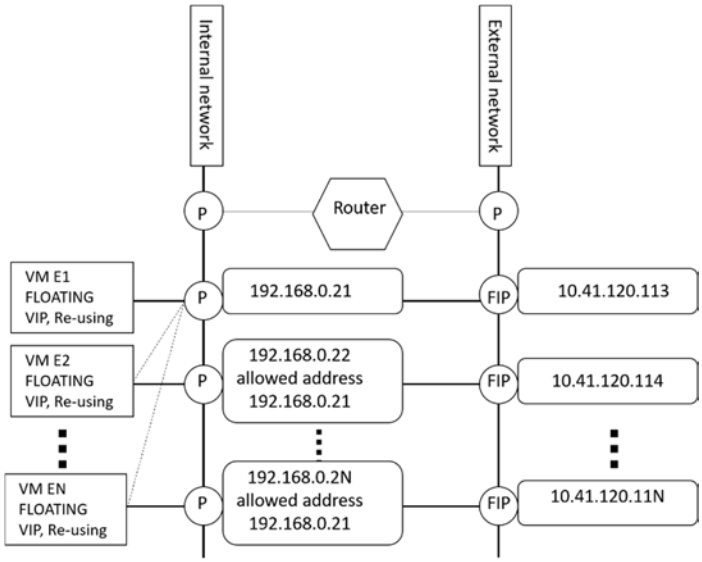


Figure A.4.7.1-1: VipCp exposed as floating IP re-uses IP address of one of the exposed VnfcCps VipCp re-uses a VnfcCp address

Figure A.4.7.1-1 shows E1, E2, ..., EN instances of a VNFC connected to an internal virtual link. The address of one of the VnfcCps is re-used as VIP address shared by all VnfcCps. Furthermore, the VnfcCps as well as the VipCp are exposed externally via floating IP addresses.

In addition to accessing the set of VnfcCps via the virtual IP address, since the individual VnfcCps (except the first one) are also exposed externally by their own floating IP addresses, each of these VNFC instances can also be accessed externally by the individual floating IP address.

A.4.7.2 VNFD representation

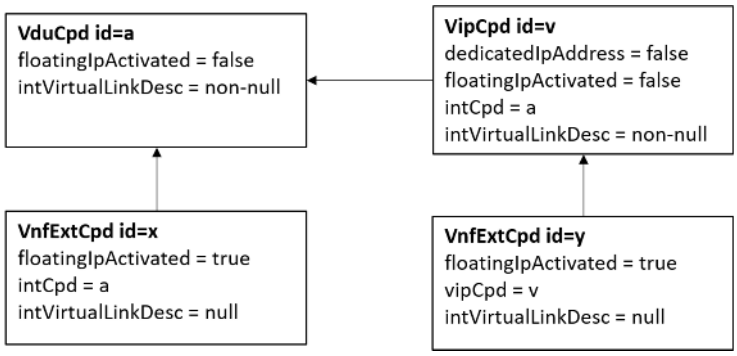


Figure A.4.7.2-1: CPDs of VduCps with one of the addresses re-used as VIP and all exposed as FIPs

Figure A.4.7.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and two VnfExtCpds exposing them, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = false: It indicates that that the VIP does not have a dedicated IP address but that it re-uses one.

A.4.7.3 Interface parameters

NFVO provides two VnfExtCpData structures as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response. One corresponds to the VipCpd and the other one to the VduCpd.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has one cpConfig entry in this example.

NOTE 1: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

NOTE 2: Based on the information from the VNFD (dedicatedIpAddress = false) the VNFM will not create a port to allocate a floating IP address for the VIP address since one of the addresses allocated to the VduCp instances is re-used a VIP address. Therefore, the floating IP address allocated to that instance is also used as the floating IP address for the VIP address.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VduCpd contains multiple cpConfig entries, one for each external CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.113)
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.113)
    cpConfig1: cpProtocolData (10.41.120.114)
     …
    cpConfigN: cpProtocolData (10.41.120.11N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM E1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (192.168.0.21)
VnfcResourceInfo: VM E2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, vnfExtCp = 22, cpProtocolInfo (192.168.0.22)
…
VnfcResourceInfo: VM EN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (192.168.0.2N)
```

VipCpInfo records:

```
vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (192.168.0.21)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.113)
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.113)
vnfExtCpInfo: cpInstanceId = 22, cpdId = x, associatedVnfcCpId = 2, cpProtocolInfo (10.41.120.114)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.11N)
```

A.4.8 UC 5-b: Variant of UC 5, only VipCp exposed

A.4.8.1 Network topology

This use case is a simplification of UC 5. Here only the VipCp is exposed with a floating IP address. The set of VnfcCps is only accessible externally with the floating IP address of the VipCp.

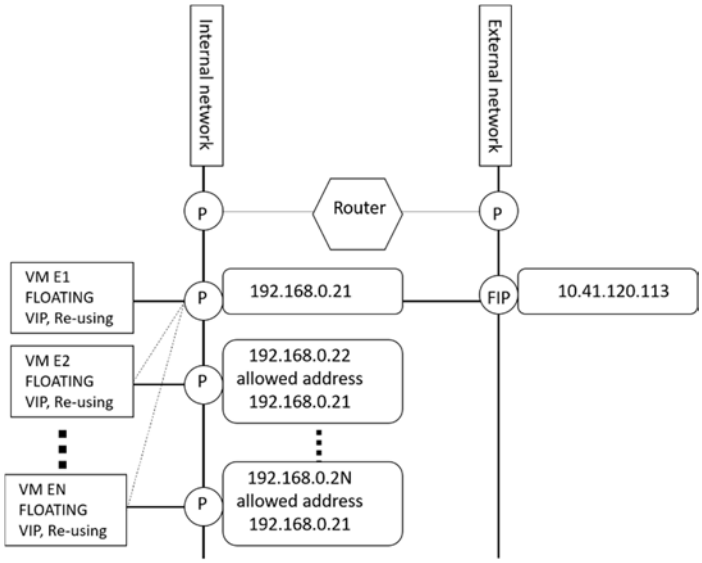


Figure A.4.8.1-1: VipCp re-uses a VnfcCp address and exposed via floating IP address

Figure A.4.8.1-1 shows E1, E2, ..., EN instances of a VNFC connected to an internal virtual link. The address of one of the VnfcCps is re-used as VIP address shared by all VnfcCps. Furthermore, the VipCp is exposed externally via a floating IP address.

A.4.8.2 VNFD representation

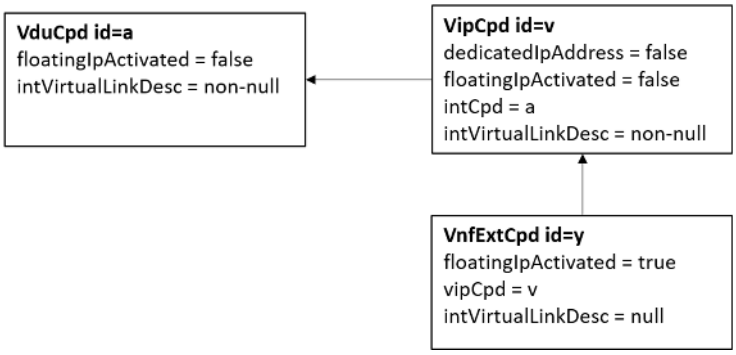


Figure A.4.8.2-1: CPDs of VduCps with one of the addresses re-used as VIP and exposed as FIP

Figure A.4.8.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and one VnfExtCpd exposing the VipCpd, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = false: It indicates that that the VIP does not have a dedicated IP address but that it re-uses one.

A.4.8.3 Interface parameters

NFVO provides one VnfExtCpData structure as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response.

The VnfExtCpData structure corresponds to the VnfExtCpd exposing the VipCpd and has one cpConfig entry in this example.

NOTE 1: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.113)

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

VnfcResourceInfo: VM E1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, cpProtocolInfo (192.168.0.21)
VnfcResourceInfo: VM E2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, cpProtocolInfo (192.168.0.22)
…
VnfcResourceInfo: VM EN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, cpProtocolInfo (192.168.0.2N)

NOTE 2: In the Or-Vnfm reference point the vnfcCpInfo structures are optional in this case, since they correspond to internal VnfcCps.

VipCpInfo records:

vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (192.168.0.21)

VnfExtCpInfo records:

vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.113)

A.4.9 UC 6: VduCps and VipCp with dedicated IP address and port exposed via floating IPs

A.4.9.1 Network topology

VnfcCps are connected to an internal link and exposed via floating IP addresses. They share a dedicated VIP address that is exposed via a floating IP address.

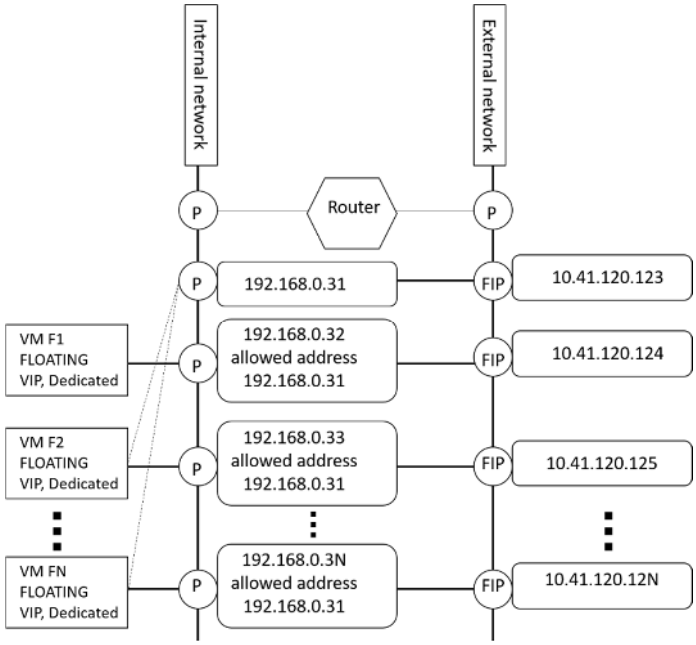


Figure A.4.9.1-1: VduCps and VipCp with dedicated IP address and port exposed via floating IPs

Figure A.4.9.1-1 shows F1, F2, ..., FN instances of a VNFC connected to an internal virtual link. A port with a dedicated VIP address shared by all VnfcCps is also connected to the internal link. Furthermore, the VnfcCps as well as the VipCp are exposed externally via floating IP addresses.

In addition to accessing the set of VnfcCps via the virtual IP address, since the individual VnfcCps are also exposed externally by their own floating IP addresses, each of these VNFC instances can also be accessed externally by the individual floating IP address.

A.4.9.2 VNFD representation

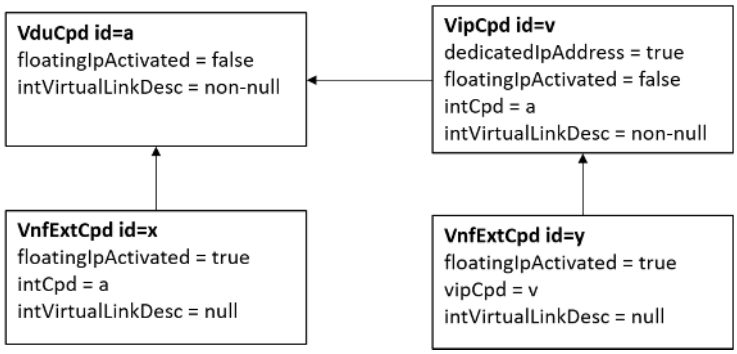


Figure A.4.9.2-1: CPDs of VduCps and dedicated VIP and all exposed as FIPs

Figure A.4.9.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and two VnfExtCpds exposing them, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = true: It indicates that that the VIP has its own dedicated IP address.

A.4.9.3 Interface parameters

NFVO provides two VnfExtCpData structures as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response. One corresponds to the VipCpd and the other one to the VduCpd.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has one cpConfig entry in this example.

NOTE: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

The VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VduCpd contains multiple cpConfig entries, one for each CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.123)
  cpdId: x
    cpConfig1: cpProtocolData (10.41.120.124)
    cpConfig1: cpProtocolData (10.41.120.125)
     …
    cpConfigN: cpProtocolData (10.41.120.12N)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM F1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, vnfExtCp = 21, cpProtocolInfo (192.168.0.32)
VnfcResourceInfo: VM F2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, vnfExtCp = 22, cpProtocolInfo (192.168.0.33)
…
VnfcResourceInfo: VM FN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, vnfExtCp = 2N, cpProtocolInfo (192.168.0.3N)
```

VipCpInfo records:

```
vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (192.168.0.31)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.123)
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfcCpId = 1, cpProtocolInfo (10.41.120.124)
vnfExtCpInfo: cpInstanceId = 22, cpdId = x, associatedVnfcCpId = 2, cpProtocolInfo (10.41.120.125)
…
vnfExtCpInfo: cpInstanceId = 2N, cpdId = x, associatedVnfcCpId = N, cpProtocolInfo (10.41.120.12N)
```

A.4.10 UC 6-b: Variant of UC 6, only VipCp exposed

A.4.10.1 Network topology

This use case is a simplification of UC 6. Here only the VipCp is exposed with a floating IP address. The set of individual VnfcCps is only accessible externally with the floating IP address of the VipCp.

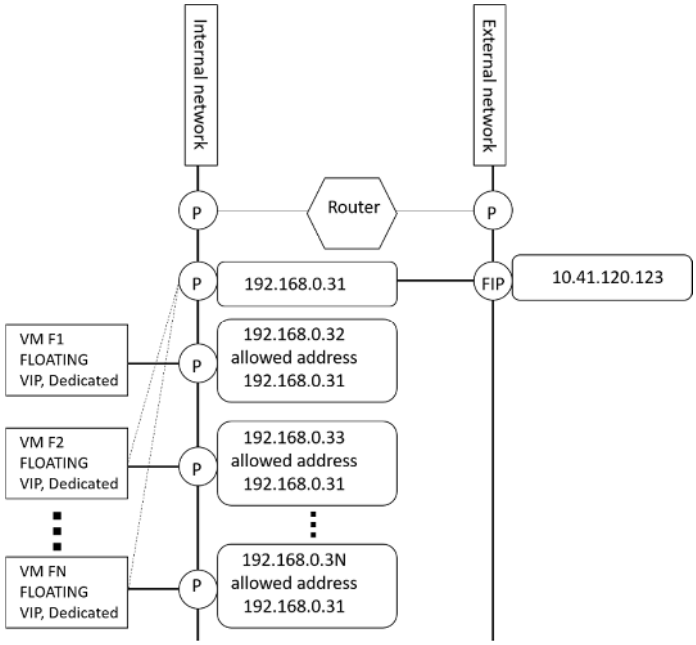


Figure A.4.10.1-1: VipCp with dedicated IP address and exposed via floating IP address

Figure A.4.10.1-1 shows F1, F2, ..., FN instances of a VNFC connected to an internal virtual link. A port with a dedicated VIP address shared by all VnfcCps is also connected to the internal link. Furthermore, that VipCp is exposed externally via a floating IP address.

A.4.10.2 VNFD representation

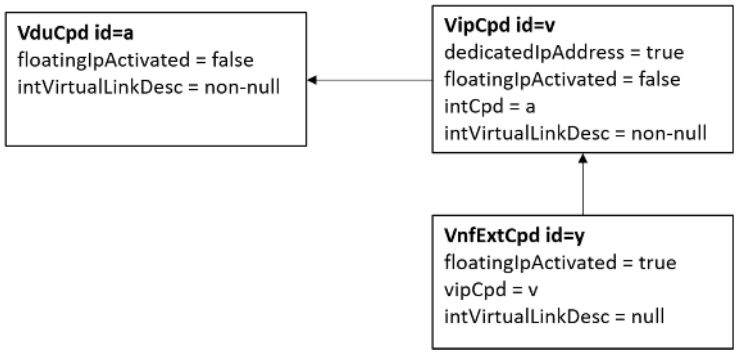


Figure A.4.10.2-1: CPDs of VduCps and dedicated VIP exposed as FIP

Figure A.4.10.2-1 shows the related CPDs in the VNFD, a VduCpd and a VipCpd and one VnfExtCpd exposing the VipCpd, with the most relevant attributes.

Additional explanation about the values assigned to some attributes:

- In the VipCpd id=v; dedicatedIpAddress = true: It indicates that that the VIP has its own dedicated IP address.

A.4.10.3 Interface parameters

NFVO provides one VnfExtCpData structure as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response.

The VnfExtCpData structure corresponds to the VnfExtCpd exposing the VipCpd and has one cpConfig entry in this example.

NOTE 1: In the more general case, it is possible to have multiple VIP CP instances exposed as external CPs based on the same VipCpd, as declared in the VipCpProfile in the VNFD. In that case, the VnfExtCpData structure that corresponds to the VnfExtCpd exposing the VipCpd has multiple cpConfig entries, one for each VIP CP instance.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.123)
```

The VNFM provides the following information related to the connection point instances as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM F1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, cpProtocolInfo (192.168.0.32)
VnfcResourceInfo: VM F2
  vnfcCpInfo: cpInstanceId = 2, cpdId = a, cpProtocolInfo (192.168.0.33)
…
VnfcResourceInfo: VM FN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, cpProtocolInfo (192.168.0.3N)
```

NOTE 2: In the Or-Vnfm reference point the vnfcCpInfo structures are optional in this case, since they correspond to internal VnfcCps.

VipCpInfo records:

```
vipCpInfo: cpInstanceId = 10, cpdId = v, vnfExtCp = 30, associatedVnfcCpId = 1,2,…,N, cpProtocolInfo (192.168.0.31)
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 30, cpdId = y, associatedVipCpId = 10, cpProtocolInfo (10.41.120.123)
```

A.4.11 UC 7: Internal VL is exposed as ExtCp

A.4.11.1 Network topology

A VnfExtCp exposes an internal VL instead of individual VnfcCps.

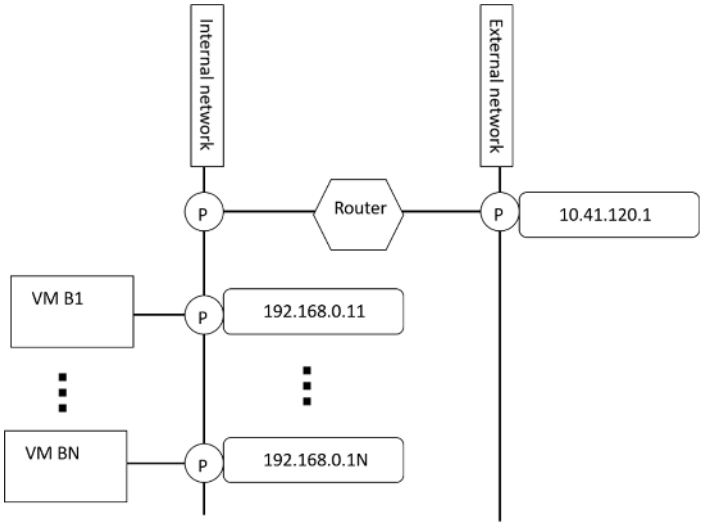


Figure A.4.11.1-1: VnfExtCp exposes an internal VL

Figure A.4.11.1-1 shows B1, ..., BN instances of a VNFC connected to an internal virtual link. A VnfExtCp exposes the internal VL. The VnfcCps remain internal, they are not exposed externally.

A.4.11.2 VNFD representation

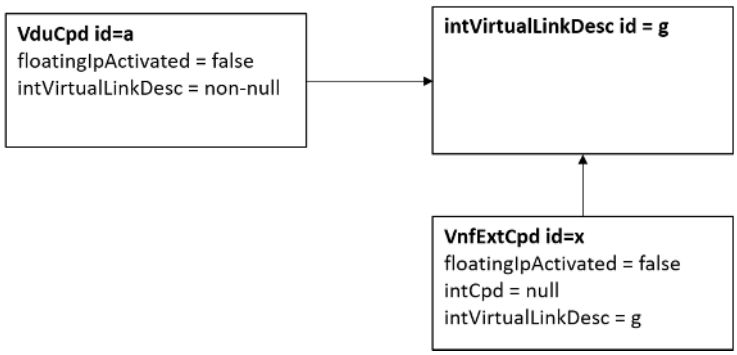


Figure A.4.11.2-1: CPDs of VduCps and internal VL exposed with a VnfExtCp

Figure A.4.11.2-1 shows the related descriptors in the VNFD, a VduCpd and an intVirtualLinkDesc and one VnfExtCpd exposing the intVirtualLinkDesc, with the most relevant attributes.

A.4.11.3 Interface parameters

NFVO provides one VnfExtCpData structure as part of the ExtVirtualLinkData structure, e.g. in the Instantiate VNF request or in a Grant response.

The VnfExtCpData structure corresponds to the VnfExtCpd exposing the intVirtualLink and has one cpConfig entry.

VnfExtCpData records:

```
VnfExtCpData:
  cpdId = y
    cpConfig1: cpProtocolData (10.41.120.1)
```

The VNFM provides the following information related to the connection point instances and virtual link instance as part of a Query VNF response:

VnfcCpInfo records:

```
VnfcResourceInfo: VM B1
  vnfcCpInfo: cpInstanceId = 1, cpdId = a, cpProtocolInfo (192.168.0.11)
…
VnfcResourceInfo: VM BN
  vnfcCpInfo: cpInstanceId = N, cpdId = a, cpProtocolInfo (192.168.0.1N)
```

NOTE: In the Or-Vnfm reference point the vnfcCpInfo structures are optional in this case, since they correspond to internal VnfcCps.

VnfVirtualLinkResourceInfo records:

```
VnfVirtualLinkResourceInfo: virtualLinkInstanceId = 40, vnfVirtualLinkDescId = g
```

VnfExtCpInfo records:

```
vnfExtCpInfo: cpInstanceId = 21, cpdId = x, associatedVnfVirtualLinkId = 40, cpProtocolInfo (10.41.120.1)
```

Annex B (informative): VNF software modification

B.1 Introduction

The present annex describes the procedures for the modification of the VNF software, both not assisted by the NFV-MANO and assisted by the NFV-MANO via change of current VNF Package of a VNF instance according to the interfaces and operations specified in the present document.

The procedures introduced in clauses B.2 and B.3 focus primarily on the interactions concerning the VNFM and on the reference points of the VNFM with other NFV-MANO or external functional blocks. Therefore, details of interactions and interface operations over other reference points are either not detailed or summarized.

B.2 VNF software modification not assisted by NFV-MANO

B.2.1 Description

In this type of VNF software modification, the VNF application is updated/upgraded by external management systems (e.g. OSS/EM) without any involvement of the NFV-MANO functional blocks. The process requires no change of the virtualised resources and resource composition of the current VNF instance. As part of the process, a new VNF Package containing the new VNF application software is made available to the NFV-MANO to be used by LCM operations after the VNF software modification.

The purpose of the NFV-MANO procedure is limited to:

a) Ensure that the information in the NFV-MANO entities is synchronized with respect to the software modification performed externally.

b) Ensure that the necessary VNF Package, artifacts and VIM assets are available within the NFV-MANO in order to handle any current or subsequent lifecycle management procedures on the affected VNF instance.

The fulfilment of the above points a) and b) allow the VNFM to use the new software images from the new VNF Package for the creation of new VNFC instances, such as during scale-out, or for the re-creation of existing VNFC instances, such as during VNF healing procedures.

NOTE: This software modification procedure was already supported in Release 2 versions of the present document as a reference and for specification completion purposes. Additional information is also present in clause B.2.3.2 of ETSI GS NFV-REL 006 [i.10].

B.2.2 Procedure

Figure B.2.2-1 shows the steps of the VNF software modification when it is not assisted by NFV-MANO.

As a pre-condition for the modification process, the VNF Package with the new VNF application software needs to be on-boarded to the NFVO according to step 1 of the procedure.

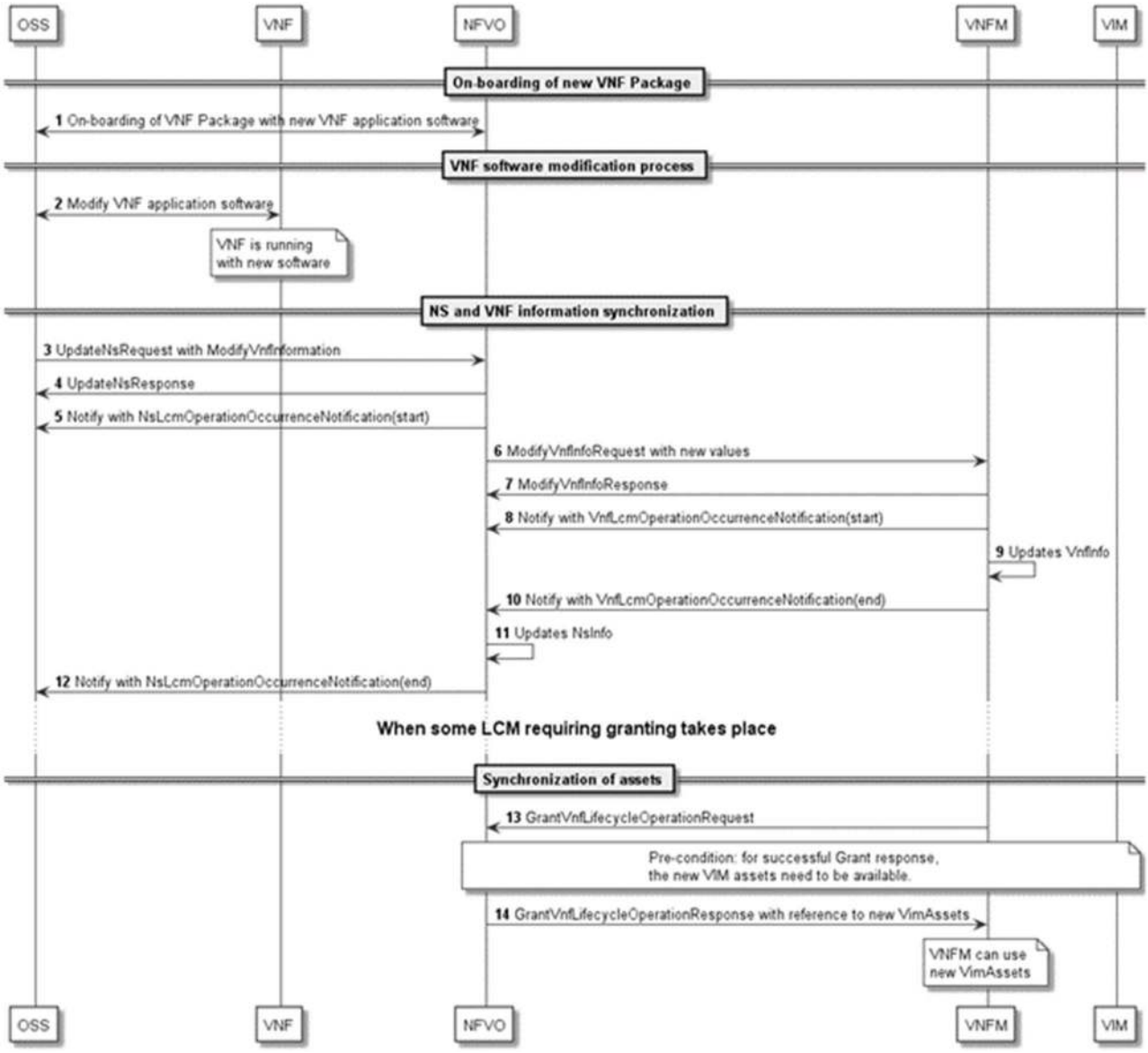


Figure B.2.2-1: Procedure of the VNF software modification not assisted by NFV-MANO

The procedure comprises the following phases and their steps:

A) On-boarding of the new VNF Package:

1) The VNF Package containing the new VNF application software is on-boarded to the NFVO as requested by the OSS. The NFVO checks the integrity and authenticity of the VNF Package and verifies that all mandatory information in the VNF Package is present and complies with the standard.

B) VNF software modification process:

2) The VNF application software modification is performed towards the VNF by the external management systems outside the NFV-MANO. As a result, at the end of this step the VNF is running the new software.

C) NS and VNF information synchronization:

3) Once the VNF application software modification is completed, the OSS requests the NFVO with an UpdateNsRequest to modify the information of the VNF so that it points to the new VNF Package, which has been on-boarded and which contains the new version of the VNF application software which also runs in the VNF instance as a result of step 2.

4) The NFVO acknowledges the request for modifying the VNF information by sending an UpdateNsResponse.

5) A notification from the NFVO is issued to subscribed consumers to notify them about the start of the NS LCM Update NS operation (see note 1).

NOTE 1: It is assumed in this procedure that the OSS has previously subscribed to the NFVO for this type of notifications.

6) The NFVO requests with a ModifyVnfInfoRequest the VNFM to modify the information of the VNF instance so that it points to the new VNF Package which has been on-boarded and which contains the new version of the VNF application software.

7) The VNFM acknowledges the request for modifying the VNF information by sending a ModifyVnfInfoResponse.

8) A notification from the VNFM is issued to subscribed consumers to notify them about the start of the Modify VNF information operation (see note 2).

NOTE 2: It is assumed in this procedure that the NFVO has previously subscribed to the VNFM for this type of notifications.

9) The VNFM updates the VNF information (VnfInfo) accordingly.

10) A notification from the VNFM is issued to subscribed consumers to notify them about the end of the Modify VNF information operation.

11) The NFVO updates the NS information (NsInfo) with the modified information of the VNF instance.

12) A notification from the NFVO is issued to subscribed consumers to notify them about the end of the NS LCM Update NS operation.

D) Synchronization of VIM assets:

The synchronization of VIM assets in between the NFVO and the VNFM takes place in the first subsequent granting exchange, whenever such granting becomes necessary.

13) The VNFM sends a GrantVnfLifecycleOperationRequest providing the new vnfdId according to the updated information in the VnfInfo.

14) The NFVO sends to the VNFM a GrantVnfLifecycleOperationResponse providing references to the new VimAssets. As a pre-condition for a successful grant response, the new VIM assets need to be available (see note 3).

NOTE 3: The creation of the new VIM assets can take place during any of the preceding steps before the VNF LCM granting response, e.g. after VNF Package on-boarding, in parallel to the VNF software modification process, in parallel during the NS and VNF information synchronization. If the creation of VIM assets is initiated only at the time of the granting request, it can significantly delay the granting operation, thus, it can adversely affect the LCM operation requiring the granting (e.g. healing, scaling).

B.3 VNF software modification assisted by NFV-MANO via change of current VNF Package

B.3.1 Overview

In this type of VNF software modification, the current VNF Package of a VNF instance is changed. In this process, the NFV-MANO functional blocks support the modification by providing and handling the necessary VNF lifecycle and virtualised resources management operations. By using the NFV-MANO functional blocks, a common handling of the VNF software modification processes can be achieved leveraging also the capabilities of NFV-MANO to create, modify and terminate virtualised resources.

As part of changing the current VNF Package of the VNF instance, modifications to the current set of virtualised resources and/or composition of the VNF instance can take place. As part of the modification process, and in support of it, temporary resources can also be created and used by the VNF to perform the change of the current VNF Package.

B.3.2 Procedure

Figure B.3.2-1 shows the steps of changing the current VNF Package of a VNF instance with resource modifications performed via the NFV-MANO functional blocks.

As a pre-condition for the modification process, the VNF Package with the new VNF application software needs to be on-boarded to the NFVO according to step 1 in the procedure.

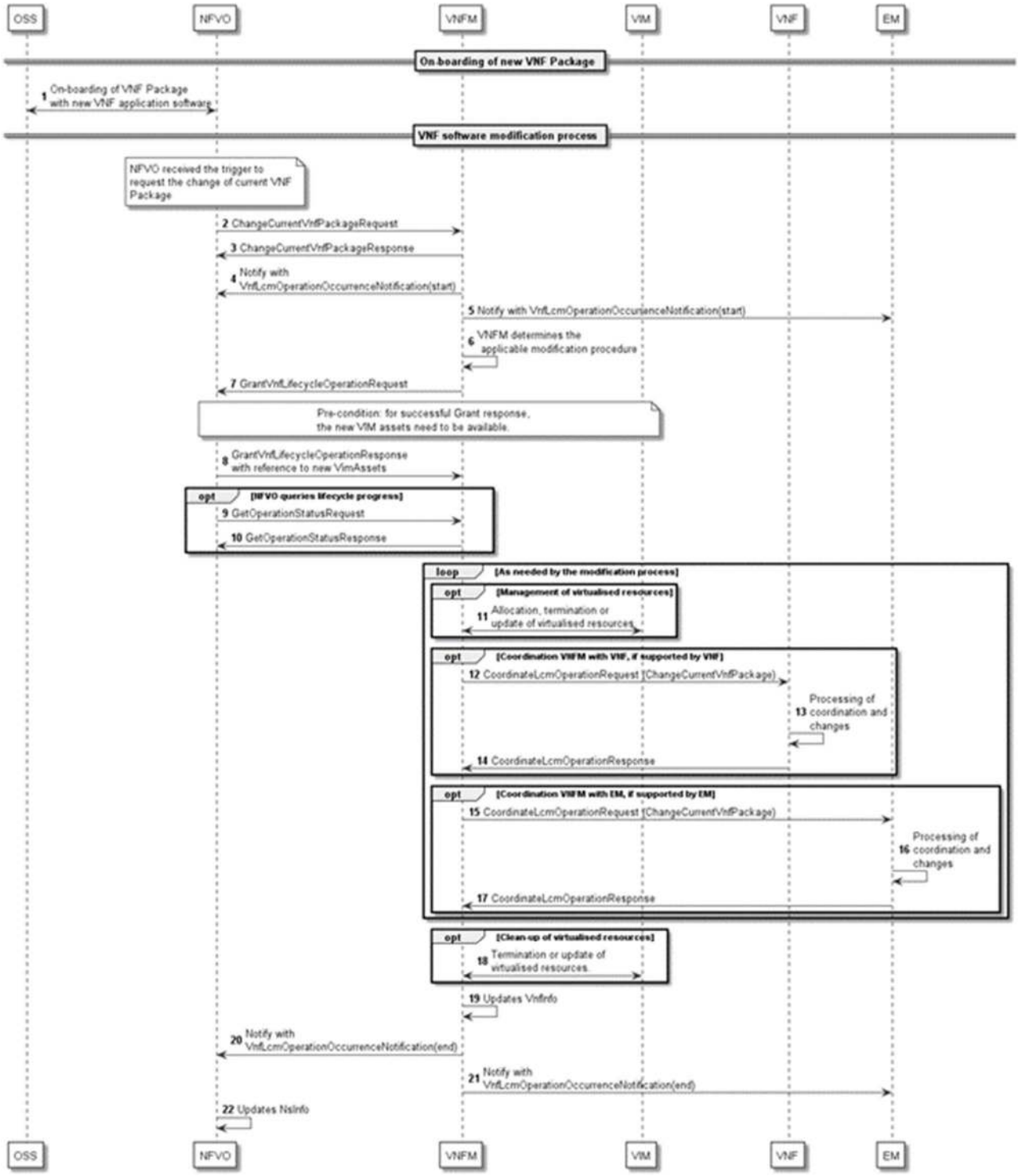


Figure B.3.2-1: Procedure of the VNF software modification performed by NFV-MANO with resource modifications

The procedure comprises the following phases and their steps:

A) On-boarding of new VNF Package:

1) The VNF Package containing the new VNF application software is on-boarded to the NFVO as requested by the OSS. The NFVO checks the integrity and authenticity of the VNF Package and verifies that all mandatory information in the VNF Package is present and complies with the standard.

B) VNF software modification process:

The NFVO has received the request from the OSS/BSS to change the current VNF Package of the VNF. The NFVO has also checked the pre-conditions for the start of the modification process (e.g. availability of the VNF Package with the new software images).

2) The NFVO sends a ChangeCurrentVnfPackageRequest with the identifier of the destination VNF Package (dstVnfdId) to be used for the change.

3) If the request has been accepted, the VNFM sends to the NFVO a ChangeCurrentVnfPackageResponse with the lifecycleOperationOccurrenceId, and the change of current VNF Package will continue with the following steps. In case of error, the modification process is stopped, and appropriate error information is returned by the VNFM to the NFVO.

4 & 5) The VNFM sends the "start" notification of the VNF lifecycle operation occurrence indicating the type of operation as "ChangeCurrentVnfPackage" to the subscribed entities of VNF LCM notification. If any error happens during the lifecycle operation, a "result" notification with appropriate error information is sent to the subscribers. See note 1.

NOTE 1: It is assumed in this procedure that the NFVO and other external management systems such as the EM have subscribed to the VNFM for this type of notifications.

6) The VNFM verifies and processes the ChangeCurrentVnfPackageRequest. This includes determining the applicable VNF Package change information (VnfPackageChangeInfo) by searching for the appropriate VersionSelector based on the input information from the operation request and the runtime information of the VNF instance (i.e. VnfInfo).

7) The VNFM determines the virtualised resources of the VNF instance that will be updated due to the change of current VNF Package and/or any virtualised resources needed for the VNF instance. The VNFM sends a GrantVnfLifecycleOperationRequest indicating the change with the following input parameters:

   a) the dstVnfdId corresponding to the destination VNF Package used for the modification;

   b) the vnfdId corresponding to the VNF Package in use before the request;

   c) the type of lifecycleOperation as "ChangeCurrentVnfPackage"; and

   d) the list of updateResource with the resources that are updated (e.g. those that will use new software images), the list of tempResource for the temporary resources, the list of addResource with the new resources to be added, and the list of removeResource with the existing resources to be terminated.

8) The NFVO sends to the VNFM a GrantVnfLifecycleOperationResponse. The response message contains information about the new VimAssets (e.g. new software images) available for the software modification. If temporary resources have been also requested, the NFVO confirms the granted temporary resources. As a pre-condition for a successful grant response, the new VIM assets need to be available (see note 2).

NOTE 2: The creation of the new VIM assets can take place during any of the preceding steps before the VNF LCM granting response, e.g. after VNF Package on-boarding.

9) (Optionally) After the ChangeCurrentVnfPackage has been activated, the NFVO can send to the VNFM, anytime during the modification process, a GetOperationStatusRequest to request status information of the ongoing VNF LCM operation.

10) If a GetOperationStatusRequest has been received by the VNFM, the VNFM processes the request and provides a GetOperationStatusResponse with information about the status of the operation.

The modification process continues in a loop of resource management and/or coordination interaction steps. In this flow, the applicable steps supported by interfaces defined in the ETSI GS NFV-IFA 006 [1] and ETSI GS NFV-IFA 008 [i.5] are listed.

NOTE 3: If virtualised resource management in indirect mode is used, virtualised resource management interactions would be supported by interfaces defined in the present document and ETSI GS NFV-IFA 005 [i.4], but these are not detailed in the present information flow.

The number of iterations and the content of each step are specific to the VNF, and moreover depend on the actual modification determined by the source VNFD, source VNF deployment flavour and target VNFD.

NOTE 4: If applicable, some steps of the loop may be executed in parallel.

11) If the current iteration step requires virtualised resource management interactions, and direct mode is applicable, the VNFM requests the allocation, termination and/or update of virtualised resources to the corresponding VIM.

If the current iteration step requires coordination with the VNF and the LCM Coordination interface is supported by the VNF, the following steps 12 to 14 are executed:

12) The VNFM sends to the VNF a CoordinateLcmOperationRequest corresponding to the ChangeCurrentVnfPackage operation.

13) The VNF processes the coordination request and performs any changes in the VNF instance as needed.

14) The VNF sends to the VNFM a CoordinateLcmOperationResponse.

If the current iteration step requires coordination with the EM (as an example of an external management system) and the LCM Coordination interface is supported by the EM, the following steps 15 to 17 are executed:

15) The VNFM sends to the EM a CoordinateLcmOperationRequest corresponding to the ChangeCurrentVnfPackage operation.

16) The EM processes the coordination request and performs any changes in the VNF instance as needed.

17) The EM sends to the VNFM a CoordinateLcmOperationResponse.

Upon completion of the modification process iterations, the virtualised resource management process continues as follows:

18) If the process requires additional virtualised resource management interactions (e.g. virtualised resource clean-up), and direct mode is applicable, the VNFM requests the termination and/or update of virtualised resources from the corresponding VIM.

Upon completion of the above steps, the procedure continues as follows:

19) The VNFM updates the VNF information (VnfInfo) as appropriate.

20 & 21) A notification from the VNFM is issued to subscribed consumers to notify them about the end of the VNF LCM operation occurrence.

22) The NFVO updates the NS information (NsInfo) according to the modified information of the VNF instance.

B.4 VNF software modification relationship to NS and NSD management

B.4.1 Introduction

The NS and the VNF are interrelated at different phases of the network design, deployment and operation, among others:

- The NSD has references to specific VNF Packages and their deployable VNF flavours via VNF profiles.

- The NS instance information maintained by the NFVO has references to the VNF instances that are part of the NS instance.

Due to the relationship between the NS and the VNF, a VNF software modification that involves handling a different VNF Package version needs additional steps for the preparation of the NS and the synchronization/update of its information.

Annex C (informative): Change history

Date	Version	Information about changes
18 December 2014	V0.0.1	Skeleton and ToC
07 January 2015	V0.0.2	Updates based on NFVIFA(14)000028r4
26 January 2015	V0.1.0	Early draft after IFA Shanghai Interim Meeting, including contributions: - NFVIFA(15)000034r3_IFA007_section_4_Overview - NFVIFA(15)000036r3_IFA0nn_Interface_WIs_section_1_Scope_small_addition - NFVIFA(15)000091r3_IFA009_section_3_Definitions
23 February 2015	V0.1.1	Editorial: Title corrected for alignment
25 June 2015	V0.1.2	Contributions included: - NFVIFA(15)000066r2_IFA007_Clause_5_VNF_Package_interface_ notification_req - NFVIFA(15)000067r3_IFA007-008_Clause_5_VNF_LC_change_interface_reqs - NFVIFA(15)000189r5_VNF_Package_management_interface_Requirements - NFVIFA(15)000252r1_IFA007_interface_requirements_VNF_LCM_Granting - NFVIFA(15)0000254r4_IFA007_interface_requirements_VNF_LCM - NFVIFA(15)000256r2_IFA007_interface_requirements_VNF_Lifecycle_ Chg_Notif - NFVIFA(15)000357r1_Change_to_conventions_for_conditional_attributes - NFVIFA(15)000523r2_IFA010-007-008_Extend_VNF_lifecycle_change_notification - NFVIFA(15)000567r1_Adding_note_from_458r3_to_all_interface_GSs - NFVIFA(15)000675r1_IFA007_Section_5_Reference_Point_Requirements - NFVIFA(15)000798r1_IFA007_Add_VNF_LCM_interface_requirements Applicability of multi-document changes checked but no changes necessary: - NFVIFA(15)000035_Blueprint_Phase_1_GSs_as_Informative_References Editorial changes: - Aligned document structure with template and IFA005 and IFA006 - Information elements clause is now clause 8 - Inserted separate "Interface requirements" and "Reference point requirements" subclauses in clauses 6 and 7 - Implemented disclaimer from NFVTSC(15)000041r3
10 July 2015	V0.1.3	Contributions included: - NFVIFA(15)000845r1_IFA007_Move_VNF_LCM_Notification_ requirement_from_798r1 Editorial changes: - Aligned labels of requirements with IFA conventions as per NFVIFA(15)000853r2
13 August 2015	V0.2.0	Contributions included: - NFVIFA(15)000526r3_IFA007_LCM_operation_granting_requirements - NFVIFA(15)000939r1_IFA007_section_5_Or- Vnfm_reference_point_requirements - NFVIFA(15)000082r8_IFA007_VNF_Lifecycle_Manager_and Lifecycle_Operation_Grantin - NFVIFA(15)000722r4_IFA007_detailed_interface_design_LCM_-_InstantiateVNF - NFVIFA(15)000723r5_IFA007_detailed_interface_design_LCM_Notification - NFVIFA(15)000838r5_IFA007_VNF_PM_interface - NFVIFA(15)000933r3_IFA007_detailed_interface_design_LCM_-_ScaleVNF - NFVIFA(15)000935r1_IFA007_detailed_interface_design_LCM_-_QueryVNF
27 August 2015	V0.2.1	Incomplete implementation of NFVIFA(15)000838r5 in v0.2.0 was fixed (clause 8 content from 838r5 was missing in v0.2.0) Editorial alignments in clause 8 (structured into subclauses as done in 838r5, text from 838 adapted to latest conventions (Parameter → Attribute, Type → Content)

Date	Version	Information about changes
02 October 2015	V0.3.0	Contributions included: - NFVIFA(15)0001141r1_IFA007-008_VNF_LCM_Healing_operation_interface_requirement - NFVIFA(15)0001197_IFA007_VNF_Package_Management_small_fix - NFVIFA(15)000837r4_IFA007_VNF_FM_interface - NFVIFA(15)0001142r2_IFA007_VNF_LCM_Healing_operation_interface_specification - NFVIFA(15)000953r4_IFA010-007-008_VNF_FM_extra_notifications - NFVIFA(15)0001199r1_IFA007_IFA013_VNF_Package_Management_Notification_Additional - NFVIFA(15)0001084r2_IFA007_IE_names_alignment - NFVIFA(15)0001221_IFA007_add_description_to_VNF_LCM_interface - NFVIFA(15)0001022r2_IFA007_FM_PM_interface_naming_alignment Editorial fixes, e.g. to align with latest interface template
09 November 2015	V0.4.0	Contributions included: - NFVIFA(15)0001154r2_IFA007_VNF_Package_interface_modify_and_query_operations - NFVIFA(15)0001139r3_IFA007_7-8_IFA008_7-9_VNF_FM_extension_for_VR_state_changes - NFVIFA(15)0001302_IFA007_Adding_VNF_performance_management_requirements - NFVIFA(15)0001152r2_IFA007_7_X_IFA008_7_X_VNFM-produced_VNF_Config_interface - NFVIFA(15)000065r4_IFA007_5_3_3_IFA008_5_2_1_1_Operate_VNF_interface_requiremen - NFVIFA(15)0001150r2_IFA007_7_2_IFA008_7_2_Operate_VNF_interface_specification - NFVIFA(15)0001151r3_IFA007_4-5_IFA008_4-5_VNFM-produced_VNF_Configuration_reqs - NFVIFA(15)000638r8_IFA007_5_2__5_3_resource_management_requirements - NFVIFA(15)0001225r1_IFA013 IFA015 Merging PM Information Elements - NFVIFA(15)0001266r2_IFA007_Exclude_Error_Cases_from_Output_IE_Cardinality - NFVIFA(15)0001347r3_IFA010_Section_7_2_Functional_requirements_for_VNF_LCM Editorial fixes: - Change of affiliation of Marc Flauw - Subclauses of Clauses 5 and 8 renumbered to keep sequence of IEs in sync with sequence of interfaces
21 December 2015	V0.5.0	Contributions included: - NFVIFA(15)0001453r3_IFA007_rapporteur_s_cleanup_of_v040 - NFVIFA(15)000934r7_IFA007_detailed_interface_design_LCM_TerminateVNF - NFVIFA(15)0001290r9_IFA007_Virtualised_Compute_Interfaces - NFVIFA(15)0001291r9_IFA007_Virtualised_Network_Interfaces - NFVIFA(15)0001292r9_IFA007_Virtualised_Storage_Interfaces - NFVIFA(15)0001455r2_IFA005_IFA006_IFA007_IFA008_IFA013_FM_PM_fixes - NFVIFA(15)0001458r1_IFA007_fixes_References_Introduction - NFVIFA(15)0001485r4_IFA007_IFA008_VNF_Scaling_Parameters - NFVIFA(15)0001495r2_IFA007_5-3-3_IFA008_5-2-1-1_VNF_LCM_extension - NFVIFA(15)0001500r1_IFA007_6_2_2_IFA013_7_7_5_Addressing_note_VNF_Package_mgmt - NFVIFA(15)0001515r3_IFA007_Non-normative_should_and_may_separated_from_1453r1 - NFVIFA(15)0001529_IFA007-008_7_X_Correction_to_subscribe_filter_for_VNF_FM_i_f - NFVIFA(15)0001596_IFA005_IFA006_IFA007_IFA008_IFA012_IFA013_Remove_section_9_S - NFVIFA(15)0001608r2_IFA005_IFA006_IFA007_IFA008_IFA013_Normative_Reference_to_IF - NFVIFA(15)0001613_IFA007_Subscribe_Notify_description_fixes Editorial fixes: - Implemented the agreement regarding table numbering - Implemented the agreement regarding the text referencing the tables for input and output parameters

Date	Version	Information about changes
February 2016	V0.6.0	Contributions included: - NFVIFA(15)0001454r5_IFA007_5_3_3_4_fixing_Virtualised_Resources_ Change_Notificat - NFVIFA(16)000072r2_IFA007_5_2_and_5_3_3_Additional_requirements_for_ indirect_RM - NFVIFA(15)0001519r5_IFA007_numberOfSteps_support_signaling - NFVIFA(16)000007r1_IFA007_referencing_IFA011 - NFVIFA(16)000042r2_IFA011_IFA007_VNF_LCM_related_information_in_ VNFD - NFVIFA(16)000106r1_IFA007_5_3_and_8_6_IFA008_5_2_and_9_4_1_ Identification_for_V - NFVIFA(16)000117r1_IFA007_8_5_4_Adding_basic_VnfInfo_attributes - NFVIFA(16)000119r1_IFA007-IFA008_7_2_Addressing_editor_note_on_VNF_operate - NFVIFA(16)000121r1_IFA007_5_3_3_Rapporteurs_fixes_indirect_RM - NFVIFA(16)000123r1_IFA007_7_2_6_2_Additional_params_in_healing - NFVIFA(16)000151r1_IFA007_Indicator_Interface, with editorial fixes (copy&paste error (replaced in change 2 "Ve-Vnfm-em by Or-Vnfm), "parameters" instead of "information elements" in operations, added "… and notifications" in 8.10 headline) - NFVIFA(15)000511r9_IFA007_6_3_detailed_interface_design_LCM_ Operation_Granting (with editorial fixes to align with the interface template) Editorial fixes: - Pre-processing done before TB approval E-mail: mailto:edithelp@etsi.org. Rapporteur's note: Had to undo the changes to front matter as this document is still intended for being made available through the open area. - Applied conventions according to NFVIFA(15)0001562r5_Interface_template_update, including removal of the editor's notes that stated the need to add UML diagrams to the IE clauses - Various small typo fixes
March 2016	V0.6.1	Re-created the ZIP archive due to a problem in the ZIP file of V 0.6.0. No changes to content.
21 March 2016	V0.7.0	Version to enter WG review Contributions included: - NFVIFA(16)000175r3_IFA007_Alarm_Cleared_Notification_and_Alarm_IE_ Update - NFVIFA(16)000183_IFA007_Referencing_IFA013_informatively - NFVIFA(16)000102r2_IFA007_Section7_2_Modification_on_Query_operation - NFVIFA(16)000142r6_IFA007_8_5_IFA008_9_3_current_scale_level_in_ VNFInfo - NFVIFA(16)000267r4_IFA007_IFA008_IFA011_scale_VNF_to_instantiation_ level - NFVIFA(16)000171r3_IFA007_C_D__8_G_H_8_I_J_Virtualised_Resources_ Performance_Ma - NFVIFA(16)000170r3_IFA007_A_B_8_A_B_8_C_D_8_E_F_Virtualised_ Resources_Fault_Man - NFVIFA(16)000176r3_IFA007_Fixing_normative_and_informative_references_ to_IFA_GS - NFVIFA(16)000197_IFA007_IFA008_instantiation_level_in_InstantiateVNF - NFVIFA(16)000217_IFA007_IFa008_Adding_description_to_VNF_Instance - NFVIFA(16)000219r2_IFA007_IFA008_resolving_editor_s_note_on_VnfInfo - NFVIFA(16)000220_IFA007_scaling_step_note_alignment_with_proposal_ from_ 779 - NFVIFA(16)000228r2_IFA007_editor_s_notes_on_externally_managed_ internal_VLs - NFVIFA(16)000231r1_IFA007_Adding_deployment_flavour_to_grant_request - NFVIFA(16)000232r3_IFA007_6_3_2_Adding_level_to_grant_request - NFVIFA(16)000234r2_IFA007_6_4_2-4_X_8_X_Y_Virtualised_Resources_Change_Notifica - NFVIFA(16)000235_IFA007_Adding_deployment_flavour_to_VnfInfo - NFVIFA(16)000239_IFA007_7_5_3_Notify_operation - NFVIFA(16)000248r2_IFA007_7_2_8_Change_VNF_Flavour - NFVIFA(16)000258r1_IFA007_5_2__5_3_3_quota_management_ requirements_in_indirect_ NFVIFA(16)000259r2_IFA007_6_x__8_x_quota_management_interfaces_in_indirect_mode NFVIFA(16)000262r1_IFA007_5_2__5_3_virtualised_resources_quota_available_notifi - NFVIFA(16)000265r4_IFA007_8_3_2__8_5_5_Adding_ResourceInfo - NFVIFA(16)000268r2_IFA007_6_2_2_and_8_2_x_IFA013_7_7_5_and_8_7_x_ accessing_VNF_

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		- NFVIFA(16)000269r1_IFA007_5_3_3_Fixing_Virtualised_Resources_ Management_interfa (also applied the pattern to the newly added indirect RM interface requirements) - NFVIFA(16)000276r1_IFA007_7_6_2_7_2_X_Clarification_on_ ModifyVnfConfiguration_a - NFVIFA(16)000277r1_IFA007_6_3_2_4_Clarification_of_rejection_ in_granting_operat - NFVIFA(16)000279_IFA007_6_3_2_LC_operation_occurrence_identifier_ in_grantin g - NFVIFA(16)000280r1_IFA007_6_4_and_8_4_Interface_spec_of_ reservation_mgmt_in_ind - NFVIFA(16)000285_IFA007_8_3_8_adding_resourceProviderId_ in_ConstraintResource - NFVIFA(16)000287r1_IFA007_Scaling_description - NFVIFA(16)000288r2_IFA007_6_4_2-3_Y_8_4_A-C__8_K_L_Virtualised_Resources_Inform - NFVIFA(16)000299r1_IFA007_8_3_4_Addressing_editor_note_in_VimInfo - NFVIFA(16)000300_IFA007_8_3_5_Addressing_editor_note_in_ZoneInfo - NFVIFA(16)000301r1_IFA007_8_5_5_IFA_011_7_1_X_Attributes_for_VnfInfo_and_VNFD (this document has inserted flavourId as well as doc 235. IFA agreed on 24 March by email to remove the duplicate variant of flavourId that was introduced by 301) - NFVIFA(16)000314r2_IFA007_5__6_and_8_Adding_VR_reservation_change_ notification_ - NFVIFA(16)000327r1_IFA007_Resolve_Editor_s_Notes_NFV002_reference - NFVIFA(16)000328r1_IFA007_Resolve_Editor_s_Notes__Functional_requirements_refer - NFVIFA(16)000329r1_IFA007_Resolve_Editor_s_Notes__Granting_ in_ScaleVnf_Descript - NFVIFA(16)000330_IFA007_Resolve_Editor_s_Notes__Indirect_ RM_IE_clause_introdu - NFVIFA(16)000331_IFA007_Resolve_quote_easy_quote_Editor_s_Notes - NFVIFA(16)000342_IFA007_8_7_VNF_PM_mirror Editorial fixes: - there were still occurrences of "input/output information element" in the GS where "input/output parameter" needs to be used. Fix as editorial - Table 7.2.4.2-1: Subscribe operation input parameters --> TerminateVnf operation input parameters - various typos - change "section" to "clause"- in FM/PM interfaces, there were still a quite few table references for input and output parameters that used the old formulation ("are listed") instead of the latest convention "shall follow the indications". Fixed. - made ToC of depth 3 instead of 4 - converted those additional Editor's Notes that were inserted during GS preparation by the rapporteur into "Rapporteur's notes". A Rapporteur's note has not been agreed by the group but represents the opinion of/tracks an action for/points out an issue detected by the rapporteur during GS preparation
20 April 2016	V0.8.0	Contributions included (review EA part 1): - NFVIFA(16)000373_IFA007_6_2_4_Add_missed_text_for_new_VNF_ package_on-boarded - NFVIFA(16)000421r2_IFA007_7_2_10_IFA008_7_2_10_review_Modify_ Vnf_fixes - NFVIFA(16)000423_IFA007_8_2_7_2_IFA013_8_6_5_2_review_ UserMetadata_mandatory - NFVIFA(16)000424r1_IFA007_6_and_7_and_8_Remove_stage3_term - NFVIFA(16)000425r2_IFA007__many__IFA008__many__review_Small_ Technical_Alignment - NFVIFA(16)000431_IFA007_7_5_3_Editorial_change_for_ AlarmClearedNotification - NFVIFA(16)000443r1_IFA007_6_3_2_2_review_Temp_Resource_in_ Notifications_delete - NFVIFA(16)000471r1_IFA007_4_1_Alignment_listing_of_interfaces - NFVIFA(16)000473_IFA007_8_2_5_VNF_Package_mgmt__correction_on_VnfPackageChange - NFVIFA(16)000481r1_IFA007_8_5_6_and_8_5_7_VNF_LCM_updates_to_VnfInfo_and_VnfRes - NFVIFA(16)000482_IFA007_7_4_2_and_7_4_5_VNF_PM_changes

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		- NFVIFA(16)000488r1_IFA007_5_2_edits_interface_naming_in_requirements_ and_titles - NFVIFA(16)000501r1_IFA007_6_2_2_Query_VNF_package_operation - NFVIFA(16)000503r1_IFA007_6_3_2_1_Policy_in_VNF_Lifecycle_Operation _Granting_in - NFVIFA(16)000506r2_IFA007_7_2_9_1_operate_VNF_operation - NFVIFA(16)000515r1_IFA007_7_2_3_Clarifications_on_Scale_VNF_operation - NFVIFA(16)000517r1_IFA007_8_3_2_and_8_3_3_Updates_to_IEs_related_ to_Granting - NFVIFA(16)000519_IFA007_8_7_2_and_8_7_3_Updates_IEs_related_to_ ObjectSelectio - NFVIFA(16)000521_IFA007_7_7_VNF_Indicator_interface_description_ alignment - NFVIFA(16)000529_IFA007_5_3_4_Correcting_req_on_query_VNF_operation - NFVIFA(16)000558r1_IFA007_7_2_3_2_7_2_4_2_7_2_6_2_8_5_6_2_ IFA008_7_2_7_2_7_2_8_
		Contributions included (review EA part 2): - NFVIFA(16)000422_IFA007_7_2_11_IFA008__sect__review_ GetOperationStatus_mandat - NFVIFA(16)000476r1_IFA007_7_2_3_and_6_3_2_moving_text_about_granting - NFVIFA(16)000357r2_IFA007_Scaling_description_delta_after_Espoo - NFVIFA(16)000444r2_IFA007__many__IFA008_9_4_2_review_Removing_ Editor_s_Notes - NFVIFA(16)000418r2_IFA007_section_7_2_3_2_IFA008_section_7_2_7_2_-_Fixing_aspec - NFVIFA(16)000408r1_IFA007_Renaming_VL_and_CP_IEs Contributions included (review EA part 3): - NFVIFA(16)000441r1_IFA007_IFA008_Remove_the_definition_of_ KeyValuePair - NFVIFA(16)000478r3_IFA007_8_6_2_VNF_LC_Change_Notification_ addressing_EN_and_co - NFVIFA(16)000495_IFA007_6_4_5_VR_PM_indirect_add_missing_ resourceProviderId - NFVIFA(16)000496r1_IFA007_5_3_5_lifecycle_change_notification_ interface_require - NFVIFA(16)000502r1_IFA007_5_3_2_6_3_2_2_VNF_instance_id_for_ granting_interface - NFVIFA(16)000523r1_IFA007_5_3_4_clarification_on_VNF_instance_ information_modif - NFVIFA(16)000546r2_IFA007_5_3_5__7_3_3__8_6_1__8_6_X_Add_ new_type_notification - NFVIFA(16)000549r1_IFA007_8_2_7_8_7_7_8_8_3_ IFA008_9_3_4_9_7_7_Use_of_time Contributions included (ATL meeting): - NFVIFA(16)000398_IFA007_8_4_2_8_4_4_8_4_6_Alignment_to_inheritance _pattern - NFVIFA(16)000419_IFA007_6_3_2_1_review_Resource_types_in_Granting - NFVIFA(16)000420r2_IFA007_7_2_3_IFA008_7_2_7_IFA011_7_1_5_3_ review_Scale_up_dow - NFVIFA(16)000445r1_IFA007_8_5_6_IFA008_9_4_2_review_VnfInfo_fixes - NFVIFA(16)000450r3_IFA007_8_5_7_IFA008_9_4_4_VnfResourceInfo_IE - NFVIFA(16)000465r3_IFA007_8_3__review_VDU_reference_duplicated Note on change in clause 8.3.2.2/resourceTemplate: The insertion of "or modification" was in the wrong place in the sentence, hinting a "modification of new resources" which is nonsense. This was corrected as an editorial action to read "modification of existing resources" - NFVIFA(16)000466r5_IFA007_6_3_2_3_review_computeFlavour_swImage_ assets_multi_VI - NFVIFA(16)000474r1_IFA007_8_3_7_and_6_3_2_2_Granting_IE_ PlacementConstraint_vs - NFVIFA(16)000504r4_IFA007_6_3_2_changes_on_VNF_Lifecycle_ Operation_Granting_int - NFVIFA(16)000513_IFA007_6_3_2_Clarifications_for_Grant_VNF_LC_ operation

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		- NFVIFA(16)000514r3_IFA007_5_2_5_3_3_10_6_4_8_quota_available_ notification_inter - NFVIFA(16)000520_IFA007_5_3_7_VNF_FM_missing_requirements - NFVIFA(16)000522r1_IFA007_7_2_9_Operate_VNF_graceful_and_forceful _stop - NFVIFA(16)000527r1_IFA007_7_2_2_and_7_2_7_VNF_LCM_QueryVNF_ filter_and_correctio Note on change in table 7.2.7.3-1: The note in the description column is not in line with the EDR. The note has been moved to the last row of the table as an editorial action - NFVIFA(16)000533r1_IFA007_8_5_3_Addressing_EN_on_ConnectionPoint - NFVIFA(16)000551r4_IFA007_7_2_3_IFA008_7_2_7_VNF_Scaling_ description - NFVIFA(16)000592_IFA007_8_5_6_IFA008_9_4_1_review_VimInfo_in_VnfInfo - NFVIFA(16)000597_IFA007_8_5_6_review_Remove_Error_from_OperateVnf - NFVIFA(16)000600r6_IFA007_IFA013_Add_support_for_Create_and_ Delete_VNF - NFVIFA(16)000652_IFA007_7_2_11_IFA008_7_2_9_GetOperationStatus _op_specific_st - NFVIFA(16)000676r1_IFA007_7_2_5_ext_VLs_in_ChangeVnfFlavour Contributions included (S1a#36 call): - NFVIFA(16)000667r1_IFA007_IFA008_small_fixes - NFVIFA(16)000721_IFA007_7_2_Adding_LCM_operation_occurrence _identifier Contributions included (S1b#50 call and EA ending 19 May): - NFVIFA(16)000453r2_IFA007_8_8_4_IFA008_9_3_4_Referencing_resources _in_alarm_IE - NFVIFA(16)000461r7_IFA007_8_6_2-5_IFA008_9_5_1-4_VnfLifecycleChangeNotification - NFVIFA(16)000720r2_IFA007_8_5_8_IFA008_9_4_5_Clarification_for_ resource_identif - NFVIFA(16)000484r9_IFA008_7_2_2_9_4_x__ IFA007_7_2_2_7_6_2_8_5_x_8_9_x_Adding_Vi Editorial fixes: - Reference i.3b renamed - "See note" harmonized - Virtualised → Virtualised - "Interface" → interface consistently - Convention enforcement: "parameter" → "attribute" in information element descriptions - "Functional requirement" → "Requirement" (table headings in Interface requirements sections) - Various minor fixes
17 June 2016	V0.9.0	Contributions included (second review EA#1, 9 June): - NFVIFA(16)000719r1_IFA007_8_5_8_Adding_back_the_reservationId - NFVIFA(16)000769r2_IFA008_7_4_2__9_2__9_8__6_2__ IFA007_7_6_2__and_IFA011_7_1_6 - NFVIFA(16)000784__IFA008_5_3_1_3_IFA007_5_3_9_renaming_VNF_ Indicator_interfac - NFVIFA(16)000786r1_IFA007_5_3_4_IFA008_5_2_1_1_Add_missing_ requirements_on_crea - NFVIFA(16)000788_IFA007_IFA008_IFA013_4_3_Removal_of_N_A_condition - NFVIFA(16)000793_IFA007_8_3_2_2nd_review_Removing_Rapp_note - NFVIFA(16)000794r1_IFA007_7_2_1_IFA008_7_2_1_2nd_review_ lcOpOccId_clarification - NFVIFA(16)000809_IFA007_Typo_Correction - NFVIFA(16)000820r1_IFA007_Resolution_of_editor_s_notes - NFVIFA(16)000835r3_IFA007_8_3_3_8_12_4_Adding_Resource_Group _Id_to_Grant_respon - NFVIFA(16)000836r2_IFA007_5_3_5_and_IFA008_5_2_1_2_Missing _req_subscription_for

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		NOTE: In the change tracked version, this was implemented using the same name tag as for 835r3, i.e. r0-835r2. - NFVIFA(16)000837_IFA007_5_3_8_7_6__8_9_and_ IFA008_5_2_1_5__7_4__9_2_on_adding - NFVIFA(16)000838r2_IFA007_8_5_x_IFA008_9_4_x_Add_Info_the_ VL_and_CP_IEs - NFVIFA(16)000839_IFA007_5_3_9__IFA008_5_2_1_4_VNF_Indicator _interface_require - NFVIFA(16)000841r1_IFA007_6_2_2__6_2_5_Query___Fetch_VNF_ Package_operation - NFVIFA(16)000852_IFA007__IFA008_Editorials_and_alignments - NFVIFA(16)000790r5_IFA007_many_IFA008_many_2nd_review_Create_ VNF_terminology_an • Rapporteur's changes when implementing this contribution: Table 8.6.8.3-1 VnfIdentifierDeletionNotification: Used past tense in Description column, instead of future as suggested by the 790r5, as notifications are about past events, not future ones. See also the entry for NFVIFA(16)0001016 - NFVIFA(16)000857r2_IFA007_7_3_2_IFA008_7_5_2_2nd_review_ Subscribe_to_Create_Del - NFVIFA(16)000860_IFA007_8_5_7_IFA008_9_8_4_2nd_review_ ResourceHandle_fix - NFVIFA(16)000862r2_IFA007_7_2_6_8_2_12_IFA008_7_2_13_9_4_12_ 2nd_review_Aligning NFVIFA(16)000864_IFA007_8_5_6_8_5_3_8_12_IFA008_9_4_3_9_4_11_9_4_9_ 2_4_2nd_ex
		- NFVIFA(16)000887r2_IFA007_Implementing_identifier_conventions_from_614r3_in _IFA (implemented under user name r2-864 same as previous contribution) • Rapporteur's changes when implementing this contribution (mostly because the IE name is different where it is declared): – Table 8.5.2.2-1 one occurrence of "Vld" replaced by "VnfVld" ("Vld" IE does not exist) – Table 8.5.7.2-1 extVirtualLink -> extVirtualLinkId not applied since another document has modified this attribute, such that it is not of type "Identifier" any longer - hence Identifier conventions do not apply – Table 8.6.3.2-1 VnfcResourceInformation -> VnfcResourceInfo – Table 8.6.4.2-1 VirtualLinkResourceInformation -> VlResourceInfo – Table 8.6.5.2-1 VirtualStorageResourceInformation -> VirtualStorageResourceInfo Contributions included (after NFVIFA#33, Sophia Antipolis): - NFVIFA(16)0001016_IFA007_many_IFA008_many_Create_VNF_ terminology_and_states_re (Rapporteur's comment: 790r5 has been superseded by 1016 which is in fact r6 of 790. The delta between 1016 and 790r5 is implemented in this revision, as 790r5 was implemented previously, effectively being equivalent to having implemented 1016 directly, instead of 790r5. Also, some instances of "VNF information element" in clauses 7.2.7.1 and 7.2.2.4 were missed to be replaced by 1016; these instances were replaced too) - NFVIFA(16)000791r5_IFA007_many_IFA008_many_2nd_review_Renaming_ Vl_and_Vld_in_IE - NFVIFA(16)000795r3_IFA007_8_5_6_7_2_3_IFA008_9_4_3_7_2_3_2nd_ review_VNF_localiz - NFVIFA(16)000889r1_IFA007_inner_grouping_of_indirect_RM_IEs (Rapporteur's comment: When used in the context of "InformationChangeNotification", replaced a few occurrences of "VirtualisedResourceWithRpChangeNotification" by "InformationWithRpChangeNotification", namely in 6.4.2.3, 6.4.3.3 and 6.4.4.3, assuming this was a copy&paste error in the original contribution, and "Information*WithRp*ChangeNotification" is the correct substitute of "InformationChangeNotification" in indirect RM) - NFVIFA(16)000919r1_IFA007_IFA008_IFA011_IFA012_IFA013_IFA014_ stage_3_data_types (Rapporteur's comment: In clauses 7.3.2.2 and 7.6.3.2, added "Filter" in the an empty content column instead of "not specified" as the parameter name is "filter", following the convention) - NFVIFA(16)000856_614bis_Conventions for Identifiers_UPDATED - NFVIFA(16)000869r1 IFA007 IFA008 IFA011 IFA012 IFA013 IFA014 Proposal for an update of the inheritance pattern convention - NFVIFA(16)000920r1_IFA007_6_3_2nd_review_operation_names_in_granting - NFVIFA(16)000983r2_IFA007_6_3_Ext_VLs_in_Granting

Date	Version	Information about changes
		- NFVIFA(16)000989_IFA007_8_11_2_3_VimInfo_in_VirtualisedResource QuotaAvailable - NFVIFA(16)0001001r3_IFA007_8_5_2_IFA008_9_4_10_Change_to_ Virtual_Lin - NFVIFA(16)0001011r2_IFA007_8_5_5_IFA008_9_4_2_Note_on_ modification_of_VnfInfo Contributions included (after EA ending 14 Jul 2016): - NFVIFA(16)0001041_IFA007_query_filter - NFVIFA(16)000806r11_IFA007_and_IFA013_identification_of_the_VNF_ Package (Rapporteur's comment: The previous changes (r0-841r1 that were applied to the Fetch VNF Package operation were moved to the "Fetch onboarded VNF Package artifacts" operation to which they apply after the change introduced by 806) - NFVIFA(16)0001032r2_IFA007_6_3_VNF_Lifecycle_Operation_Granting_ interface_8_3_6_ - NFVIFA(16)001020r3 IFA013 Abort VnfPackage Deletion - NFVIFA(16)000922r6 Conventions for the use of abbreviations Contributions included (after S1a#43 on Jul 20): - NFVIFA(16)0001066r1_IFA007_IFA008_move_extension_and_ vnfConfigurableProperty_to_VnfInfo - NFVIFA(16)0001063r1_IFA007_IFA008_IFA013_vnfInstanceName_in_ ModifyVnfConfig Editorial fixes: - Minor typos (flavour → flavour, identifier → identifier, etc.) - Renamed extVirtualLinkLink to extVirtualLink - Changed the filename convention to use six digit version string - Replaced "GrantLifecycleOperation" by "GrantVnfLifecycleOperation" in captions in line with the name of the operation and in related message names - Corrected wrong references to IFA006 from clauses 6.4 and 8.4 (indirect resource management) - In the body of clause 8.4.7.4.2, replaced "AlarmNotification" by "AlarmClearedNotification as the whole clause uses "AlarmClearedNotification" elsewhere, so this is assumed a copy&paste error - Using plural in the description of attributes and parameters of 0..N/1..N cardinality - Aligned operation names usage (single words, all uppercase) in the table captions and clause headline - Rapporteur action #1 from 489r1: ensure consistent use of "VNF Package" -> s/VNF package/VNF Package/
29 July 2016	V0.9.1	Contributions included (S1b call with approval power on 25 July 2016): - NFVIFA(16)0001077_IFA007 IFA008 IFA013 IFA015 ExtCP and LinkPort fixes - NFVIFA(16)0001078r2_IFA013_8_3_3__IFA007_8_5__IFA008_9_4_ Alignment_of_VnfInfo - NFVIFA(16)0001088r1_IFA007_IFA008_IFA013_virtualStorage_Alignment _with_IFA011 Contributions included (S1a call with approval power on 27 July 2016): - NFVIFA(16)001094r1-NFV-IFA007v000901-cb Editorials: - Cross-checked references to IFA011 and removed related rapporteur's notes - 4.2 using proper interface name: s/VNF Configuration/VNF Configuration Management/ - Table 6.3.2.2-1: Renamed vnfDescId --> vnfdId in line with the changes done in 806r11

Date	Version	Information about changes
01 August 2016	V 0.9.2	Editorials: - Restructured the sequence of sub-clauses of clause 8.5 to align with IFA008 (i.e. start with VnfInfo and InstantiatedVnfInfo) - Fixed some typos and editorial inconsistencies - Fixed references in clause 7.4.1 Description to be: • PerformanceInformationAvailableNotification (see clause 8.7.8) • PerformanceReport information element (see clause 8.7.5)
03 August 2016	V 0.9.3	Contributions included: - NFVIFA(16)001126r3_IFA007_IFA008_inconsistency_fixes Editorials as documented in NFVIFA(16)0001129: - "Change VNF Deployment Flavor" replaced by the generally-used term "Change VNF Flavour" - Applied convention for notifications - There are some references left to VirtualLinkDesc but in fact the IE is named VnfVirtualLinkDesc in IFA011. Fixed
04 August 2016	V 0.9.4	Contributions included: - NFVIFA(16)000770_Replace_primitive_type_TimeStamp_by_DateTime - Extended the implementation of change 5 in NFVIFA(16)001126r3_IFA007_IFA008_inconsistency_fixes to all places where the text is applicable (Description of input/output parameters of type ExtVirtualLink and ExtManagedVirtualLink starts with "Information about", rather than "Reference to")
19 August 2016	V0.9.4b	Alignment of the Operation Result clauses: result of an operation use past tense and return parameter use passive present tense and avoid future tense (will be). Output parameter mentioned in attribute descriptions are also changed to use "is returned" or "are returned" Other editorial bugs fixed Replaced many occurrences of NVFO with NFVO
05 September 2016	V0.9.5	Including NFVIFA(16)0001215
October 2016	V2.1.1	Publication
23 January 2017	V2.1.2	Specification maintenance begins CRs included: - NFVIFA(17)000010r2_IFA007ed221_Merging_LCCN_with_LCM_interface_ and_adding_subsc Editorial changes: - Changed page header to "Draft GS", added NFV's DRAFT GS disclaimer
03 April 2017	V2.1.3	CRs included: - NFVIFA(17)000056r1_IFA007ed221_IFA013ed221_VNF_Package_Management_ modifications - NFVIFA(17)000062r4_IFA007ed221_IFA008ed221_VimInfo_fixes_without_ VimId_changes - NFVIFA(17)000094r3_IFA007ed211_Update_the_content_and_description_of_the _alarm_ - NFVIFA(17)000103r3_IFA007ed221_ModifyVnfConfig_Split_and_Merge - NFVIFA(17)000116r2_IFA007ed211_Various_small_bugfixes - NFVIFA(17)000155_IFA007ed221_LifecycleChangeNotification_terminology_ - NFVIFA(17)000176r2_IFA007ed221_IFA008ed221_VimId_changes_separated_fr om_62r3 - NFVIFA(17)000193_IFA007ed221_ThresholdCrossedNotification_trigger_conditio n_f - NFVIFA(17)000236r1_IFA007ed221_8_5_3_clarify_description_of_ MonitoringParameter - NFVIFA(17)000257r2_IFA007ed221_IFA008ed221_IFA013ed221_Fix_to_dynami c_addresses
25 May 2017	V2.1.4	CRs included: - NFVIFA(17)000274r2_IFA007ed221_IFA008ed221_VNF_FM_Acknowledge_ Alarm_operation - NFVIFA(17)000275r1_IFA007ed221_IFA008ed221_VNF_FM_Alarm_List_ Rebuilt_operation - NFVIFA(17)000355_IFA007_Fix_inconsistencies_in_the_FaultyResourceInfo_IE - NFVIFA(17)000438_IFA007ed221_removing_two_attributes_from_ SoftwareImageInform - NFVIFA(17)000454r1_IFA007ed221__IFA008ed221__Add_notes_to_the_Delete_ PM_Jobs_op - NFVIFA(17)000455r1_IFA007ed221__Add_notes_to_the_Delete_Thresholds_ operation_fo

Date	Version	Information about changes
		- NFVIFA(17)000458r3_IFA007ed221_Clarify_the_results_of_operations_implicitly_upd - NFVIFA(17)000460_IFA007ed221_IsAutomaticInvocation_flag_for_autoscale_and_aut - NFVIFA(17)000462r1_IFA007ed221_ChangeExtVLs_fixes - NFVIFA(17)000469_IFA007ed221_resource_metadata - NFVIFA(17)000473_IFA007ed221_identifier_changes_related_to_IFA_document_256r1
13 June 2017	V2.1.5	CRs included: - NFVIFA(17)000390r2_IFA007ed221_CR_add_error_handling_operations - NFVIFA(17)000427r2_IFA007ed221_VL_and_CP_consistency - NFVIFA(17)000450r4_IFA007ed221_ChangeExtVLs_support_status - NFVIFA(17)000468r2_IFA007ed221_Notifications_triggered_by_ModifyVnf - NFVIFA(17)000470r1_IFA007ed221_Problem_with_storage_resources_in_AffectedVnfc (rapporteur changed "VnfInstance" to "VnfInfo", as this is a leftover from the original contribution having been targeted towards SOL003) - NFVIFA(17)000471r1_IFA007ed221_additionalParameters_missing_from_TerminateVnfRe - NFVIFA(17)000520r3_IFA007ed221_VimConstraint_for_resourceGroup - NFVIFA(17)000531_IFA007ed221_Improvement_of_attribute_usage_discription (implemented on top of the changes from NFVIFA(17)000427r2, as intended by this CR) - NFVIFA(17)000535r1_IFA007ed221_Add_VimConnectionInfo_input_parameter_to_Change
21 June 2017	V2.1.6	Final draft for approval after NFVIFA#57 CRs included: - NFVIFA(17)000525_IFA007ed221_VimConnectionInfo_inter_stages_consistency - NFVIFA(17)000547r3_IFA007_ed221_CR_Align_the_usage_of_VNF_instantiation_state - NFVIFA(17)000580r1_IFA007ed231_ChangedInfo_fix_of_cardinality - NFVIFA(17)000597_IFA007ed231_Small_fix_leftover_from_renaming_to_vimConnectio Editorial fixes (table formatting, empty table rows removed)
07 November 2017	V 2.3.2	Contributions incorporated that were approved at NFVIFA#73: - NFVIFA(17)000919_IFA007ed241_miscellaneous_small_fixes - NFVIFA(17)000923r1_IFA007ed241_resource_metadata_in_AffectedC_N_S_resources (change by rapporteur when implementing this CR: replaced "VnfVirtualStorageResourceInfo" by "VirtualStorageResourceInfo" in the new text to reflect the correct name of that pre-existing IE) - NFVIFA(17)000970r1_IFA007ed241_VnfcCps_in_AffectedVnfc Editorials: - Added draft disclaimer box - Fixed small typos
21 December 2017	V 2.3.3	Contributions incorporated that were approved at NFVIFA#79: - NFVIFA(17)0001037r4_IFA007ed241_CR_from_IFA_1029 (CR conflict handling by the rapporteur: the change in the description of the "address" attribute in CR 1037r4 was be applied in the new "CpProtocolInfo" IE that was introduced by 1114r1, as the "address" attribute has been moved by CR 1114r1 to this new IE) - NFVIFA(17)0001089r1_IFA007ed241_Align_query_VNF_package_operation_with_IFA013 - NFVIFA(17)0001090r1_IFA007ed241_Align_fetch_VNF_package_operation_with_IFA013 - NFVIFA(17)0001091_IFA007ed241_Align_VNF_package_notification_IE_with_IFA013 - NFVIFA(17)0001101_IFA007ed241_Clarification_of_ExtManagedVirtualLink_and_ExtVi - NFVIFA(17)0001108_IFA007ed241_remove_redundant_description_of_vnfConfigurableP

Date	Version	Information about changes
		- NFVIFA(17)0001111r1_IFA007ed241_LCM_operation_response_and_notification_in_opera (The Rapporteur has fixed the following mis-alignment: The first pattern in change 1 includes the text 'trigger the sending of the "start" LCM notification' whereas the second pattern in change 3 includes the text 'trigger the "start" notification'. As the first pattern is more precise, the rapporteur has aligned the second pattern with the first one when implementing the CR). - NFVIFA(17)0001114r1_IFA007ed241___Corrections_related_to_multiple_layer_protocol - NFVIFA(17)0001120r1_IFA007ed241_fixing_VNF_connectivity_figure
February 2018	V 2.4.1	Publication
01 March 2018	V 2.4.2	Contributions incorporated that were approved at NFVIFA#89: - NFVIFA(18)000143_IFA007ed251_align_filters_in_FM_interface_Subscribe_operation
May 2018	V 3.0.0	Release 3 baseline version created from draft v2.4.2, as agreed in NFVIFA#98
June 2018	V 3.0.1	Contributions incorporated that were approved at NFVIFA#101 and NFVIFA#102: - NFVIFA(18)000421r1_IFA007_MegaCR_FEAT04_Compute_Host_Reservation - NFVIFA(18)000507r1_IFA007ed311_-_Mirror_-_Linking_VNFC_CP_and_VnfExt CP - NFVIFA(18)000508r1_IFA007ed311_-_Mirror_-_Clarifying_association_from_VnfLinkPort_to_VnfcCp_and_VnfExtCp
June 2018	V 3.0.2	Contributions incorporated that were approved at NFVIFA#104 and NFVIFA#105: - NFVIFA(18)000430r1_FEAT07_IFA007_MegaCR_Support of policy_management_interface - NFVIFA(18)000561r3_IFA007_MegaCR_FEAT15_VNF_Snapshot - NFVIFA(18)000564_IFA007ed311_Remove_current_values_of monitoringParameter_attribute_from_VNF_LCM interface - NFVIFA)18)000612r1_IFA007ed311_–_Rel3Mirror_-_Fixing_sentence_related_to_PM_delivery_mechanism - NFVIFA(18)000619_IFA007ed311_–_Rel3Mirror_-_Fixing_note_in_VnfLinkPortInfo - NFVIFA(18)000628_IFA007ed311_–_Rel3Mirror_-_different_names_for_virtual_link_descriptor_ids - NFVIFA(18)000637_IFA007ed311_Rel3Mirror_of_435r2_metadata_for_CP_IEs - NFVIFA(18)000651_IFA007ed311_Rel3Mirror_of 477_Fixing_cardinality_of_ConstraintResourceRef
September 2018	V 3.1.2	Contribution incorporated that was approved at NFVIFA#118: - NFVIFA(18)000806_IFA007_Support_for_partial_VNF_Snapshot_Packages - NFVIFA(18)000834_IFA007ed321_Mirror_for_SOL_contribution_on_making_the_API_su rface_consistent_for_bootData Editorial fixes
November 2018	V 3.1.3	Contributions incorporated that were approved at NFVIFA#124 and NFVIFA#125: - NFVIFA(18)000859r2_IFA007ed321_Metadata_Extension_ConfigurableProps_clarification for VnfInfo - NFVIFA(18)000887r2_IFA007ed321_update_of_IEs_related_to_PM_interface_for_IFA027_alignment - NFVIFA(18)000922_IFA007_Clause_7_2_Corrections_additional_params_for_create_and revert snapshot operations - NFVIFA(18)000923r1_IFA007_Clause_8_5_Correction_cardinality_VnfcSnapshotInfo_and description VnfSnapshotInfo
January 2019	V 3.1.4	Contributions incorporated that were approved at NFVIFA#128 and NFVIFA#129: - NFVIFA(18)000983r4_IFA007ed321_Add_best_effort_in_PlacementContraint - NFVIFA(18)0001002_IFA007ed321_CR_add_policy_associate_disassociate_operations - NFVIFA(18)0001032_IFA007_VNF_snapshot_createdAt_and_userDefinedData - NFVIFA(18)0001067r1_IFA007ed321_Fix_for_condition_in_VnfLcmOperation OccurrenceNotification - NFVIFA(18)0001068r1_IFA007ed321_declaration_of_metadata_and_extensions_ - NFVIFA(18)0001070r1_IFA007ed321_Aligning_conditions_for_vduId_and_resourceTemplateId_in_Granting_interface
February 2019	V 3.1.5	Contributions incorporated that were approved at NFVIFA#137: - NFVIFA(18)0001112r4_IFA007_MegaCR_FEAT010_Or-Vnfm_ref_point_interface_specification_for_Multi-Site_Service - NFVIFA(19)000059r4_FEAT02_IFA007_MegaCR

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February 2019	V 3.1.6	Contribution incorporated that was approved at NFVIFA#139: - NFVIFA(19)000142r2_FEAT02_IFA007_Review_add_missing_parameters_to_ChangeCurrent
April 2019	V3.2.1	Version update for publication
May 2019	V3.2.2	Base line version for Release 3 Drop 3 created from published version 3.2.1
June 2019	V3.2.3	Contributions incorporated that were approved at NFVIFA#155: - NFVIFA(19)000506_IFA007_Alignment_with_Stage_3_work_on_VNF_snapshot_feature - NFVIFA(19)000532r1_IFA007_8.5.17_Correction_of_NOTE
July 2019	V3.2.4	Update with CRs: - NFVIFA(19)000653: IFA007ed331 Rel3Mirror 8.10.3 IndicatorInformation IE - NFVIFA(19)000649: IFA007ed331 Rel3Mirror 8.6.2 VnfLcmOperationOccurrenceNotification IE - NFVIFA(19)000643: IFA007ed331 Rel3Mirror VnfPkgInfo, VnfPackageOnboardingNotification, VnfPackageSoftwareImageInformation IEs - NFVIFA(19)000640: IFA007ed331 PerformanceValueEntry IE - NFVIFA(19)000634: IFA007ed331 Rel3Mirror VnfcResourceInfo, VnfVirtualLinkResourceInfo, VirtualStorageResourceInfo IE - NFVIFA(19)000621: IFA007ed331 Rel3Mirror AffectedVnfc, AffectedVirtualLink, AffectedVirtualStorage - NFVIFA(19)000619: IFA007ed331 Rel3Mirror 8.8.4 Alarm IE - rephrase "legal values" - NFVIFA(19)000614r1: IFA007ed331_Rel3_mirror_Initial_configurable_properties_values
September 2019	V3.3.1	Version update for publication
October 2019	V3.3.2	First draft for ed341
October 2019	V3.3.3	Update with CRs: - NFVIFA(19)000760: IFA007ed341 measurementContext in ThresholdCrossedNotification - NFVIFA(19)000818: IFA007ed341 modifying VNF package references - NFVIFA(19)000841: IFA007ed341 relaxing PM subscriptions - NFVIFA(19)000825r1: IFA007ed341 exposing maxScaleLevels Rapporteur action: Removing Annex on Authors & Contributors
December 2019	V3.3.4	Update with CRs: - NFVIFA(19)000923r5: IFA007ed341 Dynamic creation and deletion of trunk subports - NFVIFA(19)000950: IFA007ed341 5.3.11 add requirement for Update VNF snapshot package - NFVIFA(19)000875r5: IFA007ed341 Enhancements in ChangeExtVnfConnectivity
February 2020	V3.3.5	Update with CRs: - NFVIFA(19)000993: IFA007 Improve wording left for protocol design stage - NFVIFA(19)0001004: IFA007ed341 adding vnfConfProps to ChangeCurrentVnfPackage - NFVIFA(20)000012: IFA007ed341 adding missing extensions and vnfConfigrableProperties to ChangeVnfFlavour - NFVIFA(20)000049r1: IFA007ed341 add missing support statements - NFVIFA(20)000069r1: IFA007ed341 sync to IFA015 work according to 942r2 part1 - NFVIFA(20)000090: IFA007ed341 FEAT15 maintenance Enhancing granting related to VNF snapshot Rapporteur action: correcting references of "supportedOperations" to "supportedOperation"
March 2020	V3.3.6	Update with CRs: - NFVIFA(20)000140r1: IFA007ed341 sync to IFA015 work according to 942r18 - NFVIFA(20)000169r2: IFA007ed341 fix Enum values - NFVIFA(20)000172r1: IFA007ed341 FEAT15 alignment with stage 3 - NFVIFA(20)000192: IFA007ed341 FEAT15 alignment with stage 3 - part 2

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May 2020	V3.3.7	Update with CRs: - NFVIFA(20)000204: IFA007ed341 FEAT15 alignment with stage 3 - part 3 - NFVIFA(20)000232: IFA007ed341 FEAT15 Moving VNF snapshot package API - NFVIFA(20)000223r2: IFA007ed341 FEAT02 Indicator changes triggered by changeCurrentVnfPkg - NFVIFA(20)000287: IFA007ed341 fixing description of GetIndicatorValue Rapporteur action: correcting description of attributes that has Reference NFVIFA(20)000318r3: IFA007ed341 FEAT15 VNF state snapshot data NFVIFA(20)000326: IFA007ed341 Mirror of NFVIFA(20)000257 alignment issue
June 2020	V4.0.1	Release 4 baseline version created from published version v3.4.1
October 2020	V4.0.2	Update with CRs: - NFVIFA(20)000502: IFA007ed411 mirror of 424r1 Adding Trunk Logical Topology between VNFC CPs - NFVIFA(20)000526: IFA007ed411 mirror of 440 Modifications to VnfLcmOperationOccurrenceNotification - NFVIFA(20)000529: IFA007ed411 mirror of 384 Modifications to operationStatus parameter content type - NFVIFA(20)000068r1: IFA007ed341 sync to IFA015 work according to NFVIFA(19)000882
November 2020	V4.0.3	Update with CRs: - NFVIFA(20)000682: IFA007ed411 Rel-4 mirror of 598 VNF external connectivity use cases - NFVIFA(20)000683: IFA007ed421 Rel-4 mirror VIPs and external connectivity related updates
November 2020	V4.0.4	Update with CRs: - NFVIFA(20)000673: IFA007ed411 Rel4 mirror of 664r1 VipCp related changes in granting LCCN and linkport referencing - NFVIFA(20)000765: IFA007ed421 FEAT15 Mirror of 639 Move VnfStateSnapshotInfo from VNF snapshot package mgmt. to VNF LCM - NFVIFA(20)000776: IFA007ed421 (forward mirror of 773) Aligning with SOL302 fixing notifying information about extLinkPort - NFVIFA(20)000803r1: IFA007ed421 Rel-4 Mirror of 783 VnfExtCpInfo update
January 2021	V4.0.5	Update with CRs: - NFVIFA(20)000796: IFA007ed421 (forward mirror) FEAT10 Correction about multi-VIM support - NFVIFA(20)000925: IFA007ed411 Re4 mirror of 858 VipCp related small fix
February 2021	V4.0.6	Update with CR: - NFVIFA(20)000843r3: ENH02.05 IFA007ed421 Introduction of scaleInfo complementing instantionLevelId
March 2021	V4.0.7	Update with CRs: - NFVIFA(21)000233r1: IFA007ed421 Mirror of 218 Avoid Reference to MAN001 - NFVIFA(21)000196r5: IFA007ed421 MegaCR FEAT17 Cloud-native VNFs - NFVIFA(21)000194: IFA007ed421 Rel4 mirror of 193 refer to standardized coordination action - NFVIFA(20)000671: IFA007ed411 Rel4 mirror of 626 clarification of passing extVLs in granting
May 2021	V4.2.1	Version update for publication
July 2021	V4.2.2	First draft for ed431 created from published version v4.2.1
September 2021	V4.2.3	Update with CRs: - NFVIFA(21) 000683: IFA007Ed431 Link ports for container cluster networks - NFVIFA(21)000689: IFA007ed431 Rel4 mirror of 688 Use of old assets after ChgCurrentVnfPkg - NFVIFA(21)000714r1: IFA007ed431 FEAT17 corrections to align with SOL367r2
November 2021	V4.2.4	Update with CRs:

Date	Version	Information about changes
		- NFVIFA(21)000964: IFA007ed431 Rel.4 mirror of 943r1 fixing ambiguous note in Grant - NFVIFA(21)000916: IFA007ed431 FEAT17 Runtime modelling of VirtualCp - NFVIFA(21)000904: IFA007ed431 Rel.4 mirror of 903r1 vnfdId in resource info elements - NFVIFA(21)000865: IFA007ed431 Rel.4 mirror of 864r4 Conflicting affinity antiaffinity rules - NFVIFA(21)000850r1: IFA007ed431 Add vnfdExtInvariantId in VnfPkgInfo - NFVIFA(21)000794: IFA007ed431 Rel.4 mirror of 793r3 Granting issue
December 2021	V4.2.5	Update with CR: - NFVIFA(21)0001060: IFA007ed431 Rel.4 mirror of AffectedExtLinkPort bugfix
March 2022	V4.2.6	Update with CRs: - NFVIFA(22)000190r1: IFA007ed431 MegaCR FEAT17 Cloud-native VNFs - NFVIFA(22)000146: IFA007ed431 corrections in Clause 6.4.5 (Rel-4 mirror of NFVIFA(22)000145) - NFVIFA(22)000054: IFA007ed431 FEAT03 Mirror of 053 statements fix related to NFVI operation and maintenance (feedback from NFVSOL(21)000579r1) - NFVIFA(21)0001104: FEAT17 IFA007ed431 AffectedVirtualCp fix
April 2022	V4.2.7	Update with CR: - NFVIFA(22)000257: FEAT17 IFA007 McioInfo alignment with stage 3
June 2022	V4.3.1	Version update for publication
July 2022	V4.3.2	First draft for ed441 created from published version v4.3.1
November 2022	V4.3.3	Update with CR: - NFVIFA(22)000676r1: IFA007ed441 VimConnectionInfo alignments
December 2022	V4.3.4	Update with CRs: - NFVIFA(22)000894: IFA007ed441 Additional VimConnectionInfo alignments - NFVIFA(22)000901: IFA007ed441_correct_netAttDefResourceNamespace
January 2023	V4.3.5	Update with CRs: - NFVIFA(22)000970: IFA007ed441_8_6_3_computeResource_modification - NFVIFA(22)000971: IFA007ed441_8_6_5_storageResource_modification - NFVIFA(22)000972r1: IFA007ed441_8_6_4_networkResource_modification - NFVIFA(23)000025r1: mirror_of_024__IFA007ed441_Policy_management_alignment_with IFA048
March 2023	V4.4.1	Version update for publication
April 2023	V4.4.2	First draft for ed451 created from published version v4.4.1
May 2023	V4.4.3	Update with CR: - NFVIFA(23)000309r1: IFA007ed441 Support for DaemonSets
July 2023	V4.4.4	Updated with CRs: - NFVIFA(22)000863r3: FEAT17 IFA007ed441 Simplifications of CP configuration for containerized VNFs - NFVIFA(23)000333r2: IFA007ed451 Handling of the EN related to the use a pod manifest for a scalable VDU - NFVIFA(23)000347r3: IFA007ed451 FEAT21 MegaCR for PaaS Services aspects - NFVIFA(23)000520: FEAT31 IFA007ed451 Mega CR for deployable modules support - NFVIFA(23)000472r5: IFA007ed451 Enh01.01 MegaCR - NFVIFA(23)000526: IFA007ed451 FEAT21 MegaCR for other enhancements
July 2023	V4.4.5	Rapporteur correction (editorial): Correction of duplicate Note number from Note 9 to Note10 in clause 6.3.2.2 and updating the same note reference in relevant attributes of clause 6.3.2.2
July 2023	V4.4.6	Update with CR: - NFVIFA(23)000602: FEAT31 IFA007ed451, Correcting the placement of the text in table 8.5.3.2-1 in attribute scalestatus, captured wrongly in Mega CR NFVIFA(23)000520
February 2024	V4.5.2	Update with CR NFVIFA(23)000682r1, NFVIFA(23)000686r1, NFVIFA(23)000701 on basis of published version 4.5.1.
March 2024	V5.0.1	Update with CRs: - NFVIFA(23)000841_FEAT31 IFA007ed511 Mega CR for dynamic capacity support - NFVIFA(24)000126_IFA007ed511_FEAT35_MegaCR_for_interface_enhancements - NFVIFA(24)000215_FEAT35_IFA007ed511_MegaCR_additional changes for Tacker.

Date	Version	Information about changes
		- For NFVIFA(24)000181r1 (part of NFVIFA(24)000215 MegaCR) used the correct clause number for the IndicatorInformation IE (8.10.3 instead of 9.6.3). - Also implemented NFVIFA(23)000008r1 (not implemented in ed441) and fixed cp dates for NFVIFA(23)000309r1 and NFVIFA(23)000333r2. Following CRs needed not be implemented since the first Release 5 draft took version v4.5.2 (as basis of published v4.6.1), which had already implemented the corresponding mirrored Release 4 CRs - NFVIFA(23)000838 (NFVIFA(23)000701 was already implemented). - NFVIFA(24)000028 (NFVIFA(23)000686r1 was already implemented). - NFVIFA(24)000074 (NFVIFA(23)000682r1 was already implemented).
April 2024	V5.0.2	Update with CR NFVIFA(24)000247_IFA007ed511 Multiple clauses Editorial and technical alignments
June 2024	V5.1.1	Version update for publication
June 2024	V5.1.2	Initial ed521 draft version created from published version 5.1.1.
September 2024	V5.1.3	Update with CRs: - NFVIFA(24)000523r1 - FEAT19a IFA007 Adding container networking recommendations - NFVIFA(24)000526 - IFA007ed521 Correction output of change external VNF connectivity operation - NFVIFA(24)000545 FEAT21 IFA007ed521 MegaCR - NFVIFA(24)000553 FEAT30 IFA007ed521 MegaCR
October 2024	V5.1.4	Update with CRs: - NFVIFA(24)000592 - IFA007ed521 Review contribution - adjusting cardinality for multiple certificates
April 2025	V5.2.2	Initial draft version for ed531 created from published version v5.2.1
April 2025	V5.2.3	Update with CRs: • NFVIFA(25)000049 - FEAT21 IFA007ed531 Add support for modifying security groups • NFVIFA(25)000081 - FEAT29 IFA007ed531 MegaCR

History

Document history		
V5.1.1	June 2024	Publication
V5.2.1	December 2024	Publication
V5.3.1	August 2025	Publication